



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

52-00-00-001

DOORS, GENERAL

Introduction

The doors give access to the aircraft for the crew, passengers and maintenance and service personnel. Emergency exits are also supplied for emergency escape. An internal door separates the flight and passenger compartments.

General Description

Refer to Figures 1 and 2.

The aircraft has two passenger and two baggage doors. A Type II emergency exit and a flight compartment escape hatch is supplied for emergency use. Service doors are supplied for ground service.

Refer to Figure 1.

The forward passenger door is on the front left side of the passenger compartment. It includes air stairs and a seal inflated by the airframe deice system. The aft passenger door is on the rear left side of the passenger compartment. This door has a passive type seal inflated by cabin pressure. As an optional configuration the aft passenger door is provided with an aft entry folding stairway.

Refer to Figure 3.

Emergency exit from the flight compartment is through a flight compartment escape hatch in the roof or the flight compartment door. Both passenger doors and the aft service door are used as

Type I emergency exit doors. One Type II emergency exit is located on the forward right side of the passenger compartment. The Type II exit has a removable upper door and a hinged lower door.

Aft baggage compartment access is through an outward opening door on the left rear fuselage. This door has a seal inflated by the airframe deice system. Forward baggage compartment external access is through a door on the right front fuselage. This door has a passive type seal.

The aft service door is located on the right rear of the passenger compartment. The aft fuselage access door is installed on the bottom of the aft fuselage.

Flight compartment access is through a lockable door in the flight compartment bulkhead.

Refer to Figure 4.

The door warning system uses the Proximity Sensor Electronic Unit (PSEU) to monitor all pressurized fuselage doors. Door status is shown on the:

- Electronic Flight Information System (EFIS)
- Caution and Warning Control Panel

The Doors System includes:

- Passenger Doors (52-10-00)
- Emergency Exits (52-20-00)
- Baggage Doors (52-30-00)
- Service Doors (52-40-00)
- Interior Doors (52-50-00)
- Door Warning System (52-70-00)

PSM 1-84-2A

EFFECTIVITY:

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Detailed Description

The forward and aft passenger doors are located on the left side of the fuselage and are also used as emergency exits. Both passenger doors can be opened from either the inside or outside using internal or external handles. The forward passenger door is used as the main access to the passenger compartment. As an optional configuration the aft passenger door is provided with an aft entry folding stairway. The aft entry folding stairway is used as the second entry door stairs for the entry and exit of passengers.

A flight compartment escape hatch is located in the flight compartment roof, aft of the pilot seats. It is completely detachable, or can be partially opened to increase the airflow in the flight compartment when the aircraft is on the ground.

There is a Type II emergency exit door located on the forward right side of the passenger compartment. The emergency exit has an upper and lower door. The upper door is a plug type door which is removed completely for opening. The lower door is hinged at the bottom, and opens outward and downward from the floor level.

All the emergency exits in the passenger compartment have both internal and external handles. The flight compartment escape hatch is the only emergency exit which does not have an external handle.

On aircraft with ModSum 4-459339 incorporated, the forward baggage door is reworked and converted into a Type I emergency exit door.

The aft baggage compartment is accessible through an outward and upward opening baggage door in the left side rear fuselage. The forward baggage compartment is accessible externally through a door in the right front fuselage. The door opens inward and upward,

and then translates outward and forward. Both baggage doors can be opened or closed manually by an external handle.

Internal access to the forward baggage compartment is through a hinged door installed in the passenger compartment divider. The door has a key operated lock. There is no internal access to the aft baggage compartment.

The service door system includes an aft service door, an aft fuselage access door an APU clamshell doors. It also includes several ground service and nose compartment doors.

The four nose compartment doors are located on the nose. The aft service door is located on the right side of the aircraft at the rear of the passenger compartment. The Aft Fuselage Access Door is located flush with the underside of the rear fuselage, through which the unpressurized equipment compartment is accessed. The APU access doors are located on the bottom of the tailcone section on either side of the fuselage centerline. The doors are hinged horizontally and kept in a locked and closed position by the two pin latches, hook latches and keepers, and the three shear pins. The Refuel/Defuel Panel door is located on the lower side of the No. 2 engine nacelle, aft of the right main landing gear rear doors. The dc electrical ground power door gives access to the 28 Vdc ground power receptacle on the left side of the aircraft nose. The ac electrical ground power door gives access to the ac ground power receptacle. The hydraulic ground power doors, on the right side of both engine nacelles, give access to the hydraulic ground power connections. The ground air conditioning door is located flush with the lower right side of the rear fuselage. The lavatory service door is located flush with the lower right side of the forward fuselage.

A flight compartment door separates the passenger compartment from the flight compartment. The door is installed to the bulkhead separating the passenger and flight compartments. The door can be

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locked to prevent passenger entry. It is hinged on the left side and swings aft to open.

Forward Passenger Door

[Refer to Figure 5.](#)

The forward passenger door is located at the left side of the fuselage at the front of the passenger compartment.

The forward passenger door is hinged at the bottom and has five steps built into the inboard side of the door structure. The steps also strengthen the door. This door has an external aluminum alloy skin braced by cross beams and intercostals. Door pressurization loads are carried by ten adjustable pressure stop bolts, five on each side. The stop bolts engage with stop brackets on the door surround structure. The door pressure seal inflates to prevent air leakage.

The forward passenger door is the semi-plug type. On opening, the initial movement of the door is upward and inward. The door then rotates outward about the door's hinge.

Aft Passenger Door

The aft passenger door is a translating type located at the left side of the fuselage at the rear of the passenger compartment.

[Refer to Figure 6.](#)

This door rotates outwards about the two hinges (main and secondary). A stabilizer rod prevents the yawing movement of the door during opening and closing. Door pressurization loads are carried by 12 stops that engage with their related stop brackets on the door surround structure. In unpressurized flight, the door is prevented from sliding upward by the cam rollers on each side of the

lift/latch shaft. If one cam roller assembly fails, the remaining cam rollers prevent the door from sliding. The lift/latch shaft is secured in the closed position by both the handle lock mechanism and over-center springs.

The aft passenger door has the components that follow:

- Stabilizer bar
- Main Hinge
- Secondary Hinge
- Vent door.

Operation of either the internal or external spring loaded handle unlocks the door. The door opens or closes with the door face parallel with the fuselage.

As an optional configuration the aft passenger door is provided with an aft entry folding stairway. The aft entry folding stairway is used as the second entry door stairs for the entry and exit of passengers.

Flight Compartment Escape Hatch

[Refer to Figure 7.](#)

The flight compartment escape hatch is opened by turning the handle counterclockwise to the open position. The hatch is opened to permit depressurization or to increase airflow in the flight compartment when the aircraft is on the ground.

The door movement is controlled by the geometry of the torque tube and rollers. The hatch pivots about the rear fittings and opens approximately one inch at the front.



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A pull force on the handle of approximately 40 lb (18.1 kg), releases the rollers against the action of the locking and release springs, and the hatch can be removed from the flight compartment roof.

Type II Exit Door

[Refer to Figure 8.](#)

The upper emergency exit door is secured by a lock pin and three adjustable stops on each vertical edge that engage with three stop brackets installed on the sides of the door to secure it against cabin pressure loads. The lower emergency exit door is installed on a piano hinge at the lower edge. It is kept in the closed position by four stops on the upper edge of the door that engage with their related stop brackets on the lower edge of the upper door. The lower door opens automatically when the upper door is removed, and the lower door handle is in the GROUND position. The lower door stays closed when the lower door handle is in the WATER position.

Aft Baggage Door

[Refer to Figure 9.](#)

The aft baggage door opens outward and upward about an upper hinge, and is held open by a telescopic strut. An external tee handle is used to lock and unlock the door from the outside. There is no internal handle for the baggage door. The door pressure seal inflates to prevent air leakage. The aft baggage door has an external, aluminum alloy skin braced by cross beams. The door warning system monitors the door status and supplies a visual indication and warning in the flight compartment.

Forward Baggage Door

[Refer to Figure 10.](#)

On aircraft with ModSum 4–459339 incorporated, the forward baggage door is reworked and converted into Type I emergency exit door.

The forward baggage door is unlocked by the operation of an external tee handle. The door opens or closes with the door face parallel with the fuselage skin. This is done by two hinges (main and secondary) and a stabilizer.

Proximity sensors monitor the position of the upper fixed rollers and the door handle. A red warning light comes on in the flight compartment when the door is open or unlocked. The door warning system monitors door status and supplies visual indication and warning in the flight compartment.

Aft Service Door

[Refer to Figure 11.](#)

The aft service door is located on the right side of the aircraft at the rear of the passenger compartment. The aft service door is classified as a Type I emergency exit. It is a translating type secured against pressurization loads by five stops on each vertical edge, that engage with five related stop brackets installed on the door surround.

Forward Type I Emergency Exit Door

[Refer to Figure 12.](#)

On aircraft with ModSum 4–459339 incorporated, the forward baggage door is reworked and converted into a Type I emergency exit door.



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The forward emergency exit Type I door gives access to the forward passenger compartment. The forward emergency exit Type I door is located on the right side of the forward fuselage. It is a translating type door and is secured against pressurization loads by five stops on each vertical edge that engage with five related stop brackets installed on the door surround structure.

The proximity sensor monitors the position of the upper fixed rollers and the door handle. A red warning light comes on in the flight compartment when the door is open and unlocked.

Door Seal Pressurization System

Refer to Figure 13.

The door seal pressurization system prevents cabin air leakage through the forward passenger and aft baggage doors. Pressurized air is supplied from the airframe deicing system, and from the system reservoir tank when the engines or APU are not operating.

The door seal pressurization system has the components that follow:

- Heated check valve
- Reservoir tank
- Control valves
- Door seals
- Drain valve
- Charging valve
- Electro pneumatic shut-off valve
- Plumbing lines.

Door Warning System

The door warning system sends door sensor outputs to the Proximity Sensor Electronic Unit (PSEU). The PSEU outputs data to the Input Output Processors (IOP 1 and IOP 2). This data is displayed in the flight compartment and is used by the Cabin Pressure Control System (CPCS).

The door warning system monitors the position of the fuselage doors that follow:

- Forward Passenger Door (52–11–00)
- Aft Passenger Door (52–12–00)
- Type II Emergency Exit Door (52–22–00)
- Forward Baggage Door (52–32–00)
- Aft Baggage Door (52–31–00)
- Aft Service Door (52–41–00)

On aircraft with ModSum 4–459339 incorporated, the door warning system monitors the position of the fuselage doors that follow:

- Forward Passenger Door (52–11–00)
- Aft Passenger Door (52–12–00)
- Forward Emergency Exit Type I Door (52–24–00)
- Aft Baggage Door (52–31–00)
- Aft Service Door (52–41–00)



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The door warning system does not monitor the flight compartment emergency hatch.

[Refer to Figure 14.](#)

When any of the fuselage doors are unlocked, both the FUSELAGE DOORS warning light and the MASTER WARNING light flash. If a fuselage door is not fully closed and locked, the FUSELAGE DOORS warning light and the MASTER WARNING switchlight flash. Three tones also sound over the flight compartment speakers. When the MASTER WARNING switchlight is pushed, the light turns off. The FUSELAGE DOORS warning light stays on until the door is fully closed and locked. (Refer to SDS52–70–00).

[Refer to Figure 15.](#)

The DOORS system page on the Multi–Function Display (MFD) shows all the fuselage doors and identifies them as:

- PAX for each of the passenger doors
- BAGGAGE for each of the baggage doors
- SERVICE for the aft service door
- EMERG EXIT for the Type II emergency exit door.

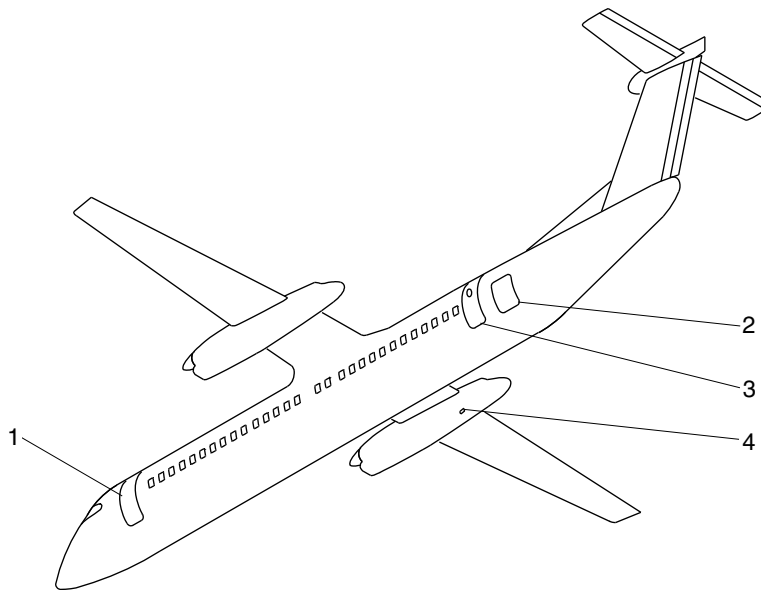
[Refer to Figure 16.](#)

If a fuselage door is not fully closed and locked, the DOORS system page identifies the door and shows it as a red filled rectangle. If the door is fully closed and locked, it is shown only as a green empty rectangle. If the data from the PSEU is invalid, the DOORS system page shows a global white INVALID DATA message.



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LEGEND

1. Forward Passenger Door.
2. Aft Baggage Door.
3. Aft Passenger Door.
4. AC Ground Power Door.

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Fuselage Doors Locator Page 1
Figure 1

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-00-00
Config 001

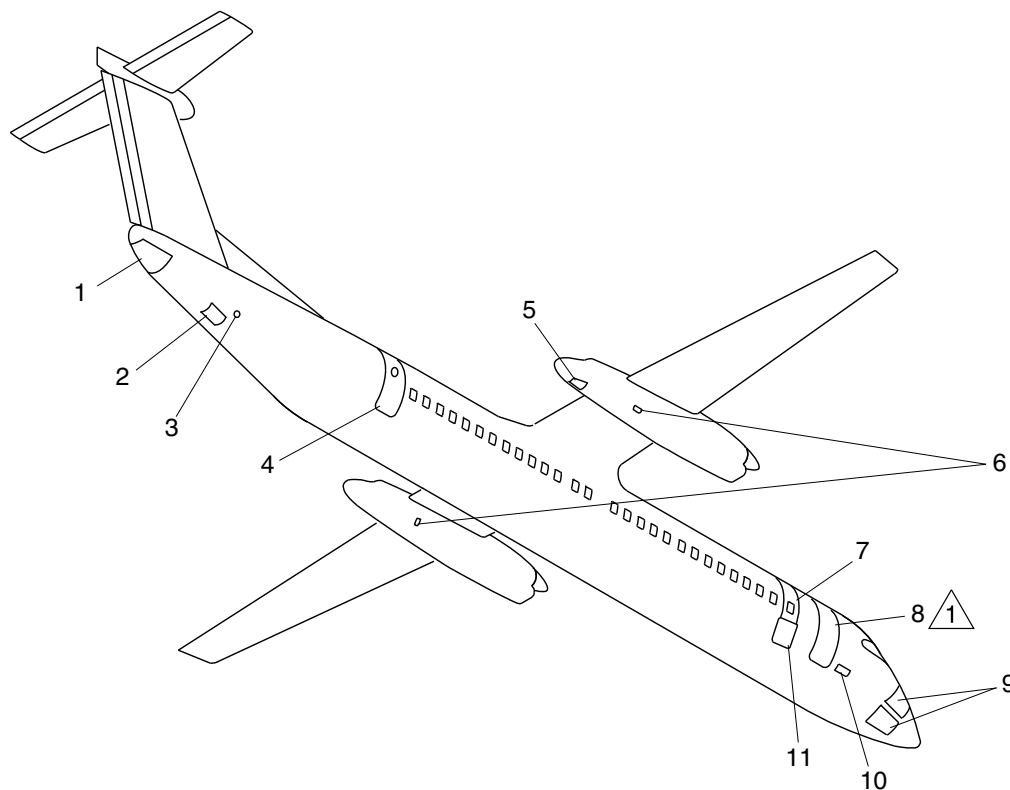
52-00-00

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LEGEND

1. APU Access Door.
2. Aft Fuselage Access Door.
3. Ground Air Conditioning Door.
4. Aft Service Door.
5. Refuel/Defuel Door.
6. Hydraulic Ground Power.
7. Upper Emergency Type II Exit Door.
8. Forward Baggage Door.
9. Nose Compartment Doors.
10. Lavatory Service Door.
11. Lower Emergency Type II Exit Door.

NOTE

- 1 On aircraft with ModSum 4-459339 incorporated, the forward baggage door is reworked and converted into Type I Emergency Exit Door.

fs045a01.dg, pr, june06/2016

Fuselage Doors Locator
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-00-00
Config 001

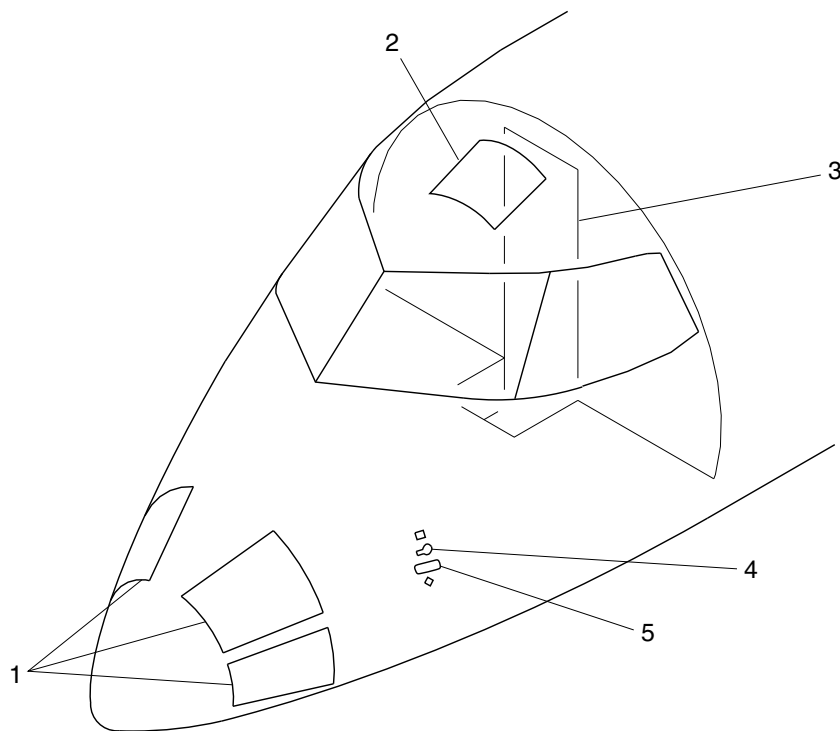
52-00-00

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LEGEND

1. Nose compartment doors.
2. Flight compartment escape hatch.
3. Flight compartment door.
4. Ground jack point door.
5. DC ground power door.

fs043a01.dg, vn, aug22/2011

Fuselage Doors Locator Page 3
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-00-00
Config 001

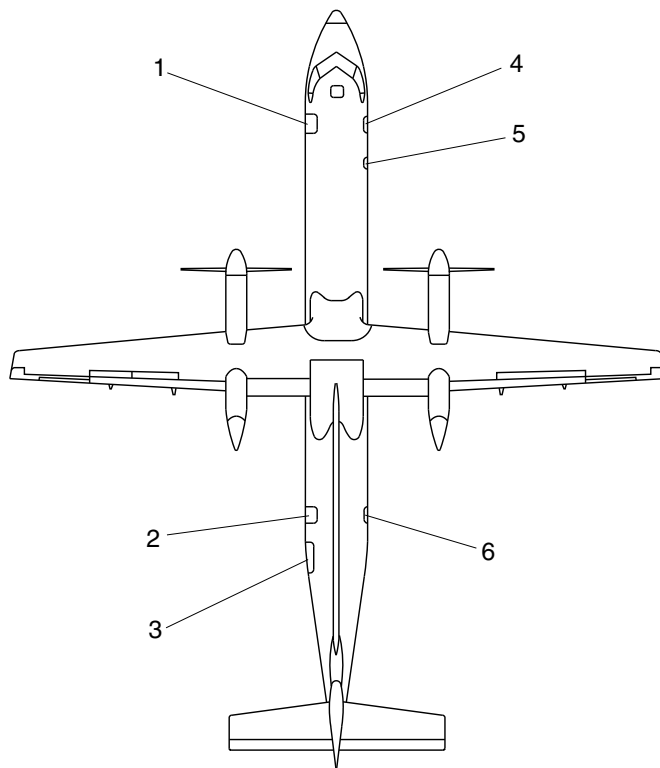
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LEGEND

1. Forward Passenger Door.
"Closed" Sensor and "Locked" Sensor.
2. Aft Passenger Door.
"Closed" Sensor and "Locked" Sensor.
3. Aft Baggage Door.
"Closed" Sensor and "Locked" Sensor.
4. Forward Baggage Door.
"Closed" Sensor and "Locked" Sensor.
5. Upper (Type II) Emergency Exit.
"Locked" Sensor.
6. Aft Service Door.
"Closed" Sensor and "Locked" Sensor.

PRE MODSUM 4-458926
PRE MODSUM 4-458969
PRE MODSUM 4-459339
PRE MODSUM 4-190614
POST MODSUM 4-309306

cg4710a01.dg, pr/cm, may03/2020

Emergency Exits Locator
Figure 4 (Sheet 1 of 2)

PSM 1-84-2A
EFFECTIVITY:
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Config 001

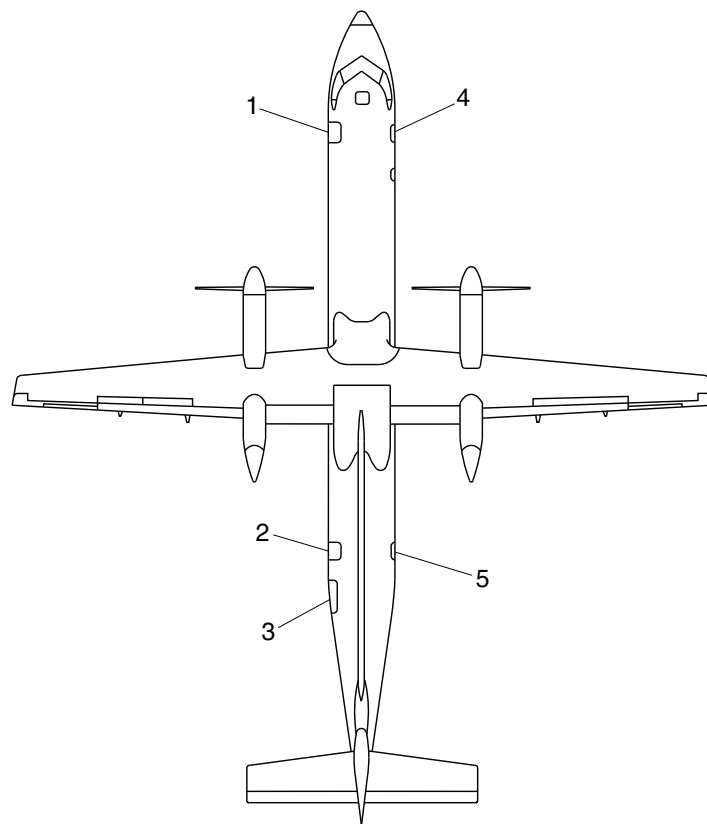
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LEGEND

1. Forward passenger door
"Closed" sensor and "Locked" sensor.
2. Aft passenger door
"Closed" sensor and "Locked" sensor.
3. Aft baggage door
"Closed" sensor and "Locked" sensor.
4. Forward RHS Type I
"Closed" sensor and "Locked" sensor.
5. Aft service door
"Closed" sensor and "Locked" sensor.

POST MODSUM 4-458926
POST MODSUM 4-458969
POST MODSUM 4-459339
POST MODSUM 4-190614
POST MODSUM 4-201647

cg4710a02.dg, cm/cm, may03/2020

Emergency Exits Locator
Figure 4 (Sheet 2 of 2)

PSM 1-84-2A
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Config 001

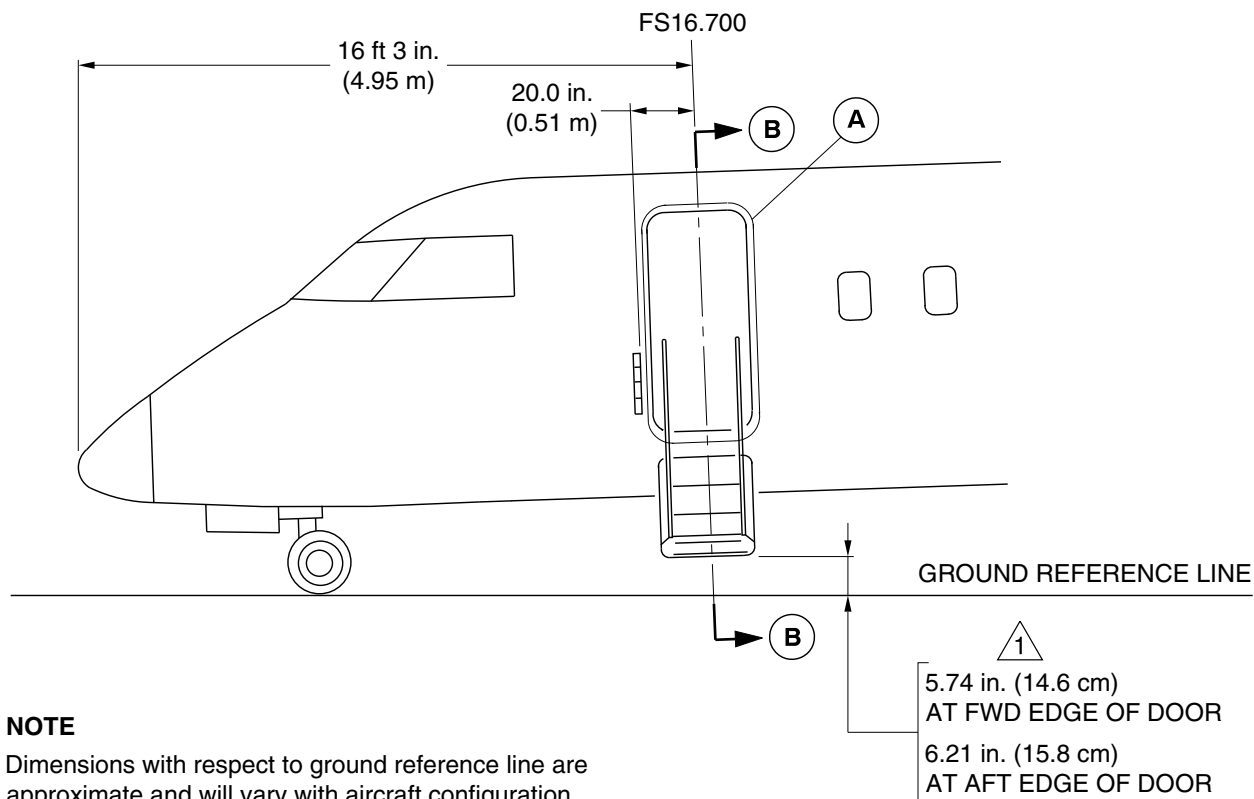
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Forward Passenger Door Dimensions
Figure 5 (Sheet 1 of 2)

PSM 1-84-2A
EFFECTIVITY:
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Config 001

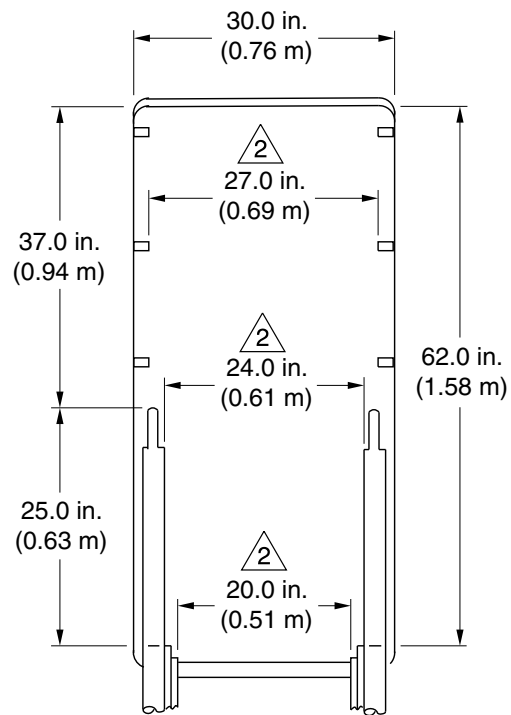
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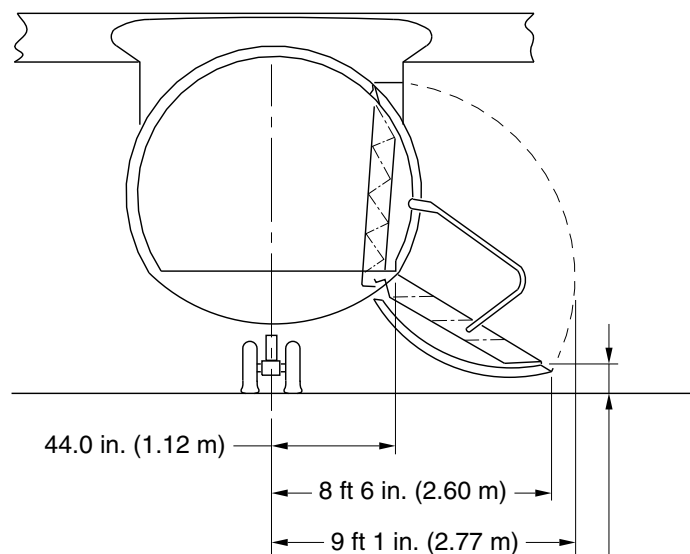
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(A) VIEW LOOKING INBOARD

NOTES

- 1 Dimensions with respect to ground reference line are approximate and will vary with aircraft configuration and loading conditions.
- 2 These dimensions are for clear openings.



(B) - (B)

VIEW LOOKING AFT

1 11.37 in. (28.9 cm)
AT DOOR CENTER LINE

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Forward Passenger Door Dimensions
Figure 5 (Sheet 2 of 2)

PSM 1-84-2A
EFFECTIVITY:
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Config 001

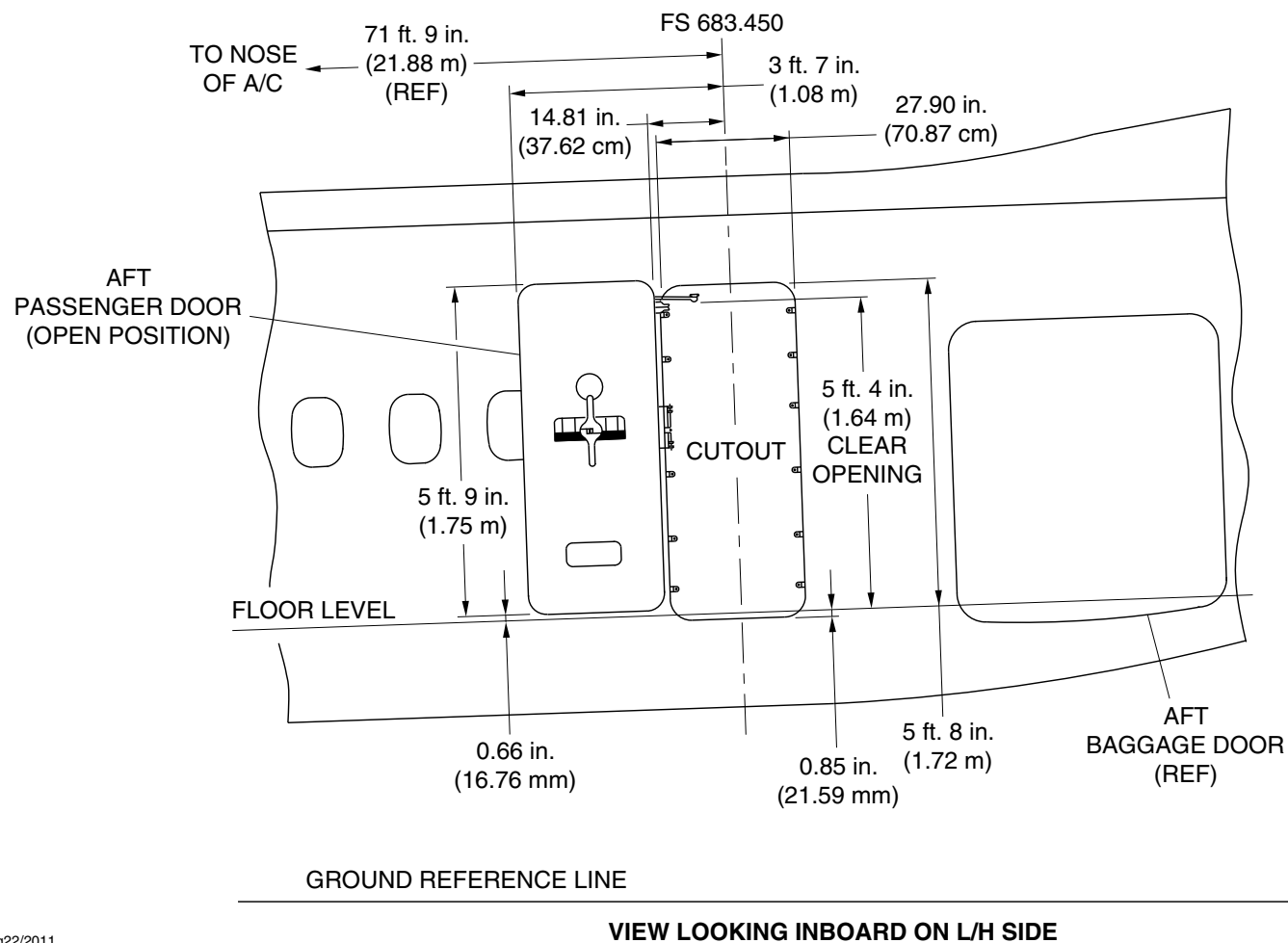
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Aft Passenger Door Dimensions
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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Config 001

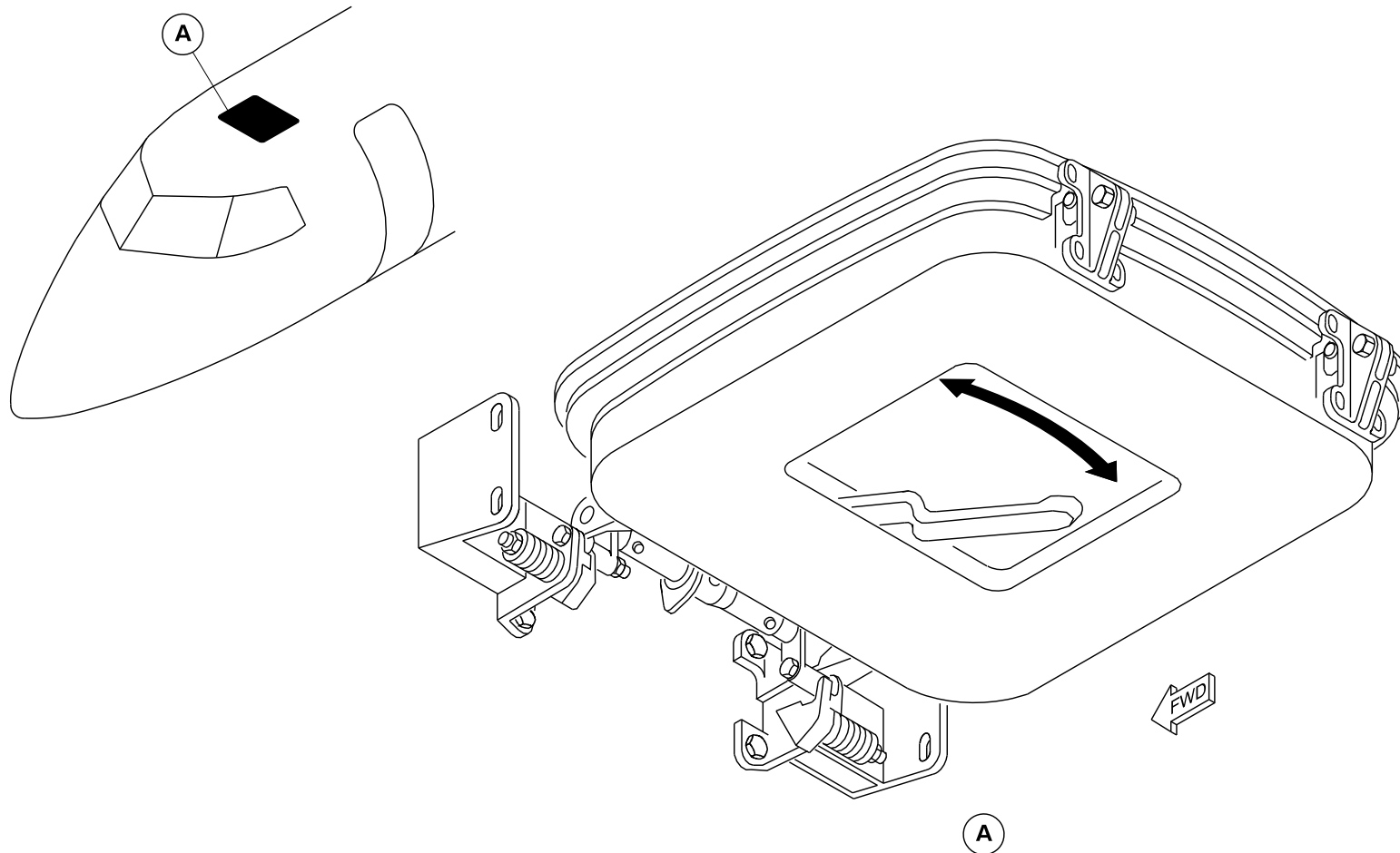
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Flight Compartment Escape Hatch Locator
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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Config 001

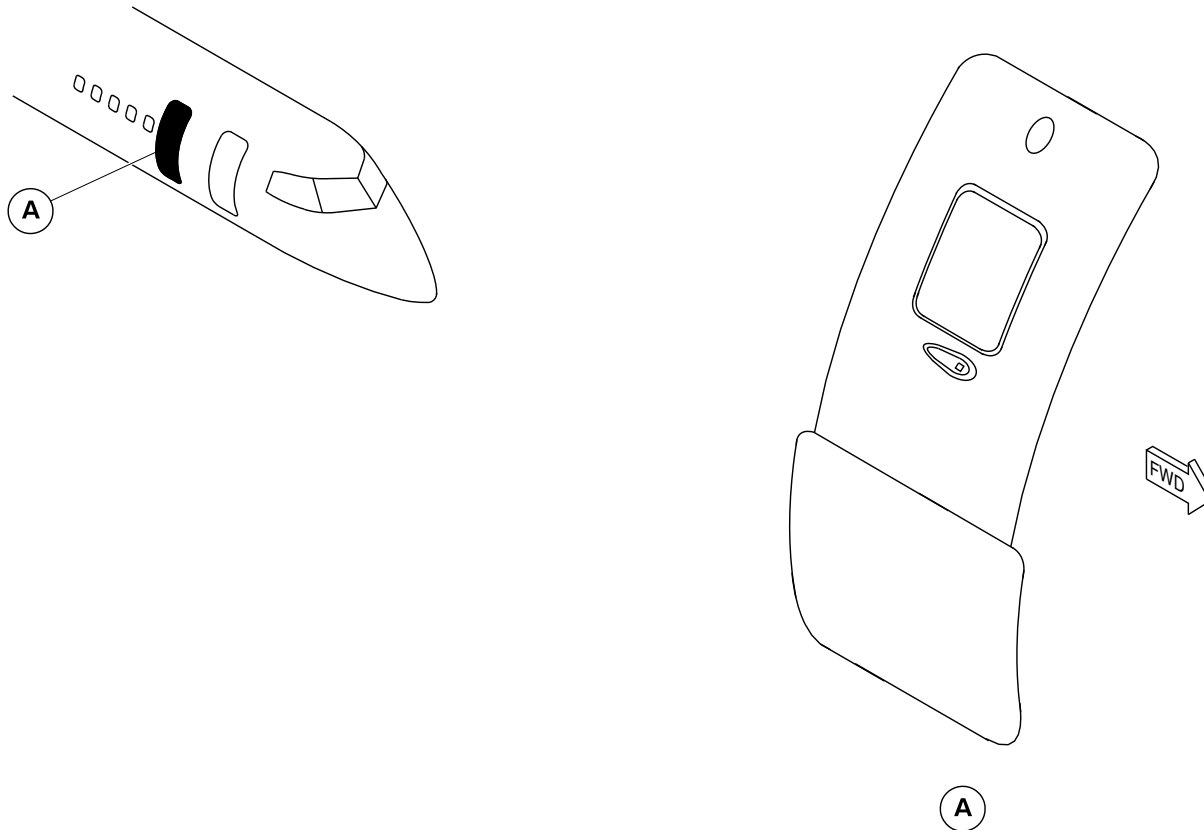
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PRE MODSUM 4-458969
PRE MODSUM 4-190561
POST MODSUM 4-309306

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Type II Emergency Exit Locator
Figure 8

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EFFECTIVITY:
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Config 001

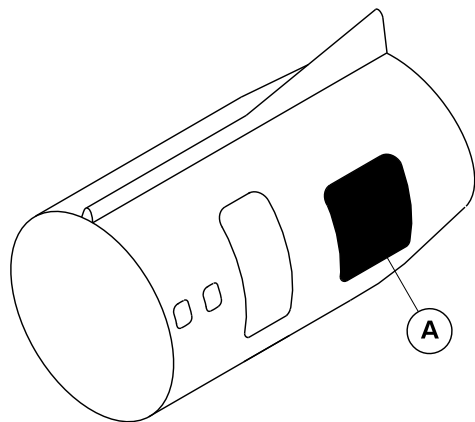
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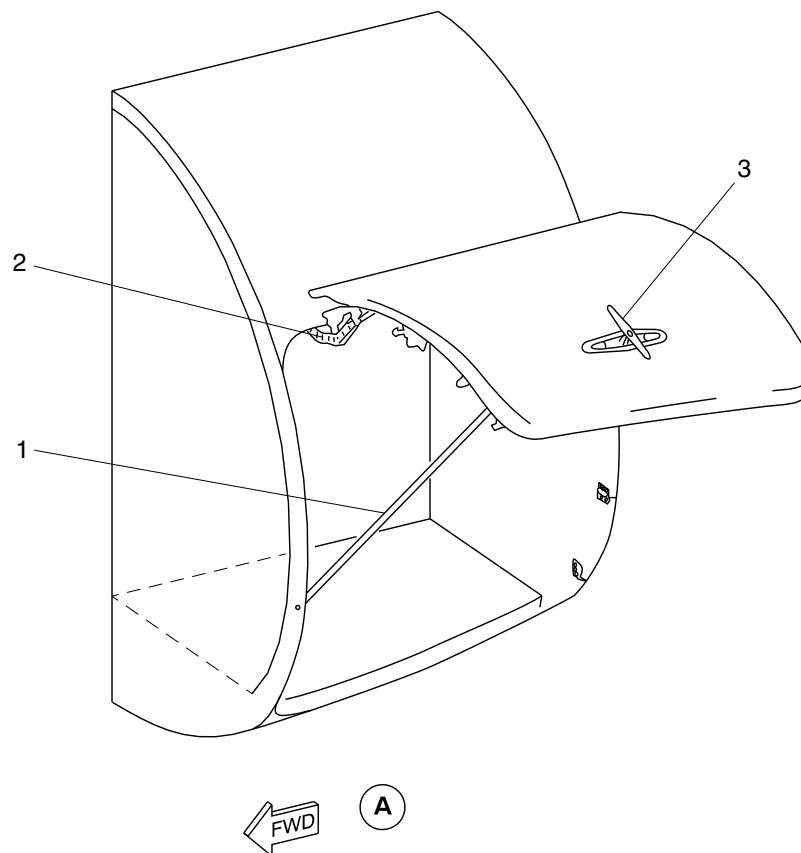
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LEGEND

- 1. Telescopic strut.
- 2. Hinge.
- 3. External handle.



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Aft Baggage Door Open
Figure 9

PSM 1-84-2A
EFFECTIVITY:
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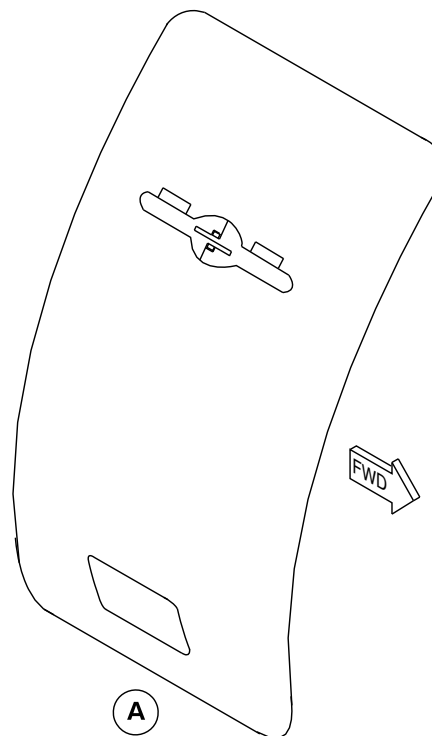
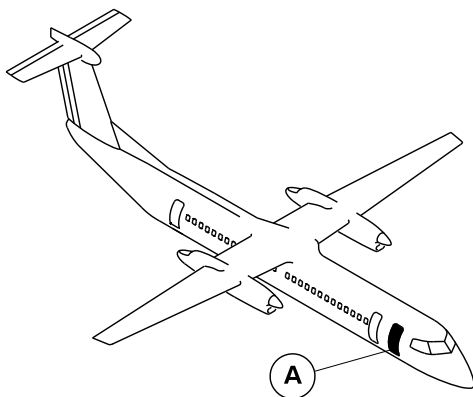
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NOTE

On aircraft with ModSum 4-459339 incorporated, the forward baggage door is reworked and converted into Type I Emergency Exit Door.

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Forward Baggage Door Locator
Figure 10

PSM 1-84-2A
EFFECTIVITY:
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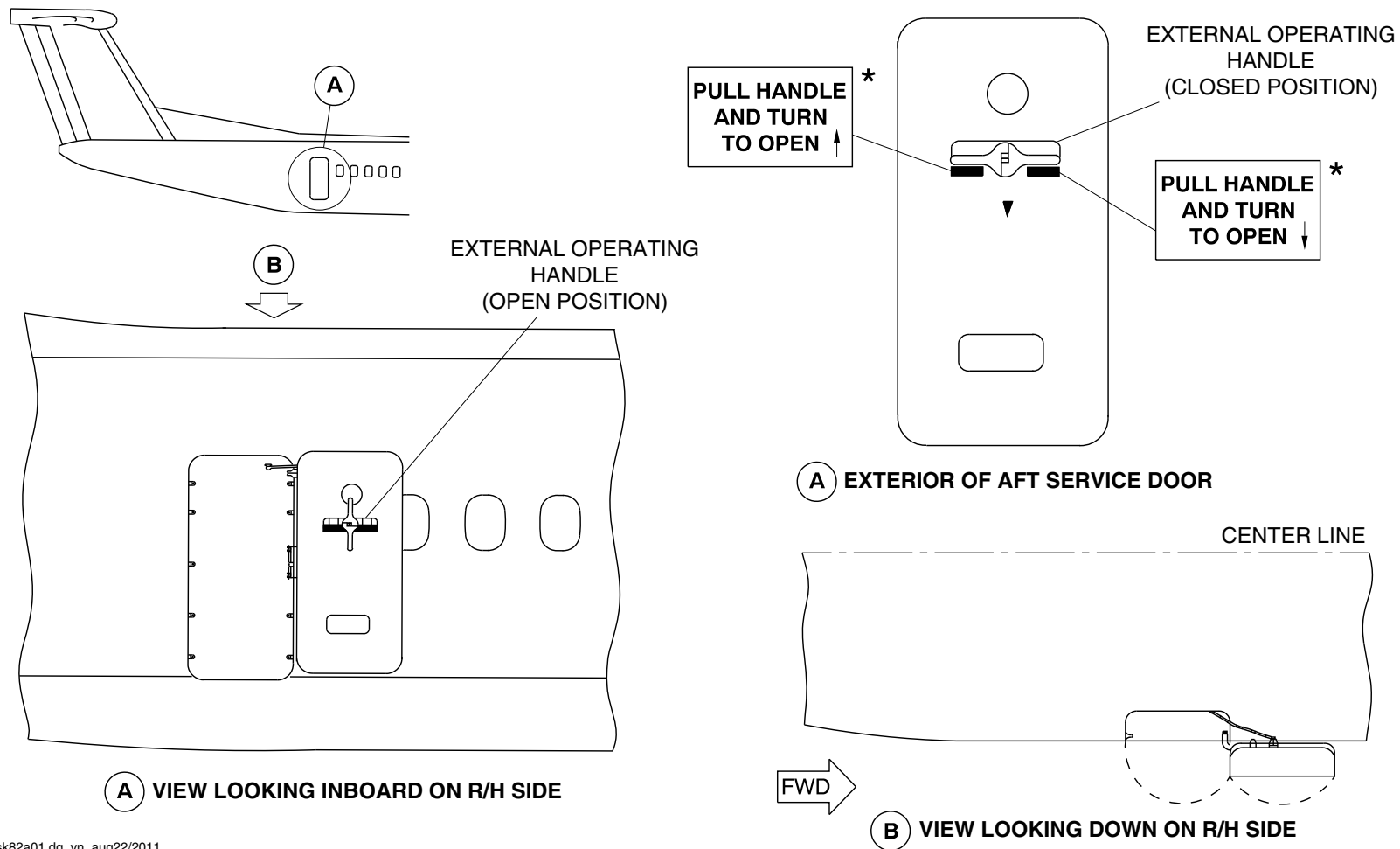
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Doors, Aft Service Door Locator
Figure 11

PSM 1-84-2A
EFFECTIVITY:
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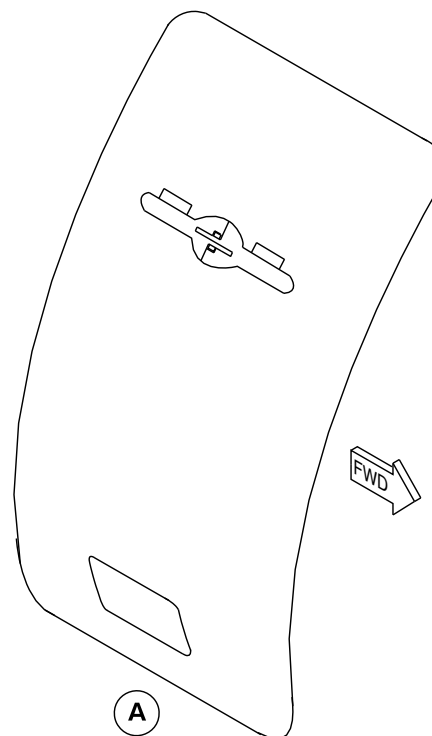
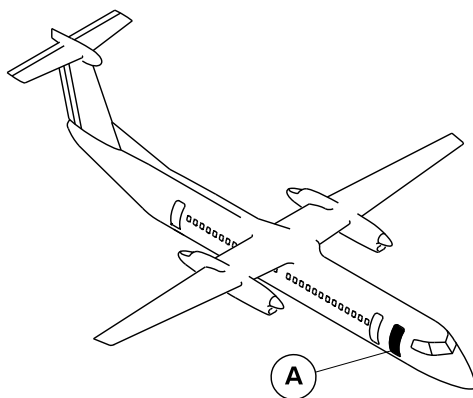
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NOTE

On aircraft with ModSum 4-459339 incorporated, the forward baggage door is reworked and converted into Type I Emergency Exit Door.

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Forward Type I Emergency Exit Door Location
Figure 12

PSM 1-84-2A
EFFECTIVITY:
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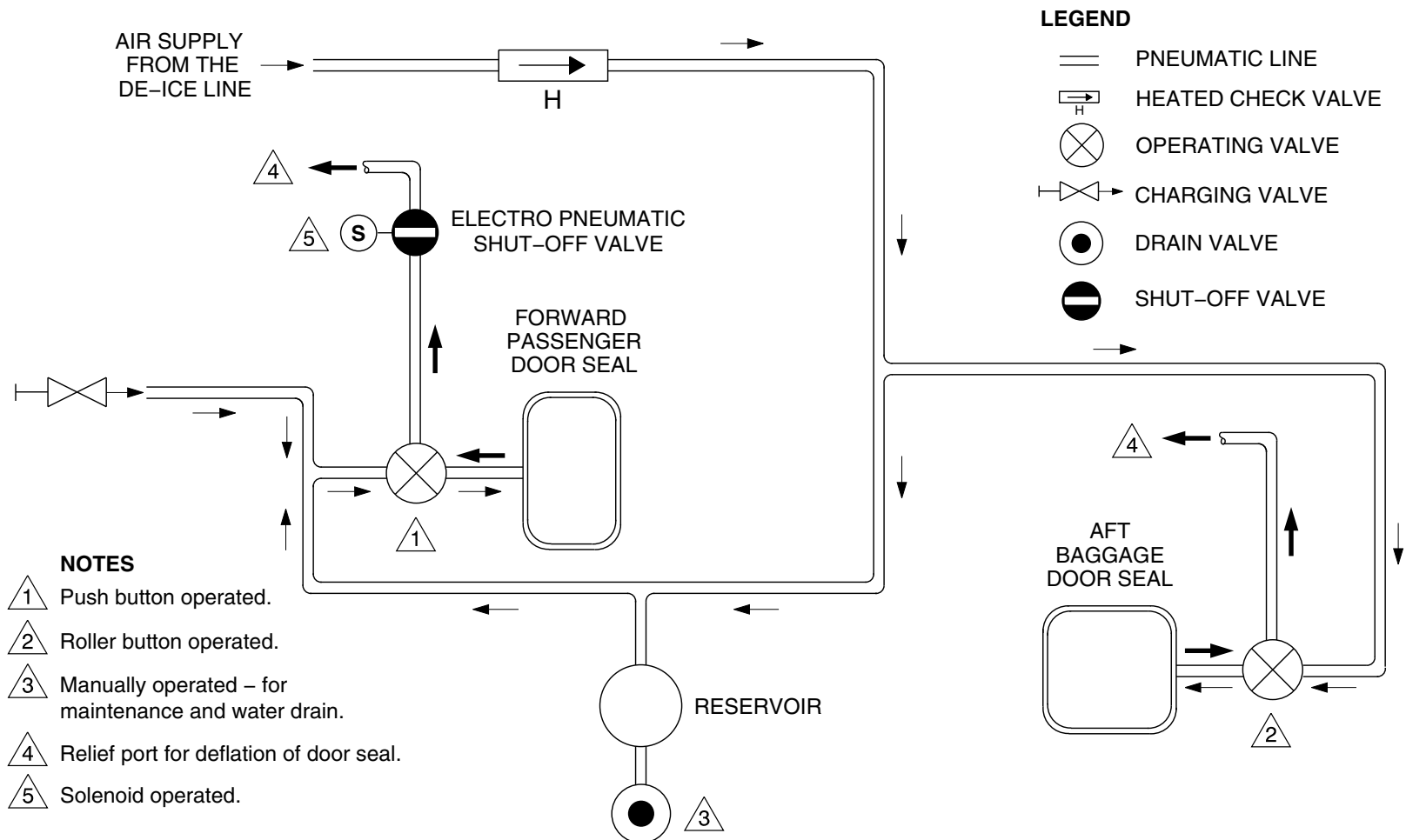
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Door Seals Pressurization System
Figure 13

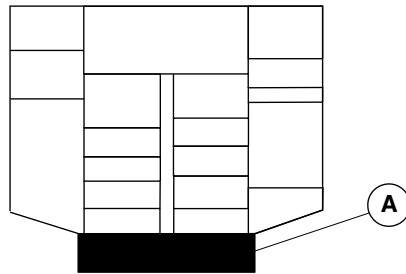
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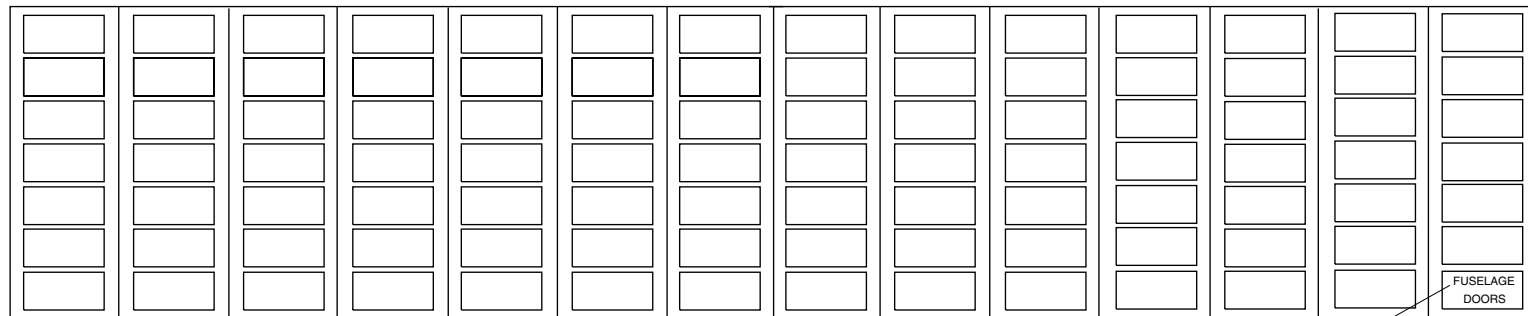
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OVERHEAD CONSOLE

LEGEND

1. Fuselage Doors (Red).



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Fuselage Doors Warning Light
Figure 14

PSM 1-84-2A
EFFECTIVITY:
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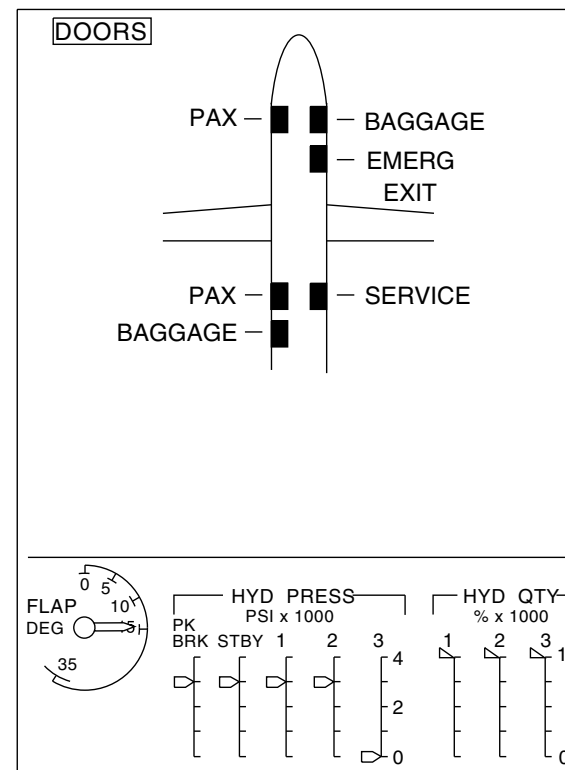
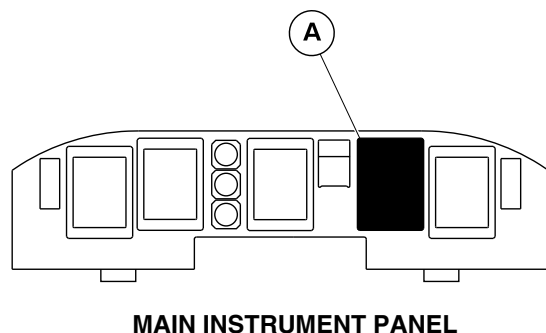
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PRE MODSUM 4-459339
PRE MODSUM 4-190614
POST MODSUM 4-309306

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Doors System Page (Doors Opened)
Figure 15 (Sheet 1 of 2)

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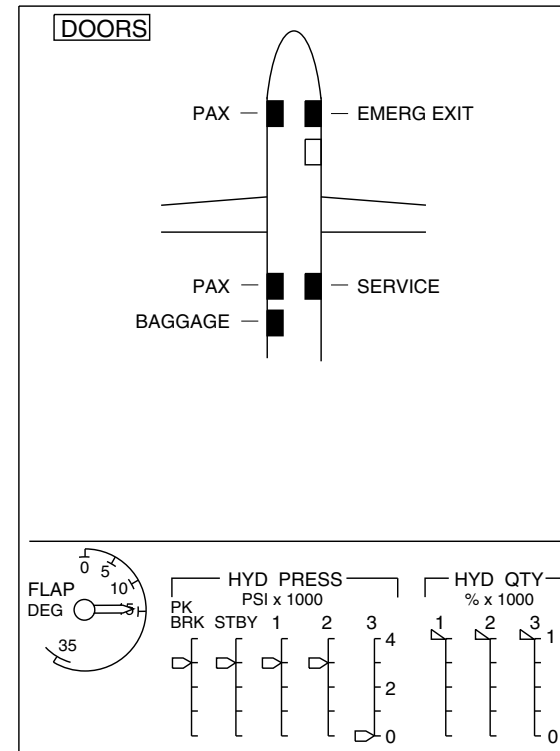
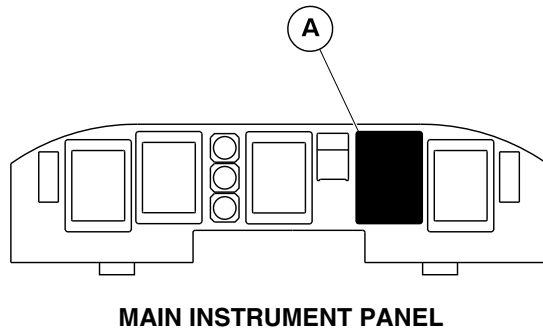
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Doors System Page (Doors Opened)
Figure 15 (Sheet 2 of 2)

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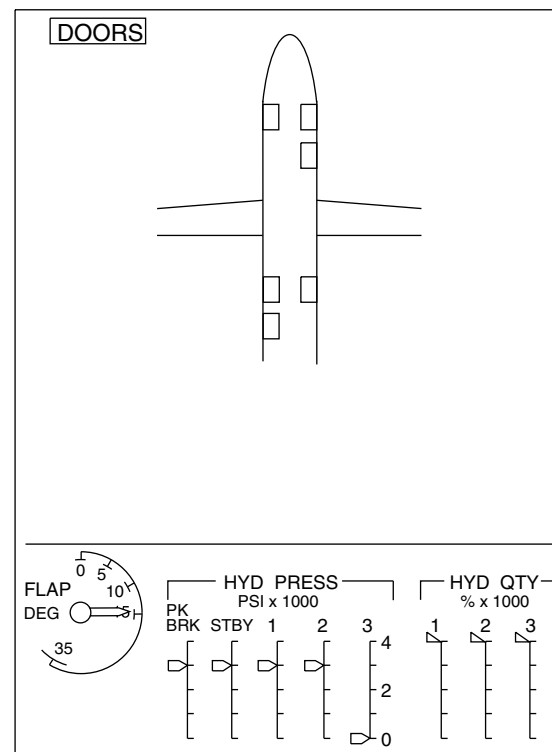
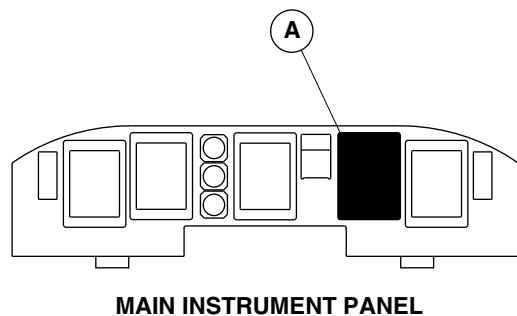
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Doors System Page (Doors Closed)
Figure 16

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PASSENGER DOORS

Introduction

Two passenger doors give access to the aircraft passenger compartment and also function as type I emergency exits.

General Description

[Refer to Figure 1.](#)

The forward and aft passenger doors are located on the left side of the fuselage and are also used as emergency exits. The doors are classified as type I emergency exits. The door warning system monitors door status and supplies visual indication and warning in the flight compartment. Automatic aircraft pressurization is prevented if either passenger door is not locked.

The Passenger Door System includes the components that follow:

- Forward Passenger Door (52-11-00)
- Aft Passenger Door (52-12-00).

The passenger doors interface with:

- Door Seal Pressurization System (52-13-00)
- Door Warning System (52-70-00)
- Proximity Sensors (52-71-00)
- Aft Entry Door Stairs – Optional Configuration (52-12-31).

Detailed Description

Both passenger doors can be opened from either the inside or outside using internal or external handles. The forward passenger door is used as the main access to the passenger compartment. As an optional configuration the aft passenger door is provided with an aft entry folding stairway. The aft entry folding stairway is used as the second entry door stairs for the entry and exit of passengers. Both the doors are classified as type I emergency exits.

[Refer to Figures 2 and 3.](#)

When either passenger door is unlocked, both the FUSELAGE DOORS warning light and the MASTER WARNING light flash. The Door Systems page on the Multi Function Display (MFD) shows the unlocked door as red filled rectangle. (Refer to SDS52-70-00). If the door is not properly closed and locked when the aircraft leaves the ground, the cabin will not pressurize automatically. The aircraft can still be pressurized manually.

Forward Passenger Door

[Refer to Figure 4.](#)

The forward passenger door is located at the left side of the fuselage at the front of the passenger compartment and classified as a type I emergency exit.

The forward passenger door is hinged at the bottom and has five steps built into the inboard side of the door structure. The steps also strengthen the door. This door has an external aluminum alloy skin braced by cross beams and intercostals. Door pressurization loads are carried by ten adjustable pressure stop bolts, five on each side. The stop bolts engage with stop brackets on the door surround



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structure. The door pressure seal inflates to prevent air leakage.
[Refer to Figure 4.](#)

The forward passenger door has the components that follows:

- An inflatable pressure seal
- Two drain valves
- An auxiliary weather seal
- Two proximity switches.

The forward passenger door is the semi–plug type. On opening, the initial movement of the door is upward and inward. The door then rotates outward about the door's hinge.

Aft Passenger Door

The aft passenger door is a translating type located at the left side of the fuselage at the rear of the passenger compartment and classified as a type I emergency exit.

[Refer to Figures 5 and 6.](#)

This door rotates outwards about the two hinges (main and secondary). A stabilizer rod prevents the yawing movement of the door during opening and closing. Door pressurization loads are carried by 12 stops that engage with their related stop brackets on the door surround structure. In unpressurized flight, the door is prevented from sliding upward by the cam rollers on each side of the lift/latch shaft. If one cam roller assembly fails, the remaining cam rollers prevent the door from sliding. The lift/latch shaft is secured in the closed position by both the handle lock mechanism and over–center springs.

The aft passenger door has the components that follow:

- Stabilizer bar
- Main Hinge
- Secondary Hinge
- Vent door.

Operation of either the internal or external spring loaded handle unlocks the door. The door opens or closes with the door face parallel with the fuselage.

[Refer to Figure 7.](#)

The aft passenger door has an optional configuration to have a provision of aft entry stairs. The stair assembly is a folding stairway and it is the second entry door stairs for the entry and exit of passengers. The stair assembly is located between fuselage Sta. X672.400 and Sta. X695.200 on the LH side of the fuselage. When not in use, the stair assembly is stowed in a stowage compartment, aft of the doorway which extends into the aft baggage area.

Door Seal Pressurization System

The door seal pressurization system prevents cabin air leakage through the forward passenger and aft baggage doors. Pressurized air is supplied from the airframe deicing system, and from the system reservoir tank when the engines or APU are not operating.

The door seal pressurization system has the components that follow:

- Heated check valve
- Reservoir tank
- Control valves



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- Door seals
- Drain valve
- Charging valve
- Electro pneumatic shut-off valve
- Plumbing lines.

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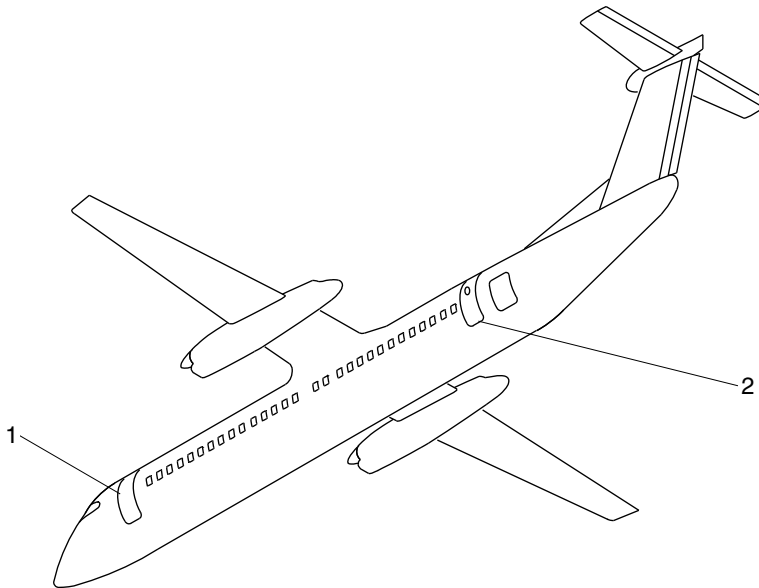


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LEGEND

- 1. Forward Passenger Door.
- 2. Aft Passenger Door.



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PASSENGER DOORS LOCATOR
Figure 1

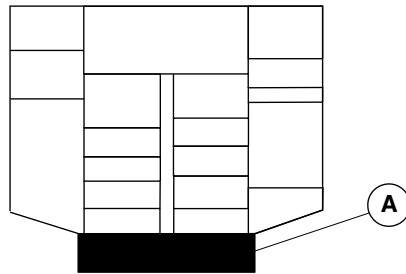
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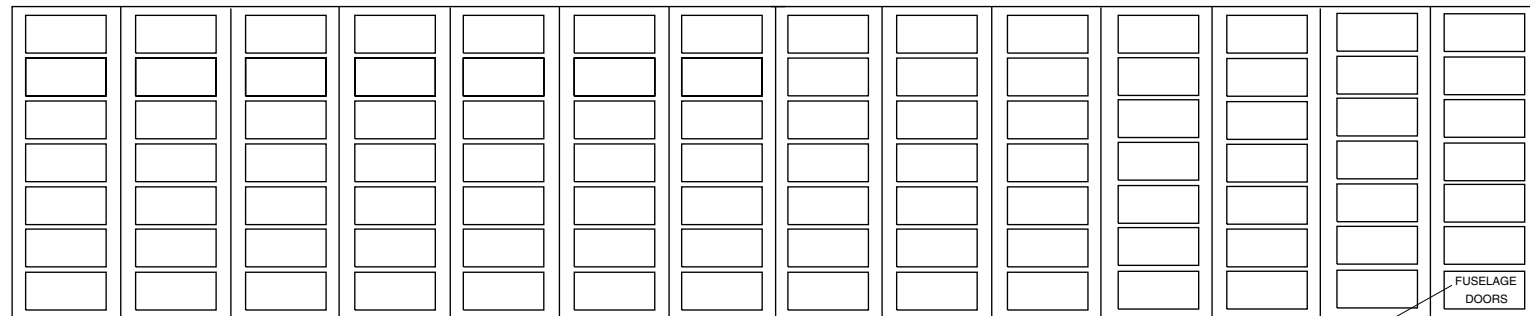
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OVERHEAD CONSOLE

LEGEND

1. Fuselage Doors (Red).



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Fuselage Doors Warning Light
Figure 2

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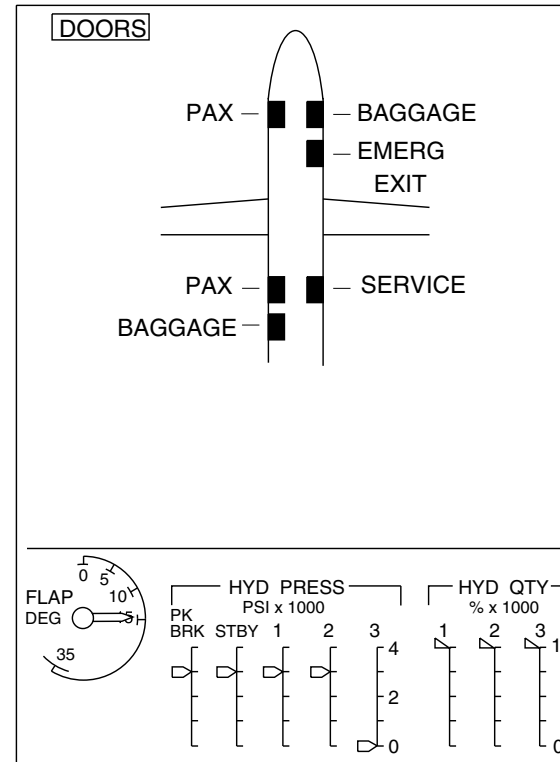
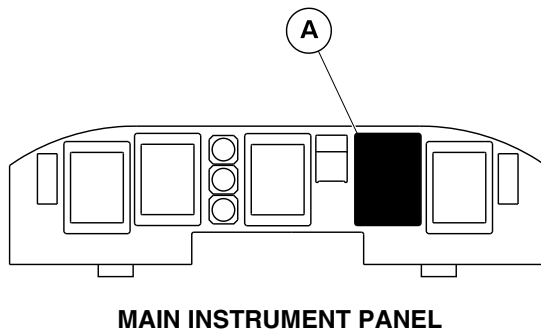
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Doors Systems Page
Figure 3 (Sheet 1 of 2)

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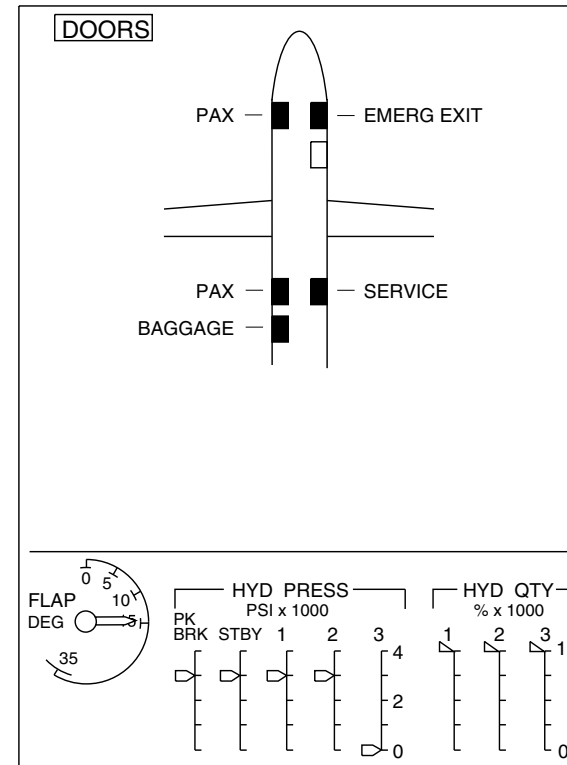
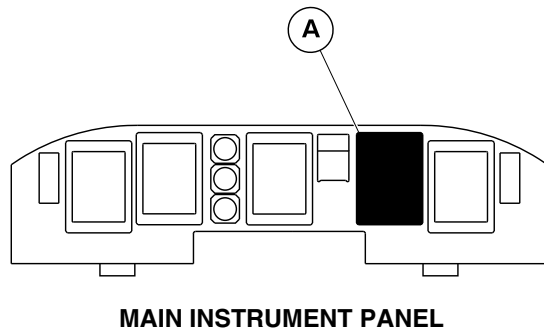
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Doors Systems Page
Figure 3 (Sheet 2 of 2)

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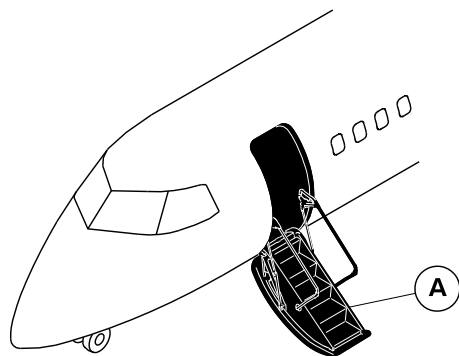
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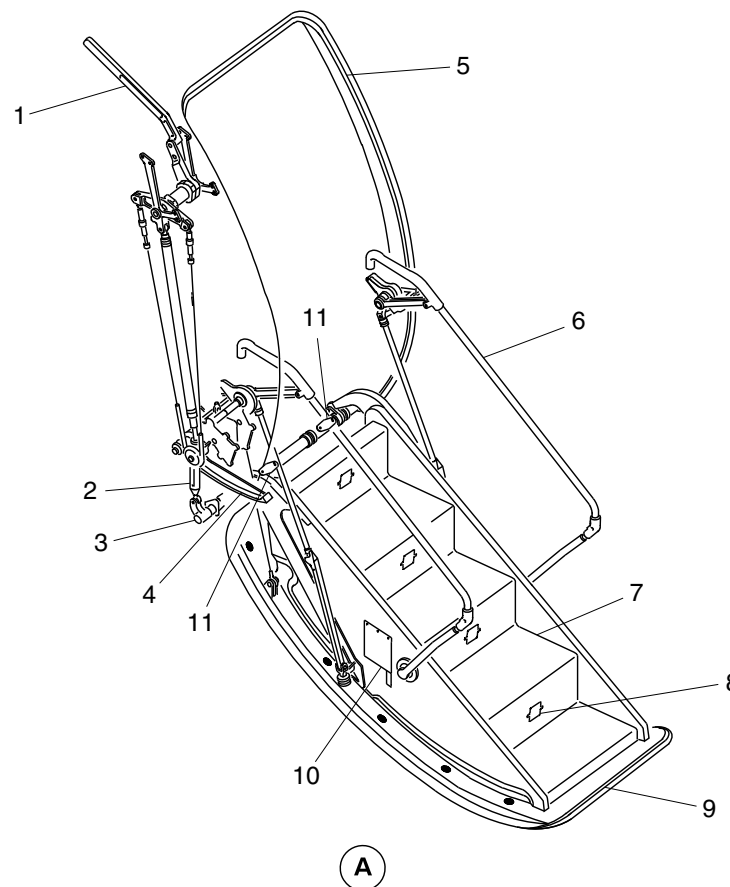
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Internal handle.
2. Door lift actuating rod.
3. Door hinge shaft.
4. External handle.
5. Inflatable seal.
6. Handrail.
7. Step.
8. Tread light.
9. Weather seal.
10. Main landing gear locking pin storage.
11. Drain plug.



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Forward Passenger Door
Figure 4

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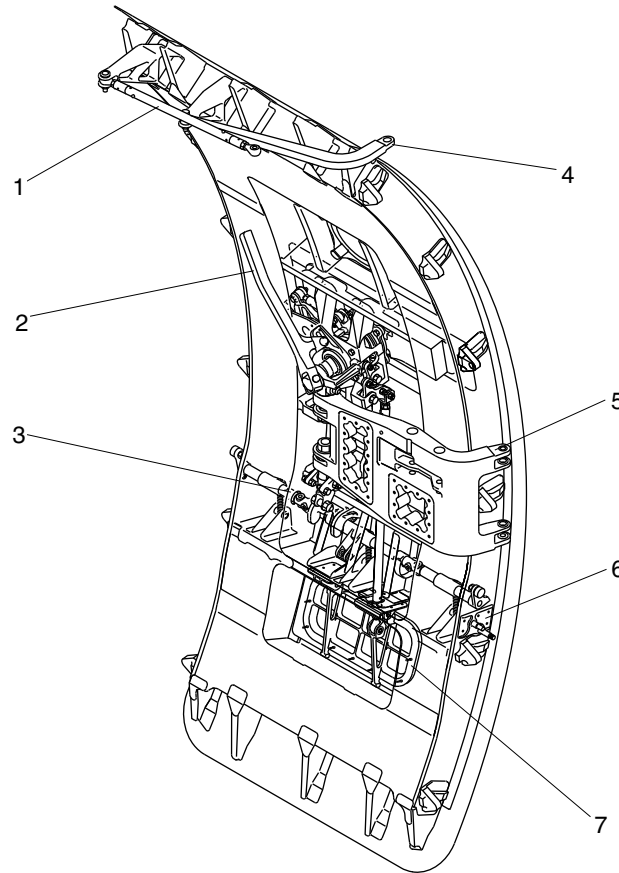
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LEGEND

1. Stabilizer Rod.
2. Internal Handle.
3. Lift And Latch Shaft.
4. Secondary Hinge.
5. Main Hinge.
6. Interlock Assembly.
7. Vent Door.

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AFT PASSENGER DOOR
Figure 5

PSM 1-84-2A
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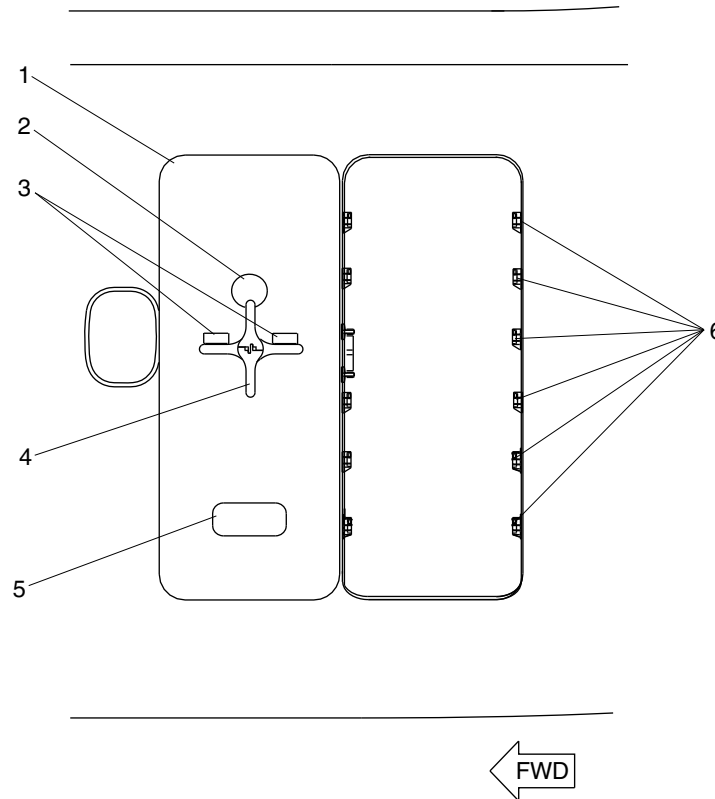
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LEGEND

- 1. Aft Passenger Door (Open Position).
- 2. Window.
- 3. Flap Cover.
- 4. External Handle (Unlocked).
- 5. Vent Door (Open).
- 6. Stops (6 Per Side).

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AFT PASSENGER DOOR EXTERNAL VIEW
Figure 6

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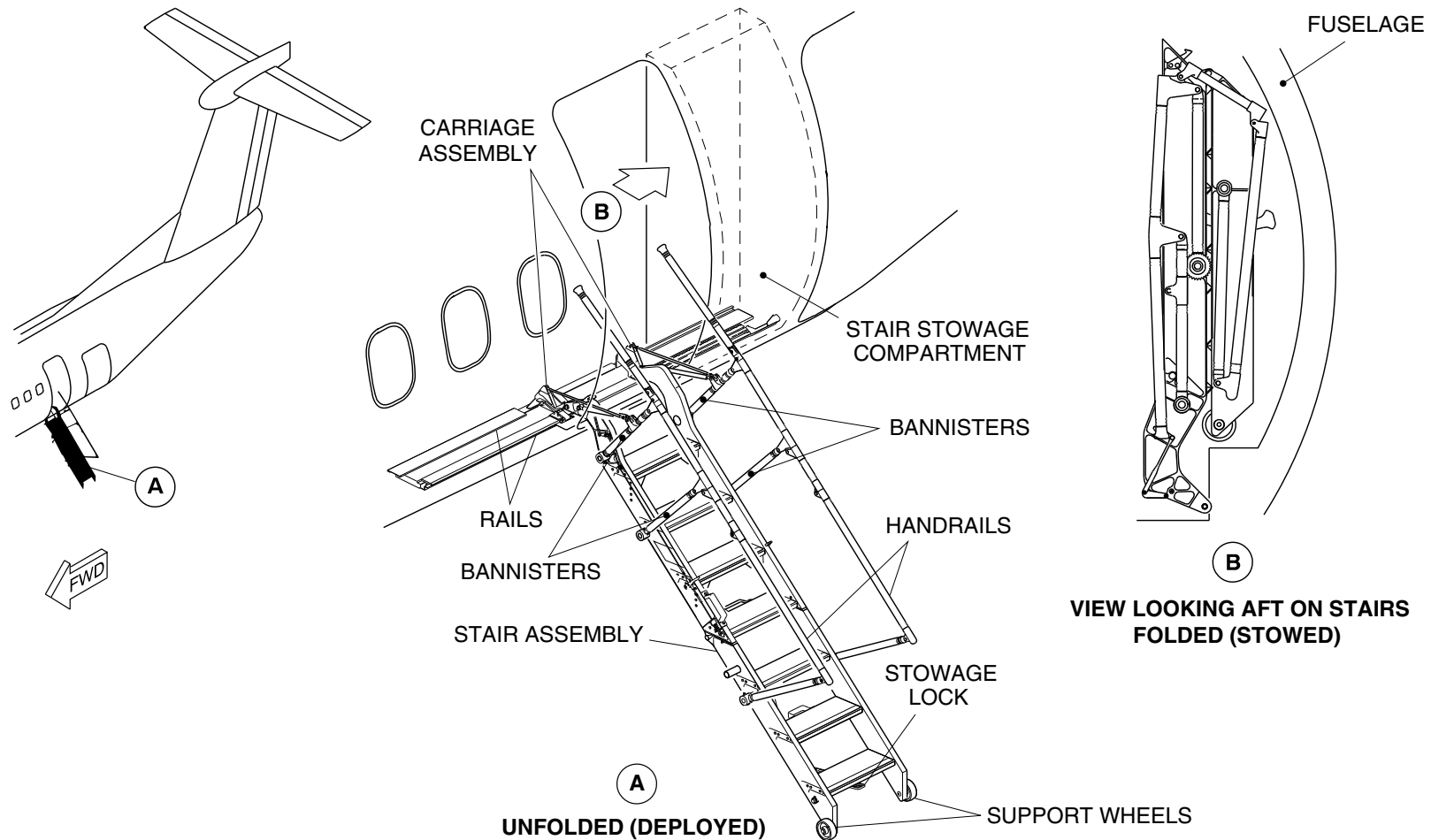
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AFT PASSENGER DOOR ENTRY STAIRS – OPTIONAL CONFIGURATION
Figure 7

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FORWARD PASSENGER DOOR

Introduction

The forward passenger door gives access to the passenger compartment and also functions as a type I emergency exit.

General Description

[Refer to Figure 1.](#)

The forward passenger door is located on the left side of the fuselage at the front of the passenger compartment, and opens outward and down. The forward passenger door is classified as a type I emergency exit.

[Refer to Figure 2.](#)

This door has five steps on the inboard side of the door structure. Internal and external door handles permit manual opening and closing from either the inside or outside of the aircraft. There are two automatically folding handrails, one on each side of the door. The door has four tread lights controlled by a switch on the flight attendant's control panel. Main landing gear lock pins are stored on the forward side of the door structure. An inflatable pressure seal prevents cabin air leakage past the door. The door warning system monitors door status and supplies visual indication and warning in the flight compartment.

The Forward Passenger Door System has the components that follow:

- Airstair (52-11-01)
- Shell, Door (52-11-06)
- Mechanism, Door Lift (52-11-11)
- Mechanism, Door Balance (52-11-16)
- Handrail, Door (52-11-21)
- Struts (52-11-26)
- Seals, Door (52-11-31)
- Valve, Control-Door Seal (52-11-36)
- Electro pneumatic shut-off valve (52-11-40)
- Drain Plugs (52-11-41)

The Forward passenger door interfaces with:

- Door Seal Pressurization System (52-13-00)
- Door Warning System (52-70-00)
- Proximity Sensors (52-71-00)

Detailed Description

[Refer to Figure 3.](#)

The forward passenger door gives access to the passenger compartment and also functions as a type I emergency exit. The forward passenger door opens and closes manually from either the inside or outside. An internal handle is located in a recess above the forward flight attendant seat.



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[Refer to Figure 4.](#)

An external handle is located forward of the door, in a recess in the fuselage.

[Refer to Figure 5.](#)

The internal and external handles are interconnected by a cable and chain system. A flush handhold, on the door outer skin, is used to pull the door open from the outside. Handrails on each side of the stairs automatically fold during door closing and unfold during opening. On opening, the initial movement is upward and inward. The door then rotates about a hinge at the bottom of the door.

[Refer to Figure 6.](#)

Two hinge arms attach the door to a door hinge shaft installed in the fuselage below the door surround bottom sill. The door hinge shaft is connected through a crank and actuating rod to a door lift mechanism. The lift mechanism has an upper crank and a pinion assembly, and is operated by the interconnected internal and external handles. Operation of either handle rotates a gear segment to drive a pinion, rotating the door hinge shaft through a pushrod, raising or lowering the door hinges.

[Refer to Figure 7.](#)

The upper crank and pinion assembly is attached to a gas spring assembly attached to the fuselage. A spring controlled overcentering trip lever and locking bar, are operated by a roller on the forward side of the door. The trip lever and spring, in conjunction with the gas

spring, gives a positive overcenter lock of the lift mechanism when the door is either in the opened or closed position.

[Refer to Figure 8.](#)

When the door is in the closed position, door pressurization loads are carried by ten adjustable stop bolts, five on each side of the door. These stop bolts engage with related stop brackets on the door surround structure. Each door stop is inclined downward, so that pressure on the door loads the door downwards towards the closed position. The geometry of the door transporting system is such that, if the door moves away from the fully closed position, cabin pressurization loading applies a downward force on the door, preventing any further movement towards the disengaged position.

[Refer to Figure 9.](#)

A door counterbalance system controls the rate the door opens and helps to close the door. The system is controlled by two balance gas springs, attached at the top to a slide block and pulley assembly and at the bottom to a fixed bracket. The fixed bracket is bolted to the counterbalance system mounting.

[Refer to Figure 10.](#)

A cable assembly attached to the fixed bracket runs up and over the slide block pulley and down to wind around a balance pulley attached to the shaft assembly. The shaft assembly is bolted to the counterbalance system mounting. The slide block and pulley assembly moves up and down in a fixed guide rail, under the action of the balance gas springs when the door is opened and closed. A



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

support pulley system on the shaft assembly is connected by a cable to a cable support attached to the forward face of the door structure.

[Refer to Figure 11.](#)

The door is supported on each side in the open position by the upper and the lower struts which fold automatically when the door is closed. Installed in each of the lower struts is a rubber bumper to protect the structure from impact damage if one of the balance gas springs is defective. Installed in each of the upper struts is an overtravel positive stop to prevent the structure from excessive bending when people are standing on the stairs. The goosenecks are installed on the forward and aft sides of the forward passenger doors. On aircraft with ModSum IS4Q2400003 incorporated, the p-clip (MS2191WDG4) attachment on gooseneck frame is replaced with a low profile Tie mount (CB3019A3N)). The Tie mount is bonded to the gooseneck frame.

[Refer to Figure 12.](#)

The door has an inflatable rubber seal installed on the fuselage structure around the edge of the door opening. Pressurized air to and from the seal is controlled by a pressurization valve operated by a rod assembly in the door lift mechanism. An auxiliary weather seal installed on the upper part of door cut-out seals the door when air pressure is not available.

Two proximity sensors monitor the forward passenger door. One sensor is located at the top aft side of the door, to make sure that the door is closed and fully down. The other sensor, located on the fuselage structure adjacent to the pressurization valve, monitors the lift link position to make sure that the door mechanism is fully overcentre and in safety.

[Refer to Figures 13 and 14.](#)

Opening of the forward passenger door is by operation of either the internal or external handle. The handle force builds up and then decreases when it passes the overcenter position. At the same time, the seal pressurization valve closes and releases seal pressure. The deflated seal vents any residual passenger compartment air pressure. The "locked" proximity switch also disconnects. Further handle travel, through the actuating rod and crank, moves the door upward and inward to clear the ten stop bolts from their stop brackets. The "closed" proximity switch disconnects as the door opens. The door is kept at the lifted position by the counterbalance system. Pushing on the handrails, or pulling on the external handhold then opens the door. The rate of opening is controlled by the gas springs of the counterbalance system. If a gas spring fails, rubber bumpers on each handrail lower strut gives free fall damage protection.

Closing the door from the inside is done by pulling up on the handrail handles, the motion being assisted by the counter balance system. Operation of the internal handle to the closed position rotates the door hinge shaft, placing the door mechanism including both the actuating rod and the mechanical spring in an overcenter position and locating the stop bolts behind their related stop brackets. Final movement of the door mechanism opens the pressurization valve and inflates the seal, and the door is fully closed and locked. Closing the door from the outside is done by first pushing the handrails inboard to break the overcenter lock, then manually raising and closing the door. Moving the external handle flush with the fuselage locks the door in the same way as the internal handle. During the opening and closing sequences, the gas springs dampen any tendency for high impact at both ends of the door travel.



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Refer to Figures 15 and 16.

If the door is unlocked, the FUSELAGE DOORS warning and the MASTER WARNING light flash. The Doors Systems page on the Multi Function Display (MFD) shows the unlocked door as a red filled rectangle. If the door is not locked, the door seal will not inflate. If the door is not properly closed and locked when the aircraft leaves the ground, the cabin will not pressurize automatically.

Airstair

The airstair has five steps and four tread lights. It is attached to the door shell by four attachment bolts.

Shell, Door

The door is made of an aluminum alloy external skin braced by cross beams and intercostals.

Mechanism, Door Lift

The door lift mechanism is operated by either the internal or external handle. The lift mechanism lifts the door on opening, and locks it on closing. At the locked position, the actuating rod engages the "locked" proximity switch and the pressurization valve. The geometry of this lift mechanism causes an over center condition in both door up and down positions, holding the door in either position. A mechanical spring attached to the crank and pinion assembly, and secured to the fuselage structure, adds positive over center locking.

Mechanism, Door Balance

The door balance mechanism balances the weight of the door using two gas springs, a shaft assembly, cables and a slide block and pulley assembly. The rate of opening and closing is controlled by the two gas springs, attached at the top to a slide block and pulley assembly and at the bottom to a fixed bracket. A cable assembly attached to the fixed bracket runs up and over the slide block pulley and down to wind around a balance pulley, mounted on a shaft assembly. The shaft assembly is bolted to the counterbalance system mounting. The slide block and pulley assembly goes up and down in a fixed guide rail, under the action of the gas springs when the door is opened and closed. A support pulley, installed on the shaft assembly, is connected by cable to a cable support attached to the forward face of the door structure.

Handrail, Door

Two handrails, one on each side of the door, fold automatically when the door is closed and unfold when the door is opened. A rubber bumper is installed in each of the lower struts to protect the structure from impact damage if one gas spring fails.

Refer to Figures 17 and 18.

On Airstair door with jetway compatibility (Modsum 4Q422100, 4Q458687, 4Q458845 or SB 84-52-79 or SB 84-52-86), there are two collapsible handrails, one on each side of the door. The handrails fold automatically when the door is closed and unfold when the door is opened. A retainer bracket on the handrail assembly prevents the removable pin from coming out of the handrail. To make the door jetway compatible, turn the retainer bracket, pull out the removable pin from the handrail assembly, and put the handrail assembly on the handrail holder assembly.

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Struts

The struts go slightly overcenter against the stops to hold the door in the fully open position. The struts include an energy absorber, to protect the structure from impact damage, if one gas spring balance fails.

Seals, Door

A door seal is installed on the door surround structure and inflates with pressurized air. An auxiliary weather seal supplies sealing when air pressure is not available.

Valve, Control–Door Seal

The door seal pressurization valve is operated when the door is closed and the lift mechanism is overcenter. The valve prevents pressurization of the seal unless the door is fully closed and locked. The valve releases seal pressure when the door is unlocked.

Electro pneumatic shut–off valve

The solenoid operated valve is installed on the relief port line of the forward passenger door seal pressurization valve. The primary function of this electro pneumatic shut–off valve is to prevent depressurization of the door seal system if airspeed is more than 100 knots (airspeed > 100 knots). This feature prevents inadvertent door seal depressurization for the forward passenger door.

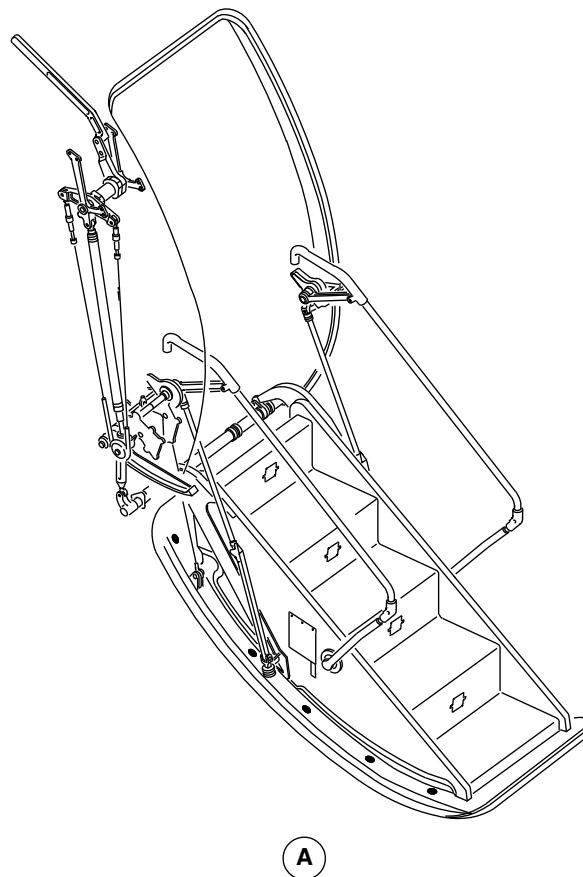
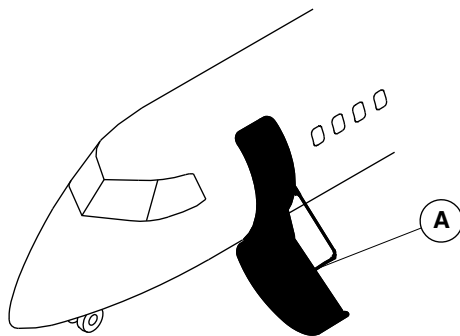
Drain plugs

Two drain plugs are installed on the channel adjacent to the hinge of the door. When the door is open, the drain plugs can be removed to drain accumulated water from the channel.



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FORWARD PASSENGER DOOR LOCATION
Figure 1

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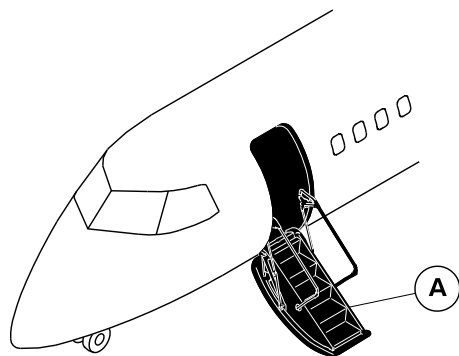
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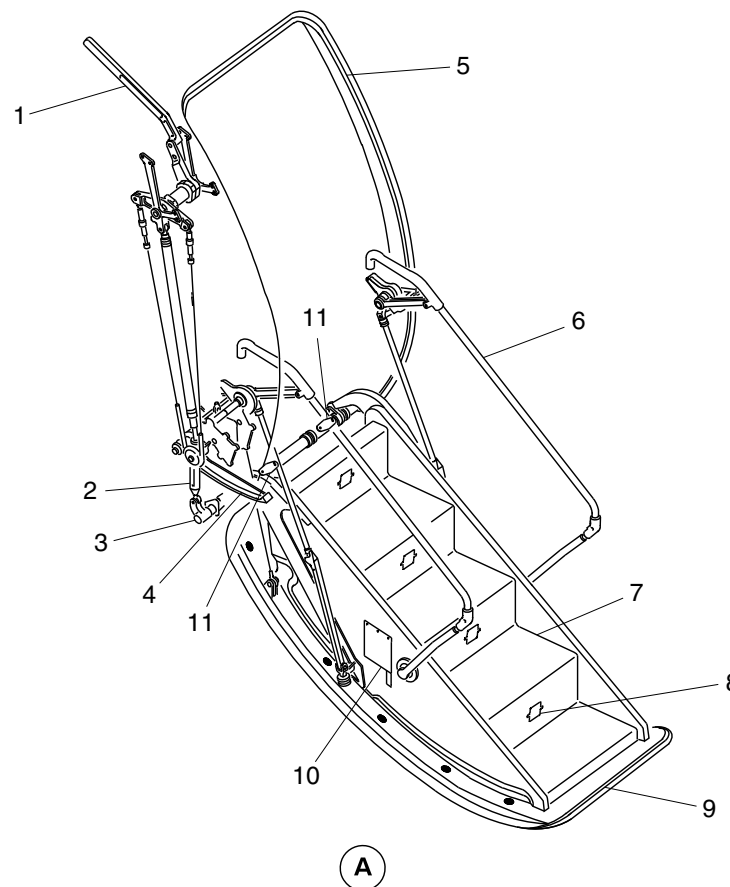
DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Internal handle.
2. Door lift actuating rod.
3. Door hinge shaft.
4. External handle.
5. Inflatable seal.
6. Handrail.
7. Step.
8. Tread light.
9. Weather seal.
10. Main landing gear locking pin storage.
11. Drain plug.



fs042a01.dg, kg/vn, aug22/2011

Forward Passenger Door
Figure 2

PSM 1-84-2A

EFFECTIVITY:

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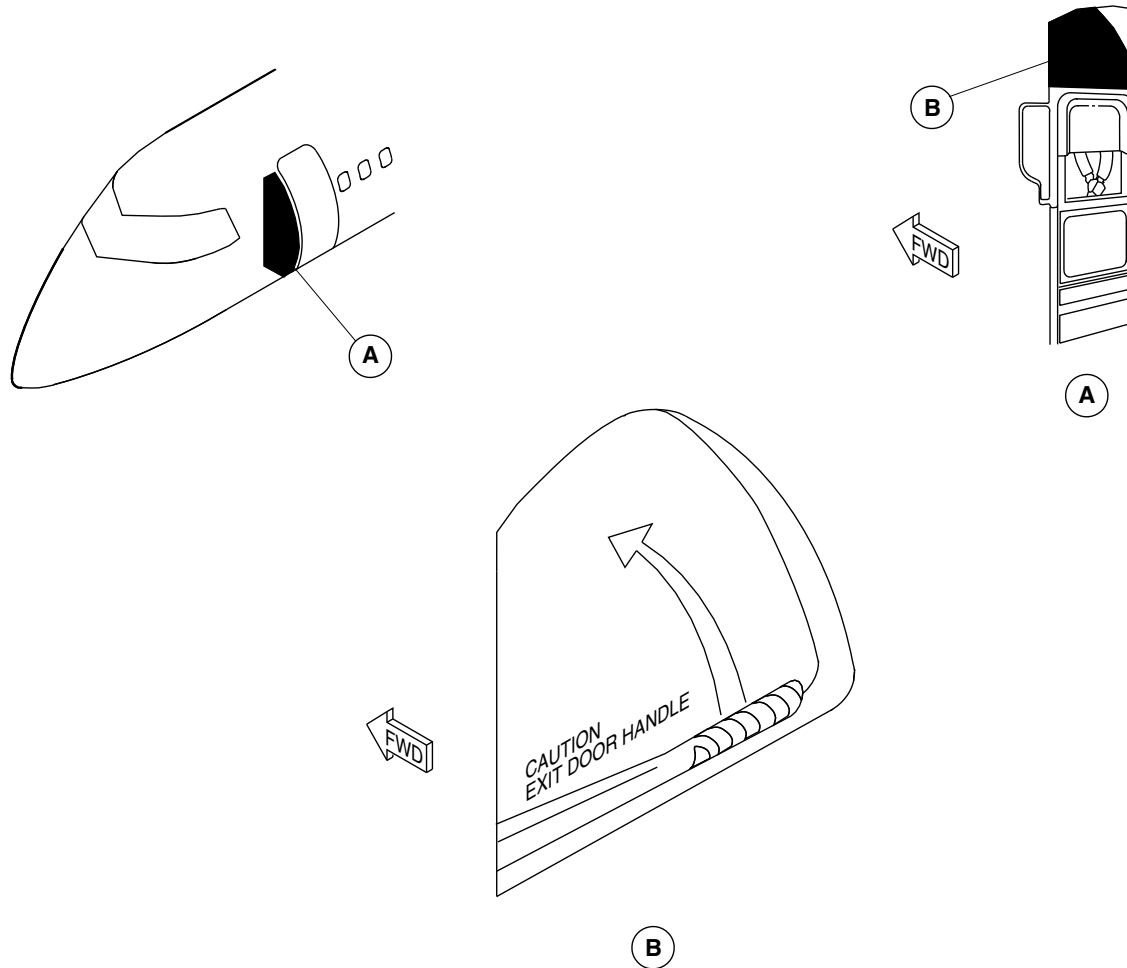
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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FORWARD PASSENGER DOOR INTERNAL HANDLE LOCATION
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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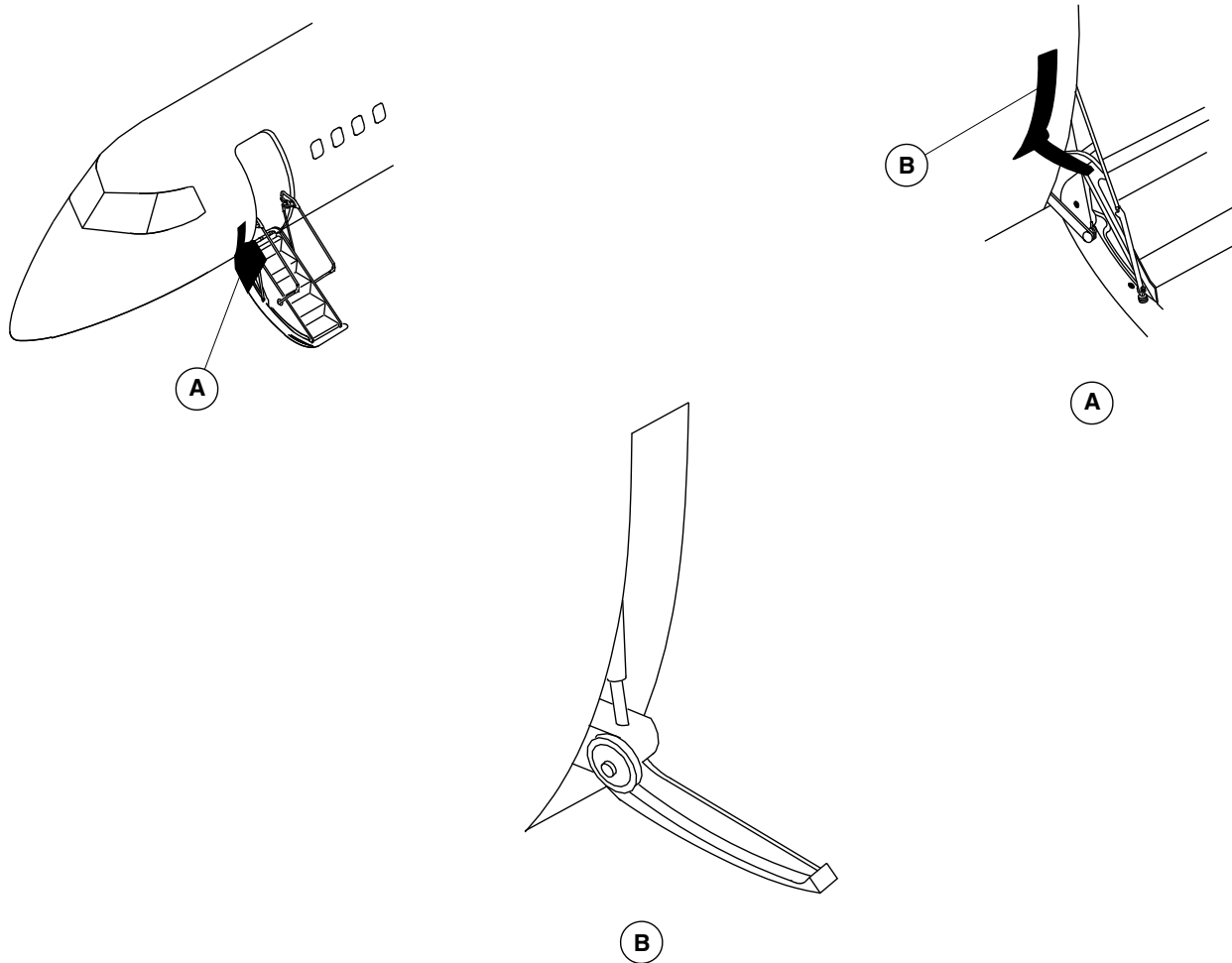
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs297a01.cgm

FORWARD PASSENGER DOOR EXTERNAL HANDLE LOCATION
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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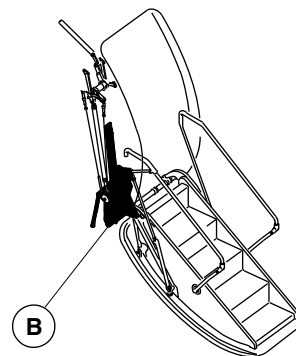
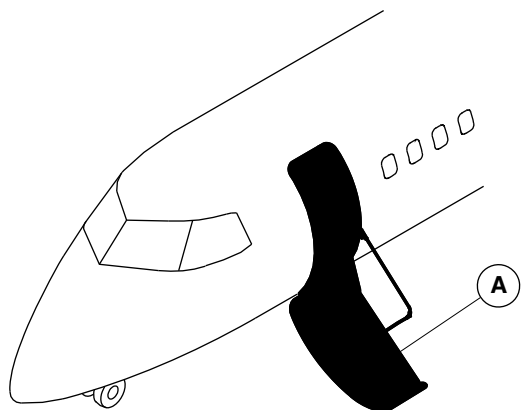
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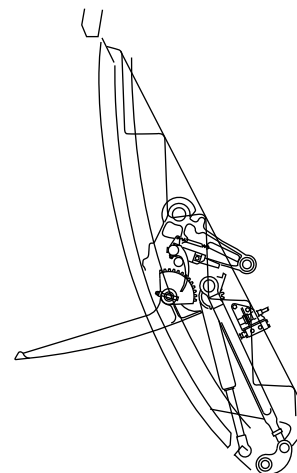
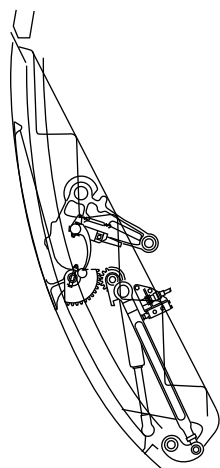


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A FORWARD PASSENGER DOOR



B DOOR CLOSED AND LOCKED

B DOOR LIFTED

fs563a01.cgm

FORWARD PASSENGER DOOR CLOSED AND LIFTED POSITIONS
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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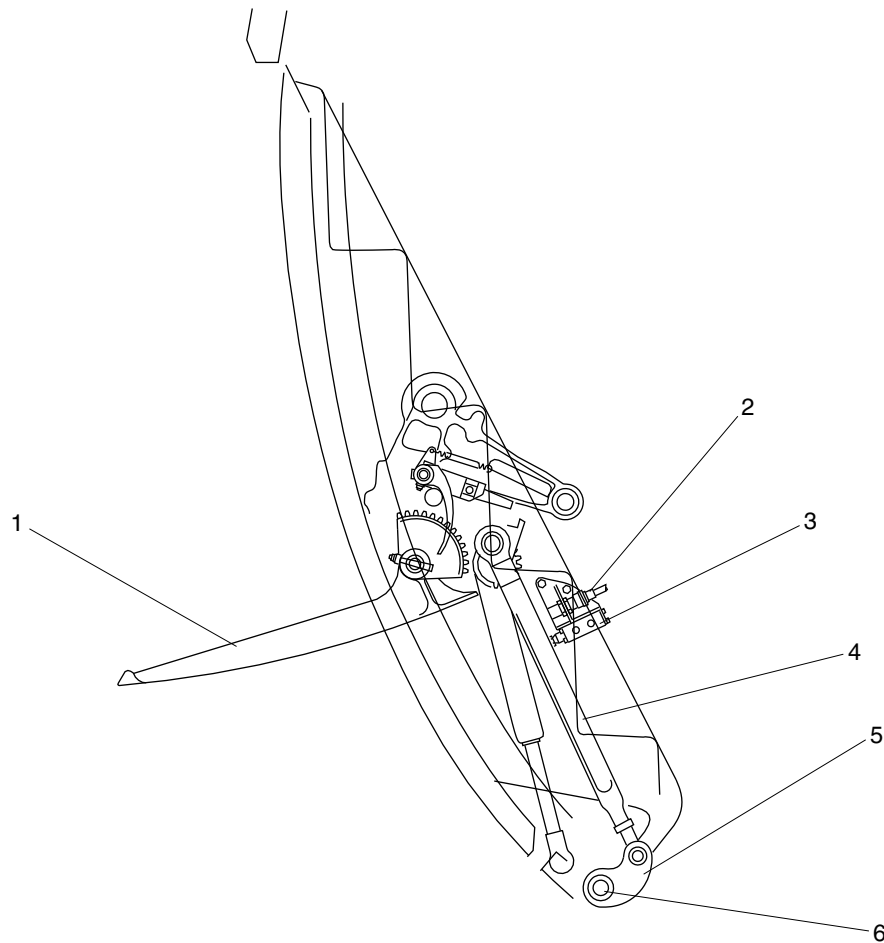
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. External Handle.
2. Proximity Sensor.
3. Pressurizing Valve.
4. Actuating Rod.
5. Crank.
6. Door Hinge Shaft.

fs564a01.cgm

FORWARD PASSENGER DOOR LIFT MECHANISM
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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Config 001

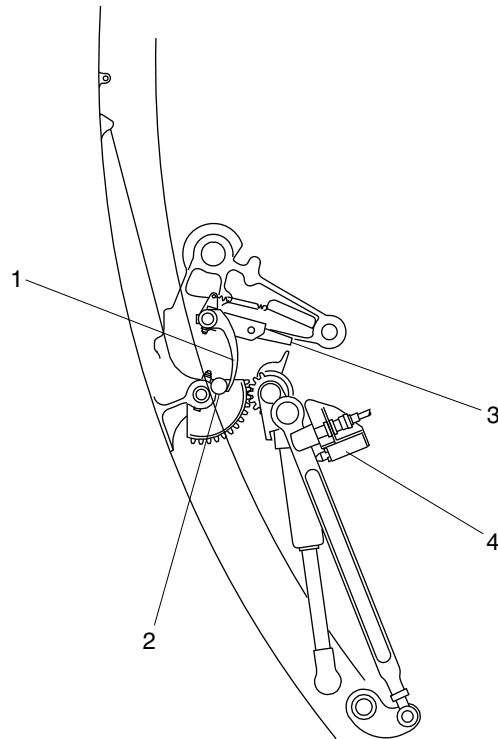
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Trip Lever.
2. Roller (Door Down Position).
3. Locking Bar.
4. Pressurization Valve.

fs561a01.cgm

FORWARD PASSENGER DOOR TRIP LEVER AND LOCKING BAR
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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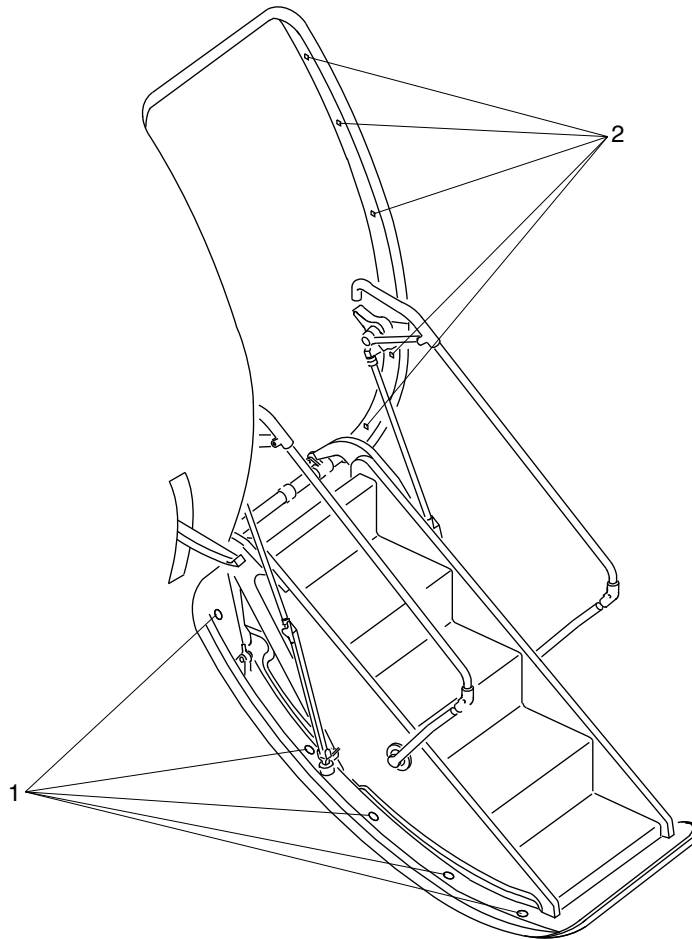
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Door Stop Bolts (5 Per Side).
- 2. Door Stop Brackets (5 Per Side).

fs583a01.cgm

FORWARD PASSENGER DOOR STOP BOLTS
Figure 8

PSM 1-84-2A
EFFECTIVITY:
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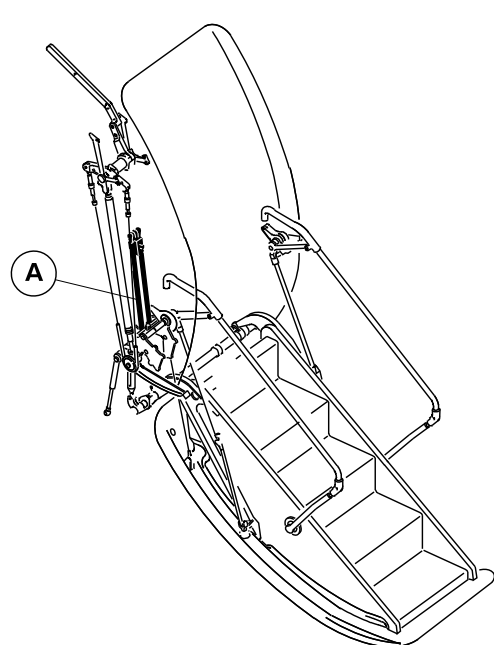
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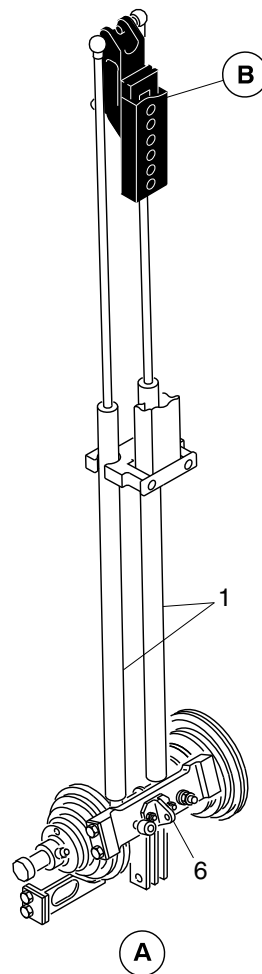


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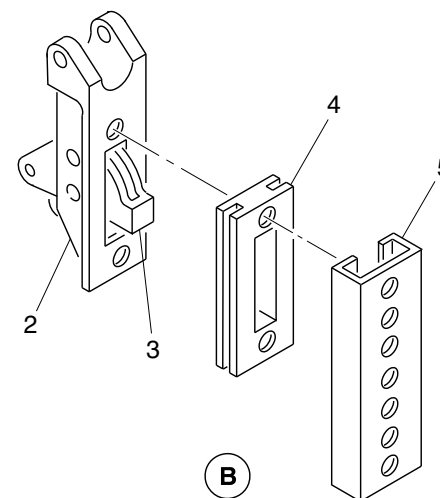


FWD PASSENGER DOOR



LEGEND

1. Gas springs.
2. Chassis.
3. Pulley.
4. Slide block.
5. Channel-pulley guide.
6. Bracket.



fs039a01.dg, vn, aug22/2011

Forward Passenger Door Balance System
Figure 9

PSM 1-84-2A
EFFECTIVITY:
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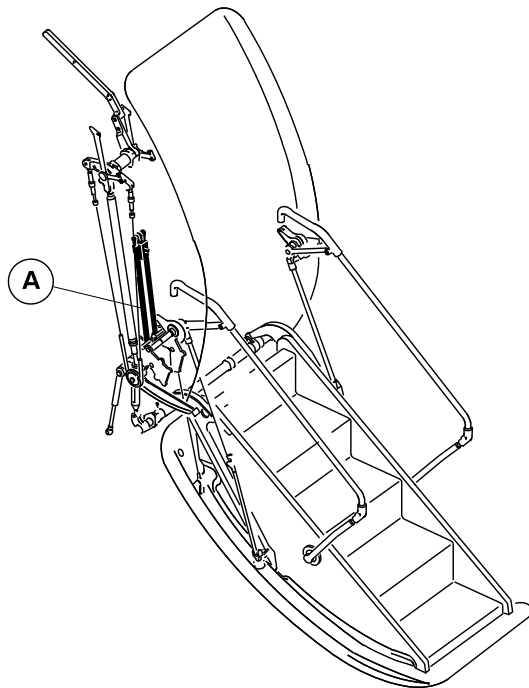
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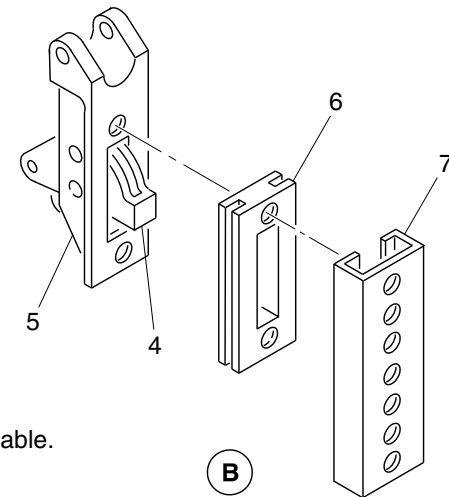
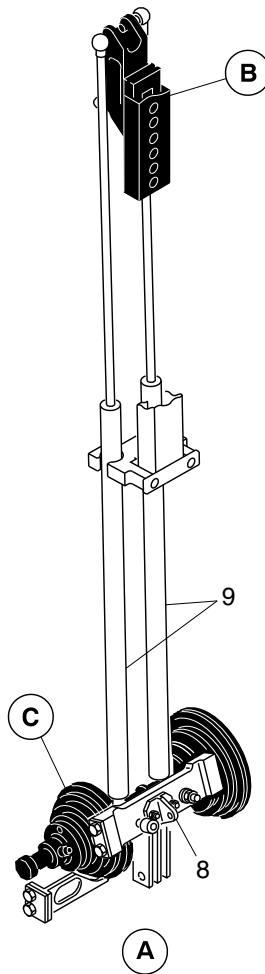


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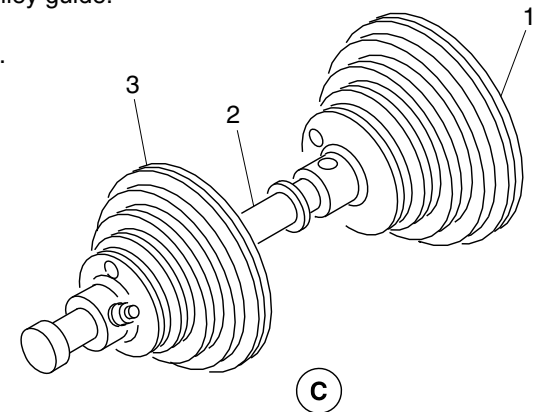


FWD PASSENGER DOOR



LEGEND

1. Drum–cable door.
2. Shaft.
3. Drum–spring strut cable.
4. Pulley.
5. Chassis.
6. Slide block.
7. Channel–pulley guide.
8. Bracket.
9. Gas springs.



fs040a01.dg, vn/gv, sep26/2011

Forward Passenger Door Balance System Detail
Figure 10

PSM 1-84-2A
EFFECTIVITY:
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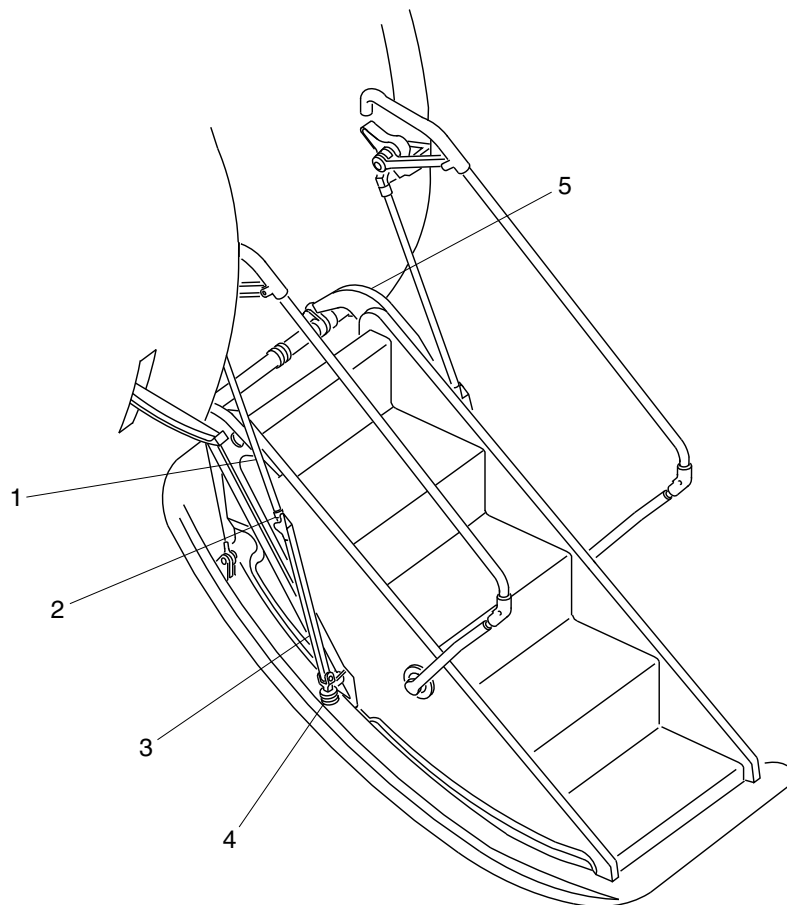
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LEGEND

- 1. Upper strut.
- 2. Overtravel positive stop.
- 3. Lower strut.
- 4. Rubber bumper.
- 5. Gooseneck.

fs562a01.dg, vn, aug22/2011

Forward Passenger Door Struts
Figure 11

PSM 1-84-2A
EFFECTIVITY:
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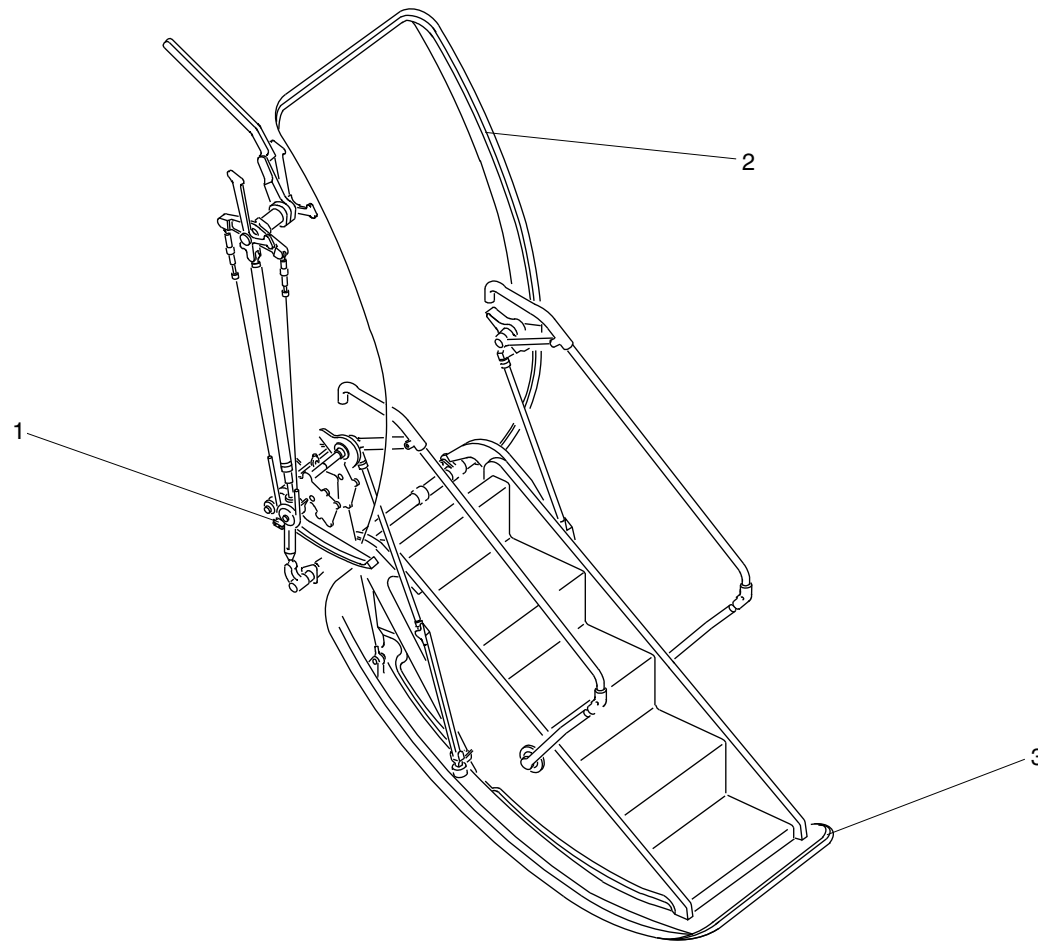
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LEGEND

- 1. Pressurization Valve.
- 2. Inflatable Seal.
- 3. Weather Seal.

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FORWARD PASSENGER DOOR SEAL
Figure 12

PSM 1-84-2A
EFFECTIVITY:
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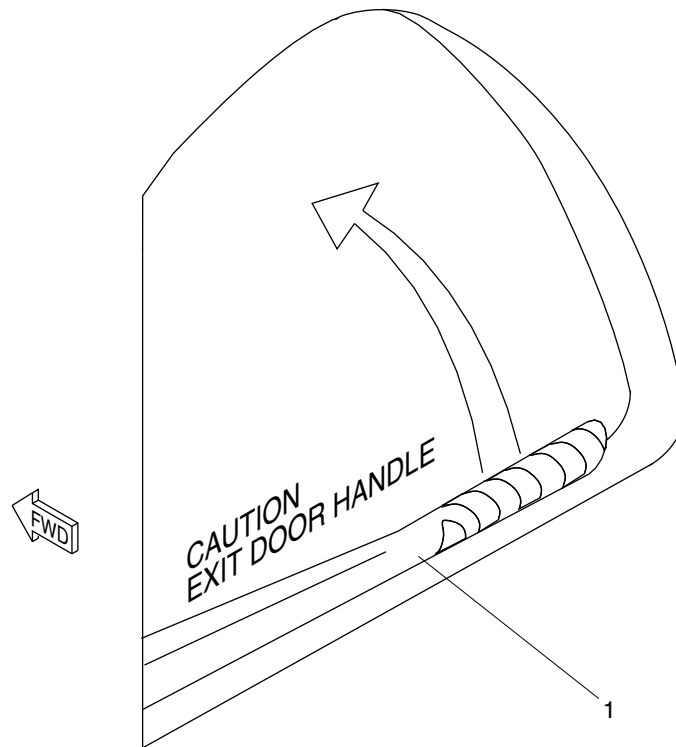
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LEGEND

1. Internal Door Handle.

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FORWARD PASSENGER DOOR INTERNAL HANDLE
Figure 13

PSM 1-84-2A
EFFECTIVITY:
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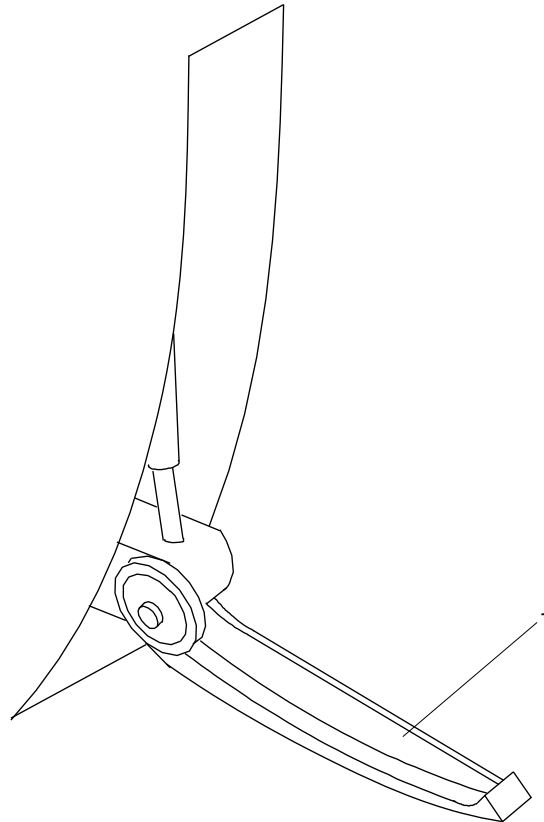
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LEGEND

1. External Door Handle.

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FORWARD PASSENGER DOOR EXTERNAL HANDLE
Figure 14

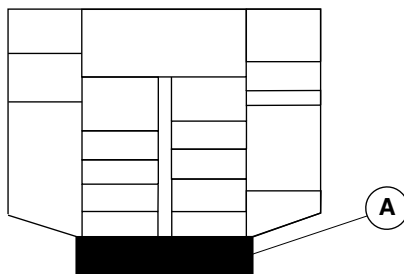
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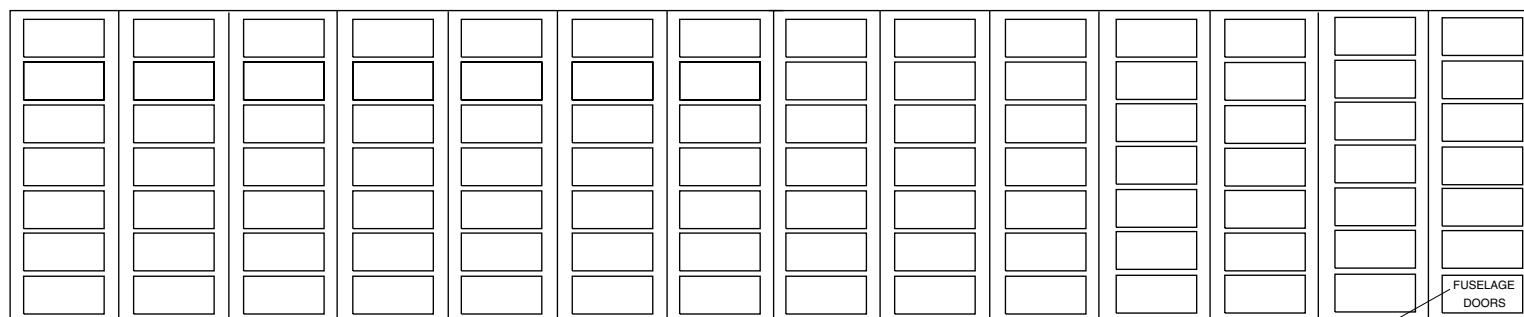
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OVERHEAD CONSOLE

LEGEND

1. Fuselage Doors (Red).



1

A

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Fuselage Doors Warning Light
Figure 15

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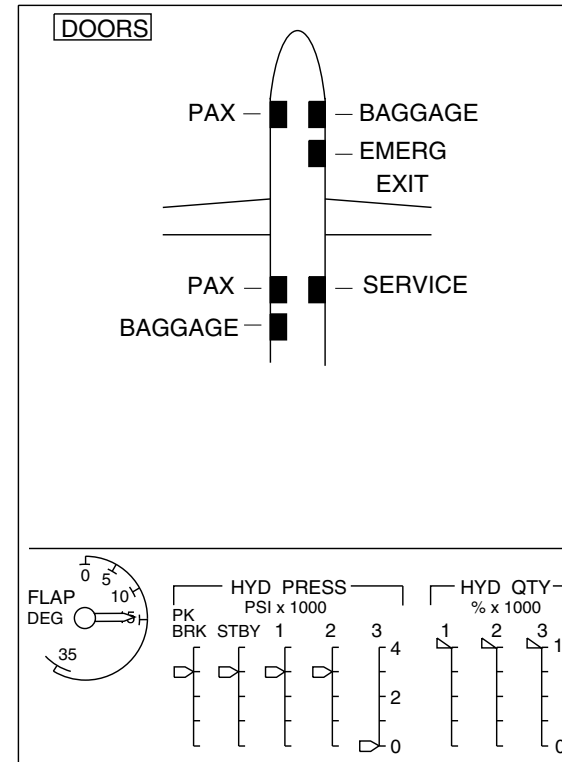
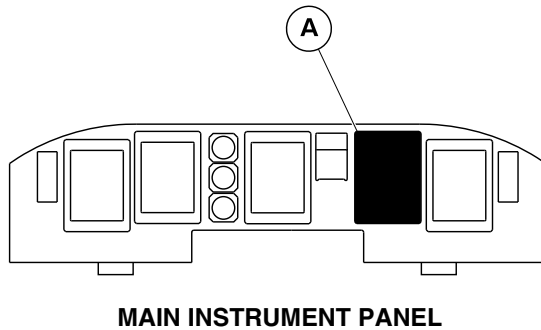
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PRE MODSUM 4-458969
PRE MODSUM 4-459339
PRE MODSUM 4-190614
POST MODSUM 4-309306**

fs747a01.dg, cm/cm, may03/2020

Doors Systems Page
Figure 16 (Sheet 1 of 2)

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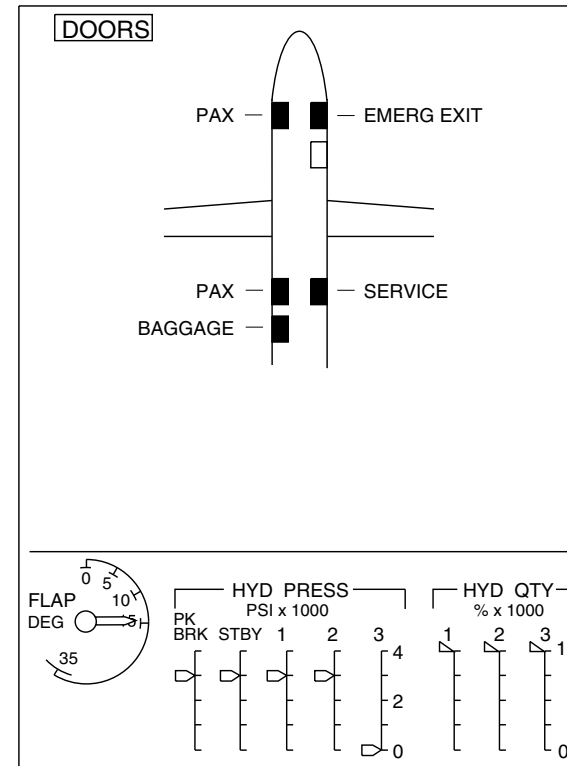
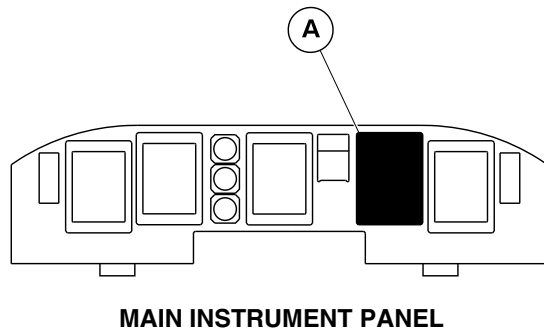
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POST MODSUM 4-201647

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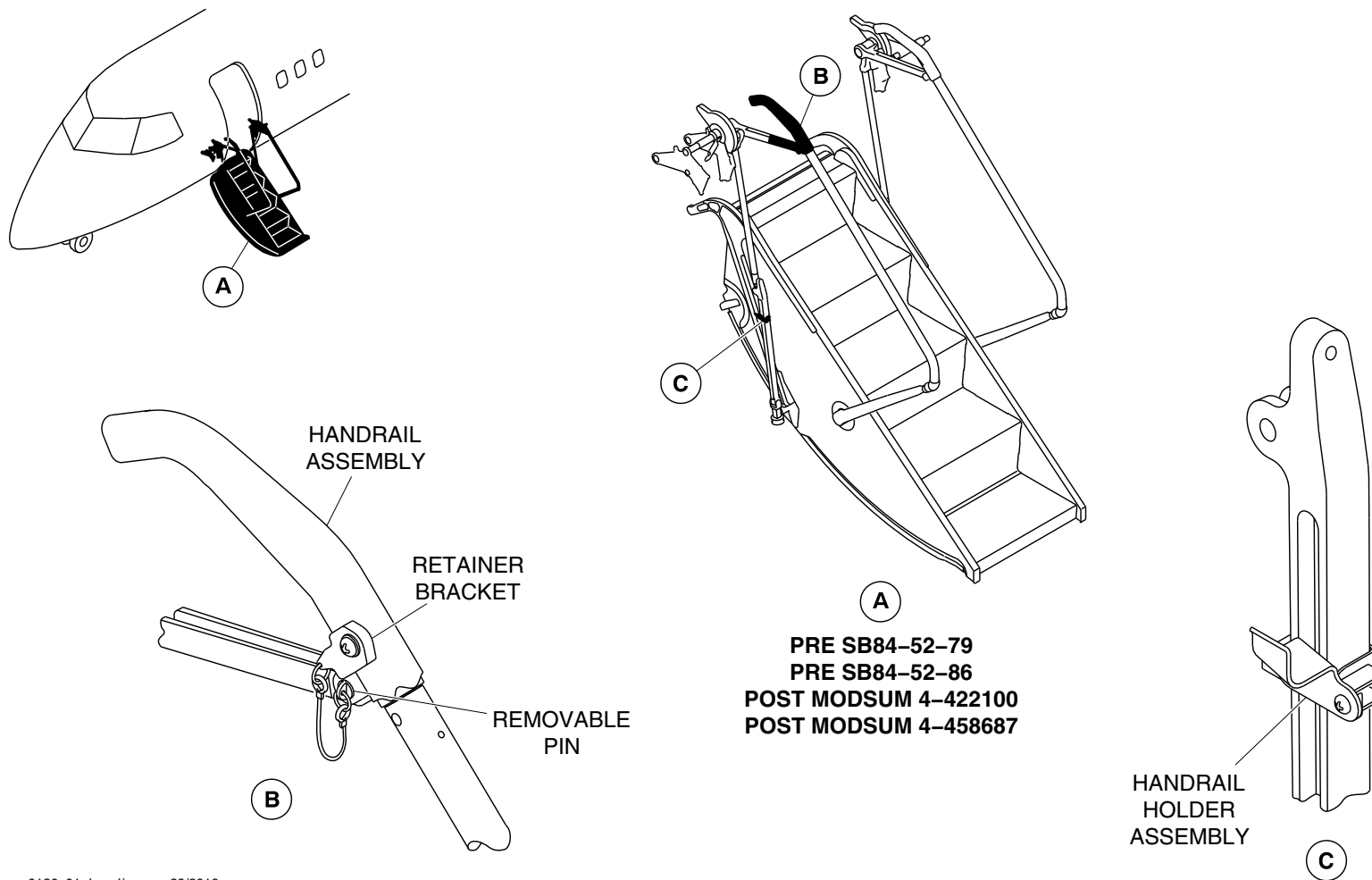
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cg0136a01.dg, sd/nn, mar29/2016

Airstair Door with Jetway – Folding Handrail
Figure 17

PSM 1-84-2A
EFFECTIVITY:
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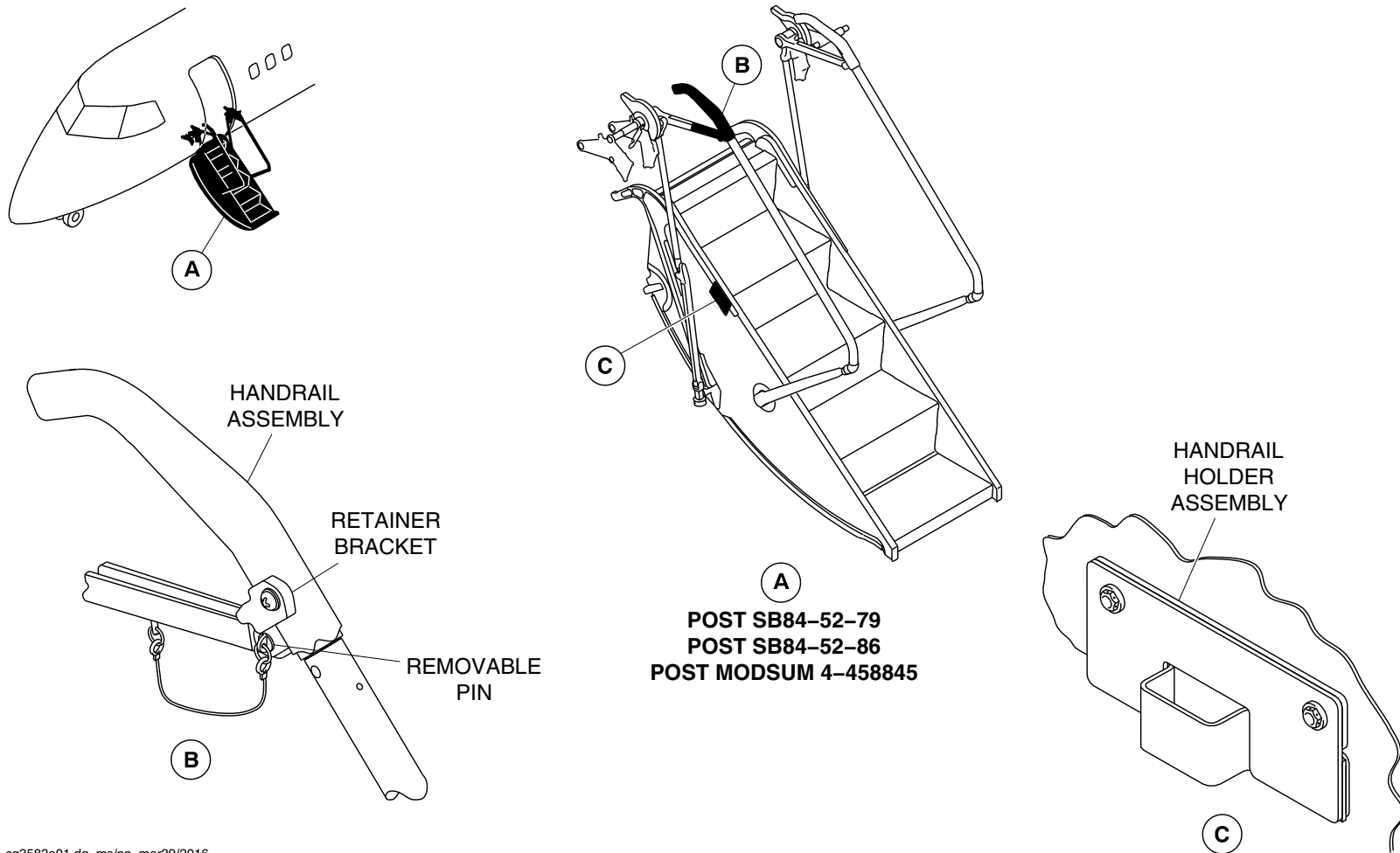
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cg3583a01.dg, ms/nn, mar29/2016

Airstair Door with Jetway – Folding Handrail
Figure 18

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

52-12-00-001

AFT PASSENGER DOOR

Introduction

The aft passenger door gives access to the passenger compartment, and also functions as a Type I emergency exit.

On aircraft with modsum 4-459324, the aft passenger door is permanently locked.

General Description

Refer to Figure 1.

The aft passenger door is located at the rear of the passenger compartment on the left side of the aircraft. The aft passenger door is classified as a type I emergency exit. It is a translating type secured against pressurization loads by six stops located on each vertical edge, that engage with twelve related stop brackets installed on the door structure.

The door is unlocked by operation of a spring loaded internal or external handle. The door opens or closes with the door face parallel with the fuselage skin. Proximity sensors monitor the position of the upper fixed rollers and the door handle. The door warning system monitors door status and supplies visual indication and warning in the flight compartment.

The Aft Passenger Door System has the components that follow:

- Mechanism, Handle (52-12-01)

- Mechanism, Vent (52-12-06)
- Mechanism, Lift and Latch (52-12-11)
- Mechanism, Hinge (52-12-16)
- Seal, Door (52-12-21).

The Aft passenger door interfaces with:

- Door Warning System (52-70-00)
- Proximity Sensors (52-71-00)
- Aft Entry Door Stairs – Optional Configuration (52-12-31).

Detailed Description

Refer to Figure 2.

The aft passenger door gives access to the passenger compartment, and also functions as a Type I emergency exit.

The door operating and retention system is operated by the internal or external handle. The internal handle is pulled inwards and turned clockwise to unlock the handle. When the door handle is pulled inwards the vent door opens to release pressure in the cabin and the proximity switch opens. When the handle is turned clockwise, the lift/latch shaft pushes the lift/latch rollers down, and the door moves inwards and upwards to clear the door stop brackets. The door movement is controlled by two hinges (main and secondary) and a stabilizer.

At the uppermost lifted position, the hold-up hook is connected to the hold-up hook pin of the door hinge, and the door is moved outward. As the door moves outward, the interlock cam connects to a latch crank on the lift/latch shaft to prevent accidental handle operation.

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The door has two fixed rollers located on each side around the upper and lower parts, running in guides on the fuselage which give tangential movement of the door. In pressurized flight, the door is retained by the stops against pressurization and vertical loads. In unpressurized flight, the door is kept from upward movement by the cam rollers on each side of the lift/latch shaft. If one of the two cam roller assemblies fails, it is backed up by the other cam roller, to prevent the door opening in flight. The lift/latch shaft is secured at the closed position by means of both the handle lock mechanism and the over-center springs.

When the door reaches the fully opened position, the door open lock hook on the hinge arm is connected to the door open lock pin on the door to keep the door at its position. An interlock assembly prevents handle operation when the door is opened, preventing the hold-up hook from being unlatched.

[Refer to Figure 3.](#)

The external handle is kept in a recess in the outer skin and is connected to the same handle shaft as the internal handle. The handle can be accessed when the flap covers are pushed in. The handle is then pulled out and turned counter clockwise to open the door.

For safety, a window in the door gives a view of the outside before the door is opened.

[Refer to Figure 4.](#)

To close the door, the lock lever adjacent to the lock hook is pulled to release the open lock, and the door is brought to the closed position. The interlock mechanism is ready for disengagement at this position.

The internal handle is turned counter clockwise to disconnect the door hold-up hook from the hold-up hook pin. The door is moved

down to the closed position, and the stops align with their related stop brackets.

The handle is moved outward to lock the door and close the proximity switch. Cabin pressurization is prevented when the vent door is not fully closed and locked. The door can also be closed by using the external handle.

The main door and vent door are sealed around their periphery. The seal is inflated by cabin pressure through holes in the seal, to keep the gap around the door closed.

[Refer to Figure 5.](#)

As an optional configuration the aft passenger door is provided with an aft entry folding stairway. The aft entry folding stairway is used as the second entry door stairs for the entry and exit of passengers

Mechanism, Handle

The internal handle operates the lift/latch shaft, and the vent door operating rod. The lower end of the internal handle is connected to the shaft of the handle mechanism. When the internal handle (or external handle) is pulled full travel, the handle shaft can be turned to operate the lift/latch through cranks, rods and an idler link. The external handle is kept flush with the door skin and locked in the closed position. Pulling and turning the handle turns the handle and lift/latch shafts.

Mechanism, Vent door

The linkage from the handle transmits the handle input to the vent door mechanism. When the door handle is pulled, the vent door opens to release pressure from the cabin. When the door is in the closed position, the final movement to lock the internal handle or



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

external handle closes the vent door. The vent door is spring loaded in the direction of the open position, so that it is kept open in case of fracture or disconnection of the vent door operating mechanism.

The primary function of the vent door is as follows:

- To release residual cabin pressure before the door is opened
- To prevent pressurization loads on the door before the door is fully closed and locked.

Mechanism, Lift and Latch

The linkage from the handle to the lift/latch shaft transmits handle input to the shaft. It locks the lift/latch shaft assembly when the handles are in their locked position.

The bias spring assembly for the latch causes an overcenter against the lift/latch shaft, and locks it at the closed or open position.

The overcenter position and linkage from the handle mechanism, gives dual locking of the lift/latch shaft.

The lift/latch shaft is installed to a cam roller assembly at both ends. When the door is opened, the lift/latch shaft turns and let the rollers push down the guide tracks, lifting the door for outward movement. The door operation sequence is reversed when the door is closed. When the door is in the closed position, the roller center is offset inward against the door lift-up plane, resulting in the rollers being kept in the door closing direction, against the upward movement of the door.

Mechanism, Hinge

The main and secondary hinge arms keeps the door face parallel with the fuselage skin during door opening or closing. The stabilizer rod helps stabilize the door during opening and closing.

Seal, Door

The door seal is of the passive type. Several holes in the seal let it be inflated by cabin pressure. This type of seal is also installed on the vent door.

Aft Entry Door Stairs – Optional Configuration

[Refer to Figure 5.](#)

The aft passenger door has an optional configuration to have a provision of aft entry stairs. The stair assembly is a folding stairway and it is the second entry door stairs for the entry and exit of passengers. The stair assembly is located between fuselage Sta. X672.400 and Sta. X695.200 on the LH side of the fuselage. When not in use, the stair assembly is stowed in a stowage compartment, aft of the doorway which extends into the aft baggage area.

Aft Entry Stairs

In the deployed position, the aft entry stair assembly is designed to accommodate different loading conditions of the aircraft. When not in use, the stair assembly is stowed in a stowage compartment, aft of the doorway which extends into the aft baggage area. The stair assembly is designed as a maintenance free item. In the construction of the stair assembly, sealed bearings and bronze bushings are used in all hinges and attachments. This feature makes the stair assembly almost lubrication free.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The aft entry stair assembly system has the components that follow:

- Assembly, Carriage (52–12–31)
- Assembly, Stair (52–12–31)
- Assembly, Handrails (52–12–31)
- System, Counterbalance (52–12–31)
- System, Locking (52–12–31).

Detailed Description

[Refer to Figure 5.](#)

The stair assembly is designed for ground use only and cannot be operated until the aft passenger door is opened. Also it must be noted that the aft entry stair assembly is not to be used in an emergency evacuation. The stair assembly is deployed from the inside of the aircraft when the aft entry door is opened. The stair assembly is attached to a carriage assembly which allows the stairs to move forward and aft into the stowed and deployed positions. The carriage assembly rides on floor level rails.

To deploy the stairs:

The release pedal on the main forward hinge of the carriage assembly is pressed to disengage the catch fitting on the floor. This unlocks the carriage assembly from the stowed position. The stair assembly now can be pulled from the stowage compartment to the door opening. The stair assembly is moved to the door opening until the carriage assembly contacts its stops. This position of the stair assembly is the stairs deployed position. The carriage assembly is locked in this deployed position. The release pedal on the main forward hinge of the carriage engages the catch fitting on the floor.

To deploy, the stowage lock from the stairs is released and the stair assembly is pushed out with the assist handles. The stair assembly unfolds on the way down. When the stair assembly is in the fully extended position with the wheels touching the ground, the lock latch is engaged with the lock pin. This locks the upper and lower stairs together to be rigid as one unit.

To retract the stairs:

The lock latch is released from the locked position when the foot lever on the forward outer stair lug is pushed down. This removes the lock between the upper and lower stairs to allow the stairway to be folded. The stair assembly now can be pulled with the handrails inboard and down to lift and retract to the stowed position. In this position the stair assembly is within the fuselage profile. Stowage lock on the upper stairs is engaged to the lower stairs. The release pedal on the main forward hinge of the carriage assembly is pressed to disengage the catch fitting on the floor. This unlocks the carriage assembly from the stairs deployed position.

The stair assembly is pushed from the door opening to the stowage compartment until the carriage assembly contacts its stops. This position of the stair assembly is the stairs stowed position. The carriage assembly is locked in this stowed position. The release pedal on the main forward hinge of the carriage engages the catch fitting on the floor.

If necessary, ground personnel can operate the stair assembly from the outside of the aircraft. Stair assembly can be deployed and retracted from the outside of the aircraft. Placards are installed on the interior of the aircraft and on the stair assembly. These placards give instructions to operate the stair assembly in different modes to the crew and maintenance personnel.



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Assembly, Carriage

The carriage assembly allows the stairs to be translated forward and aft into the stowed and deployed positions. The carriage assembly is designed to ride on floor level rails. The rails and the necessary support structure are added to the aircraft product structure as part of the optional configuration. The stair is attached to the forward and aft hinge brackets of the carriage assembly. The hinge brackets are joined as an assembly via the inboard and outboard carriage tie members. The outboard tie member is fitted with wheels and slider blocks. These allow the carriage assembly to slide along the aircraft mounted fixed rails. The release pedal installed on the main forward hinge bracket engages two respective catch fittings on the floor. These fittings lock the carriage assembly in the stowed and deployed positions of the stairs. All rails and the necessary support structure are protected against contamination and debris with covers and nylon seals.

Assembly, Stair

The stair assembly consists of upper and lower stairs. Upper stair includes two upper stringboards and three steps. The lower stair includes two lower stringboards and four steps. All horizontal step surfaces are covered with non-skid material. The upper stair is attached to the lower stair with two hinges. They are the forward and aft hinges. The other end of the upper stair is connected to the main hinges of the carriage assembly. They are the outboard hinges of the carriage assembly. Two support wheels attached to the lower end of the lower stair keep the stair structure off the ground. The stair is fitted with a spring loaded stowage lock. This holds the stair assembly in the stowed configuration. The stowage lock engages the upper step and can be released from aircraft and ground.

Assembly, Handrails

Handrails assembly includes six bannisters and two handrails. The handrails and bannisters are magnaformed assemblies. They consist of machined 7075 aluminum fittings and 2024 aluminum tubing. Every handrail includes four tubes and four fittings. Each bannister is connected to the stringboard with a hinge. The stair assembly design permits the stairs and the handrails to fold and extend during deployment and retraction.

System, Counterbalance

Motion and balance during retraction and deployment of the stairs is controlled via a gas spring and a tension rod mounted on either side of the stair assembly. The tension rod is mounted between the upper bannister and the carriage assembly secondary hinge. This is the inboard hinge on the carriage assembly. The gas spring connects the upper bannister and the upper stringboard. The gas springs are compressed to give resistance when the stair is lowered. During retraction the gas springs are extended to help lift the stairs.

System, Locking

The locking system engages automatically when the stair assembly is fully extended. This is when the stair assembly is fully deployed. The locking system has the components that follow:

- a foot lever (foot release step) located at floor level inside the aircraft
- a ground release lever
- a rigid push rod and crank arm assembly
- a cable and pulley assembly



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- a spring and lock latch.

Also the stair is fitted with a spring loaded stowage lock, which holds the stair assembly in the stowed configuration. The stowage lock engages the upper step and can be released from the inside of the aircraft or the outside from the ground.

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EFFECTIVITY:
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Config 001

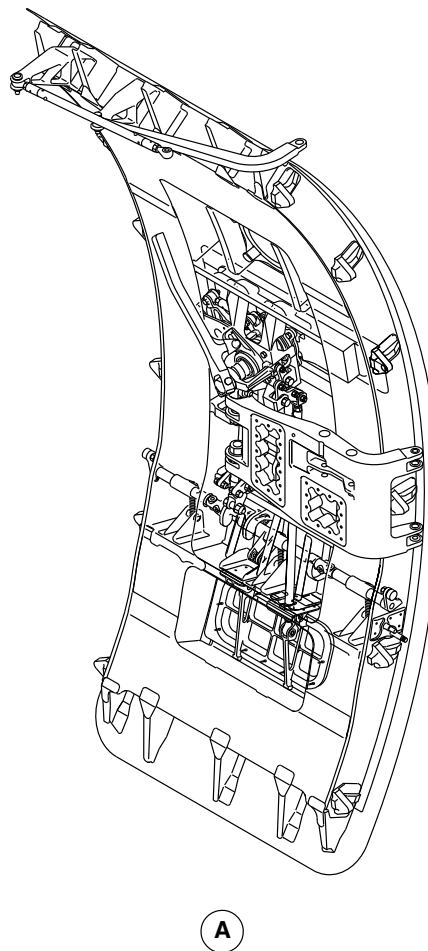
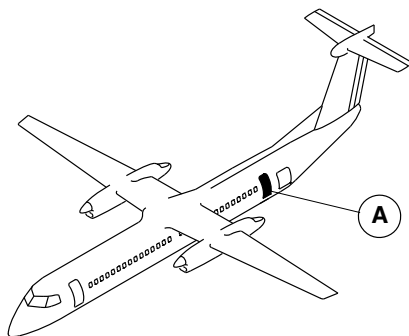
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AFT PASSENGER DOOR LOCATION
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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Config 001

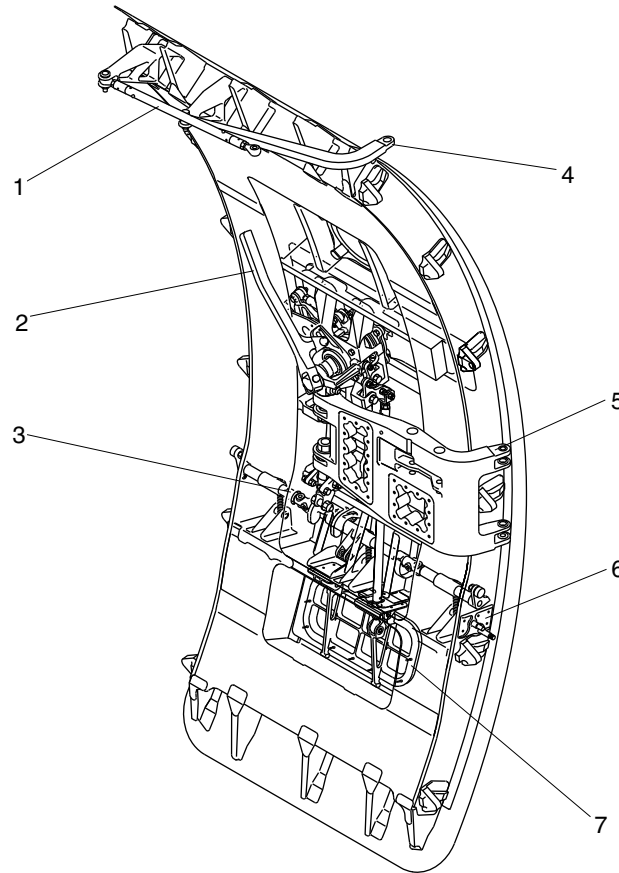
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Stabilizer Rod.
2. Internal Handle.
3. Lift And Latch Shaft.
4. Secondary Hinge.
5. Main Hinge.
6. Interlock Assembly.
7. Vent Door.

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AFT PASSENGER DOOR
Figure 2

PSM 1-84-2A
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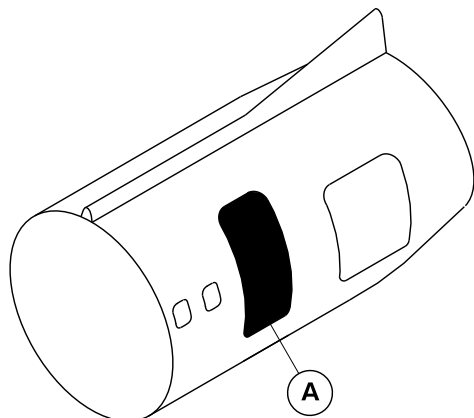
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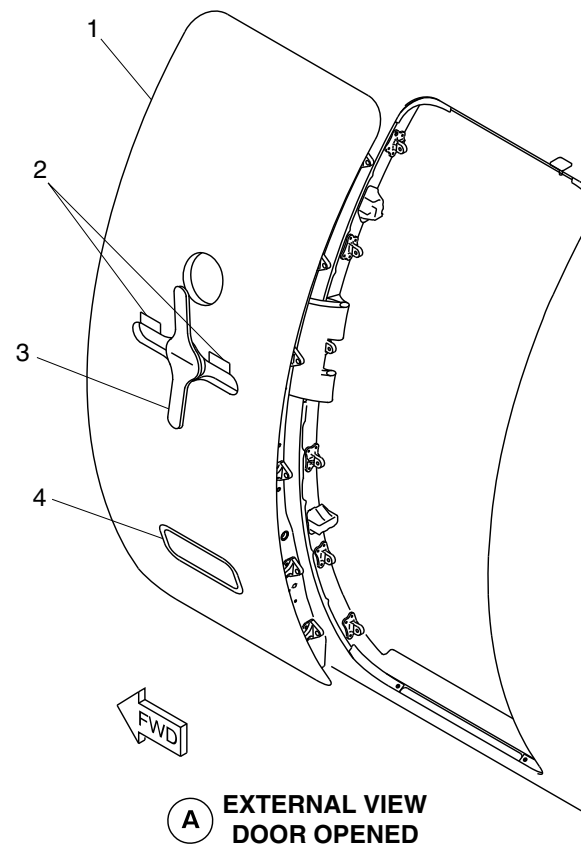
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Aft passenger door (open position).
- 2. Flap cover.
- 3. External handle (unlocked).
- 4. Vent door.



fs862a01.dg, vn, aug22/2011

Aft Passenger Door Exterior
Figure 3

PSM 1-84-2A
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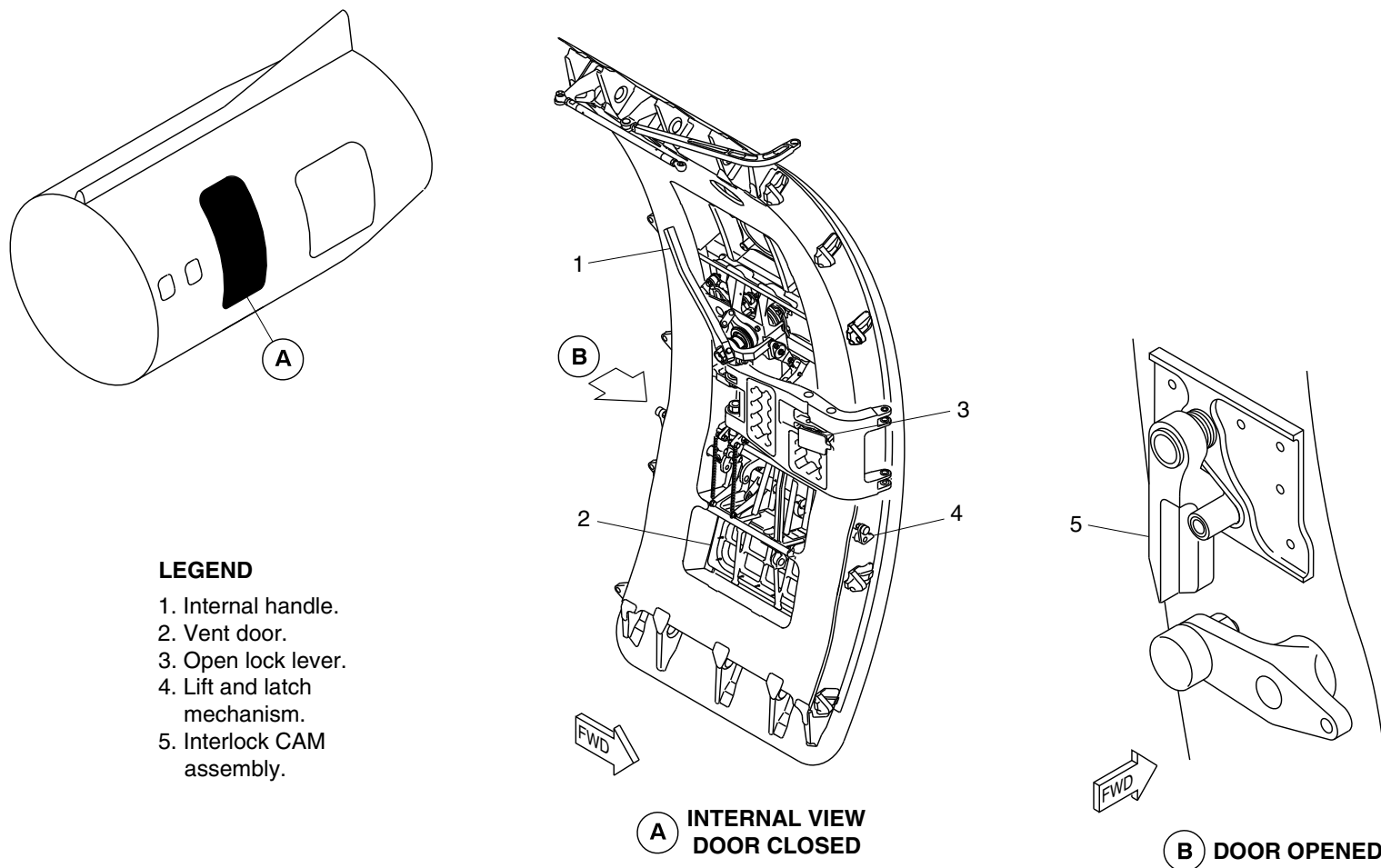
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Aft Passenger Door Internal Detail
Figure 4

PSM 1-84-2A
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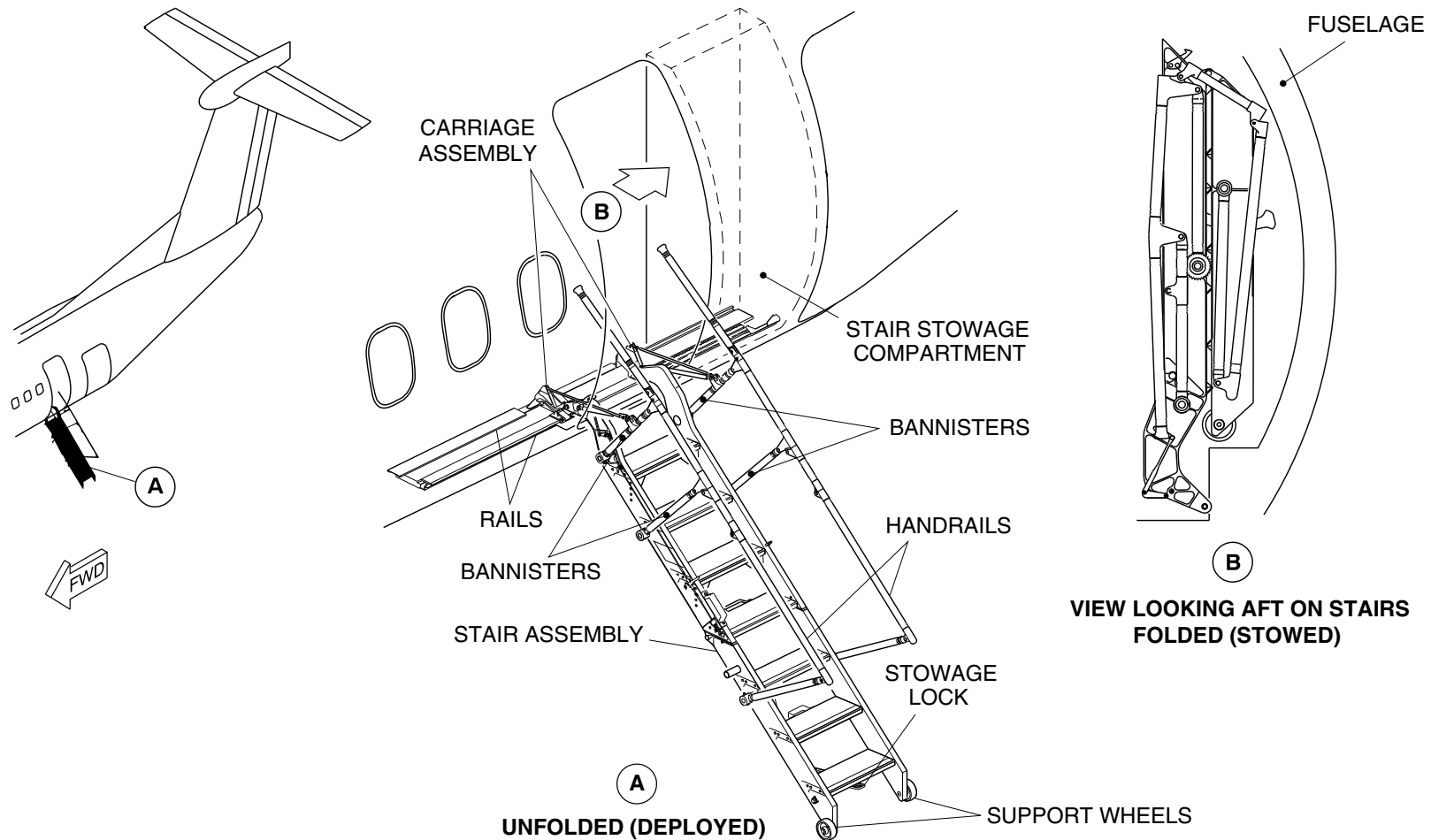
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AFT PASSENGER DOOR ENTRY STAIRS – OPTIONAL CONFIGURATION
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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DOOR SEAL PRESSURIZATION SYSTEM

Introduction

The door seal pressurization system seals the forward passenger and aft baggage doors to prevent loss of cabin pressure.

General Description

The door seal pressurization system uses airframe deicing air to inflate seals on the forward passenger door and the aft baggage door. The seals inflate when the door is closed and the door handle is in the closed position. The seals deflate when the door handle is turned to the open position. If the aircraft engines are not operating, the air is supplied from a pressurized reservoir tank. When the aircraft is on the ground, the tank is pressurized through a charging valve located in the wardrobe.

The door seal pressurization system has the components that follow:

- Tank, Reservoir (52-13-01)
- Valve, Drain (52-13-06)
- Check Valve, Heated (52-13-11)
- Valve, Charging (52-13-16)
- Lines (52-13-21)
- Filter, Door Seal (52-13-26)

- Valves, Control (52-11-36, 52-31-31)
- Electro pneumatic shut-off valve (52-11-40).

Detailed Description

[Refer to Figure 1.](#)

The door seal pressurization system is connected to the airframe deicing system. When the deicing system is pressurized, air is routed through the heated check valve to pressurize the reservoir tank and the door seals. When the engines are not operating, the reservoir tank supplies the door seals with pressurized air through one outlet. A pipe connected to this outlet divides into two lines, one for each door seal. Each door seal inflates when the door handle is moved to the locked position. Air is distributed to the seal through the control valve. When the handle is moved to the unlocked position, the seal is depressurized through the control valve.

The inflatable seals are located on the door surround of the passenger door and on the aft baggage door itself. The seals for the two doors have the same cross-section. When the seals are pressurized, they extend up to 1 inch and form a seal by contacting the facing structure. Each seal has a fabric core wrapped in silicon material. The passenger door seal has a polyester core. The aft baggage door seal has a NOMEX core for increased fire protection.

A drain valve is fitted to the system to allow depressurization and water draining of the system, for maintenance action. [Refer to Figure 4 and 5.](#) After maintenance the reservoir can be pressurized on the ground, through the charging valve located in the wardrobe on aircraft with PRE MODSUM 4-153583. On aircraft with POST MODSUM 4-153583 through the charging valve located below the floor at the forward passenger door entrance.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Tank, Reservoir

Refer to Figures 2 and 3.

The reservoir tank holds enough air for five door opening and closing sequences. It has a volume of 652 cubic inches (10.7 litres) when not pressurized by the deice system. It is located in the pressurized part of the centre fuselage under the cabin floor.

Valve, Drain

Refer to Figure 3.

The manually operated drain valve is used to depressurize the system before maintenance on the door seal system. It is also used to drain any water that may be in the reservoir. The valve is operated from the outside of the aircraft under the fuselage.

Check Valve, Heated

The valve allows airframe deicing pressure to flow to the door seal system and prevents reverse flow of air, back to the deicing system. The valve is electrically heated to prevent moisture, which may be in the deicing air supply, from freezing inside the valve. The Timer Monitor Unit (TMU) of the deicing system controls the heater, with inputs from Static Air Temperature sensors 1 and 2. If both SAT sensors indicate that the temperature is greater than 9 °C, the heater is deactivated. The heater is activated when one SAT sensor indicates that the temperature is less than 5 °C. Failure of the heater or failure to activate the heater is detected by the TMU and the DEICE TIMER caution light comes on.

Valve, Charging

Refer to Figures 4 and 1.

The manually operated charging valve allows the system to be pressurized from an external source. This allows opening and closing of the two doors without operating an engine.

Lines

All the components in the system are connected by the aluminum plumbing lines.

Filter, Door Seal

On aircraft with MODSUM 4-113726 or SB 84-52-71 incorporated, a door seal filter, which contains a desiccant, is installed. The desiccant removes the moisture if present in the bleed air, before it enters the inflatable door seal of the aft baggage compartment and the air stair door.

On aircraft with SB84-52-91 or MODSUM 4-113942 incorporated, a door seal filter, which contains a desiccant, is not installed.

Valves, Control

The control valves allow inflation or deflation of the passenger and aft baggage door seals. Each control valve is installed on the appropriate doorframe, within the door mechanism. The control valve is a three way mechanical valve. One port is connected to the pressure line, while the second port is connected to the seal. The third port is connected to the ambient air and acts as a relief port to deflate the seal.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The door handle mechanisms on the passenger and aft baggage doors operate their related control valves. When the handle is in the locked position, the seal port is connected to the pressure port and the seal is inflated. When the handle is turned to open the door, the seal port is connected to the relief port and the seal deflates. The passenger door has a pushbutton which activates its control valve (refer to SDS 52–11–00). The aft baggage door has a roller which activates its control valve (refer to SDS 52–31–00).

Electro pneumatic shut-off valve

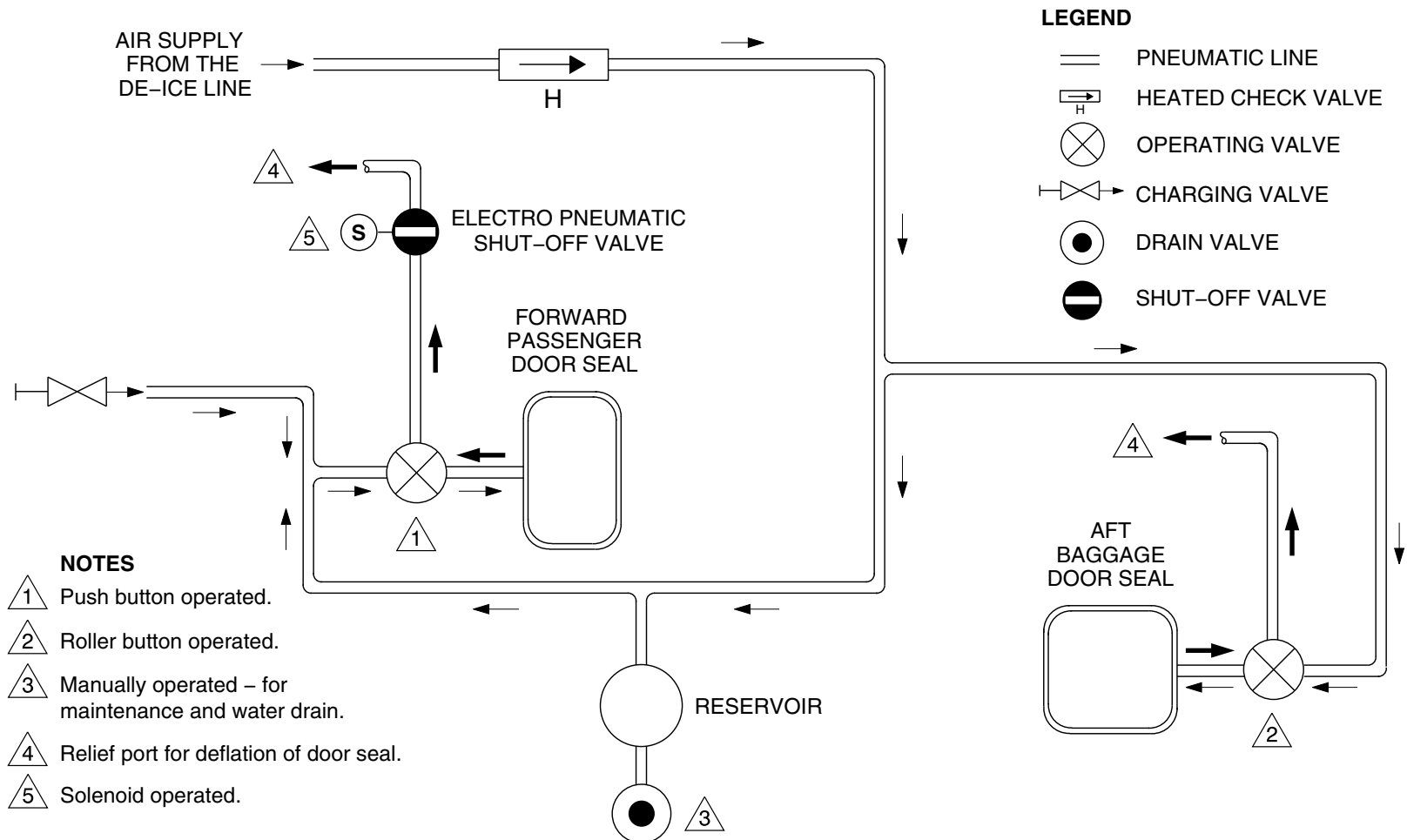
Refer to Figures 1 and 2.

The solenoid operated valve is installed on the relief port line of the forward passenger door seal pressurization valve. On the ground when the door is unlocked, the handle mechanism operates the control valve and connects the door seal port to the relief port inside the valve and the seal deflates. The seal deflation pressure passes through this shut-off valve to the ambient air.

The primary function of this electro pneumatic shut-off valve is to prevent depressurization of the door seal system if airspeed is more than 100 knots (airspeed > 100 knots). If the airspeed is 90 knots or on the ground the valve will allow the door seal system to be depressurized. This feature prevents inadvertent door seal depressurization for the forward passenger door. The airspeed signals to the valve are provided by the aircraft air data system.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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Door Seal Pressurization System
Figure 1

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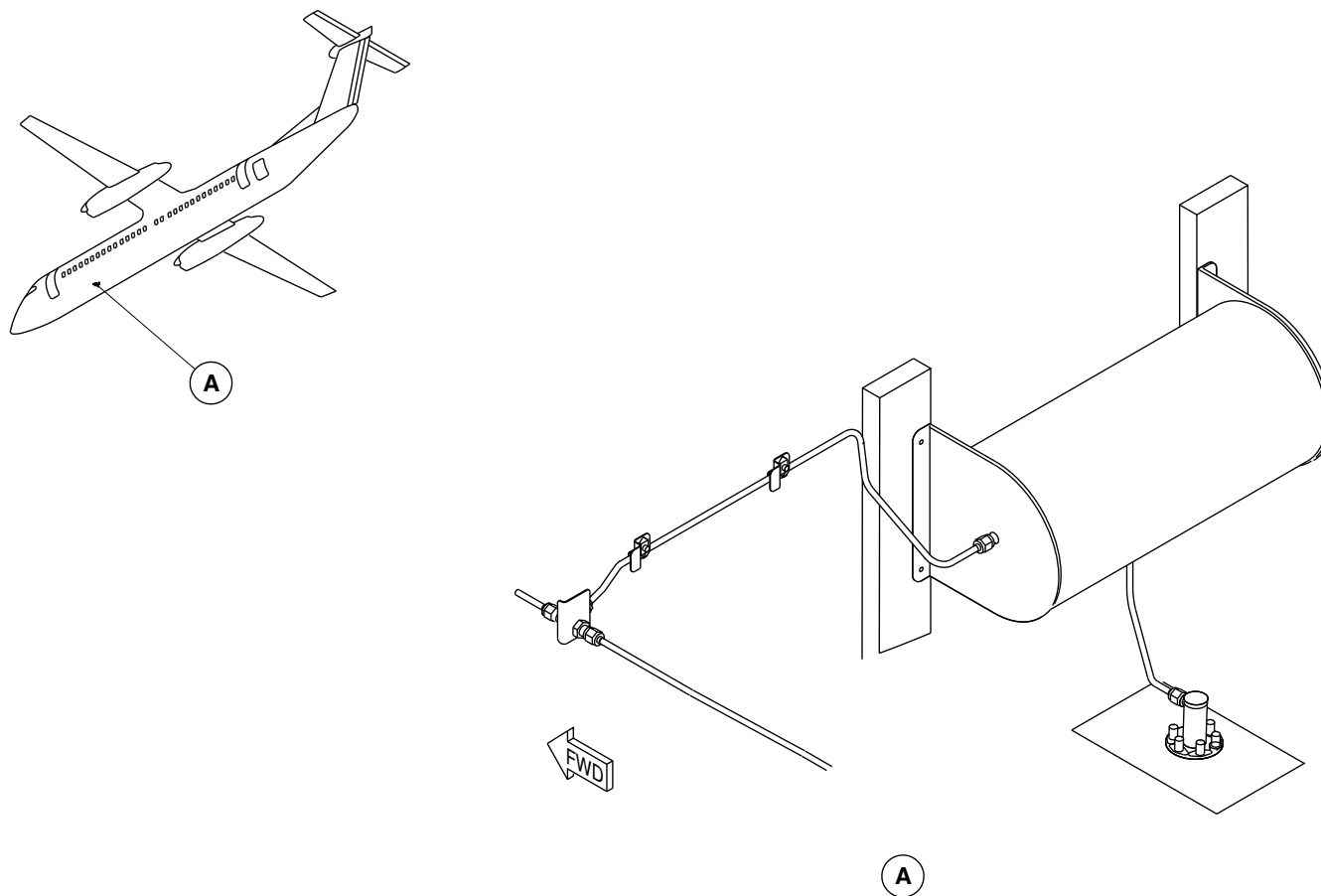
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DOOR SEAL PRESSURIZATION RESERVOIR LOCATOR
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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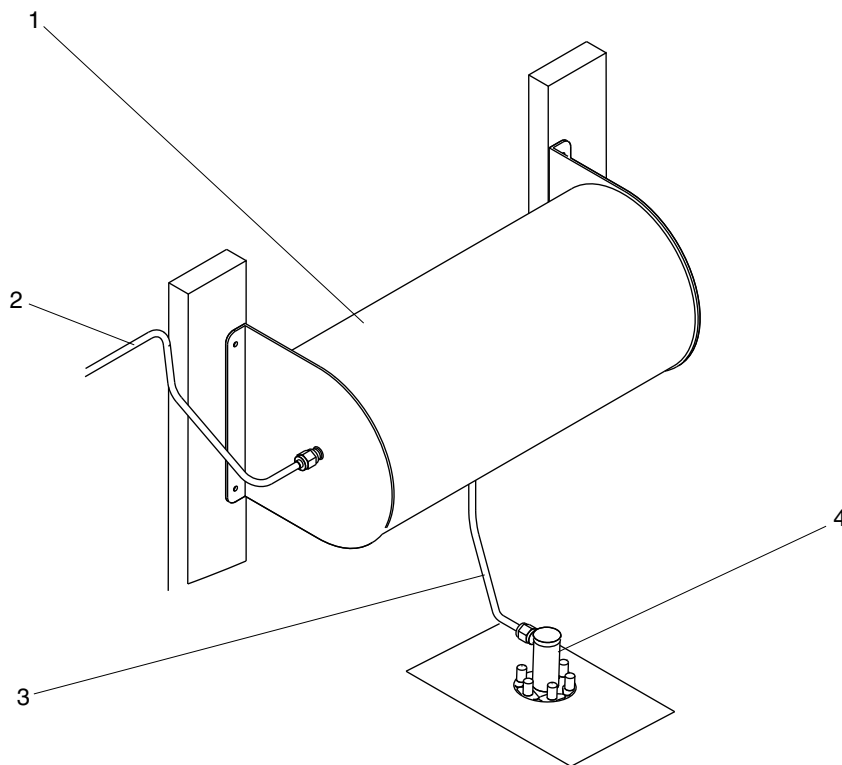
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LEGEND

- 1. Reservoir Tank.
- 2. Door Seal Line.
- 3. Drain Line.
- 4. Drain Valve.

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DOOR SEAL PRESSURIZATION RESERVOIR DETAIL
Figure 3

PSM 1-84-2A
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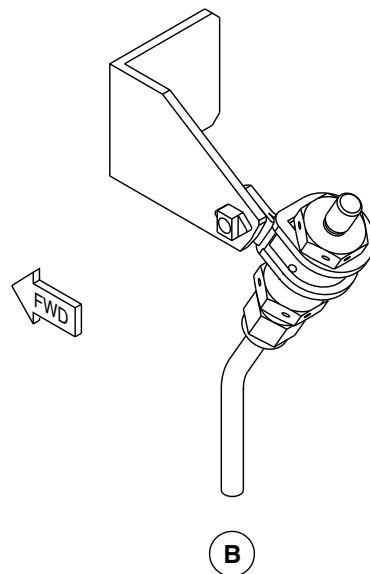
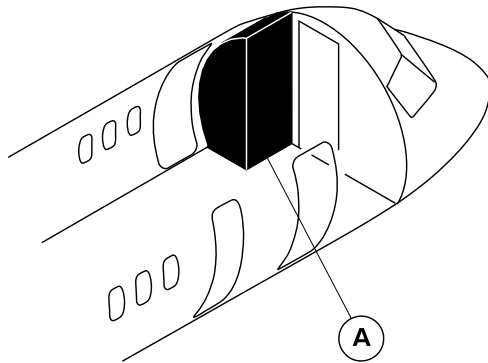
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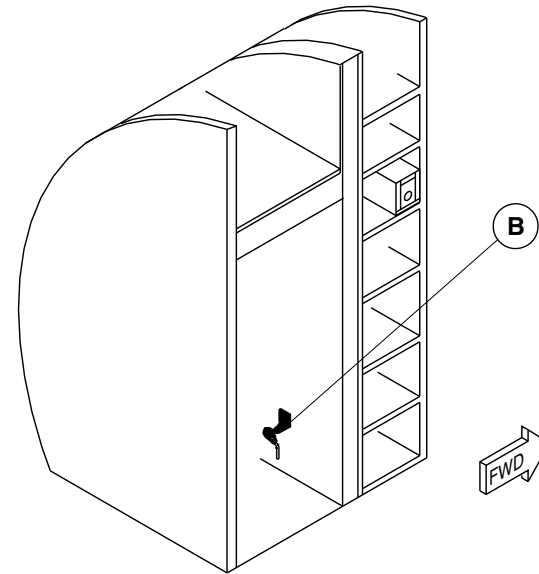


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PRE MODSUM 4-153583



A AVIONICS RACK ELECTRICAL
INSTALLATION

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DOOR SEAL PRESSURIZATION CHARGING VALVE (PRE MODSUM)
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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LEGEND

- 1. Inflatable seal.
- 2. Charging valve.
- 3. Pneumatic line.

**BELOW THE FLOOR INSTALLATION
FORWARD PASSENGER DOOR ENTRANCE**

B POST MODSUM 4-153583

DOOR SEAL PRESSURIZATION CHARGING VALVE (POST MODSUM)
Figure 5

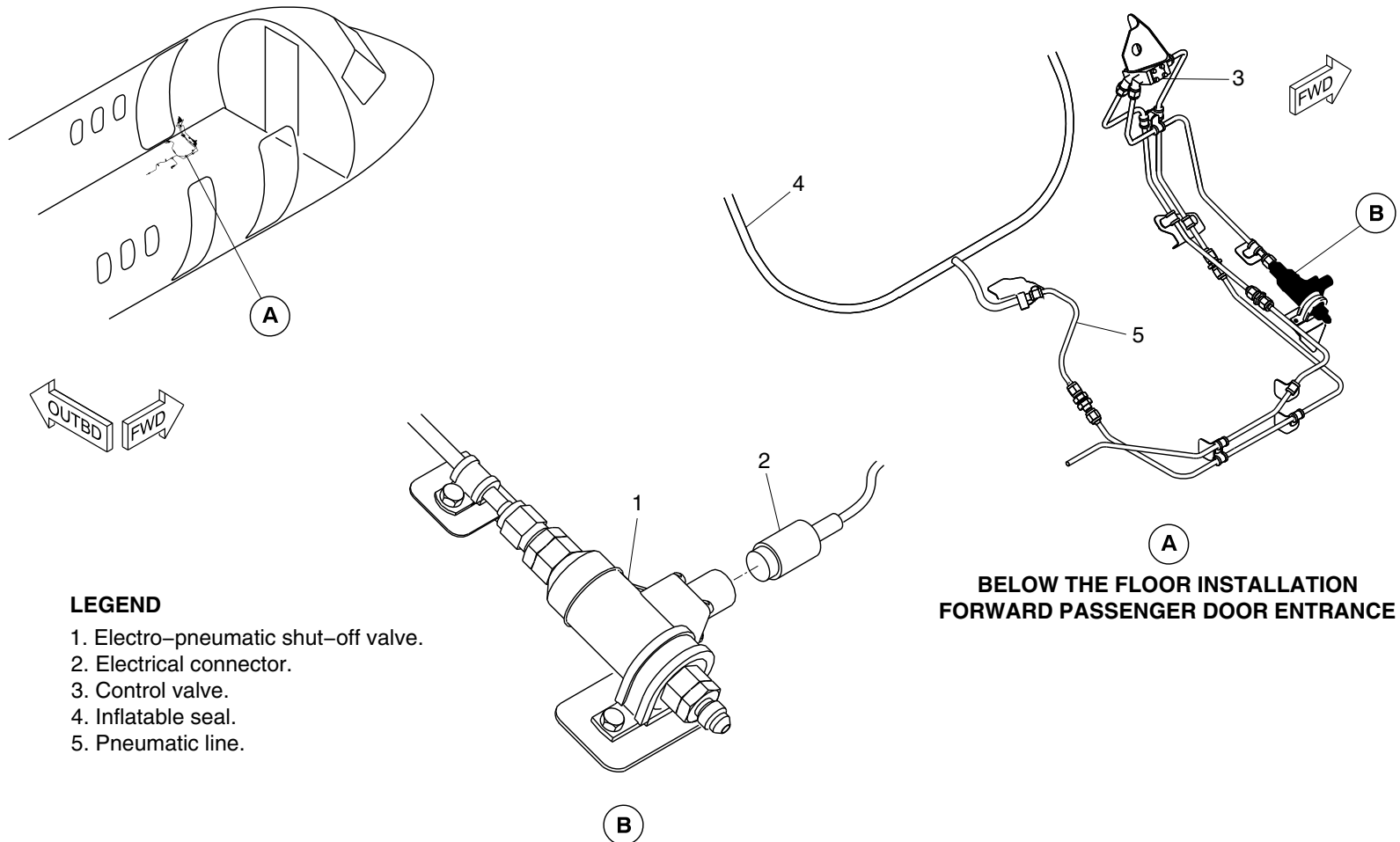
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DOOR SEAL PRESSURIZATION – ELECTRO PNEUMATIC SHUT-OFF VALVE
Figure 6

PSM 1-84-2A
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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52-20-00-001

EMERGENCY EXITS

Introduction

The emergency exits gives access to and from the passenger and flight compartments for emergency evacuation.

General Description

Refer to Figure 1.

There are four emergency exits located in the passenger compartment, three Type I and one Type II. On aircraft with Modsum 4-190561 incorporated, the Type II emergency exit door is eliminated. The door warning system monitors emergency exit door status and supplies visual indication and warning in the flight compartment. There is an emergency exit located in the flight compartment roof.

The three Type I exits are:

- Forward Passenger door
- Aft Passenger door
- Aft Service door.

The emergency exits include the components that follow:

- Flight Compartment Escape Hatch (52-21-00)
- Upper Forward Emergency Type II Exit Door (52-22-00)

- Lower Forward Emergency Type II Exit Door (52-23-00).

On aircraft with Modsum 4-458926, 4-458969, 4-459339, 4-190614 or 4-201647 incorporated there are four Type I emergency exits located in the passenger compartment. The door warning system monitors emergency exit door status and supplies visual indication and warning in the flight compartment. There is an emergency exit located in the flight compartment roof.

The four Type I exits are:

- Forward Passenger door
- Aft Passenger door
- Aft Service door
- Forward Emergency Exit Type I door.

On aircraft with Modsum 4-459334 incorporated there are two Type III emergency exits located in the passenger compartment on each side of the fuselage at Sta. X479.70 (Refer to 52-25-00).

Detailed Description

All the Type I, Type II and Type III emergency exits have both internal and external handles. The flight compartment escape hatch is the only emergency exit which does not have an external handle.

Type of Exits

The types of exits are defined as follows:

- Type I (52-24-00)



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Floor level exit with a rectangular opening of not less than 24 in. (610 mm) wide by 48 in. (1219 mm) high.

- Type II (52–22–00/52–23–00)

Rectangular opening of not less than 20 in. (508 mm) wide by 44 in. (1118 mm) high. Type II exits must be floor level unless located over wing.

- Type III (52–25–00)

Rectangular opening of not less than 20 in. (508 mm) wide by 36 in. (914 mm) high. The step up inside the airplane may not be more than 20 in. (508 mm) .

Emergency Exits

[Refer to Figures 2 and 3.](#)

The forward and aft passenger doors are located on the left side of the fuselage and are also used as emergency exits. The forward passenger door opens outwards and down. The aft passenger and aft service door open outward and forward. The aft service door is located on the right side of the aircraft at the rear of the passenger compartment.

[Refer to Figure 4.](#)

The flight compartment escape hatch is located in the flight compartment roof, aft of the pilot seats. It is completely detachable, or can be partially opened to increase the airflow in the flight compartment when the aircraft is on the ground. A rope helps the

flight crew exit the flight compartment aircraft when using the flight compartment emergency escape hatch.

[Refer to Figure 5.](#)

The Type II emergency exit door is located on the forward right side of the passenger compartment. The emergency exit has an upper and lower door. The upper door is a plug type door which is removed completely for opening. The lower door is hinged at the bottom, and opens outward and downward from the floor level.

[Refer to Figure 9.](#)

On aircraft with Modsum 4–458926, 4–458969, 4–459339, 4–190614 or 4–201647 incorporated, a Type I emergency exit door is located on the forward right side of the passenger compartment. The emergency exit opens outward and forward.

Forward Passenger Door

The forward passenger door is located at the left side of the fuselage at the front of the passenger compartment and is classified as a Type I emergency exit. The forward passenger door is hinged at the bottom and has five steps built into the inboard side of the door structure. The steps also strengthen the door. This door has an external aluminum alloy skin braced by cross beams and intercostals.

The door seal pressurization system prevents cabin air leakage through the forward passenger door. Door pressurization loads are carried by ten adjustable pressure stop bolts, five on each side. The stop bolts engage with stop brackets on the door surround structure.

The forward passenger door is the semi-plug type. On opening, the initial movement of the door is upward and inward. The door then rotates outward about the door's hinge.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Aft Passenger Door

Refer to Figure 7.

The aft passenger door is a translating type located at the left side of the fuselage at the rear of the passenger compartment and is classified as a Type I emergency exit. Operation of either the internal or external spring loaded handle unlocks the door. The door opens or closes with the door face parallel with the fuselage.

Aft Service Door

Refer to Figure 8.

The aft service door is located on the right side of the aircraft at the rear of the passenger compartment and is classified as a Type I emergency exit. It is a translating type secured against pressurization loads by five stops on each vertical edge, that engage with five related stop brackets installed on the door surround.

Forward Emergency Exit Type I Door

Refer to Figure 6.

On aircraft with Modsum 4–458926, 4–458969, 4–459339, 4–190614 or 4–201647 incorporated a forward emergency exit Type I door is located on the right side of the forward fuselage. The forward emergency exit door is classified as a Type I emergency exit. It is a translating type door and is secured against pressurization loads by five stops on each vertical edge that engage with five related stop brackets installed on the door surround structure.

Flight Compartment Escape Hatch

Refer to Figure 10.

The flight compartment escape hatch is opened by turning the handle counterclockwise to the open position. The hatch is opened to permit depressurization or to increase airflow in the flight compartment when the aircraft is on the ground.

The door movement is controlled by the geometry of the torque tube and rollers. The hatch pivots about the rear fittings and opens approximately one inch at the front.

A pull force on the handle of approximately 40 lb (18.1 kg), releases the rollers against the action of the locking and release springs, and the hatch can be removed from the flight compartment roof.

Type II Exit Door

Refer to Figure 11.

The upper emergency exit door is secured by a lock pin and three adjustable stops on each vertical edge that engage with three stop brackets installed on the sides of the door. to secure it against cabin pressure loads. The lower emergency exit door is installed on a piano hinge at the lower edge. It is kept in the closed position by four stops on the upper edge of the door that engage with their related stop brackets on the lower edge of the upper door. The lower door opens automatically when the upper door is removed, and the lower door handle is in the GROUND position. The lower door stays closed when the lower door handle is in the WATER position.

On aircraft with Modsum 4–458969 incorporated, the Type II door is disabled. The internal handles are removed and the external handle is locked to make the door inoperable.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

On aircraft with Modsum 4–190561 incorporated, the Type II exit door is eliminated.

Door Warning System

The door warning system monitors the position of the emergency exit doors that follow:

- Forward Passenger Door (52–11–00)
- Aft Passenger Door (52–12–00)
- Type II Emergency Exit Door (52–22–00) (52–23–00)
- Aft Service Door (52–41–00).

On aircraft with Modsum 4–458926, 4–458969, 4–459339, 4–190614 or 4–201647 incorporated, the door warning system also monitors the emergency exit doors that follow:

- Forward Passenger Door (52–11–00)
- Aft Passenger Door (52–12–00)
- Forward Emergency Exit Type I Door (52–24–00)
- Aft Service Door (52–41–00).

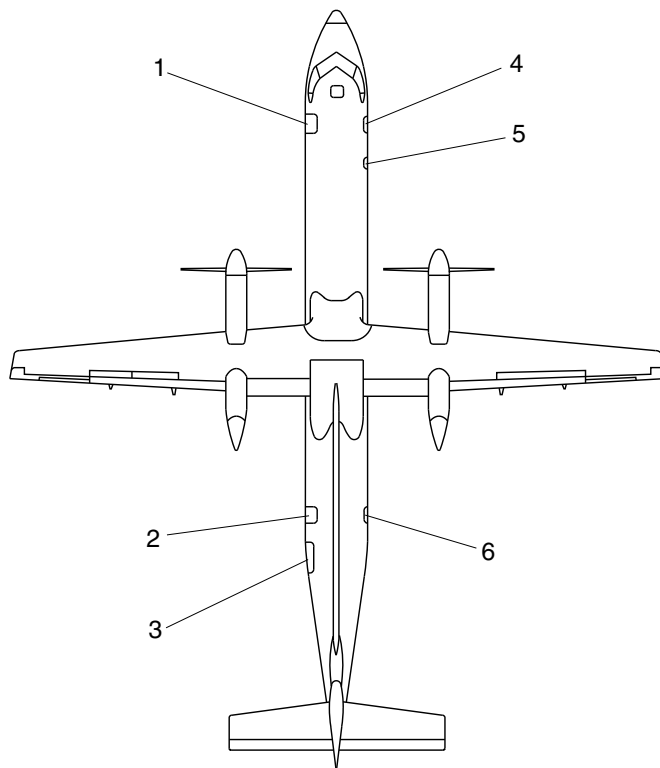
[Refer to Figures 12 and 13.](#)

When any of the Type I or TYPE II emergency exit doors are unlocked, both the FUSELAGE DOORS warning light and the MASTER WARNING light flash. The Door Systems page on the Multi Function Display (MFD) shows the unlocked door as red filled rectangle with a related legend. (Refer to SDS52–70–00). There is no visual warning if the flight compartment hatch is unlocked.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Forward Passenger Door.
"Closed" Sensor and "Locked" Sensor.
2. Aft Passenger Door.
"Closed" Sensor and "Locked" Sensor.
3. Aft Baggage Door.
"Closed" Sensor and "Locked" Sensor.
4. Forward Baggage Door.
"Closed" Sensor and "Locked" Sensor.
5. Upper (Type II) Emergency Exit.
"Locked" Sensor.
6. Aft Service Door.
"Closed" Sensor and "Locked" Sensor.

PRE MODSUM 4-458926
PRE MODSUM 4-458969
PRE MODSUM 4-459339
PRE MODSUM 4-190614
POST MODSUM 4-309306

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Emergency Exits Locator
Figure 1 (Sheet 1 of 2)

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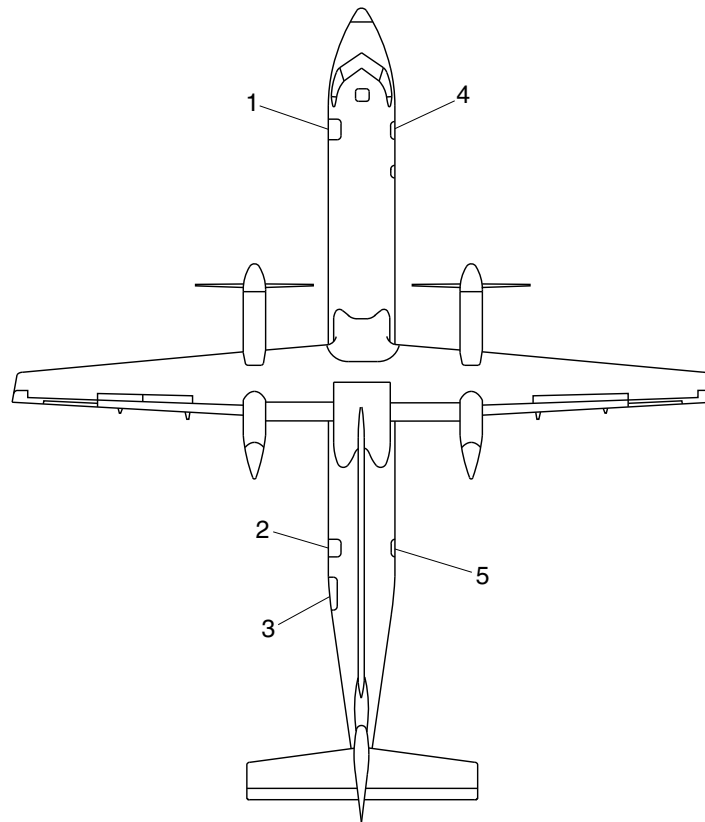
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Forward passenger door
"Closed" sensor and "Locked" sensor.
2. Aft passenger door
"Closed" sensor and "Locked" sensor.
3. Aft baggage door
"Closed" sensor and "Locked" sensor.
4. Forward RHS Type I
"Closed" sensor and "Locked" sensor.
5. Aft service door
"Closed" sensor and "Locked" sensor.

POST MODSUM 4-458926
POST MODSUM 4-458969
POST MODSUM 4-459339
POST MODSUM 4-190614
POST MODSUM 4-201647

cg4710a02.dg, cm/cm, may03/2020

Emergency Exits Locator
Figure 1 (Sheet 2 of 2)

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-20-00
Config 001

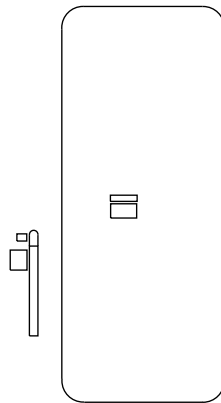
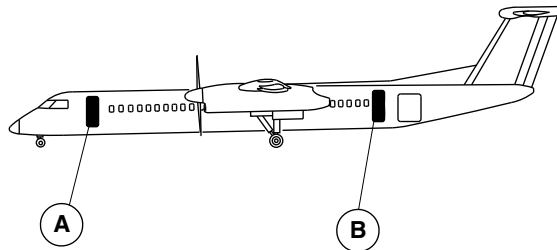
52-20-00

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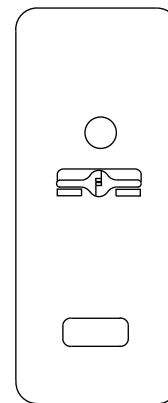


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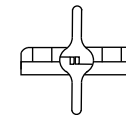
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A EXTERIOR OF FWD PASSENGER DOOR



B EXTERIOR OF AFT PASSENGER DOOR



HANDLE IN OPEN
POSITION
(TYPICAL)

fsn28a01.cgm

EMERGENCY EXITS, PASSENGER DOORS LOCATOR Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-20-00
Config 001

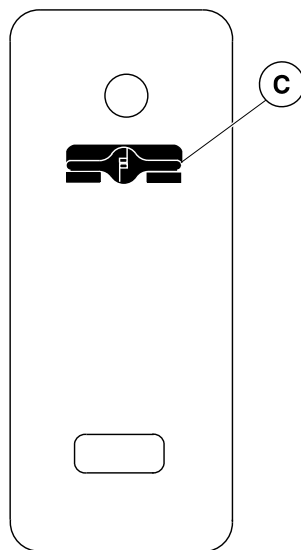
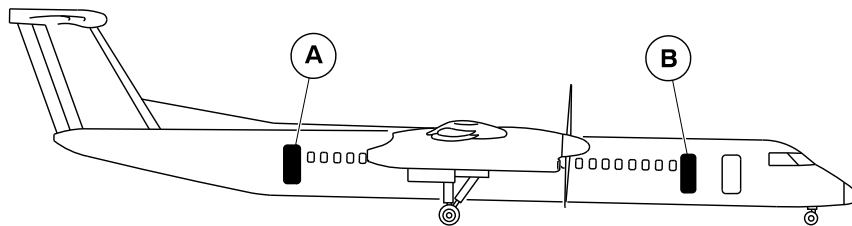
52-20-00

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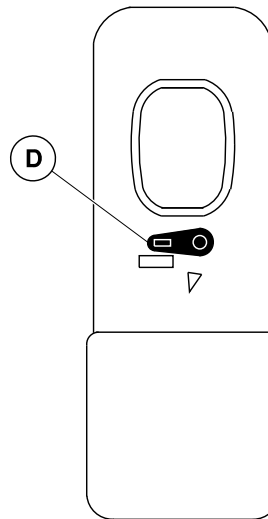


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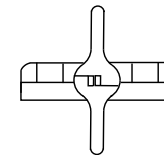
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



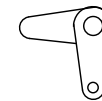
(A)
EXTERIOR OF AFT SERVICE DOOR



(B)
EXTERIOR OF TYPE II EMERGENCY EXIT
PRE MODSUM 4-458926
PRE MODSUM 4-458969
PRE MODSUM 4-190614
POST MODSUM 4-309306



(C)
**HANDLE IN OPEN
POSITION
(TYPICAL)**



(D)
**HANDLE IN OPEN
POSITION
(TYPICAL)**

fsn29a01.dg, vn/gv, sep24/2020

Aft Service, Type I and Type II Door Locator
Figure 3 (Sheet 1 of 2)

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-20-00
Config 001

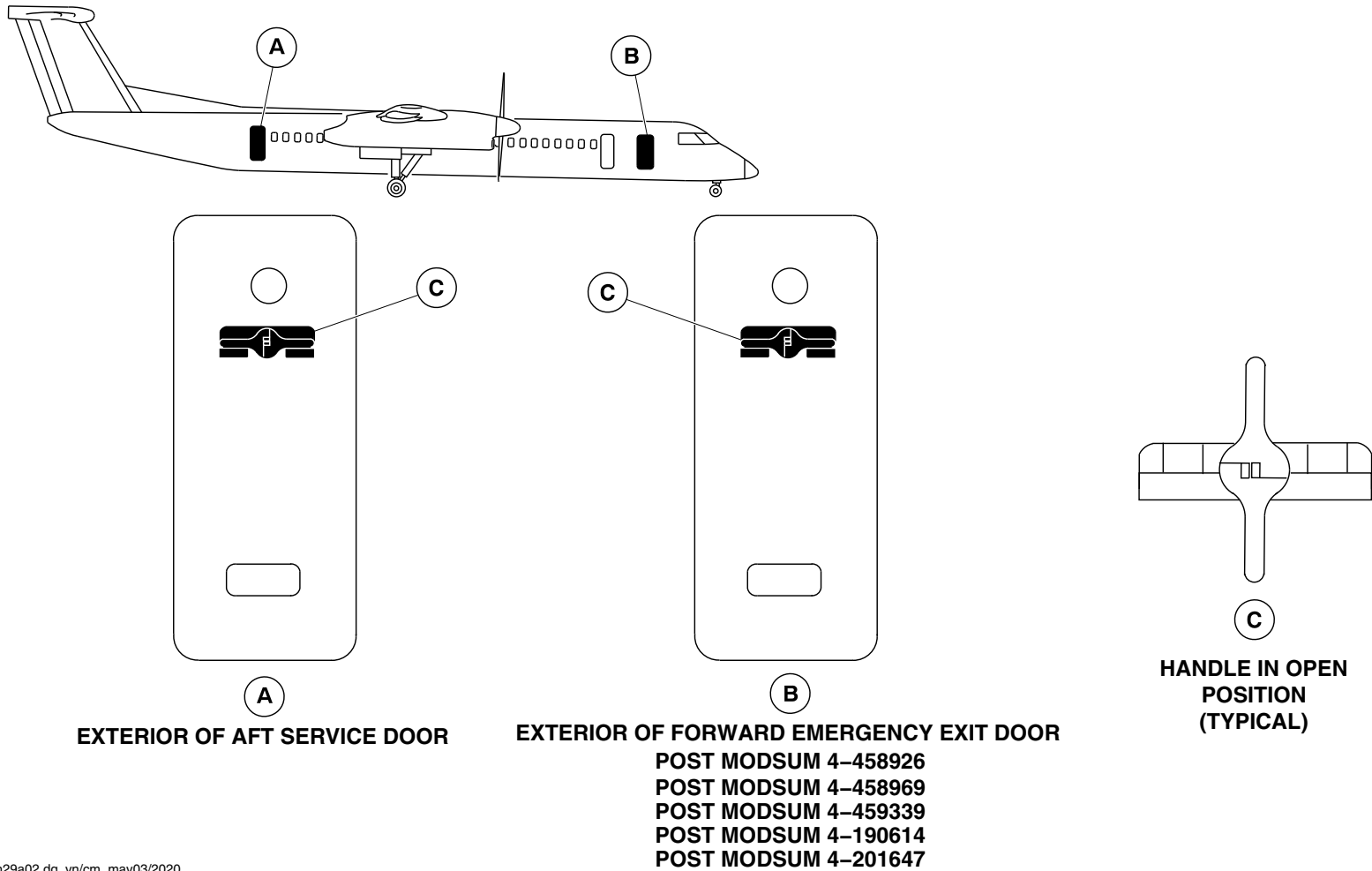
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fsn29a02.dg, vn/cm, may03/2020

Aft Service, Type I and Type II Door Locator
Figure 3 (Sheet 2 of 2)

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-20-00
Config 001

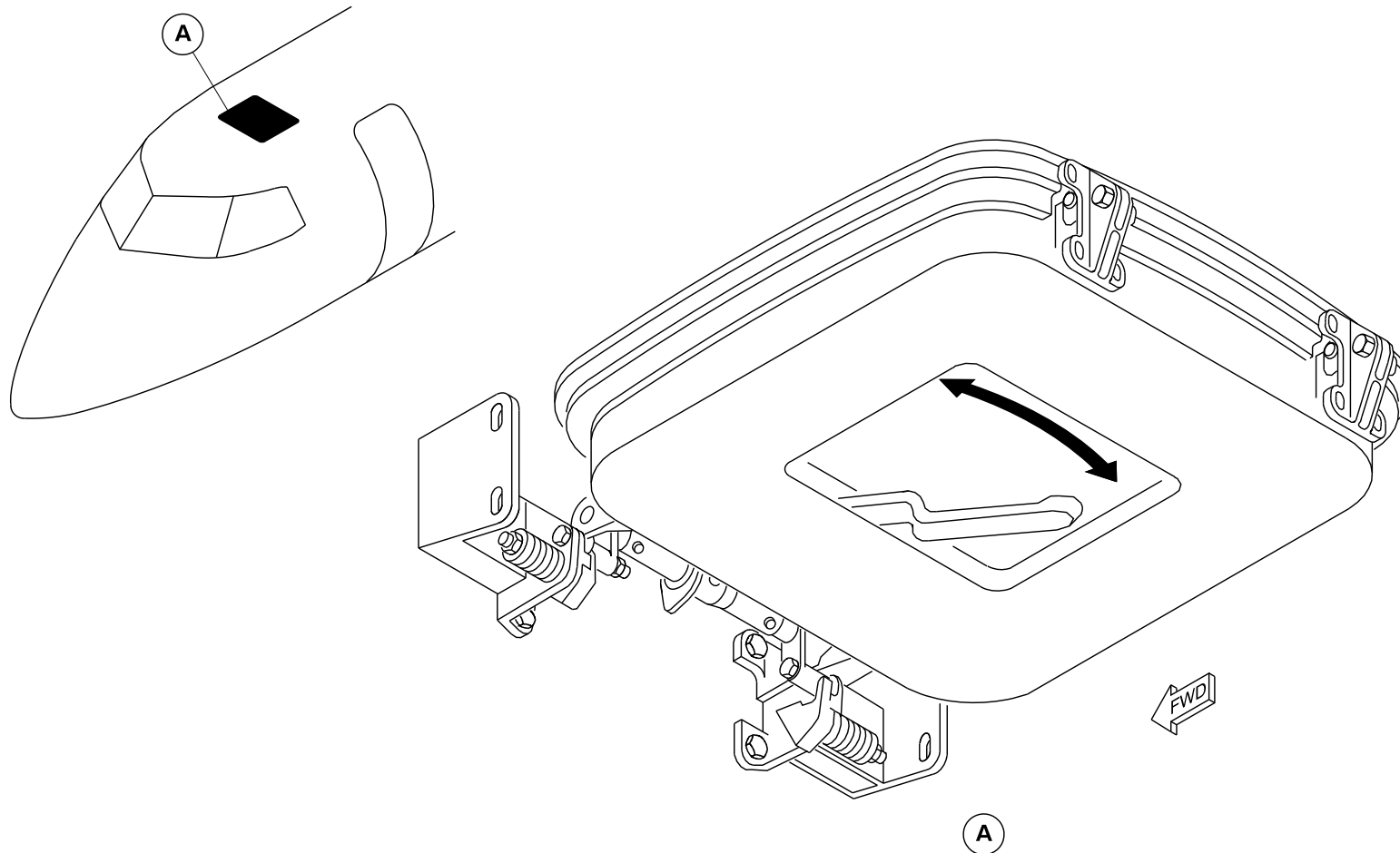
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fs421a01.dg, vn, aug22/2011

Flight Compartment Escape Hatch Locator
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-20-00
Config 001

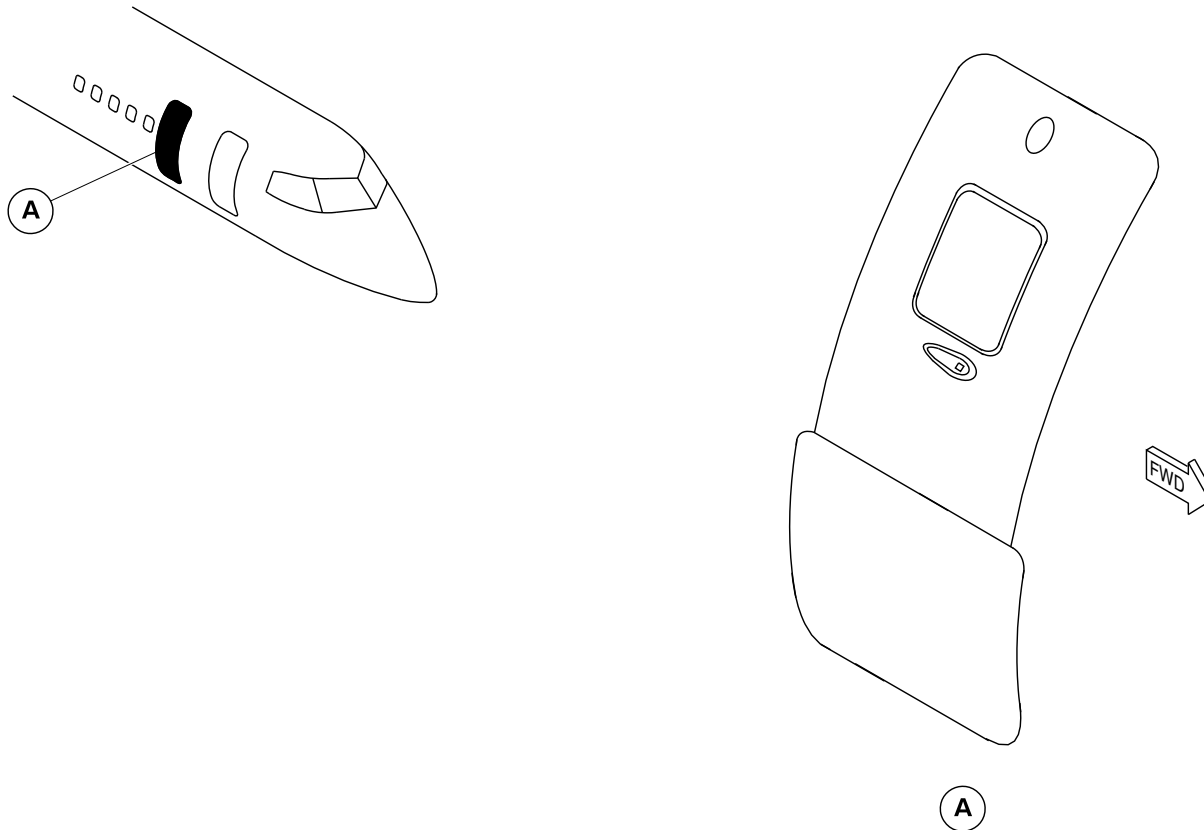
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PRE MODSUM 4-458969
PRE MODSUM 4-190561
POST MODSUM 4-309306

fs850a01.dg, cm/cm, may03/2020

Type II Emergency Exit Locator
Figure 5

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-20-00
Config 001

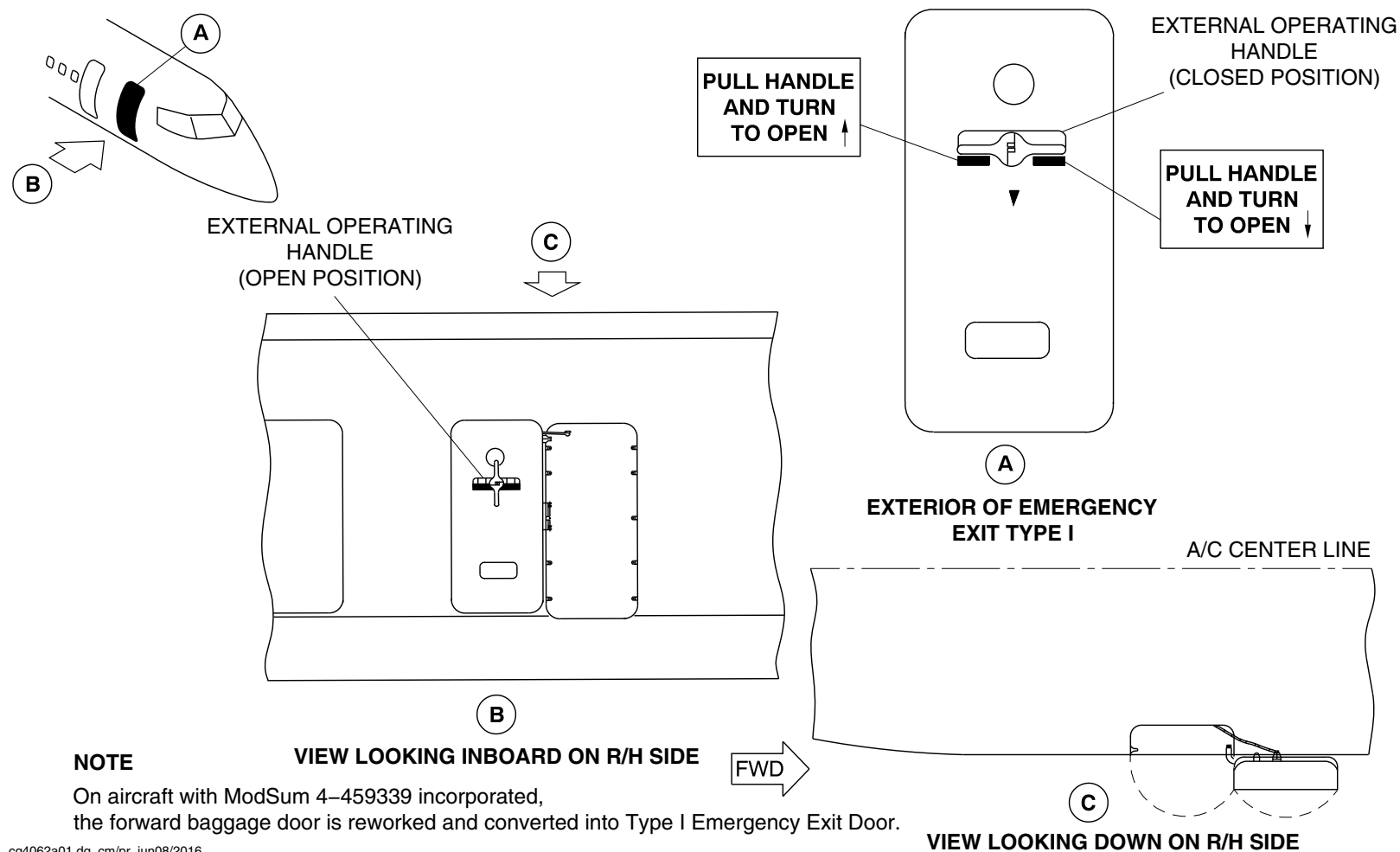
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Forward Emergency Exit Type I
Figure 6

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-20-00
Config 001

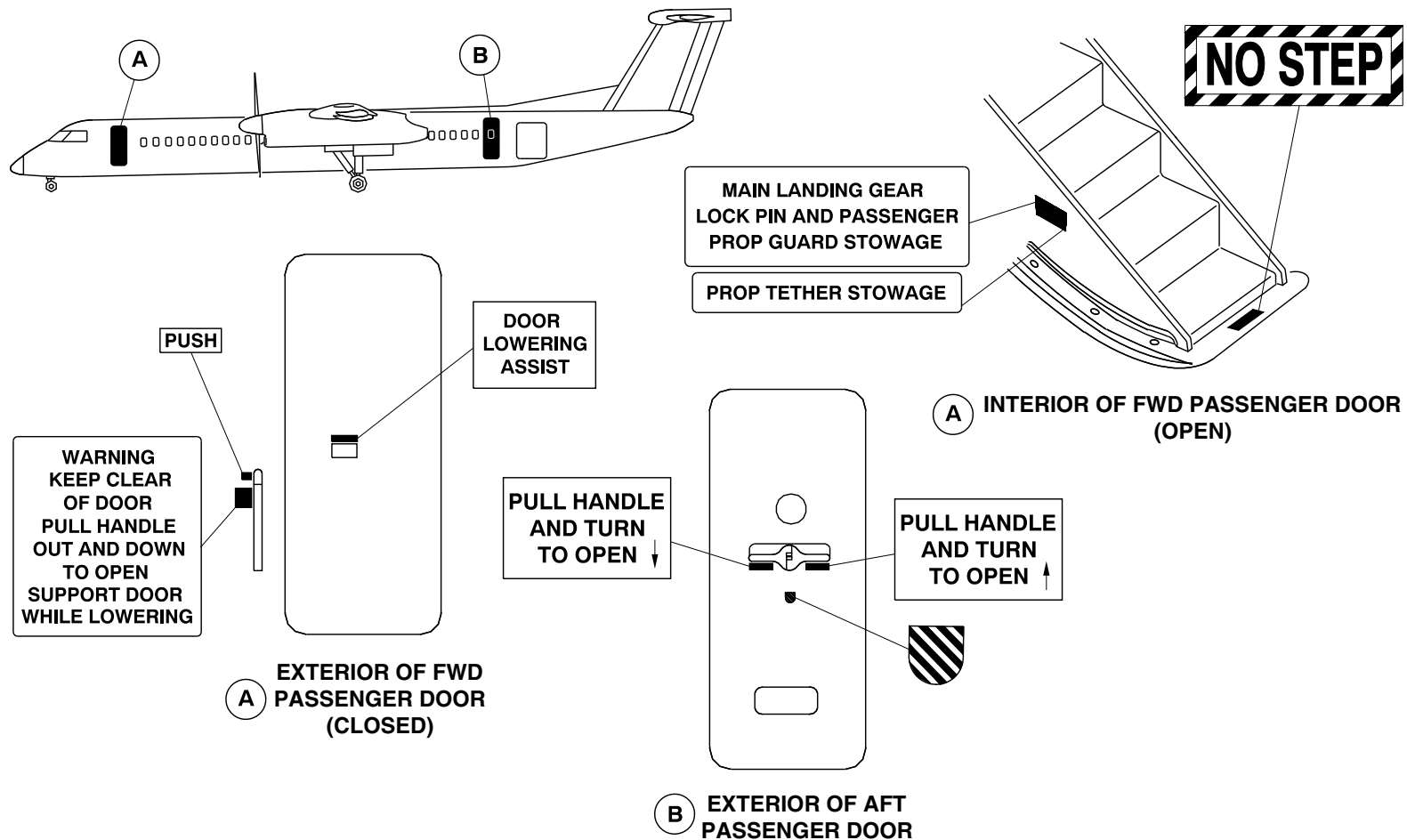
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fsm73a01.dg, vn, aug22/2011

Passenger Doors Operation
Figure 7

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-20-00
Config 001

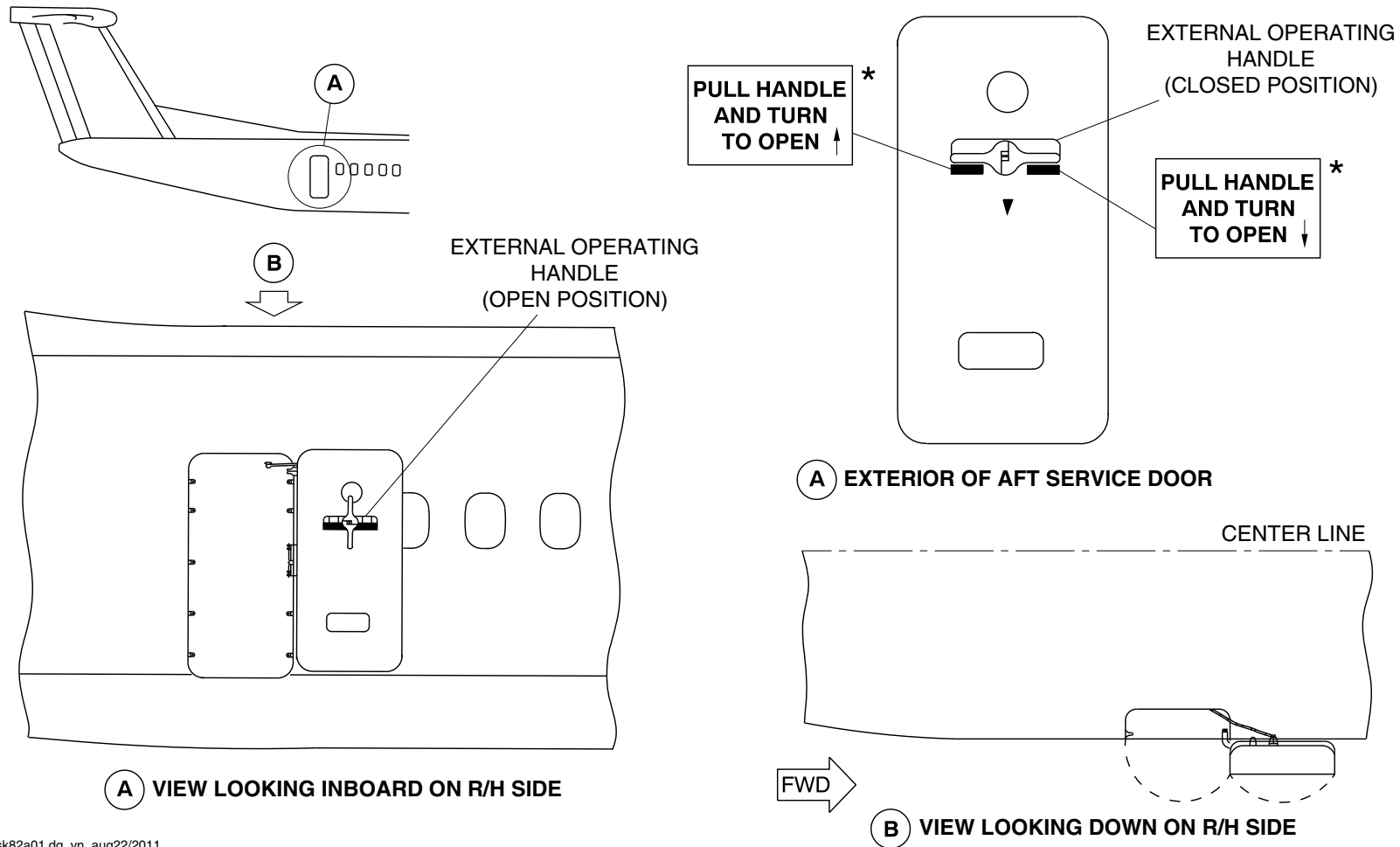
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fsk82a01.dg, vn, aug22/2011

Aft Service Door Operation
Figure 8

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-20-00
Config 001

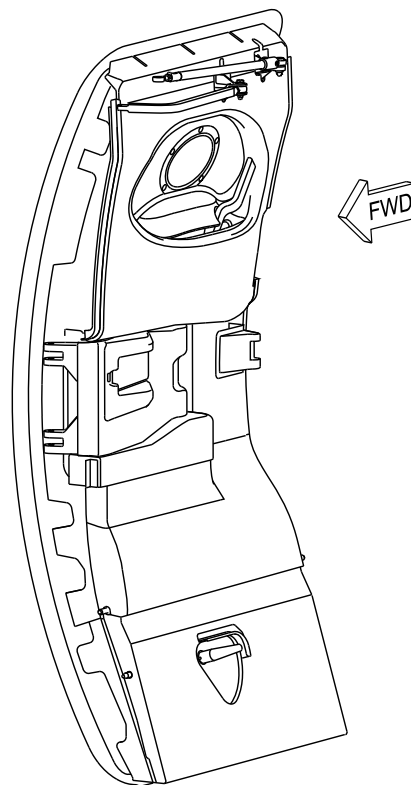
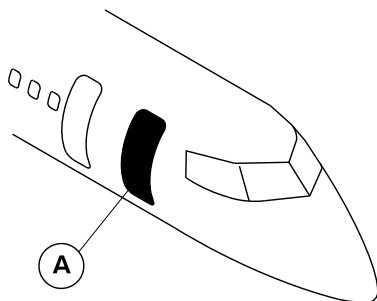
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A

VIEW LOOKING
OUTBOARD

NOTE

On aircraft with ModSum 4-459339 incorporated,
the forward baggage door is reworked and
converted into Type I Emergency Exit Door.

cg4698a01.dg, cm/pr, jun08/2016

Forward Type-I Emergency Exit Location
Figure 9

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 2 of 52-20-00

Config 001

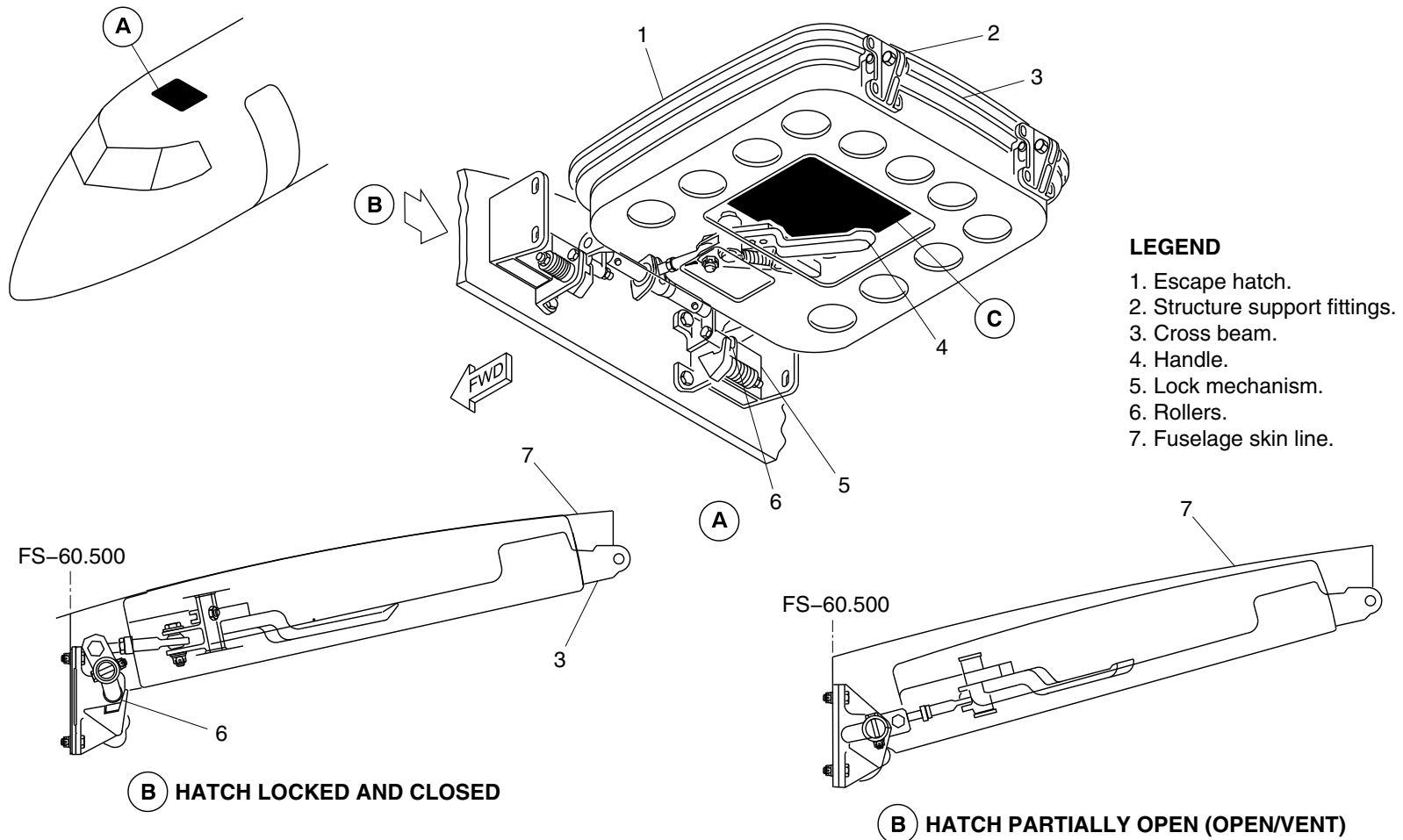
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fsi35a01.dg, vn, aug22/2011

Emergency Escape Hatch Operation
Figure 10 (Sheet 1 of 2)

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-20-00
Config 001

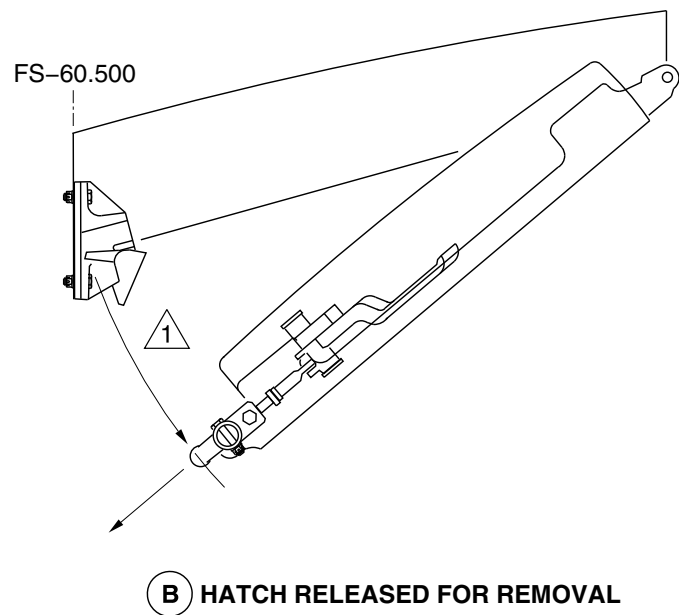
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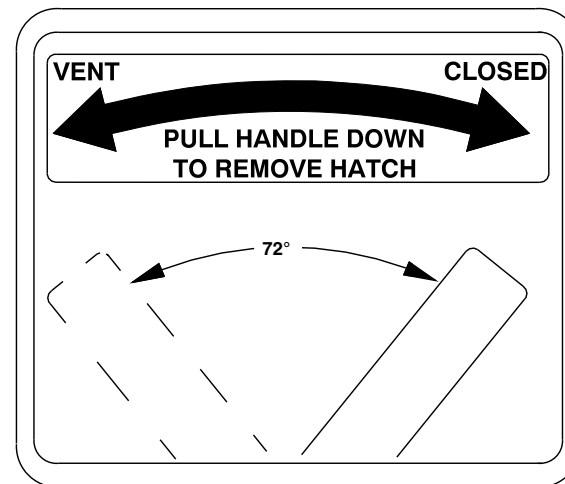
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NOTE

- 1 With the escape hatch in vent position – a force of approximately 40 lb (18.1 kg) on the handle is necessary to open the escape hatch.



fsi35a02.dg, vn, aug28/2011

Emergency Escape Hatch Operation
Figure 10 (Sheet 2 of 2)

PSM 1-84-2A
EFFECTIVITY:
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Config 001

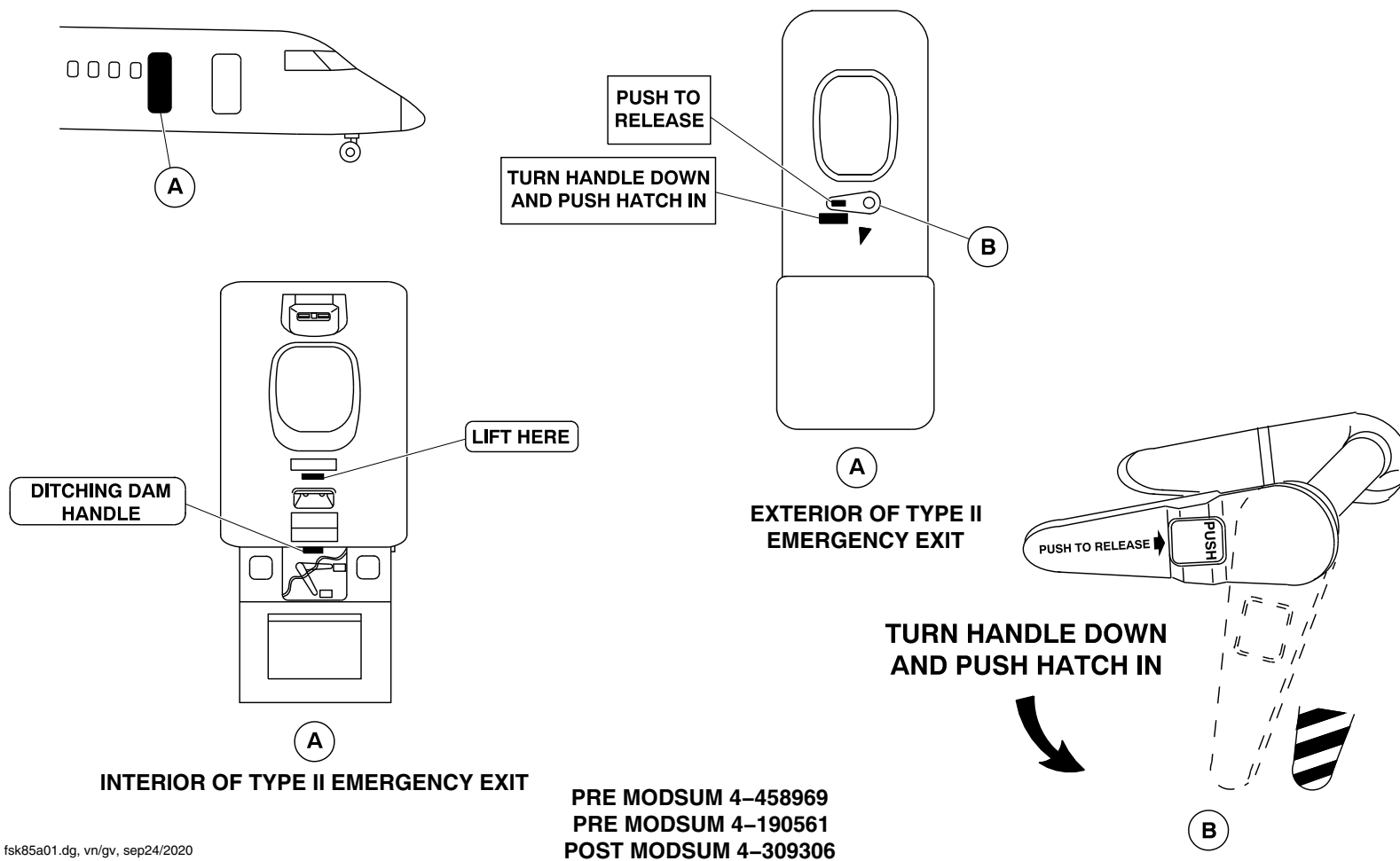
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fsk85a01.dg, vn/gv, sep24/2020

Type II Door Operation
Figure 11

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-20-00
Config 001

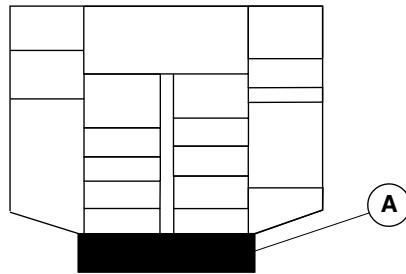
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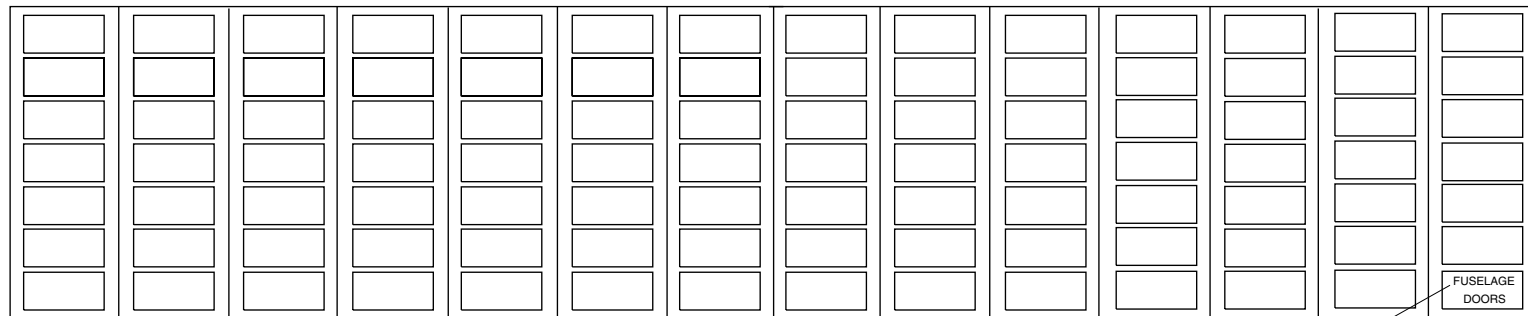
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



OVERHEAD CONSOLE

LEGEND

1. Fuselage Doors (Red).



fs805a01.cgm

Fuselage Doors Warning Light
Figure 12

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-20-00
Config 001

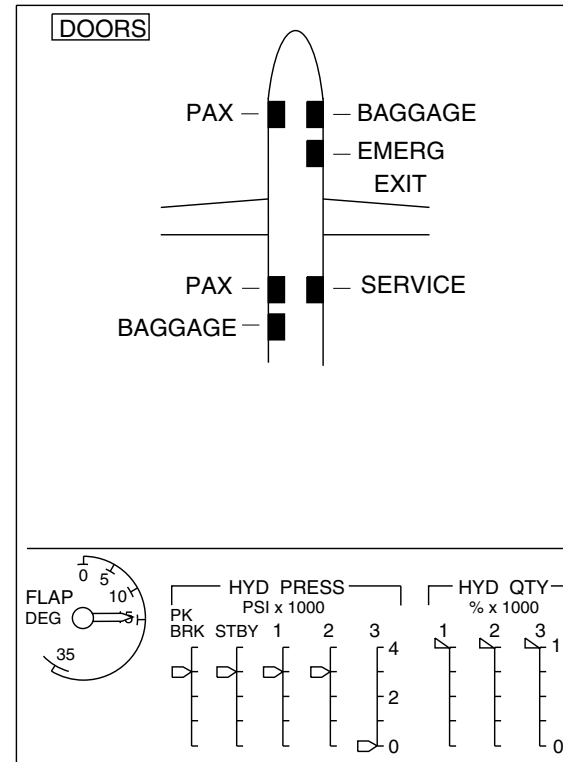
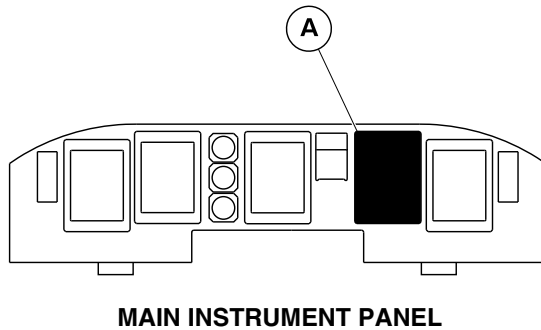
52-20-00

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A

**PRE MODSUM 4-458926
PRE MODSUM 4-458969
PRE MODSUM 4-459339
PRE MODSUM 4-190614
POST MODSUM 4-309306**

fs747a01.dg, cm/cm, may03/2020

Doors Systems Page
Figure 13 (Sheet 1 of 2)

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-20-00
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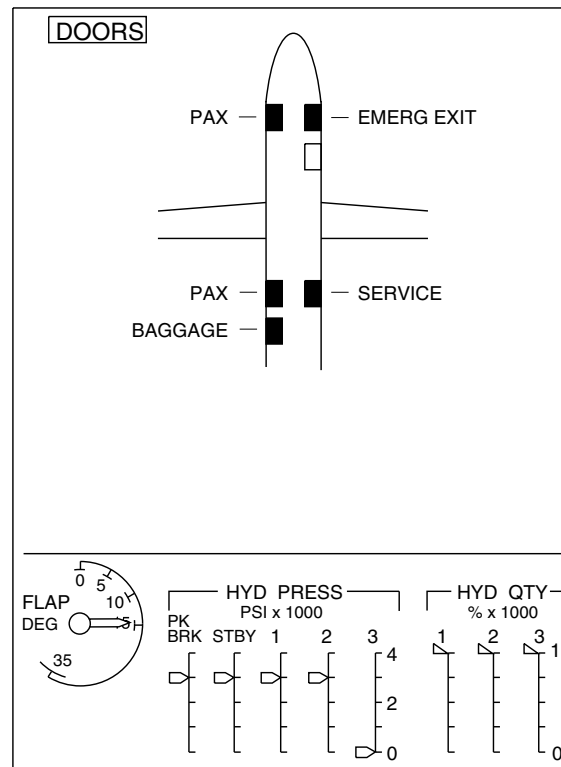
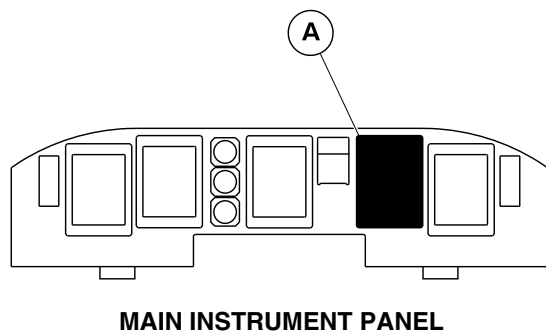
52-20-00

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A

POST MODSUM 4-458926
POST MODSUM 4-458969
POST MODSUM 4-459339
POST MODSUM 4-190614
POST MODSUM 4-201647

fs747a02.dg, cm/cm, may03/2020

Doors Systems Page
Figure 13 (Sheet 2 of 2)

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

52-21-00-001

FLIGHT COMPARTMENT ESCAPE HATCH

Introduction

The flight compartment escape hatch gives emergency access and ventilation to the flight compartment.

General Description

Refer to Figure 1.

The flight compartment escape hatch is located in the flight compartment roof, aft of the pilot seats. It is completely detachable, or can be partially opened to increase the airflow in the flight compartment when the aircraft is on the ground.

The hatch is installed in the flight compartment roof by two support fittings at the rear, and is secured at the front by two locking and release arms. An operating handle, located in the center of the hatch, is kept in an open or closed position by an overcentre spring. A seal installed around the edge of the hatch to prevent pressurization leakage.

The flight compartment escape hatch includes the components that follow:

- Latch Mechanism (52-21-01)
- Door Seal (52-21-06).

Detailed Description

The handle is turned counterclockwise to open the hatch. The hatch is opened to permit depressurization or to increase airflow in the flight compartment when the aircraft is on the ground.

Refer to Figure 2.

The door movement is controlled by the geometry of the torque tube and rollers. The hatch pivots about the rear fittings and opens approximately 1 in. (25.4 mm) at the front.

A pull force on the handle of approximately 40 lb (18.1 kg), releases the rollers against the action of the locking and release springs, and the hatch can be removed from the flight compartment roof.

Latch Mechanism

The latch mechanism controls opening and closing of the hatch. When the handle is moved to the open position, the mechanical linkage connected to the handle turns the torque tubes. The rollers then engage in detents on the structure to let the hatch door partially open.

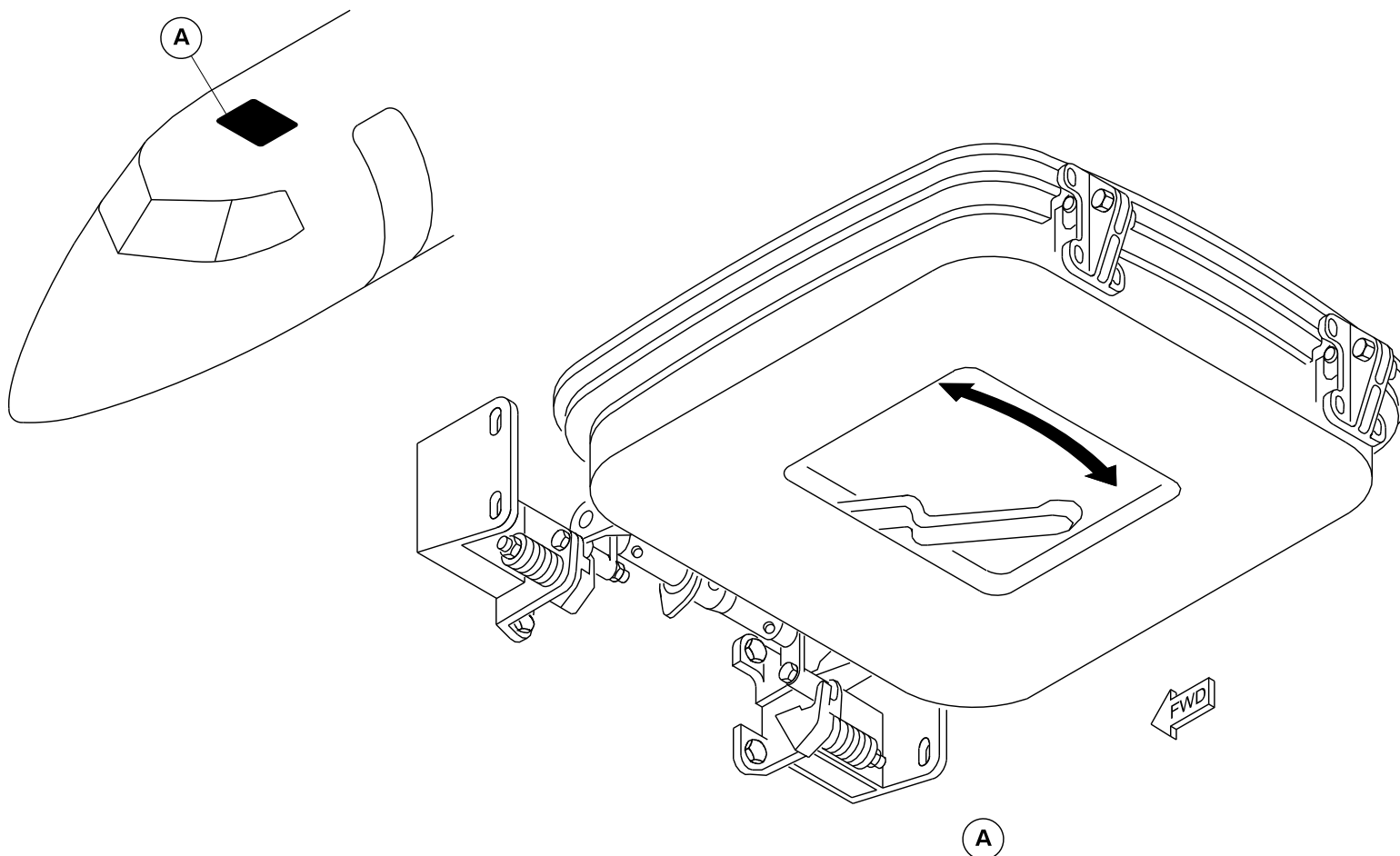
Door Seal

The door seal helps to prevent pressurization leakage when the hatch is closed. The seal is installed around the edge of the hatch.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs421a01.dg, vn, aug22/2011

Flight Compartment Escape Hatch Location
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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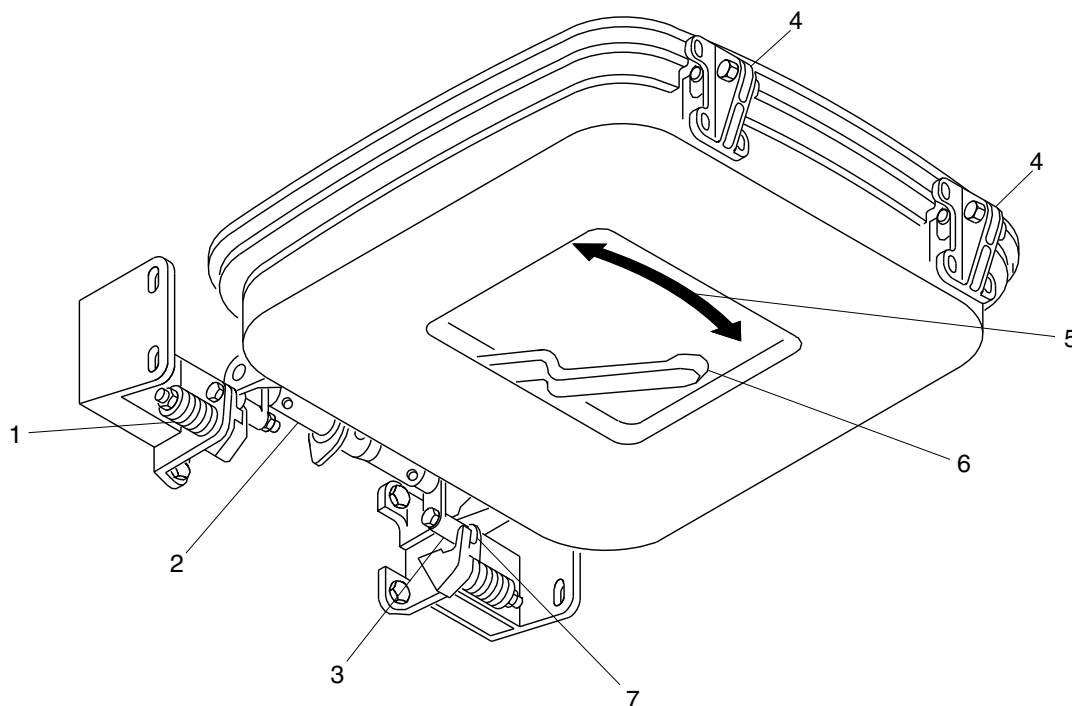
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Arm/spring detent mechanism.
- 2. Torque shaft assembly.
- 3. Arm.
- 4. Rear support fittings.
- 5. Instructional label.
- 6. Operating handle.
- 7. Roller.

fs385a01.dg, vn, aug22/2011

Flight Compartment Escape Hatch
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

52-22-00-001

FORWARD EMERGENCY TYPE II EXIT DOOR

Introduction

The type II emergency exit doors supply an exit for emergency evacuation of the passenger compartment.

General Description

[Refer to Figure 1.](#)

On aircraft with Modsum 4-458969 incorporated, the type II/III door is disabled. The internal handles are removed and the external handle is locked to make the door inoperable.

The type II emergency exit doors are located on the forward right side of the passenger compartment. The emergency exit has an upper and lower door. The upper door is a plug type door which is removed completely for opening. The lower door is hinged at the bottom, and opens outward and downward from the floor level.

The door warning system monitors door status and supplies visual indication and warning in the flight compartment. The indications are shown in:

- Door Warning System (52-70-00)

The upper and lower emergency Type II exit doors has the components that follow:

- Latch mechanism (52-22-01) (52-23-01)

- Door seal (52-22-06) (52-23-06).

Detailed Description

[Refer to Figure 2.](#)

The upper door is secured by a lock pin and three adjustable stops on each vertical edge that engage with three stop brackets installed on the sides of the door. to secure it against cabin pressure loads. The lower door is installed on a piano hinge at the lower edge. It is kept in the closed position by four stops on the upper edge of the door that engage with their related stop brackets on the lower edge of the upper door. The lower door opens automatically when the upper door is removed, and the lower door handle is in the GROUND position. The lower door stays closed when the lower door handle is in the WATER position.

[Refer to Figure 3.](#)

The upper door is unlocked by operation of the internal or external handle. Both handles are spring-loaded and interconnected by cables and pulleys. When the internal handle of the upper door is pulled, the vent dish opens and pressure in the aircraft is released to the atmosphere. The locking pins retract and the door can then be removed.

[Refer to Figure 4.](#)

To remove the upper door using the external handle, operate the pushbutton and the handle springs out. The handle is turned counterclockwise to operate the cable and shaft quadrant to open



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

the vent dish and retract the locking pin. The door is pushed inward and removed.

[Refer to Figure 5.](#)

The lower door handle has two positions, GROUND and WATER. The handle is normally in the GROUND position, and lets the lower door open automatically when the upper door is removed. When the handle is moved to the WATER position, the two shoot bolts engage with the fittings on the door structure to prevent the opening of the lower door. The lower door handle has an overcentre spring to hold the handle in the engaged, or disengaged position. A stop cable, made of nylon rope, is installed between the forward face of the lower door and the door cut-out structure, to keep the door from opening too far and damaging the structure. The cable can also be used to pull the door up to the closed position.

The upper and lower doors are sealed by a passive type seal. A neoprene sponge seal is installed around the vent dish.

[Refer to Figures 6 and 7.](#)

A proximity switch, installed on the upper edge of the upper door, monitors the lock position of the door. If the door is unlocked, the FUSELAGE DOORS warning and the MASTER WARNING light flash. The Doors Systems page on the Multi Function Display (MFD) shows the unlocked door as a red filled rectangle. If the door is not properly closed and locked when the aircraft leaves the ground, the cabin will not pressurize automatically.

Mechanism, Latch (Upper door)

The lower end of the internal handle is connected to the handle shaft which is spring loaded to the locking direction of the lock pin. When the internal handle is pulled, the vent dish is opened and the lock pin

is retracted, the upper door can now be removed. The lock pin is kept at the locked position by two independent bias springs. The internal handle has a cover to prevent the door from being accidentally opened.

Mechanism, Latch (Lower door)

The lower door handle is installed with an overcenter spring, to hold the handle at the engaged or disengaged position. Two shoot bolts engage to fittings on the door structure to keep the door closed.

Door seal (Upper and lower door)

The upper and lower door seals are a passive type. Several holes in the seals let them be inflated by cabin pressure.

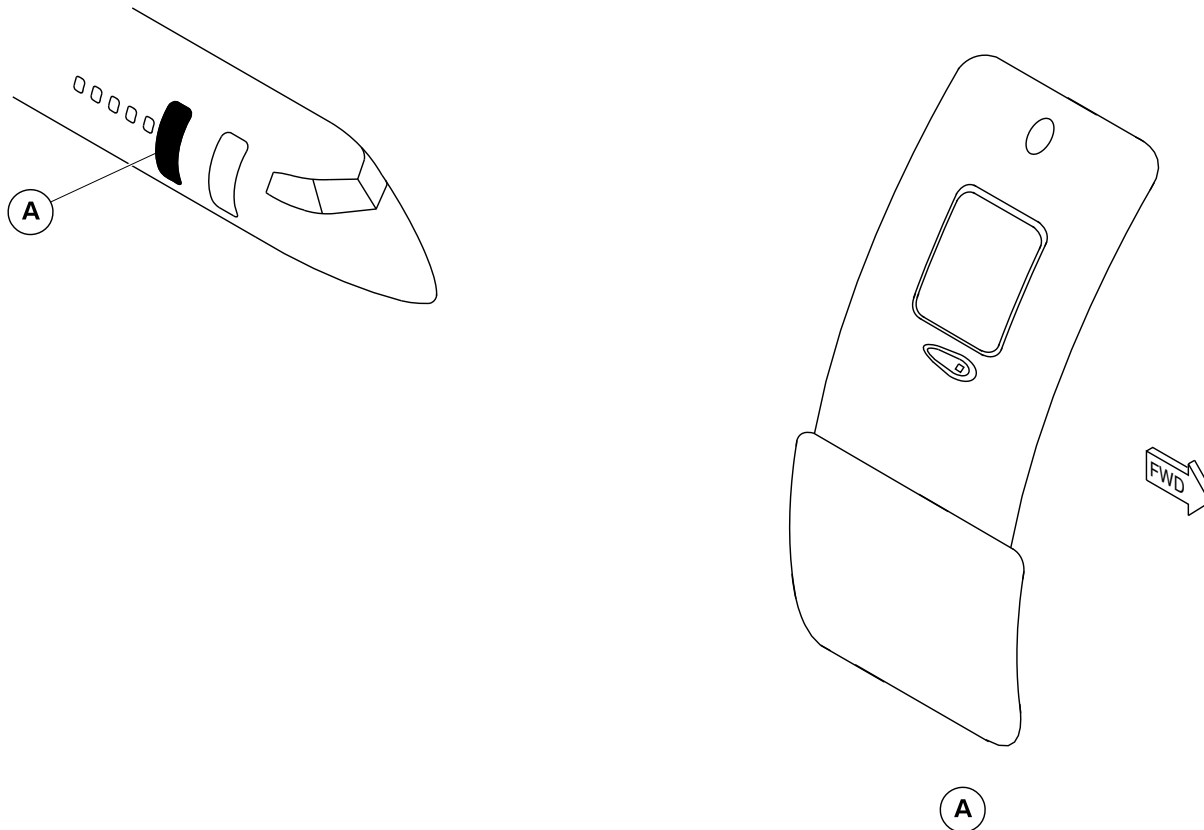
Vent Dish seal (Upper door)

This seal is a compression seal made of neoprene sponge and fits around the periphery of the vent dish.



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PRE MODSUM 4-458969
PRE MODSUM 4-190561
POST MODSUM 4-309306

fs850a01.dg, cm/cm, may03/2020

Type II Emergency Exit Locator
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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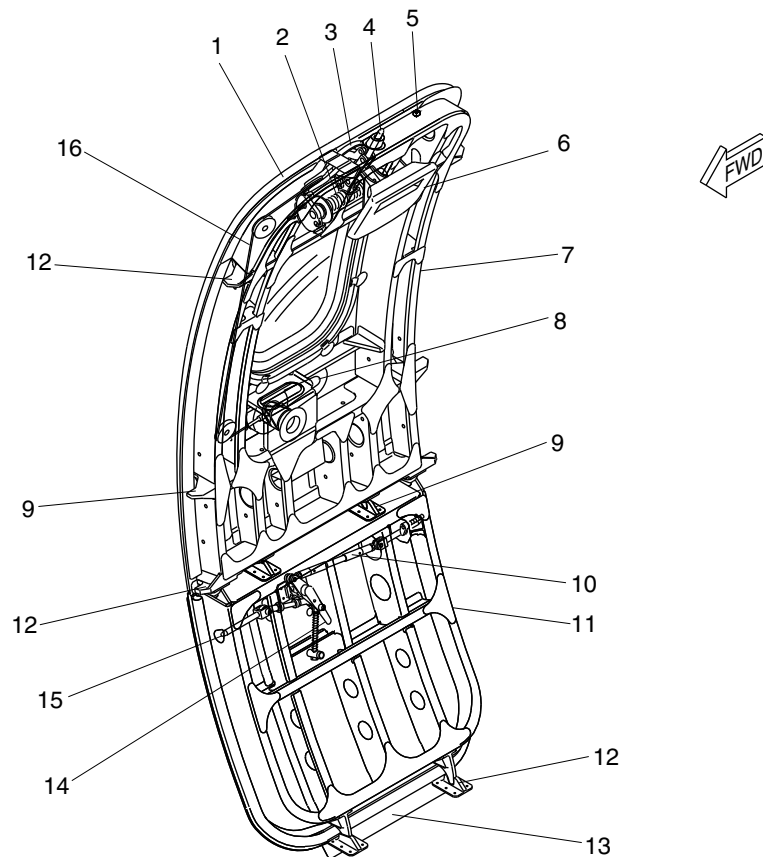
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LEGEND

1. Door Seal.
2. Bias Spring.
3. Vent Dish.
4. Lock Pin.
5. Proximity Switch.
6. Internal Handle.
7. Upper Door.
8. External Handle.
9. Backup Stop.
10. Linkage at Lower Door
11. Lower Door.
12. Main Stop.
13. Hinge.
14. Lower Door Handle.
15. Shoot Bolt.
16. Cable Assembly.

fs354a01.cgm

UPPER AND LOWER TYPE II EMERGENCY EXIT DOORS
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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Config 001

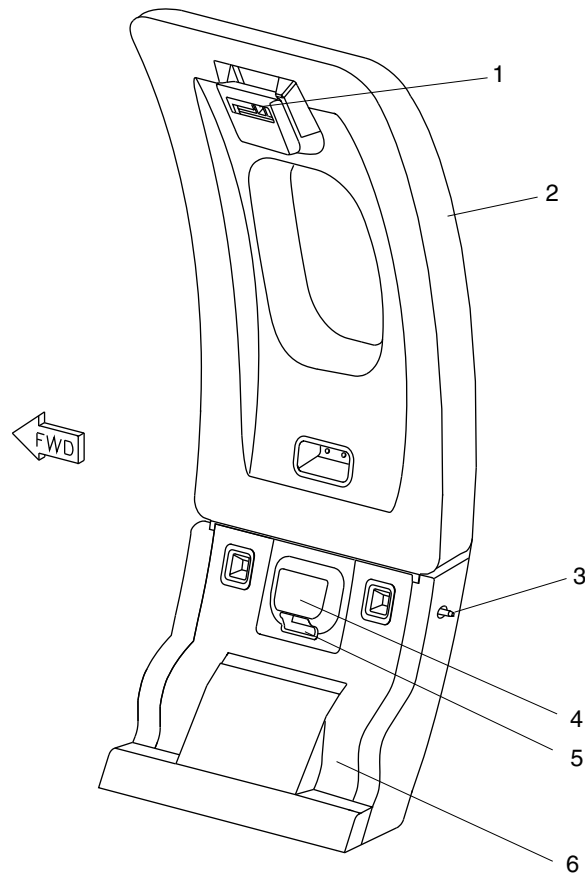
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LEGEND

- 1. Upper Door Internal Handle.
- 2. Upper Door.
- 3. Lower Door Shoot Bolt.
- 4. Lower Door Handle Cover.
- 5. Lower Door Handle.
- 6. Lower Door.

fs858a01.cgm

INTERNAL DETAIL TYPE II EMERGENCY EXIT DOORS
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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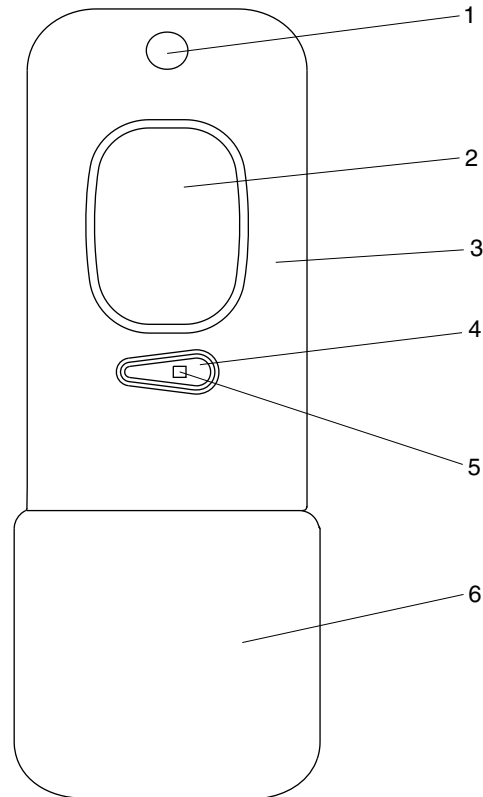
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Vent Dish.
- 2. Window.
- 3. Upper Door.
- 4. External Handle.
- 5. Push Button.
- 6. Lower Door.

fs856a01.cgm

EXTERNAL DETAIL TYPE II EMERGENCY EXIT DOORS
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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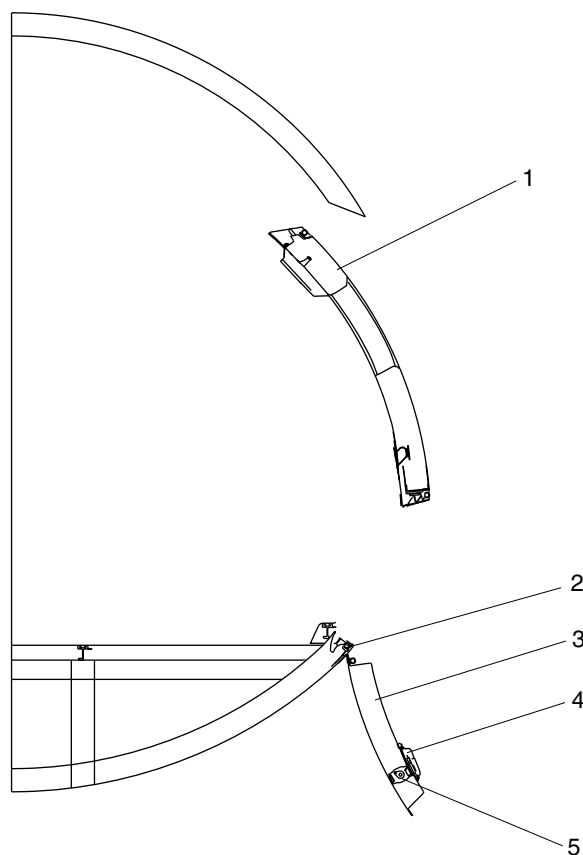
52-22-00

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Upper Door (Open Position).
2. Hinge.
3. Lower Door.
4. Lower Door Handle.
5. Lower Door Shoot Bolt.

NOTE

Upper door detaches from aircraft when opened.

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TYPE II EMERGENCY EXIT DOORS OPEN
Figure 5

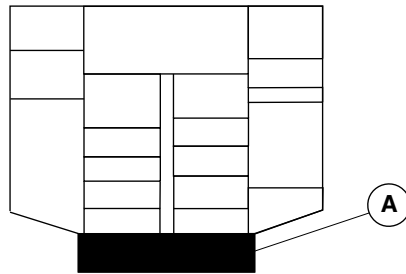
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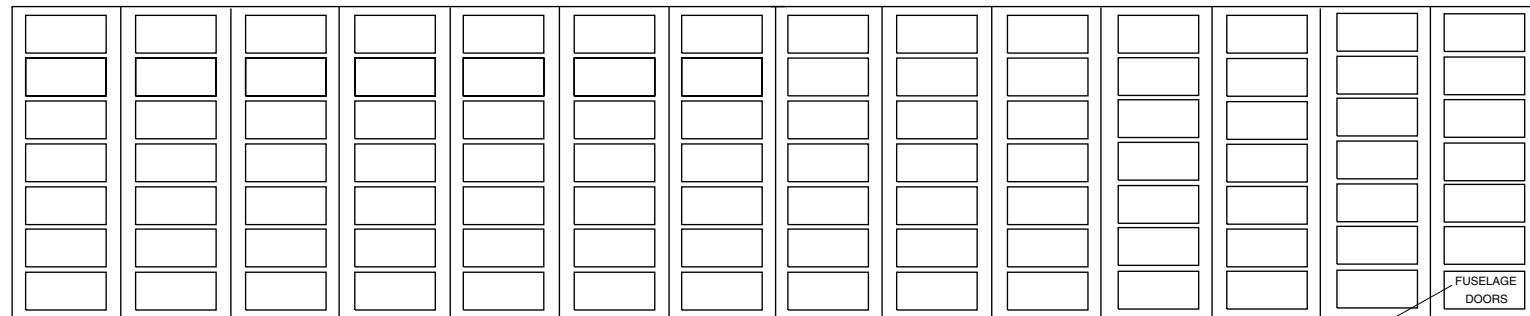
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OVERHEAD CONSOLE

LEGEND

1. Fuselage Doors (Red).



fs805a01.cgm

Fuselage Doors Warning Light
Figure 6

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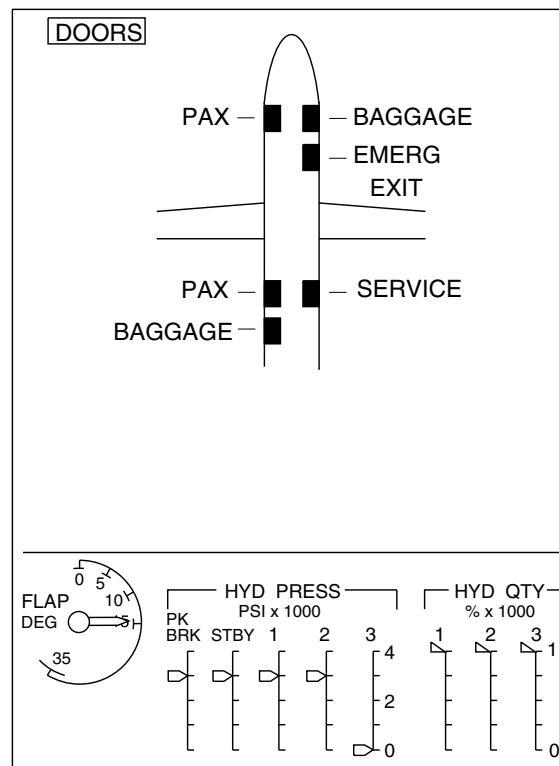
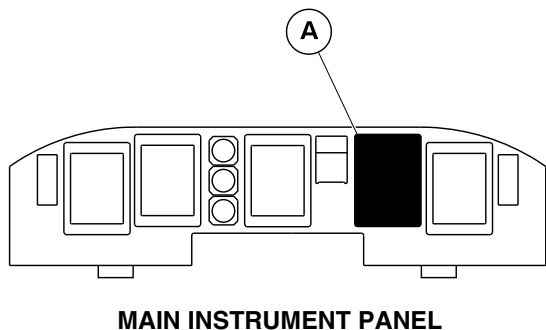
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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A

**PRE MODSUM 4-458926
PRE MODSUM 4-458969
PRE MODSUM 4-459339
PRE MODSUM 4-190614
POST MODSUM 4-309306**

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Doors Systems Page
Figure 7 (Sheet 1 of 2)

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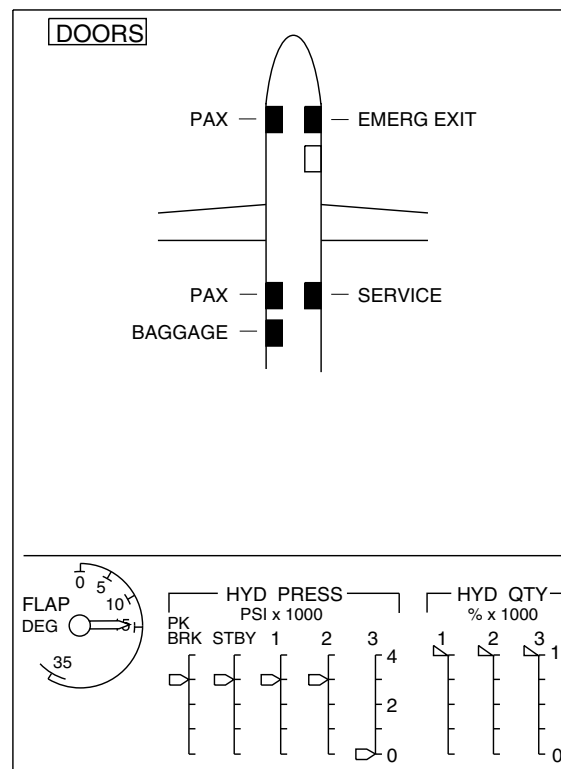
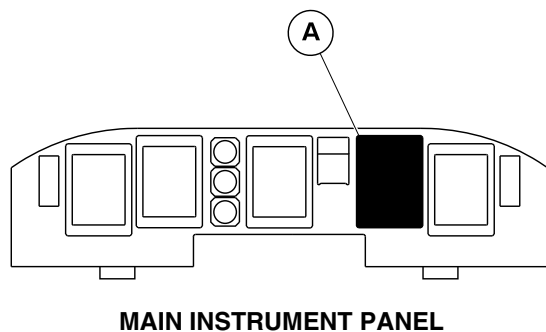
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POST MODSUM 4-458926
POST MODSUM 4-458969
POST MODSUM 4-459339
POST MODSUM 4-190614
POST MODSUM 4-201647

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Doors Systems Page
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

52-24-00-001

FORWARD EMERGENCY EXIT TYPE I DOOR

Introduction

On aircraft with the ModSum 4-458926, 4-458969, 4-190614 or 4-201647 incorporated a forward emergency exit Type I door is installed. The forward emergency exit Type I door gives access to the forward passenger compartment.

On aircraft with ModSum 4-459339 incorporated, the forward baggage door is reworked to add an internal handle, a window and ditching dam guide pin bracketry to convert into a Type I emergency exit door. The forward emergency exit Type I door gives access to the forward passenger compartment.

General Description

[Refer to Figure 1.](#)

The forward emergency exit Type I door is located on the right side of the forward fuselage. It is a translating type door and is secured against pressurization loads by five stops on each vertical edge that engage with five related stop brackets installed on the door surround structure. The proximity sensor monitors the position of the upper fixed rollers and the door handle. A red warning light comes on in the flight compartment when the door is open and unlocked.

The forward emergency exit Type I door system has the components that follow:

- Mechanism, Handle (52-24-01)
- Mechanism, Vent door (52-24-06)
- Mechanism, Lift and Latch (52-24-11)
- Mechanism, Hinge (52-24-16)
- Seal, Door (52-24-21)
- Dam, Ditching (25-62-02).

The forward emergency exit Type I door interfaces with:

- Door Warning System (52-70-00)
- Proximity Sensors (52-71-00).

Detailed description

[Refer to Figure 2.](#)

The door operating and retention system is operated by the internal or external handle. The internal handle is pulled inwards and turned counter clockwise to unlock the handle. When the door handle is pulled inwards, the vent door opens to release the pressure in the cabin and the proximity switch opens. When the handle is turned counter clockwise, the lift/latch shaft pushes the lift/latch rollers down, and the door moves inwards and upwards to clear the door stops. At the uppermost lifted position, the hold-up hook is connected to the hold-up hook pin of the door hinge, and the door is moved outward. As the door moves outward, the interlock cam connects to a latch crank on the lift/latch shaft, to prevent the accidental handle operation.

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The door has two fixed rollers located on each side around the upper and lower parts running in guides on the fuselage, which gives tangential movement of the door. In pressurized flight, the door is held by the stops against the pressurization and vertical loads. In unpressurized flight, the door is kept from upward movement by cam rollers on each side of the lift/latch shaft. If one cam roller assembly fails, the second cam roller assembly prevents the door from opening in flight. The lift/latch shaft is secured at the closed position by means of both the handle lock mechanism and the overcenter springs.

Refer to Figure 3.

The door is unlocked by operation of the spring loaded internal or external handle. The door opens or closes with the door face in parallel with the fuselage skin (this is done by the main and secondary hinges and a stabilizer). When the door reaches the fully opened position, the door open lock hook on the hinge arm is connected to the door open lock pin on the door to keep the door at its position. An interlock assembly prevents the handle operation when the door is opened, preventing the hold-up hook from being unlatched.

Refer to Figure 4.

The external handle is kept in a recess in the outer skin and is connected to the same handle shaft as that of the internal handle. The handle can be accessed when the flap covers are pushed in. The handle is then pulled out and turned clockwise to open the door.

For safety, a window in the door gives a view of the outside before the door is opened.

Refer to Figure 4.

To close the door, the release lever adjacent to the lock hook is pulled to release the open lock, and the door is brought to the closed position. The internal handle is turned clockwise to disconnect the door hold-up hook from the hold-up hook pin. The door is moved to the closed position, and the stops align with their related stop bracket lugs. The handle is moved outward to lock the door and closes the proximity switch. The door can also be closed externally by using the external handle. The main door and vent door are sealed around their periphery. The seal is inflated by cabin pressure through holes in the seal to keep the gap around the closed door. Cabin pressurization is prevented when the vent door is not fully closed and locked.

Mechanism, Handle

The internal handle operates the lift/latch shaft, and the vent door operating rod. The lower end of the internal handle is connected to the shaft of the handle mechanism. When either the internal or external handle is operated, the handle shaft can be turned to operate the lift/latch through cranks, rods and an idler link. The external handle is kept flush with the door skin and locked in the closed position.

Mechanism, Vent

The linkage from the handle transmits the handle input to the vent door mechanism. When the door handle is pulled, the vent door opens to release the pressure from the cabin. When the door is in the closed position, the final movement to lock the internal or



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

external handle closes the vent door. The vent door is spring-loaded in the direction of the open position, so that it is kept open in case of fracture or disconnection of the vent door operating mechanism.

The primary functions of the vent door are as follows:

- Release residual cabin pressure to the atmosphere before the door is opened in order to make outward load on the door minimal
- Prevents the pressurization loads on the door in case of manual pressurization before the door is fully closed and locked.

Mechanism, Lift and Latch

The linkage from the handle to the lift/latch shaft transmits handle input to the shaft and locks the lift/latch shaft assembly when the handles are in their locked position. The bias spring assembly for the latch causes an overcenter against the lift/latch shaft, and locks it at the closed or opened position. The overcenter position and linkage from the handle mechanism gives dual locking of the lift/latch shaft.

The lift/latch shaft is installed to a cam roller assembly at both ends. When the door is opened, the lift/latch shaft turns and let the rollers push down the guide tracks, lifting the door for outward movement. The door operation sequence is reversed at door closing. When the door is in the closed position, the roller center is offset inward against the door lift-up plane, resulting in the rollers being kept in the door closing direction, against the upward movement of the door.

Mechanism, Hinge

The main and secondary hinge arms and the door makes a parallelogram, which keeps the door face in parallel with the fuselage skin during door opening or closing. During the travel of the door around the furthest point from the fuselage, the yawing movement of the door is suppressed by the stabilizer rod.

Seal, Door

The door seal is of the passive type which has several holes in the seal and is inflated by cabin pressure through the holes. This type of seal is installed on both the main and vent doors.

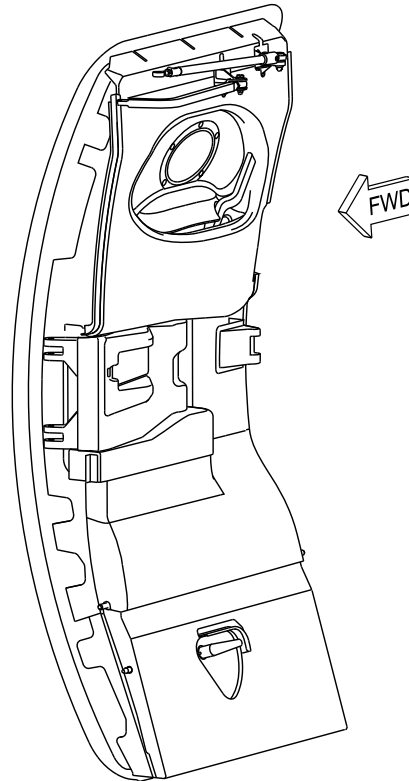
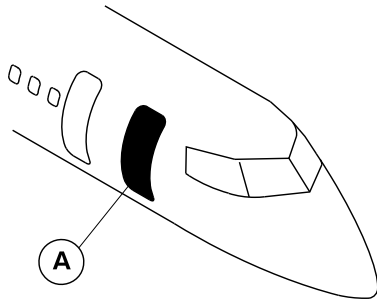
Dam, Ditching

The ditching dam is a composite panel installed in the bottom part of the Type I emergency exit door. For ditching dam evacuation, the handle on the ditching dam rotated 60° clockwise to engage the lock pins on either side of the door and guide pins are retracted. This action will lock the ditching dam with the door surround structure and release it from the Type I emergency exit door.



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**VIEW LOOKING
OUTBOARD**

cg4045a01.dg, cm, aug07/2015

Forward Type-I Emergency Exit Location
Figure 1

PSM 1-84-2A
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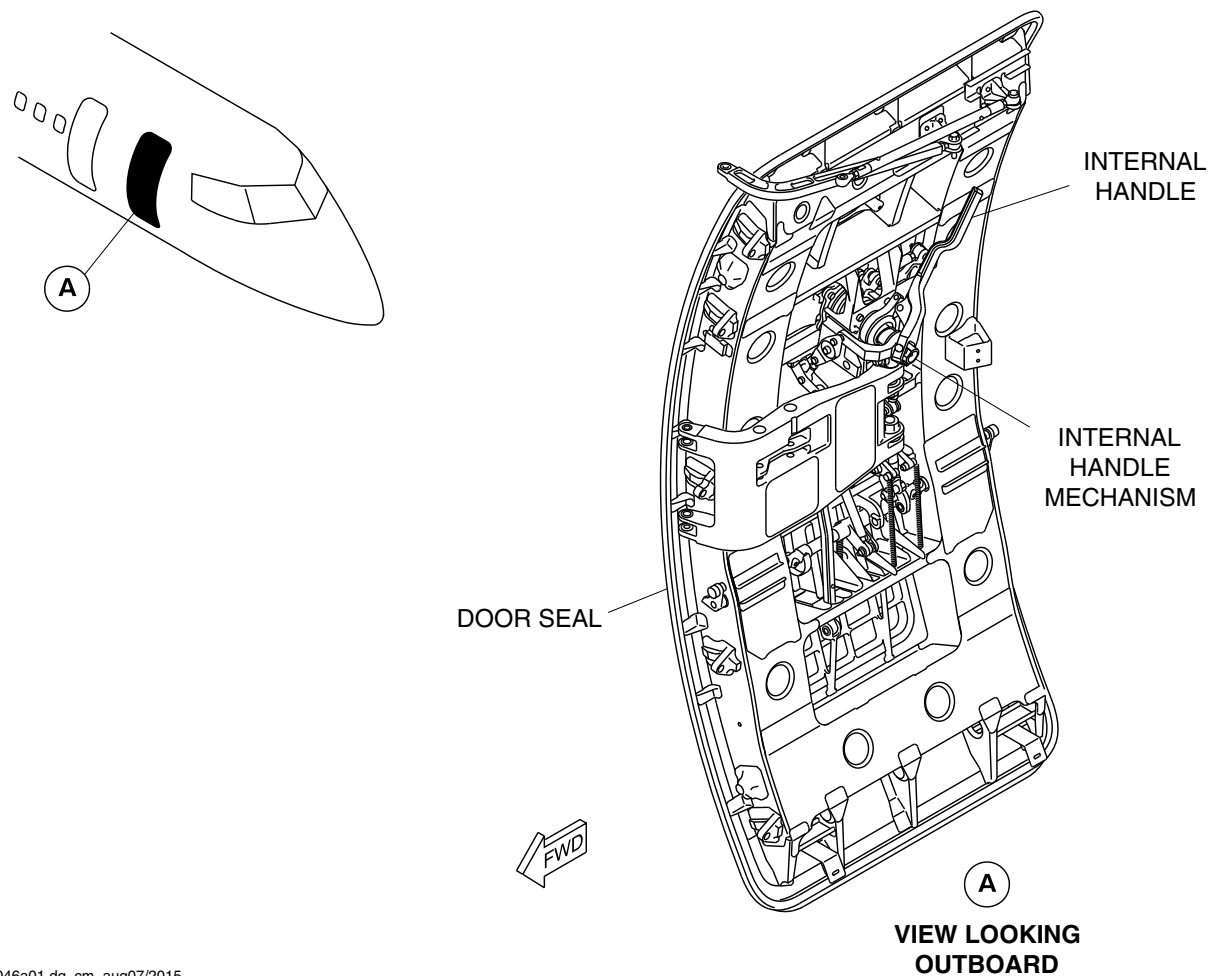
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Forward Type-I Emergency Exit Detail Page 1
Figure 2

PSM 1-84-2A
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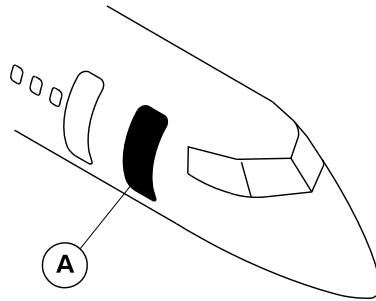
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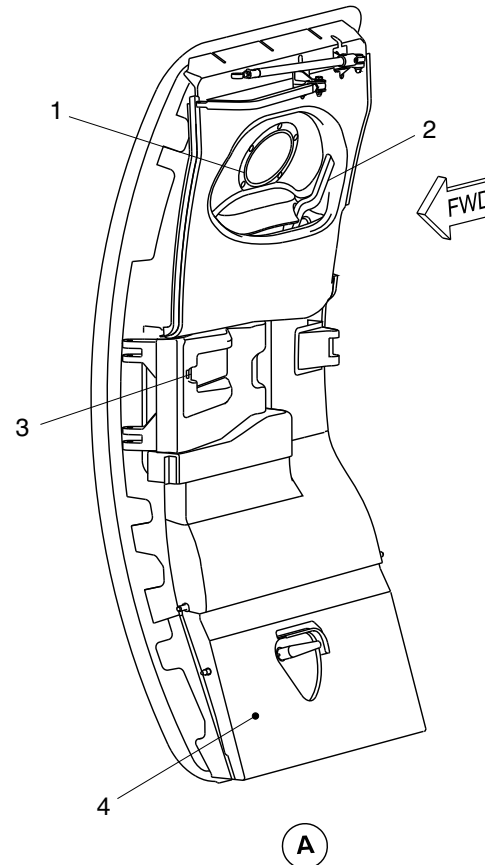
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LEGEND

- 1. Window.
- 2. Internal handle.
- 3. Open lock lever
- 4. Ditching dam.



**VIEW LOOKING
OUTBOARD**

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Forward Type-I Emergency Exit Internal View
Figure 3

PSM 1-84-2A
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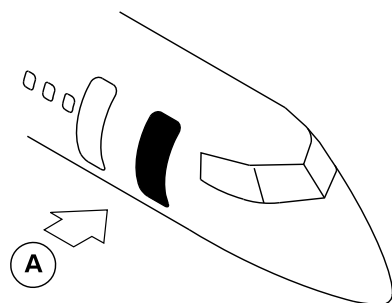
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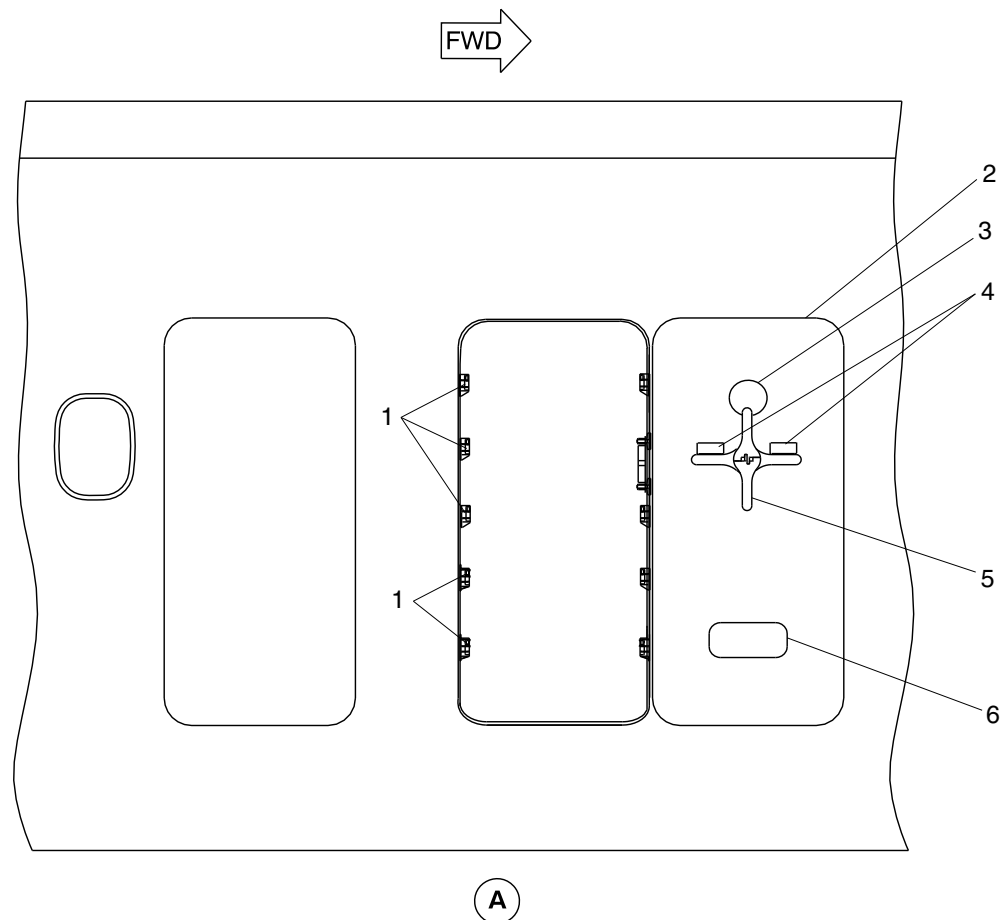
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Stops (5 per side).
2. Forward type I emergency exit. exit (open position)
3. Window.
4. Flap cover.
5. External handle (unlocked).
6. Vent door (open).



VIEW LOOKING INBOARD

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Forward Type-I Emergency Exit Detail Page 2
Figure 4

PSM 1-84-2A
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52-25-00-001

EMERGENCY EXIT TYPE III DOOR

Introduction

On aircraft with the Modsum 4-459334 incorporated, the two emergency exit type III doors are installed. The two emergency exit type III doors supply an exit for emergency evacuation of the passenger compartment.

General Description

Refer to Figure 1.

There are two type III emergency exit doors, one on each side of the fuselage, at Sta. X479.70. The doors are the plug type which have to be removed completely for opening.

The emergency exit type III doors have the components that follow:

- Mechanism, Latch (52-25-01)
- Door seal (52-25-21)

The door warning system monitors door status and supplies visual indication and an aural warning in the flight compartment. The type III emergency exit door interfaces with:

- Door Warning System (52-70-00)
- Proximity Sensors (52-71-01)

Detailed Description

Refer to Figure 2.

The door is secured by a lock pin to the upper side, two stops at the bottom side and three adjustable stops on each vertical edge. The three adjustable stops engage with three stop brackets installed on the sides of the door to secure it against cabin pressure loads.

The door is unlocked by the operation of the internal or external handle. Both handles are spring-loaded and interconnected by cables and pulleys. When the internal handle of the door is pulled, the vent dish opens and pressure in the aircraft is released to the atmosphere. The locking pin retracts and the door can then be removed.

Refer to Figures 3 and 4.

To remove the door with the external handle, quick release button is operated to release the handle. The handle is turned counterclockwise to operate the cable and shaft quadrant to open the vent dish and retract the locking pin. The door is pushed inward and removed.

Refer to Figures 5 and 6.

A proximity switch, installed on the upper edge of the door, monitors the lock position of the door. If the door is unlocked, the FUSELAGE DOORS warning and the MASTER WARNING light flash. The Doors Systems page on the Multi Function Display (MFD) shows the unlocked door as a red filled rectangle. If the door is not properly closed and locked the cabin will not pressurize.



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Mechanism, Latch (Door)

The lower end of the internal handle is connected to the handle shaft which is spring loaded to the locking direction of the lock pin. When the internal handle is pulled, the vent dish is opened and the lock pin is retracted, the door can now be removed. The lock pin is kept at the locked position by two independent bias springs. The internal handle has a cover to prevent the door from being accidentally opened.

Door seal

The door seals are passive type installed around the doors. The seals contain several holes which let the seal inflate under cabin pressure.

Vent Dish seal (Door)

This seal is a compression seal made of neoprene sponge and fits around the periphery of the vent dish.

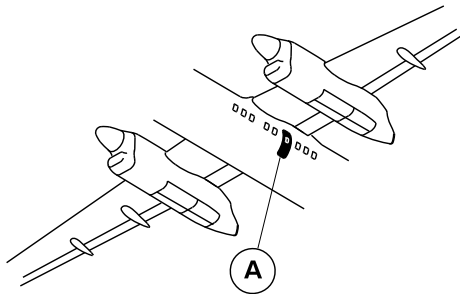
Assist handle

On aircraft with ModSum 4-459476 or SB84-25-183 incorporated, the assist handle is attached with the door liner.



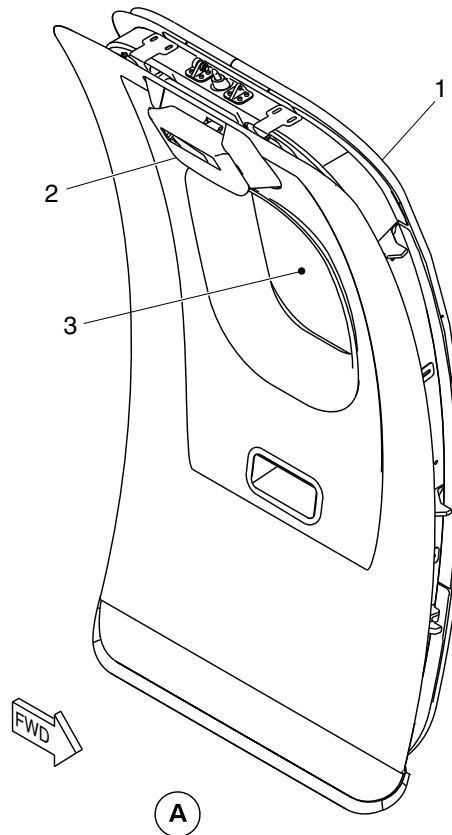
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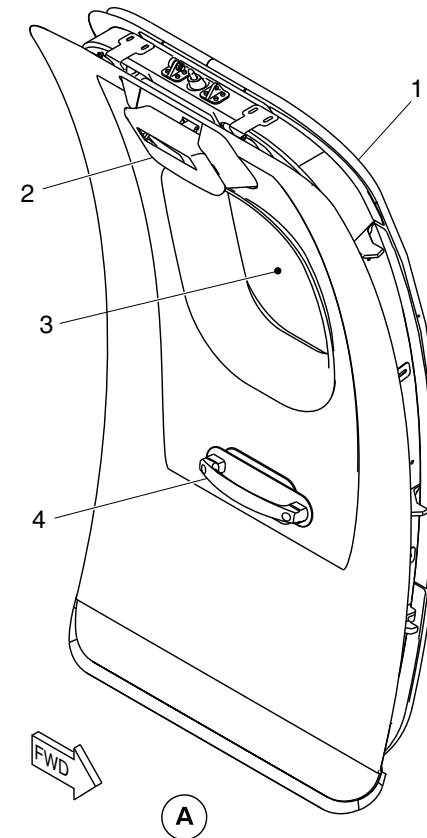


LEGEND

1. Type III emergency exit door.
2. Internal handle.
3. Window.
4. Assist handle.



PRE MODSUM 4-459476
PRE SB84-25-183
LEFT DOOR SHOWN
RIGHT DOOR SIMILAR



POST MODSUM 4-459476
POST SB84-25-183
LEFT DOOR SHOWN
RIGHT DOOR SIMILAR

cg4300a01.dg, rs/gw, jan4/2017

Type III Emergency Exit Door Locator
Figure 1

PSM 1-84-2A
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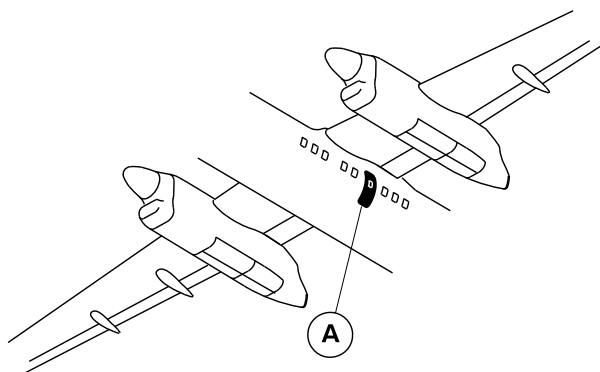
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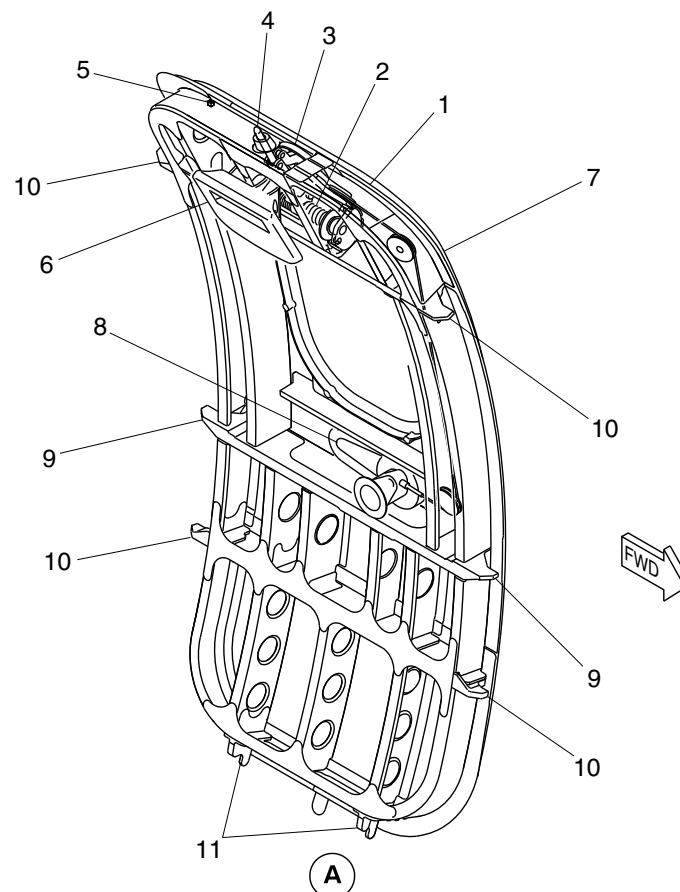
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LEGEND

1. Door seal.
2. Bias spring.
3. Vent dish.
4. Lock pin.
5. Proximity switch.
6. Internal handle.
7. Door.
8. External handle.
9. Backup stop.
10. Main stop.
11. Bottom door stop.



**LEFT DOOR SHOWN
RIGHT DOOR SIMILAR**

cg4301a01.dg, rs, dec21/2015

Type III Emergency Exit Door
Figure 2

PSM 1-84-2A
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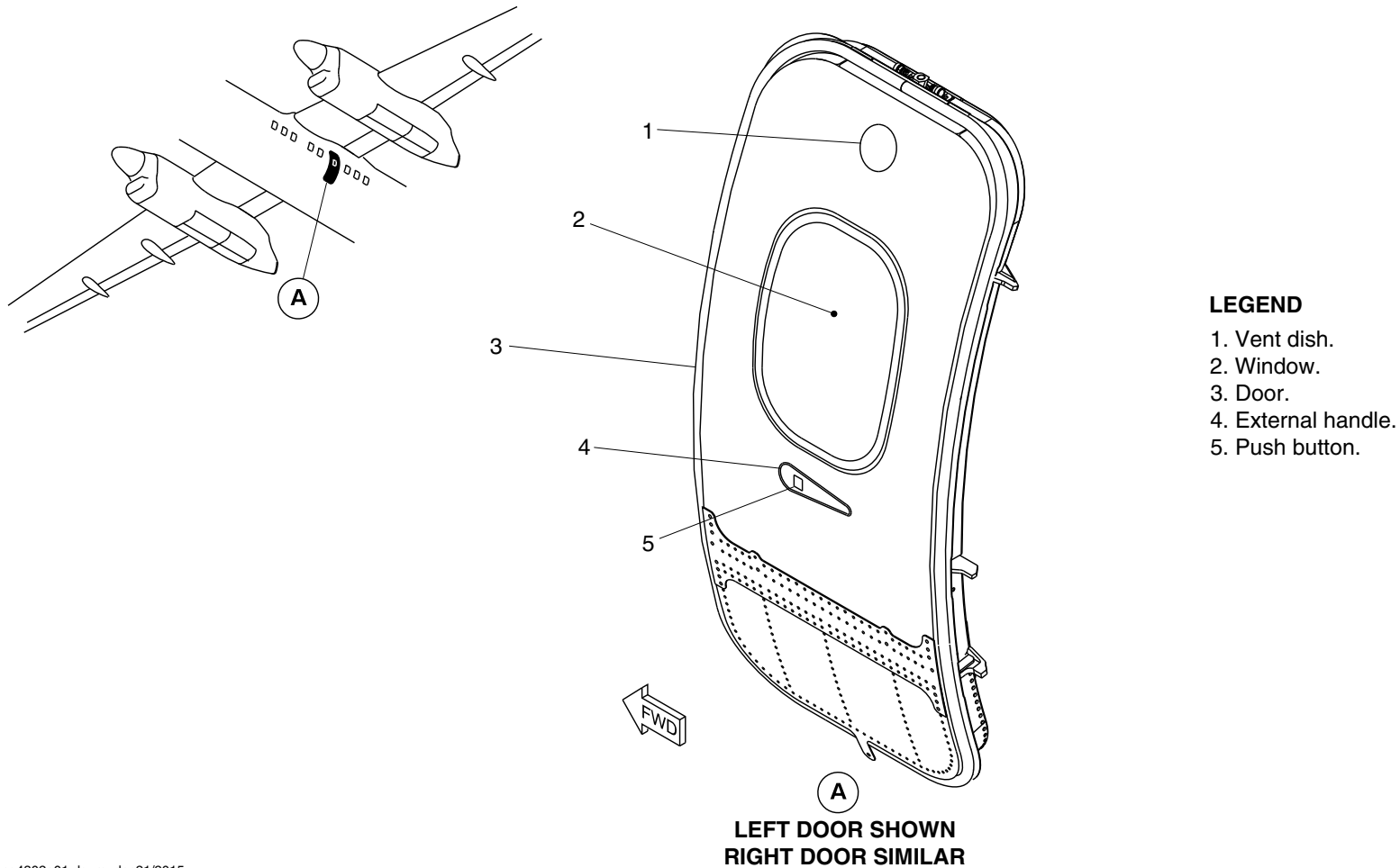
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External Detail Type III Emergency Exit Door
Figure 3

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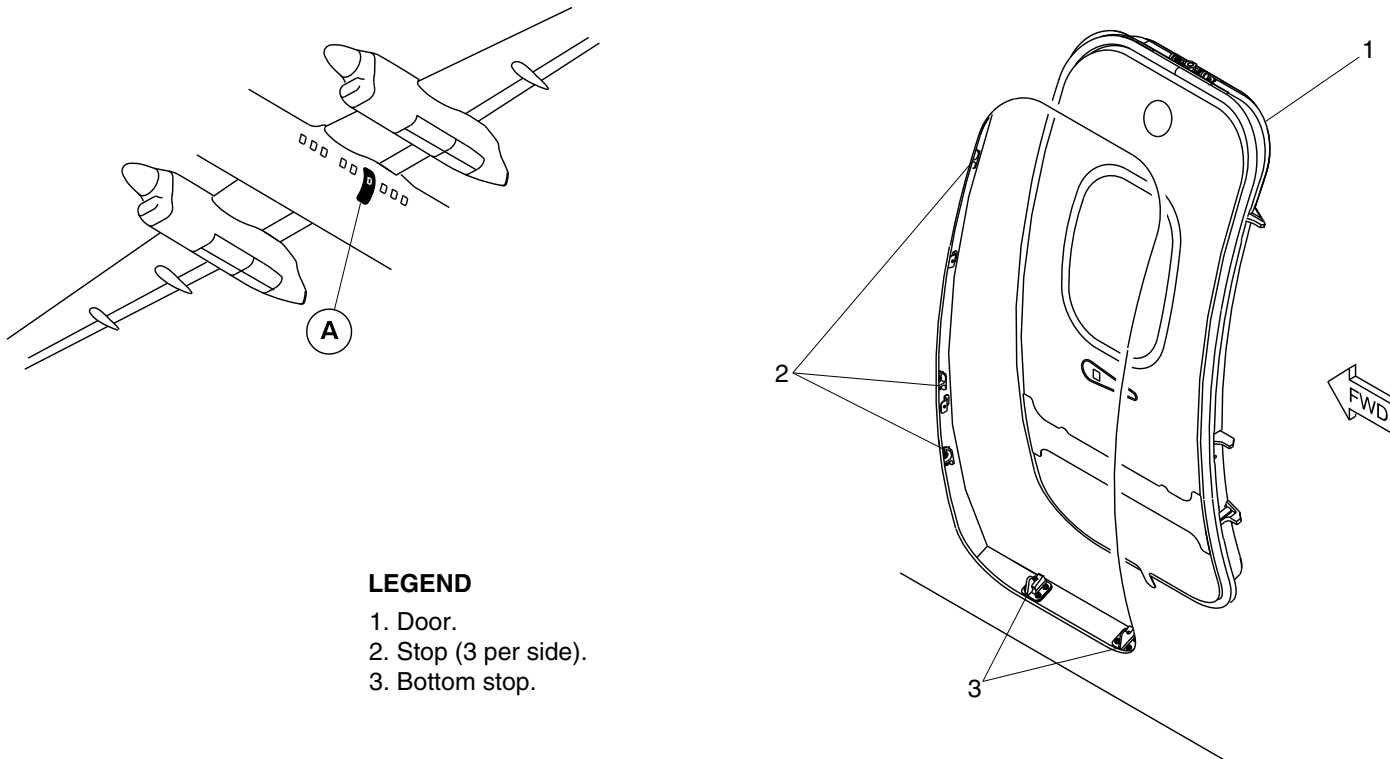
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LEGEND

- 1. Door.
- 2. Stop (3 per side).
- 3. Bottom stop.

A

LEFT DOOR SHOWN
RIGHT DOOR SIMILAR

cg4303a01.dg, rs, dec21/2015

Type III Emergency Exit Door Open
Figure 4

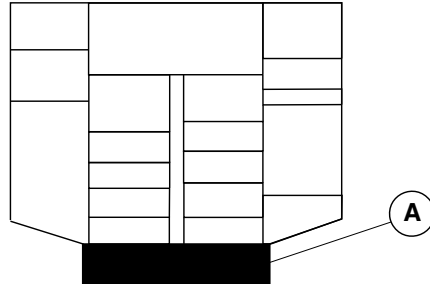
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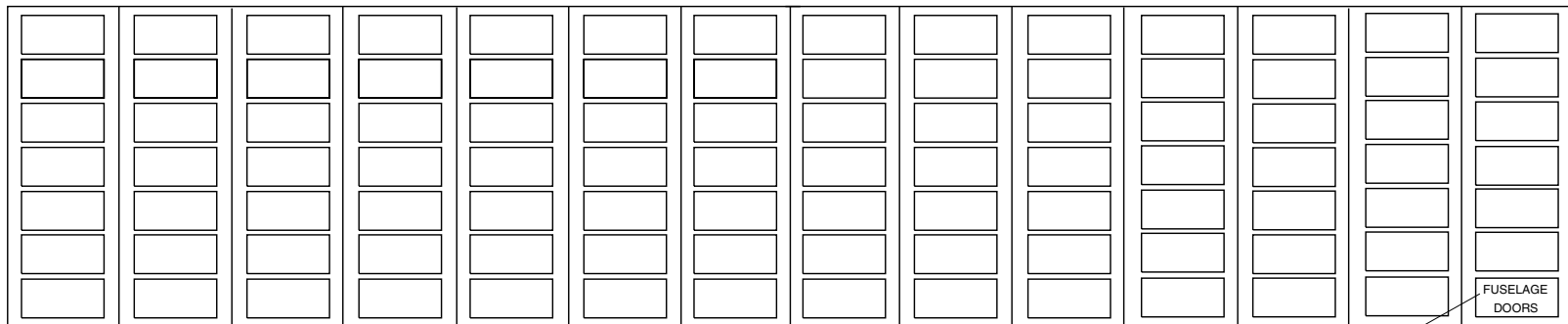
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OVERHEAD CONSOLE

LEGEND

1. Fuselage doors (Red).



cg4304a01.dg, rs, dec21/2015

Fuselage Doors Warning Light
Figure 5

PSM 1-84-2A
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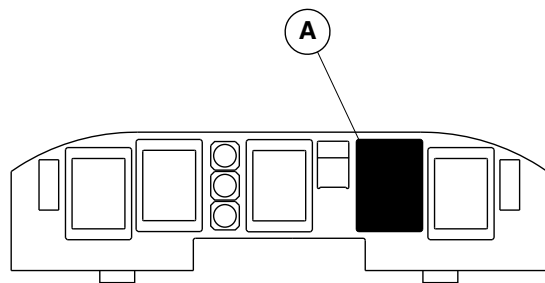
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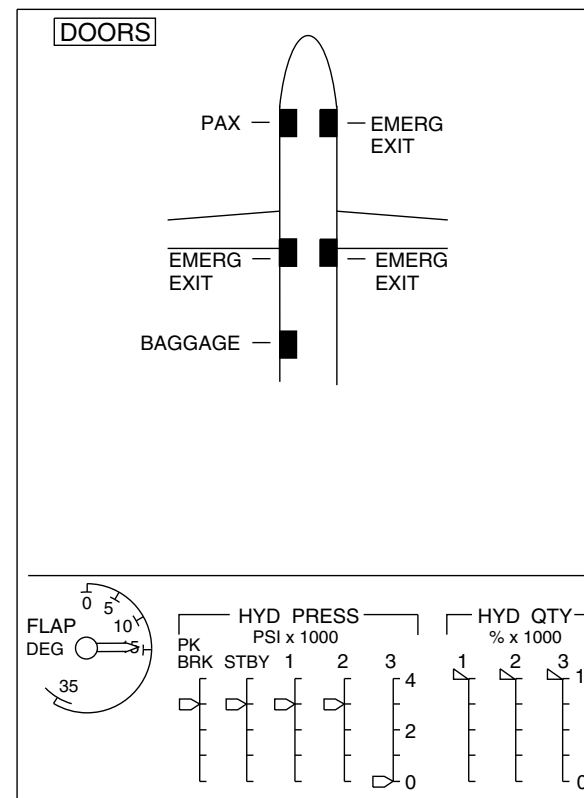


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MAIN INSTRUMENT PANEL



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Doors System Page
Figure 6

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**ON A/C ALL

52-30-00-001

BAGGAGE DOORS

Introduction

The baggage doors give access to the forward and aft baggage compartments.

General Description

Refer to Figure 1.

The aft baggage/cargo compartment is accessible externally, through an outward and upward opening baggage door in the left side rear fuselage.

The forward baggage compartment is accessible externally through a door in the right front fuselage. The door opens inward and upward, and then translates outward and forward. The doors can be opened or closed manually by an external handle.

On aircraft with Modsum 4-459324 or 4-458926 or 4-190614 incorporated, the forward baggage door is removed and replaced with a Type 1 emergency exit door.

On aircraft with Modsum 4-201647 incorporated, a Type 1 emergency exit door is fitted at the location of the forward baggage door.

On aircraft with Modsum 4-459339 incorporated, the forward baggage door is reworked to add an internal handle, a window and

ditching dam guide pin bracketry to convert into a Type I emergency exit door.

Internal access to the forward baggage compartment is through a hinged door installed in the passenger compartment divider. The door has a key operated lock.

There is no internal access to the aft baggage compartment.

The aft baggage door incorporates an inflatable door seal, and the forward baggage door a passive type seal.

Detailed Description

The baggage doors includes the components that follow:

- Aft Baggage Door (52-31-00)
- Forward Baggage Door (52-32-00)

Aft Baggage Door

Refer to Figures 2 and 3.

The aft baggage door opens outward and upward about an upper hinge, and is held open by a telescopic strut. An external tee handle is used to lock and unlock the door from the outside. There is no internal handle for the baggage door. The door pressure seal inflates to prevent air leakage. The aft baggage door has an external, aluminum alloy skin braced by cross beams. The door warning system monitors the door status and supplies a visual indication and warning in the flight compartment.



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Forward Baggage Door

Refer to Figures 4 , 5 and 6.

On aircraft with Modsum 4–459324 or 4–458926 or 4–190614 incorporated, the forward baggage door is removed and replaced with a Type 1 emergency exit door.

On aircraft with Modsum 4–201647 incorporated, a Type 1 emergency exit door is fitted at the location of the forward baggage door.

On aircraft with Modsum 4–459339 incorporated, the forward baggage door is reworked to add an internal handle, a window and ditching dam guide pin bracketry to convert into a Type I emergency exit door.

The forward baggage door is unlocked by the operation of an external tee handle. The door opens or closes with the door face parallel with the fuselage skin. This is done by two hinges (main and secondary) and a stabilizer.

Proximity sensors monitor the position of the upper fixed rollers and the door handle. A red warning light comes on in the flight compartment when the door is open or unlocked. The door warning system monitors door status and supplies visual indication and warning in the flight compartment.

PSM 1–84–2A

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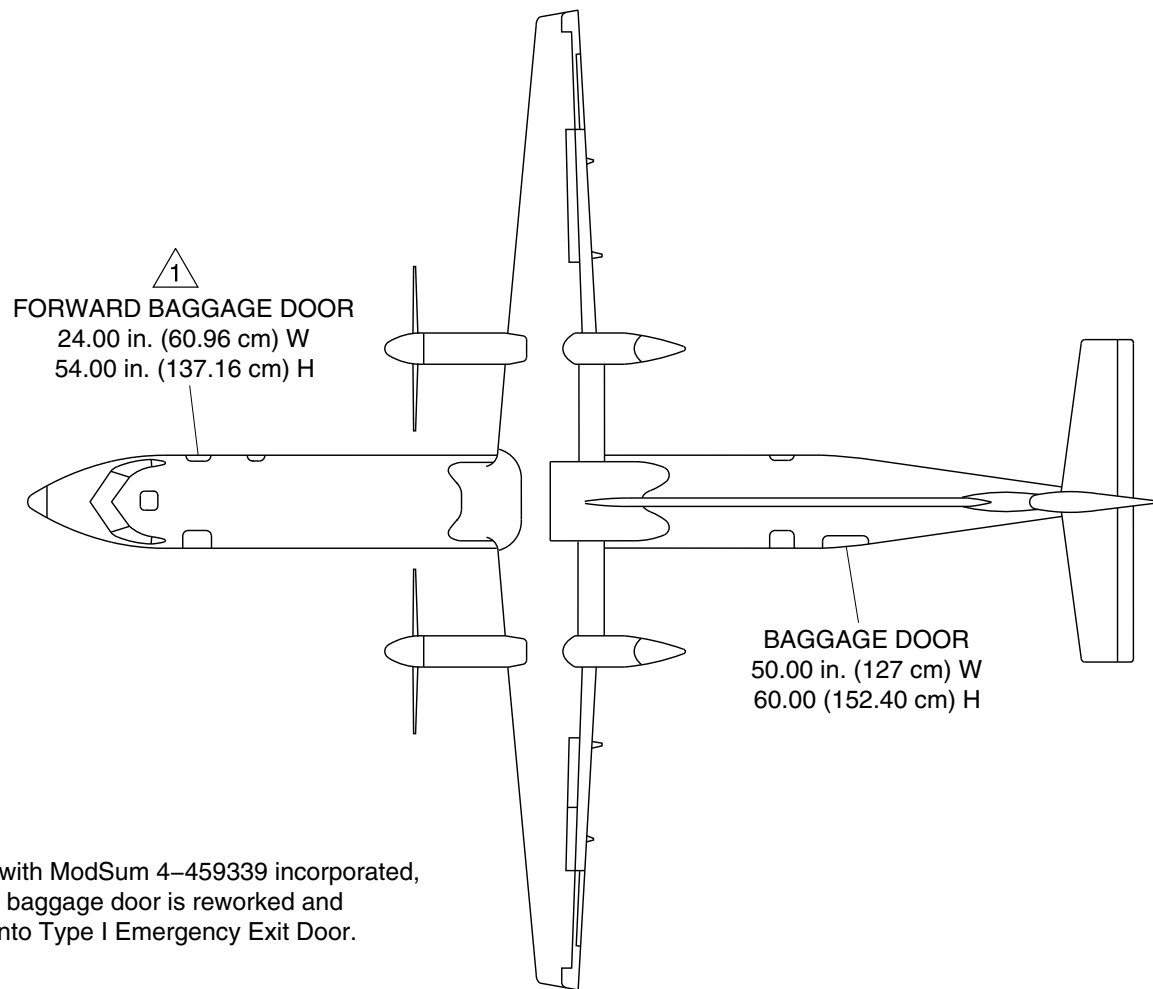
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NOTE



On aircraft with ModSum 4-459339 incorporated, the forward baggage door is reworked and converted into Type I Emergency Exit Door.

fsh59a01.dg, vn/pr, jun06/2016

Baggage Door Locations
Figure 1

PSM 1-84-2A

EFFECTIVITY:

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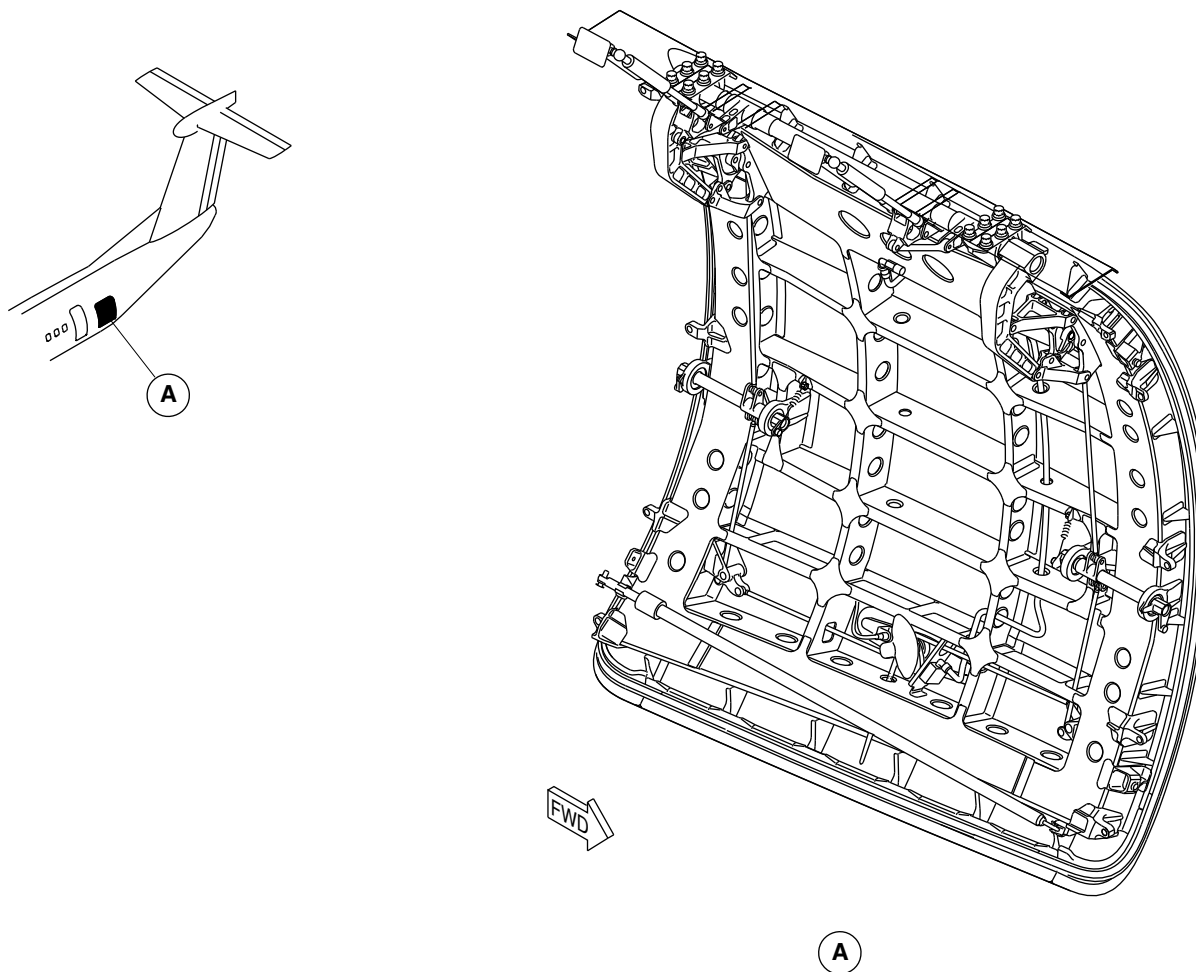
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DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AFT BAGGAGE DOOR LOCATION
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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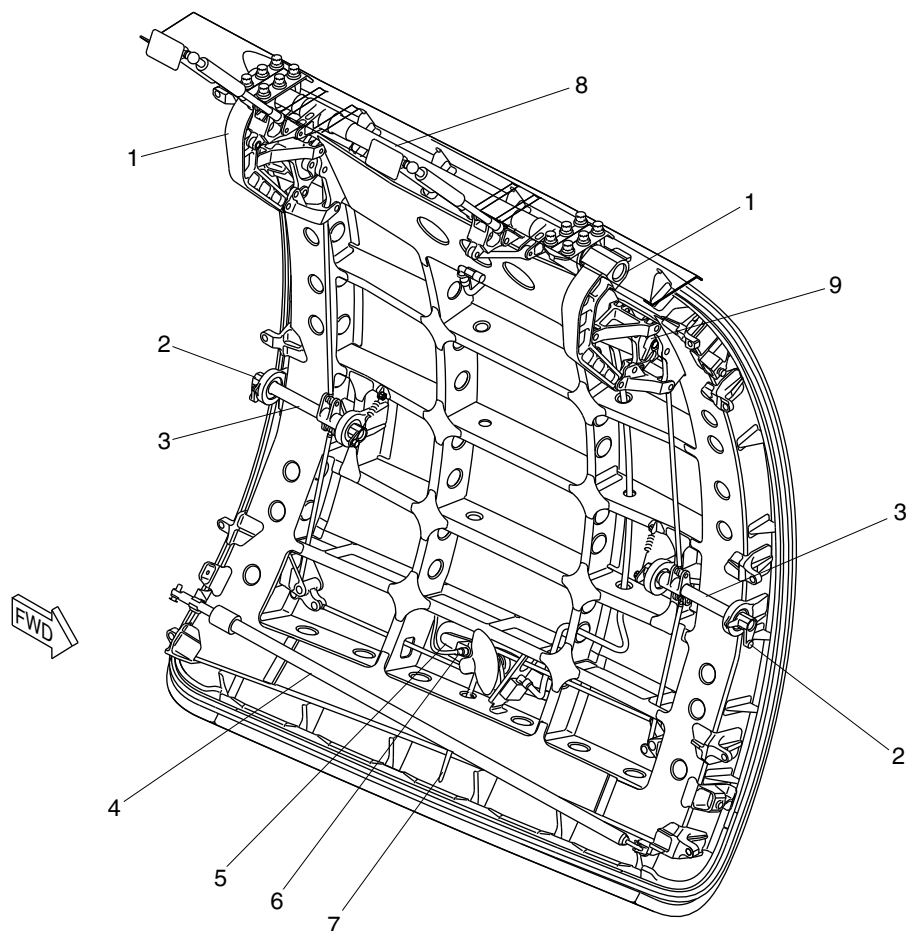
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Upper Hinge Arms.
2. Latch Roller.
3. Latch Shaft.
4. Telescopic Strut.
5. External Tee Handle.
6. "Locked" Proximity Switch.
7. Handle Lock.
8. Upper Hinge Shaft.
9. Lockout Hook (Aft Side Similar).

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AFT BAGGAGE DOOR DETAIL
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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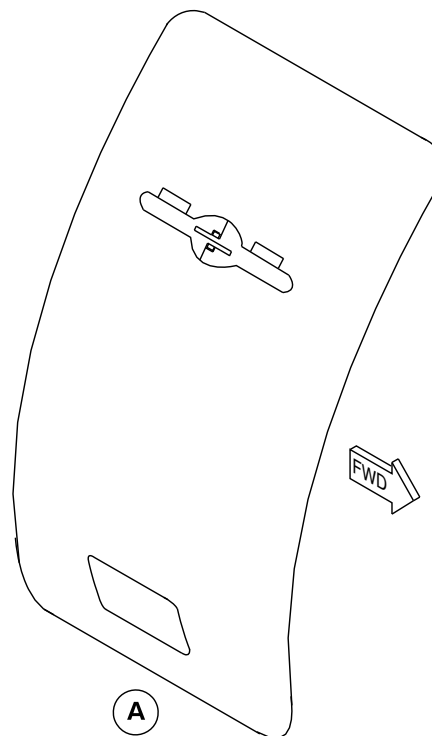
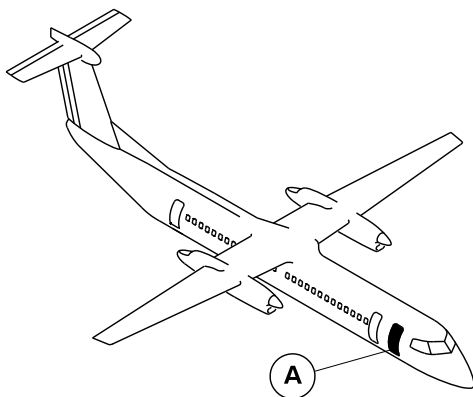
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



NOTE

On aircraft with ModSum 4-459339 incorporated, the forward baggage door is reworked and converted into Type I Emergency Exit Door.

cg4702a01.dg, pr, june06/2016

Forward Baggage Door Location
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-30-00
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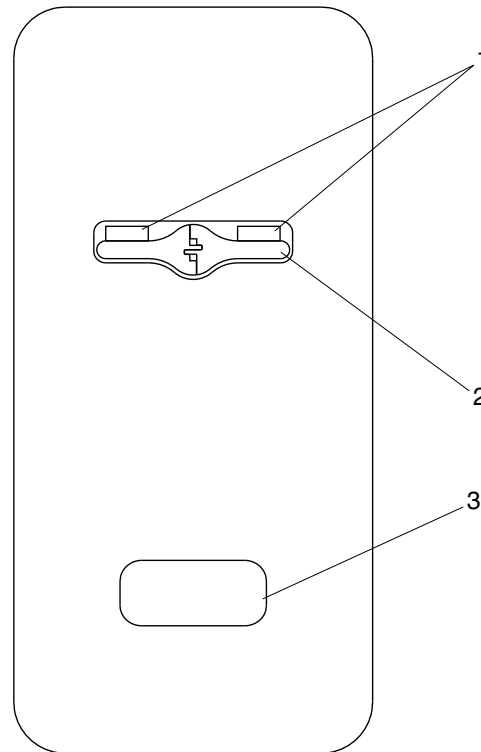
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Flap Cover.
- 2. External Handle.
- 3. Vent Door.

NOTE

On aircraft with ModSum 4-459339 incorporated,
the forward baggage door is reworked and
converted into Type I Emergency Exit Door.

cg4711a01.dg, pr, jun06/2016

Forward Baggage Door External Detail Page 1
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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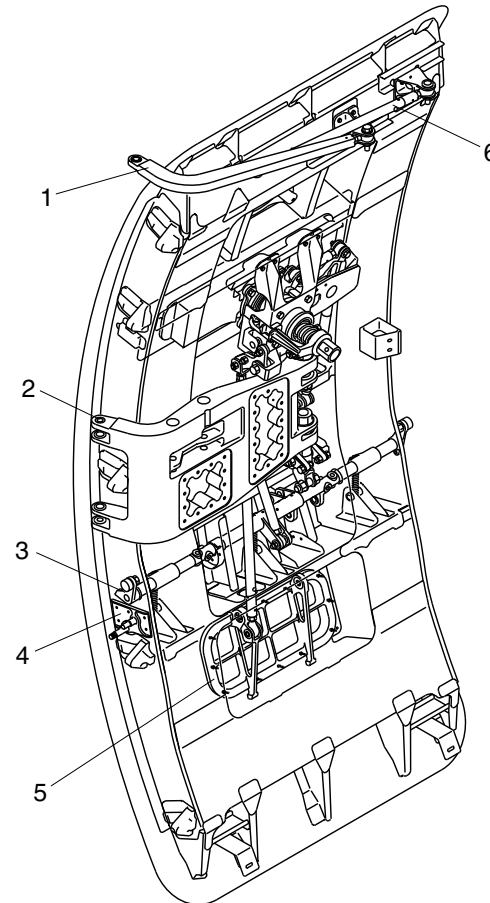
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Secondary Hinge.
2. Main Hinge.
3. Lift And Latch Shaft.
4. Interlock Assembly.
5. Vent Door.
6. Stabilizer Rod.

NOTE

On aircraft with ModSum 4-459339 incorporated, the forward baggage door is reworked and converted into Type I Emergency Exit Door.

cg4712a01.dg, vn/pr, jun26/2016

Forward Baggage Door External Detail Page 2
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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52-31-00-001

AFT BAGGAGE DOOR

Introduction

The aft baggage door gives access to the aft baggage compartment.

General Description

[Refer to Figures 1 and 2.](#)

The aft baggage door is located on the left side of the fuselage at the rear baggage compartment. The door opens outward and upward about an upper hinge, and is held open by a telescopic strut. An external tee handle is used to lock and unlock the door from the outside. There is no internal handle for the baggage door. The door pressure seal inflates to prevent air leakage. The aft baggage door has an external, aluminum alloy skin braced by cross beams. The door warning system monitors the door status and supplies a visual indication and warning in the flight compartment.

The Aft Baggage Door System has the components that follow:

- Mechanism, Handle (52-31-01)
- Mechanism, Hinge (52-31-06)
- Mechanism, Latch (52-31-11)
- Strut (52-31-16)
- Mechanism, Door Balance (52-31-21)

- Seal, Door (52-31-26)
- Valve, Control-Door Seal (52-31-31).

Detailed Description

[Refer to Figures 3 and 4.](#)

The external tee handle, located in a recess in the lower part of the door, locks and unlocks the door. A push button in the handle releases it from its flush stowage position. The initial movement of the door is inward and downward, with roller guides guiding the latch rollers on each end of the latch shafts. The door then turn outwards about an upper hinge at the top of the door.

[Refer to Figure 5.](#)

Two upper hinge arms attach to an upper hinge shaft installed in the fuselage above the door. Roller guides on the door surround structure guide the latch rollers as the door moves to the fully lowered position. When the door reaches the fully lowered position, lockout hooks engage in the upper hinge arms to keep the door at a constant distance from the fuselage. A handle lock prevents operation of the tee handle, and keeps the lockout hook engaged. A telescopic strut helps hold the door in the open position.

[Refer to Figure 6.](#)

When the door is in the closed position, door pressurization loads are held by ten pressure stops, five on each side of the door. These stops engage with ten stop brackets on the door surround structure. The latch rollers also hold the aft baggage door and prevent vertical movement.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Refer to Figures 7 and 8.

A balance system helps to balance the weight of the door as the door is opened. The system is controlled by four gas springs, which apply a balance force to the upper hinge shaft through levers and rod assemblies. These springs also dampen and control the rate of door movement as the door is opened or closed.

Refer to Figure 9.

The door has an inflatable rubber seal installed around the edge of the door. Pressurized air is supplied to the seal and is controlled by a pressurization valve. The pressurization valve operates when the tee handle is extended or retracted. When the handle is extended, the seal deflates. When the handle is retracted, the seal inflates.

Two proximity sensors monitor the status of the aft baggage door. One sensor is located at the forward fixed-roller guide to monitor the open and closed positions. The other sensor, located in the door handle housing, monitors the handle retracted and extended position.

Refer to Figures 10, 11 and 12.

The door operating and retention system is linked by cams to the external tee handle, and controls the door movement. The external handle push-button is pushed to release the handle from the recessed position. The handle is pulled out fully to release the lock and to disconnect the seal pressurization valve and the "locked" proximity switch. The handle is turned 90 degrees counterclockwise to operate the latch rollers. The door then moves inward and downward. The "closed" proximity switch opens, and the stops clear their corresponding stop brackets on the fuselage. At the fully lowered position, the lockout hooks latch to the upper hinge arms. The door then opens outward and upward about the door hinge,

helped by the counter balance assembly. As the door opens, the tee handle locks in the open position by a handle lock mechanism. The gas springs, located above the door hinge, helps to open and hold the door open. The telescopic strut is then used to keep the door in the opened position.

Refer to Figures 11 and 12.

To close the aft baggage door, the telescopic strut is disengaged and pulled down. The door movement is helped by the counter balance system. When the door reaches the lowered position, the latch rollers move into the roller guides, and the handle lock mechanism is released. The external handle is turned clockwise to the horizontal position to push the latch rollers down, and the door is lifted outward and upward. This disconnects the lockout hook and puts the stop bolts behind their related pressure stops, and connects the "closed" proximity switch. The tee handle is pushed in to connect the "locked" proximity switch and the pressurization valve inflates the door seal.

Mechanism, Handle

The external handle mechanism is turned to operate both the latch shaft and roller assemblies. The mechanism also operates the movement of the linkages to the lockout hooks.

Mechanism, Hinge

The hinge mechanism is located above the aft baggage door and is attached to the fuselage by two hinge support fittings. The balance system rod assemblies are attached to the hinge. The two hinge arms are attached to the door by upper and lower links.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Mechanism, Latch

The latch mechanism has latch shaft and roller assemblies located at the forward and aft sides of the door. The tee handle is turned to cause the latch rollers to move along the roller guides. In the fully closed position, the latch roller is offset inward from the door movement direction. The overcenter geometry keeps the roller at that position, and against downward movement of the door. The latch shafts are kept in the closed position by both the handle mechanism and overcenter springs

Strut

The telescopic strut keeps the door in the open position, and is used to close the door. One end of the strut is attached to a bracket on the aft baggage door, and the other end clamps to door when it is closed.

Mechanism, Door Balance

The door balance mechanism has four gas springs, triangular levers and rods, and helps to balance the door weight as the door turns about the hinge shaft. This reduces the force necessary to raise the door, and overbalances the door weight in the fully raised position.

Seal, Door

A door seal is installed on the door surround structure to prevent air leakage through the aft baggage door. The seal inflates with pressurized air from a reservoir tank or the deicing system.

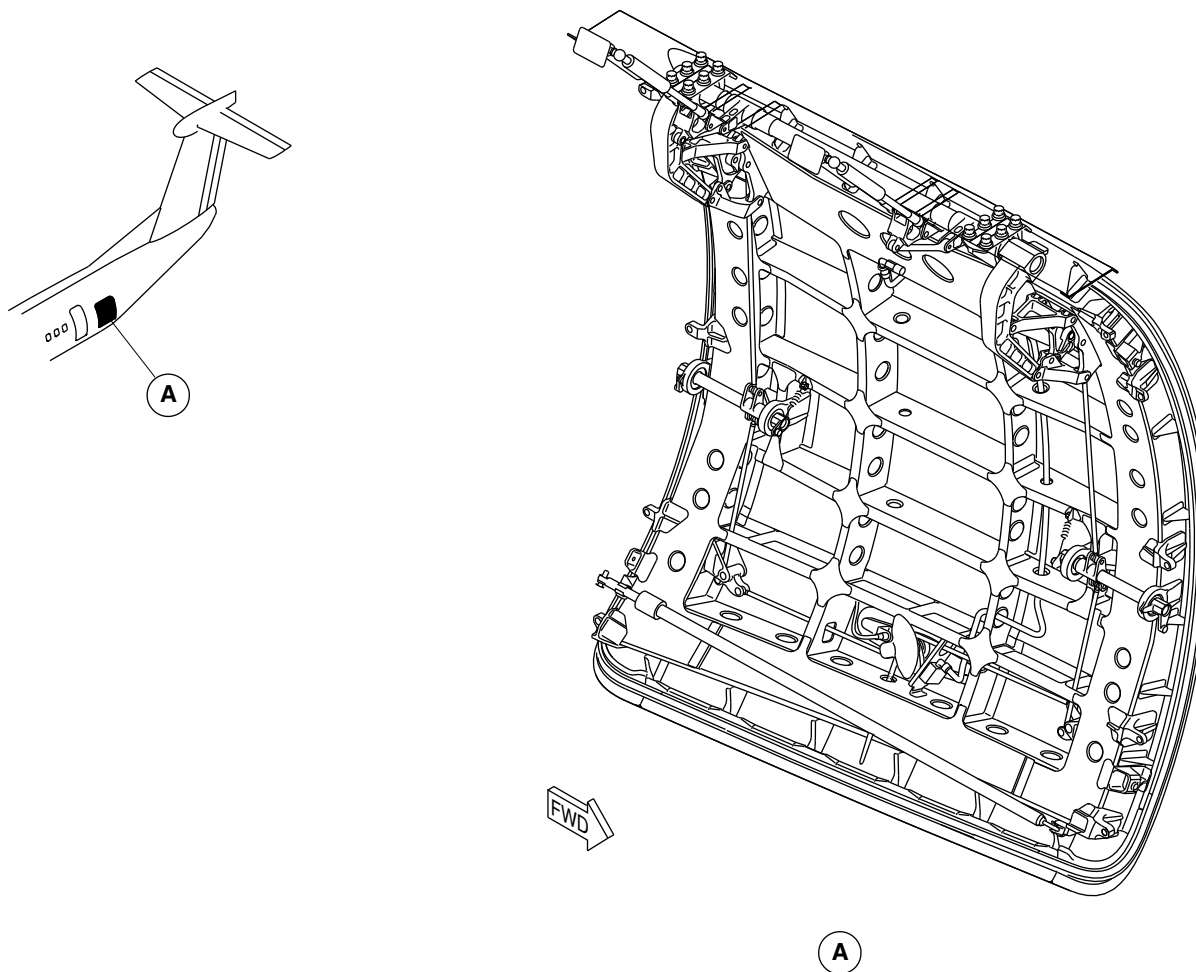
Valve, Control–Door Seal

The door seal pressurization valve operates when the door handle is pushed in and the door is in the locked position. If the door is not fully closed and locked, the pressurization valve will not inflate the seal. The valve deflates the door seal when the door is unlocked by the external handle.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AFT BAGGAGE DOOR LOCATION
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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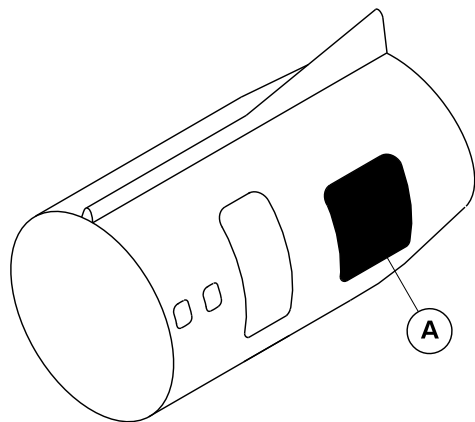
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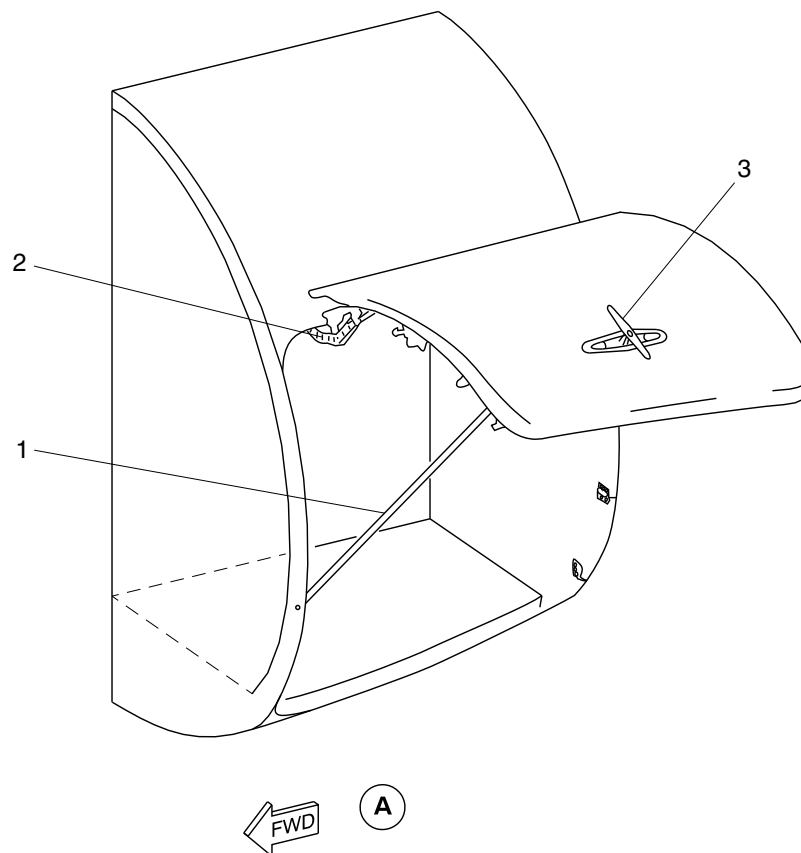
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Telescopic strut.
- 2. Hinge.
- 3. External handle.



fs382a01.dg, vn, aug22/2011

Aft Baggage Door Open
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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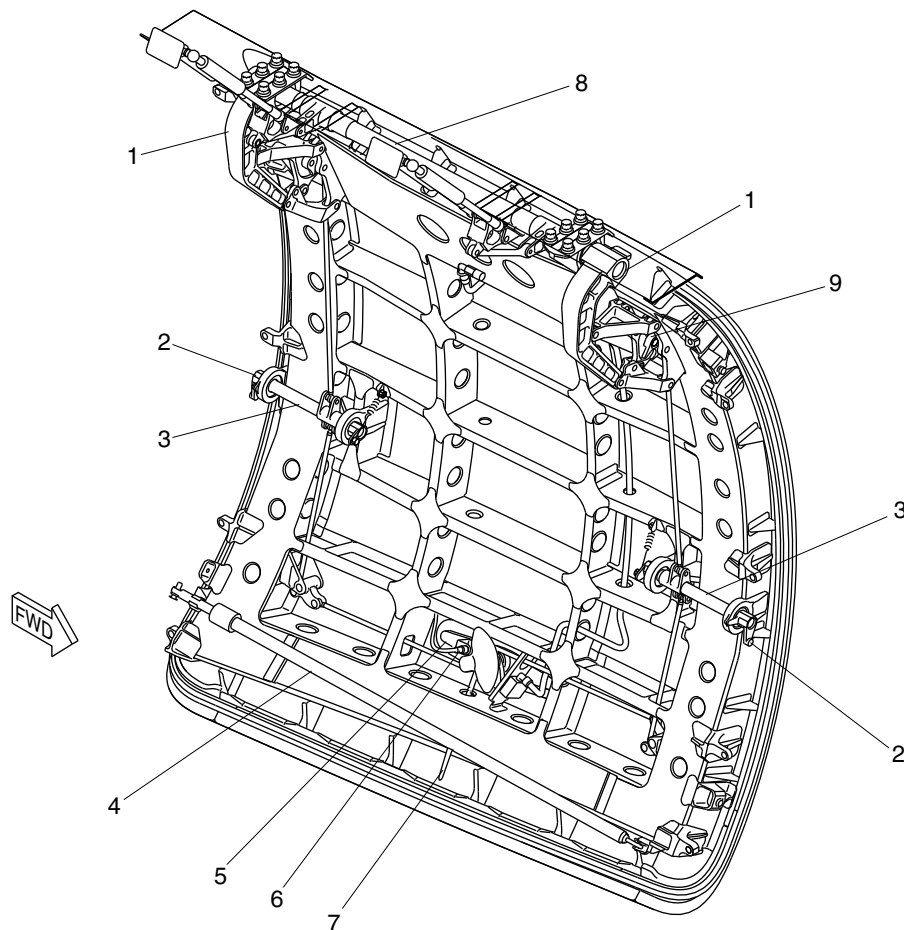
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Upper Hinge Arms.
2. Latch Roller.
3. Latch Shaft.
4. Telescopic Strut.
5. External Tee Handle.
6. "Locked" Proximity Switch.
7. Handle Lock.
8. Upper Hinge Shaft.
9. Lockout Hook (Aft Side Similar).

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AFT BAGGAGE DOOR DETAIL
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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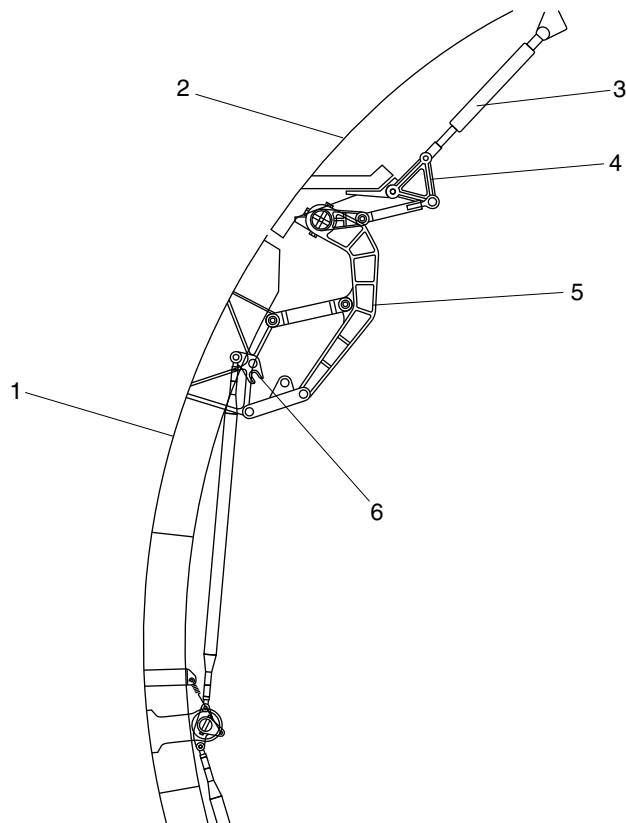
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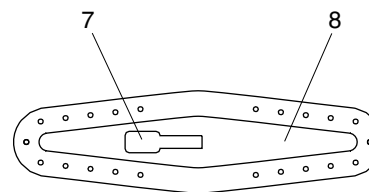
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Aft Baggage Door "Closed".
- 2. Fuselage.
- 3. Gas Spring.
- 4. Triangular Lever.
- 5. Hinge Arm.
- 6. Lockout Hook.
- 7. Push Button.
- 8. Aft Baggage Door Handle "Closed".



fs466a01.cgm

AFT BAGGAGE DOOR CLOSED
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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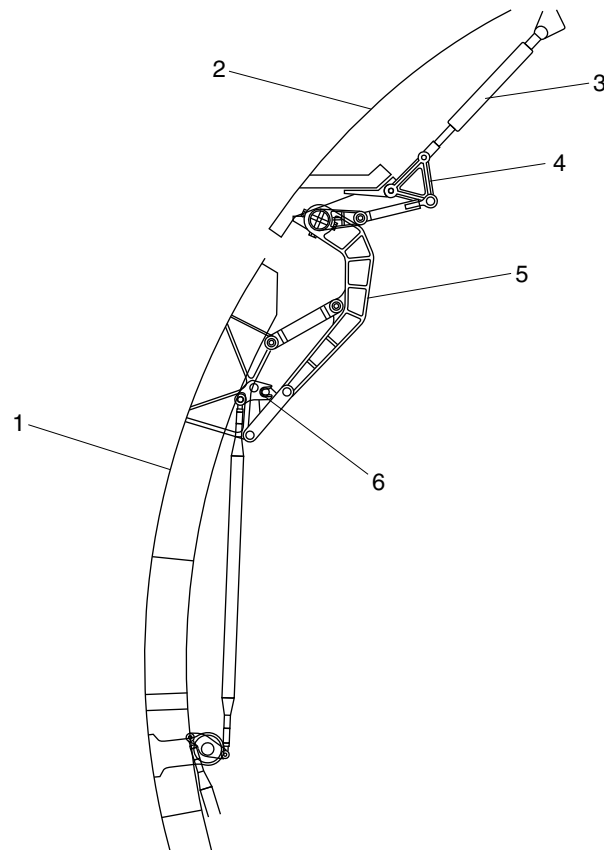
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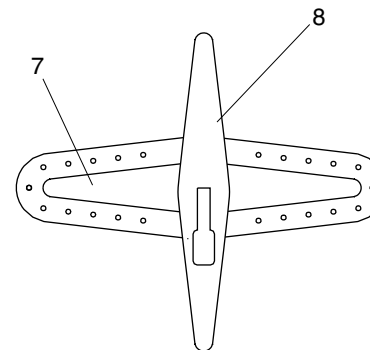
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Aft Baggage Door "Lowered".
2. Fueslage.
3. Gas Spring.
4. Triangular Lever.
5. Hinge Arm.
6. Lockout Hook Engaged.
7. External Door Handle Recess.
8. Aft Baggage Door Handle "Opened".



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AFT BAGGAGE DOOR LOWERED
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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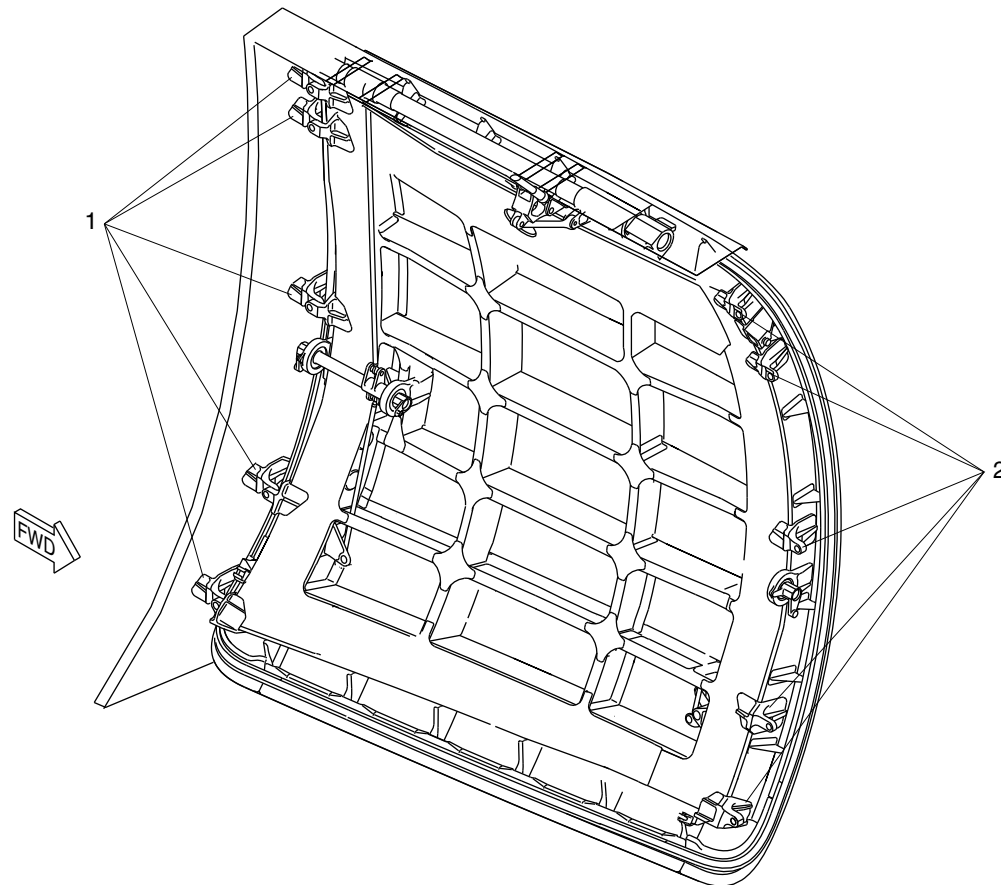
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Pressure Stop Brackets (Forward Side Similar).
2. Pressure Stops (Five Per Side).

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AFT BAGGAGE DOOR STOPS
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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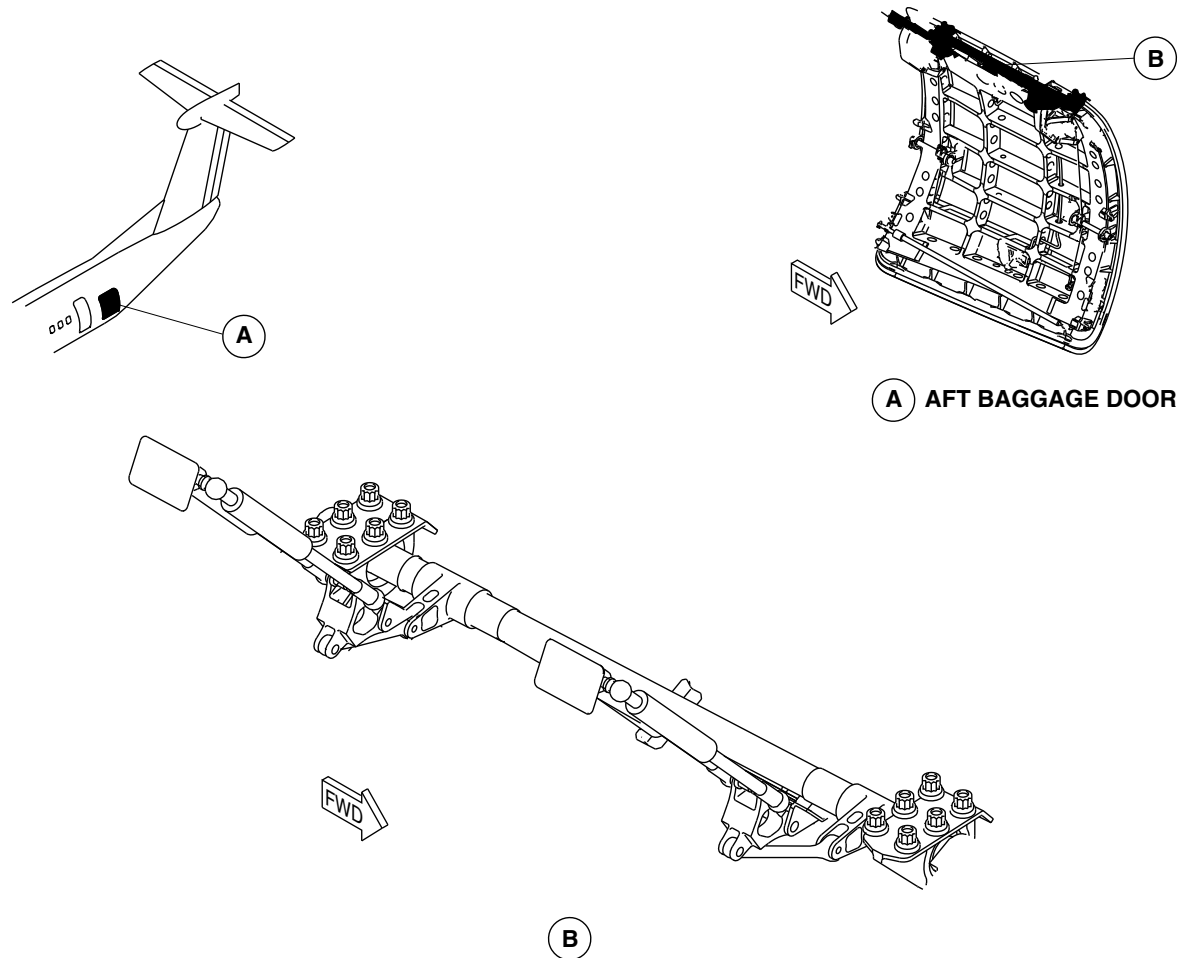
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AFT BAGGAGE DOOR BALANCE SYSTEM LOCATION
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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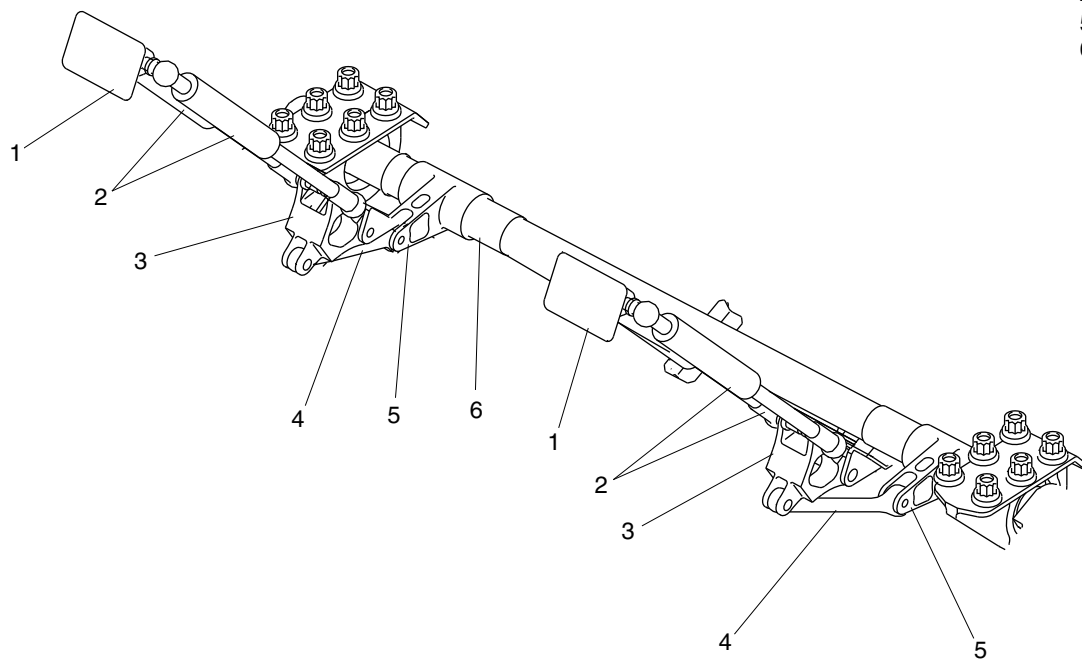


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

1. Support Bracket.
2. Gas Springs.
3. Triangular Lever.
4. Rod Assembly – Balance Spring.
5. Hinge Lever.
6. Upper Hinge Shaft.



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AFT BAGGAGE DOOR BALANCE SYSTEM
Figure 8

PSM 1-84-2A
EFFECTIVITY:
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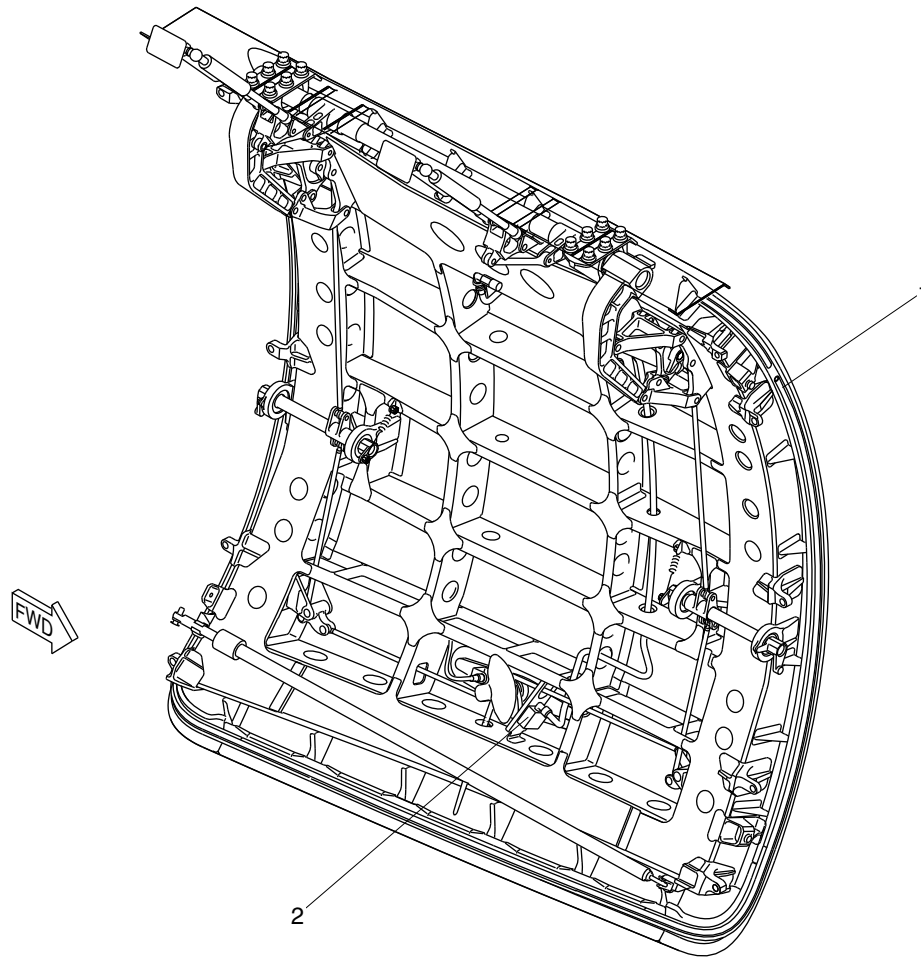
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Inflatable Seal.
- 2. Pressurizing Valve.

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AFT BAGGAGE DOOR SEAL AND PRESSURIZING VALVE
Figure 9

PSM 1-84-2A
EFFECTIVITY:
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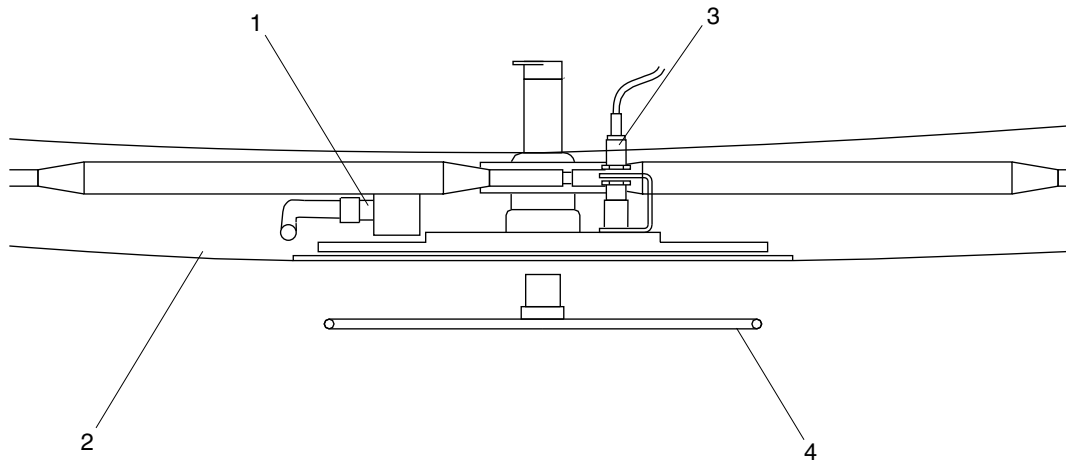


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

1. Pressurizing Valve.
2. Aft Baggage Door.
3. "Locked" Proximity Sensor.
4. External Handle.



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AFT BAGGAGE DOOR EXTERNAL HANDLE
Figure 10

PSM 1-84-2A
EFFECTIVITY:
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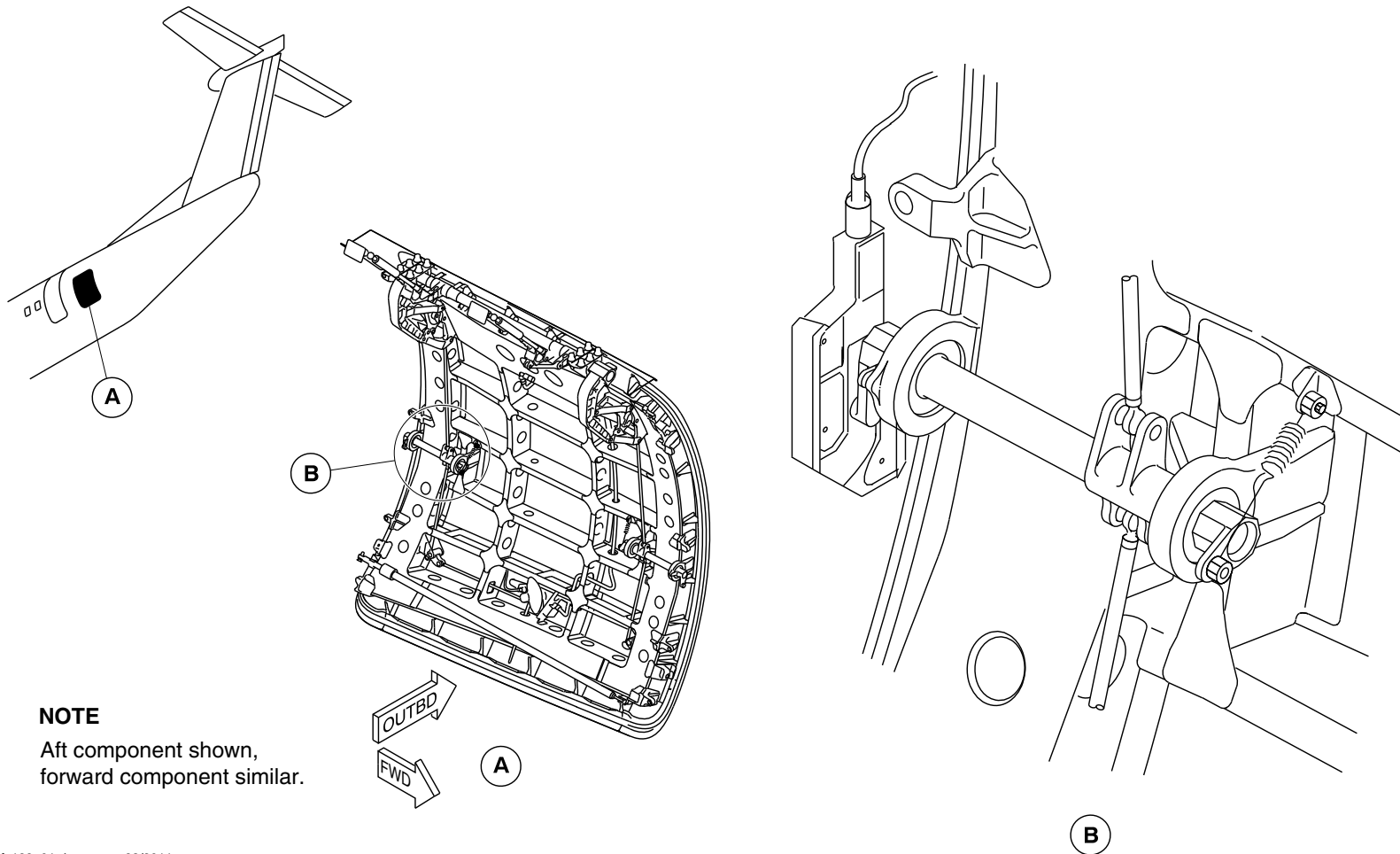
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



NOTE

Aft component shown,
forward component similar.

fs129a01.dg, vn, aug28/2011

Aft Baggage Door Roller Guide Location
Figure 11

PSM 1-84-2A
EFFECTIVITY:
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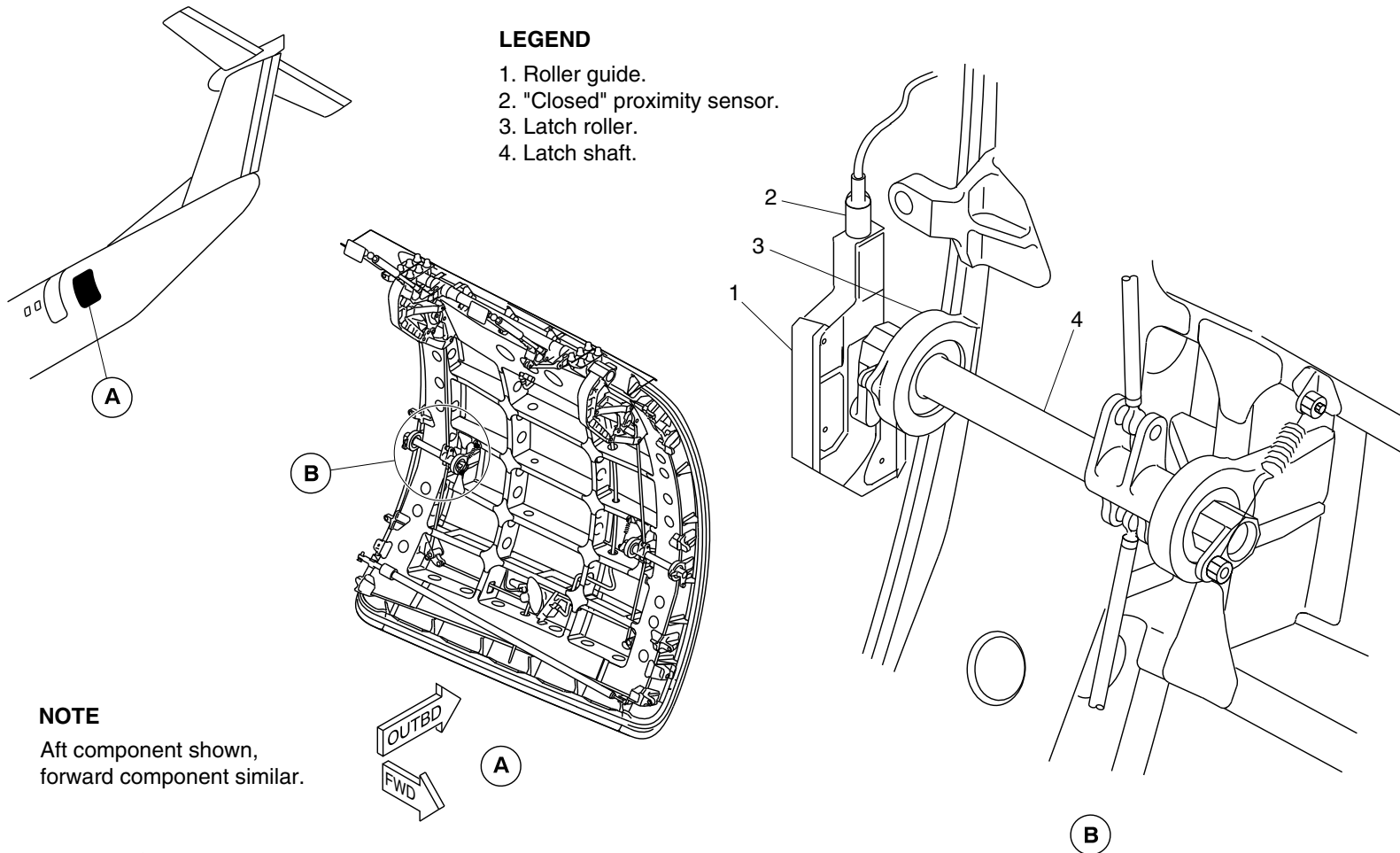


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

1. Roller guide.
2. "Closed" proximity sensor.
3. Latch roller.
4. Latch shaft.



NOTE

Aft component shown,
forward component similar.

fs120a01.dg, vn, aug28/2011

Aft Baggage Door Roller Guide
Figure 12

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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52–32–00–001

FORWARD BAGGAGE DOOR

Introduction

The forward baggage door gives access to the forward baggage compartment.

On aircraft with Modsum 4–459324 or 4–458926 or 4–190614 incorporated, the forward baggage door is removed and replaced with a Type 1 emergency exit door.

On aircraft with Modsum 4–201647 incorporated, a Type 1 emergency exit door is fitted at the location of the forward baggage door.

On aircraft with Modsum 4–459339 incorporated, the forward baggage door is reworked to add an internal handle, a window and ditching dam guide pin bracketry to convert into a Type I emergency exit door.

General Description

[Refer to Figure 1.](#)

The forward baggage door is located on the forward right side of the aircraft.

The door is unlocked by the operation of an external tee handle. The door opens or closes with the door face parallel with the fuselage skin. This is done by two hinges (main and secondary) and a stabilizer.

Proximity sensors monitor the position of the upper fixed rollers and the door handle. A red warning light comes on in the flight compartment when the door is open or unlocked. The door warning system monitors door status and supplies visual indication and warning in the flight compartment. The indications are shown in:

- Door Warning System (52–70–00)

The forward baggage door system has the components that follow:

- Handle mechanism (52–32–01)
- Vent mechanism (52–32–06)
- Lift and latch mechanism (52–32–11)
- Hinge mechanism (52–32–16)
- Door seal (52–32–21).

Detailed Description

The forward baggage door is of a translating type secured against pressurization loads by five stops located on each vertical edge that engages with five related stop brackets installed on the door structure.

[Refer to Figure 2.](#)

The door has two fixed rollers located on each side around the upper and lower parts, that move in guides on the fuselage that give the door tangential movement. In pressurized flight, the door is held by the stops against pressurization and vertical loads. In unpressurized flight, the door is kept from upward movement by the cam rollers on each side of the lift/latch shaft. If one cam roller assembly fails, the second cam roller assembly prevents the door from opening in flight.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The lift/latch shaft is secured at the closed position by means of both the handle lock mechanism and the overcenter springs.

Refer to Figure 3.

The door opening and retention system is operated by the external handle. The external handle is kept in a recess in the outer skin and is connected to the handle shaft. The handle can be accessed when the flap covers are pushed in. The handle is then pulled out and turned counter clockwise. The first pulling operation opens the vent door to release pressure in the cabin and the proximity switch opens. Further operation makes the lift/latch turn pushing the lift/latch roller down, and the door moves inwards and upwards to clear the door stops.

At the uppermost lift position, the hold-up hook is connected to the hold-up hook pin of the door hinge and the door is moved outward. As the door moves outward, the interlock cam connects to a latch crank on the lift/latch shaft, to prevent accidental handle operation. When the door reaches the fully open position, the door open lock hook on the hinge arm is connected to the door open lock pin to keep the door in the open position.

To close the door, the release handle adjacent to the lock hook is pulled to release the interlock mechanism. The door handle is turned clockwise to disconnect the door hold-up hook from the hold-up hook pin. The door is lowered to the closed position aligning the stops behind their related stop bracket. Final inward movement of the handle locks the door and closes the proximity switch. Automatic cabin pressurization is prevented when the door is not fully closed and locked.

The baggage door and vent door have seals around their periphery. The seals are inflated by cabin pressure through holes in the seals.

Mechanism, Handle

The external handle is connected to the shaft of the handle mechanism. When the handle is pulled full travel, the handle shaft turns the lift/latch shaft through cranks, rods, and an idler link.

Mechanism, Vent door

The linkage from the handle transmits the handle input to the vent door mechanism. When the door handle is pulled, the vent door opens to release pressure from the cabin. When the door is in the closed position, the final movement to lock the handle closes the vent door. The vent door is spring-loaded in the direction of the open position, so that it is kept open in case of fracture or disconnection of the vent door operating mechanism.

The primary function of the vent door is as follows:

- To release residual cabin pressure to the atmosphere before the door is opened
- To prevent pressurization loads on the door before the door is fully closed and locked.

Mechanism, Lift and Latch

The linkage from the handle to the lift/latch shaft transmits handle input to the shaft, and locks the lift/latch shaft assembly when the handle is in the locked position.

The bias spring assembly for the latch causes an overcenter against the lift/latch shaft, and locks it at the closed or open position. The overcenter position and linkage from the handle mechanism, gives dual locking of the lift/latch shaft.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

A cam roller assembly is installed at both ends of the lift/latch shaft. When the door is opened, the lift/latch shaft turns to let the rollers push down the guide tracks, lifting the door for outward movement. The door operation sequence is reversed at door closing. In the door closed position, the roller center is offset inward against the door lift-up plane, resulting in the rollers being kept in the door closing direction, against the upward movement of the door.

Mechanism, Hinge

The main and secondary hinge arms keeps the door face parallel with the fuselage skin when the door is opened or closed. During the travel of the door around the furthest position from the fuselage, the yawing movement of the door is suppressed by the stabilizer rod.

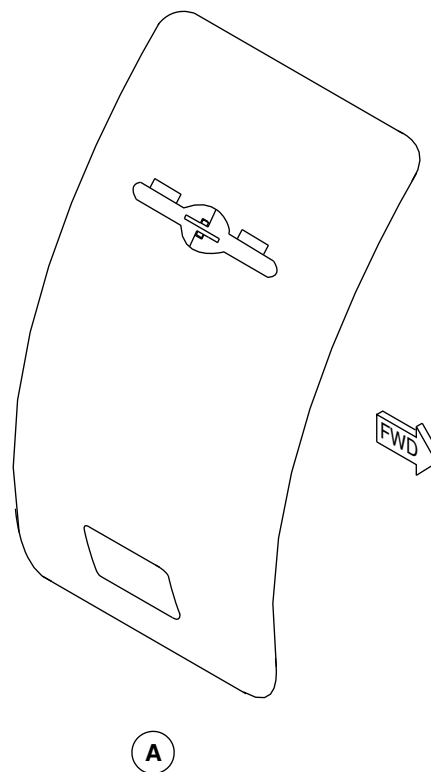
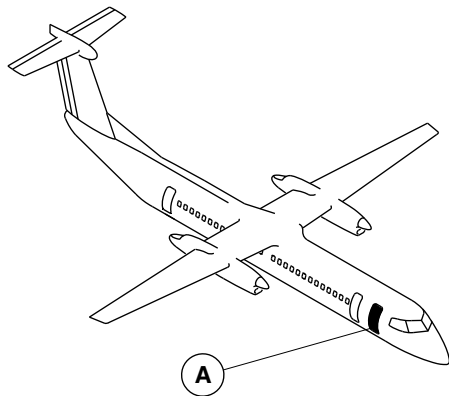
Door Seal

The door seal is of the passive type. Several holes in the seal let it be inflated by cabin pressure. This type of seal is also installed on the vent door.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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FORWARD BAGGAGE DOOR LOCATION
Figure 1

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-32-00
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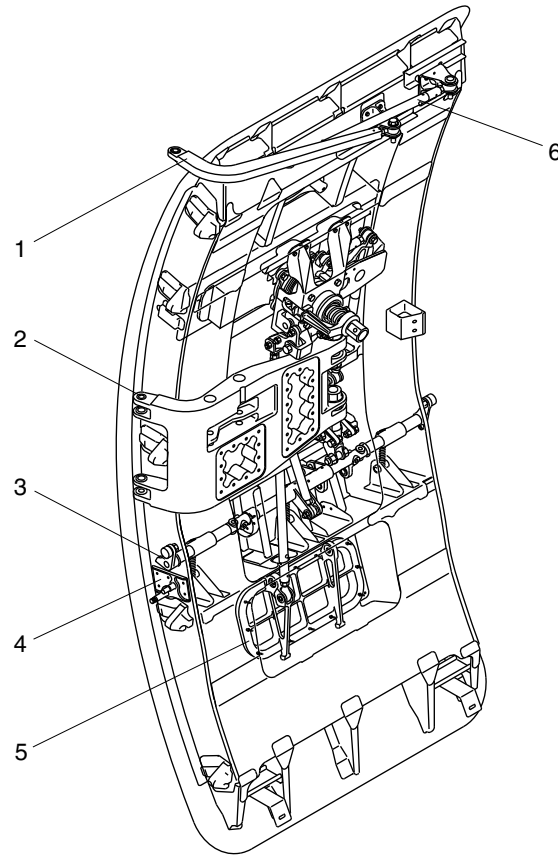
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Secondary Hinge.
- 2. Main Hinge.
- 3. Lift And Latch Shaft.
- 4. Interlock Assembly.
- 5. Vent Door.
- 6. Stabilizer Rod.

fs351a01.cgm

FORWARD BAGGAGE DOOR
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-32-00
Config 001

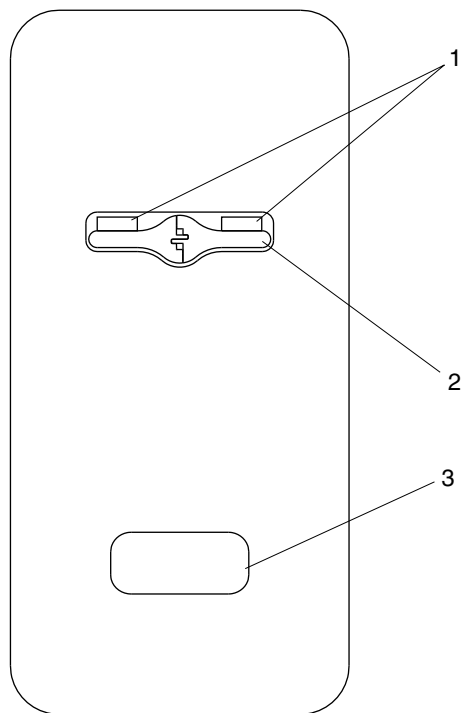
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Flap Cover.
- 2. External Handle.
- 3. Vent Door.

fs848a01.cgm

FORWARD BAGGAGE DOOR EXTERNAL DETAIL
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

52-40-00-001

SERVICE DOORS

Introduction

The aft service door gives access to the rear passenger compartment, and also functions as a Type I emergency exit. The aft fuselage access door gives access to equipment in the unpressurized area of the aft fuselage. The Auxiliary Power Unit (APU) access doors give access for ground maintenance of the optional APU. The ground service and nose compartment doors give access for maintenance and servicing purposes.

General Description

[Refer to Figures 1 and 2.](#)

Service doors are installed at various locations on the aircraft.

[Refer to Figures 3 and 4.](#)

The four nose compartment doors are located on the nose.

[Refer to Figure 5.](#)

The aft service door is located on the right side of the aircraft at the rear of the passenger compartment.

[Refer to Figure 6.](#)

The Aft Fuselage Access Door is located flush with the underside of the rear fuselage, through which the unpressurized equipment

compartment is accessed. The door is opened or closed from the exterior of aircraft.

[Refer to Figure 7.](#)

The APU access doors are located on the bottom of the tailcone section on either side of the fuselage centerline. The doors are hinged horizontally and kept in a locked and closed position by the two pin latches, hook latches and keepers, and the three shear pins.

The Refuel/Defuel Panel door is located on the lower side of the No. 2 engine nacelle, aft of the right main landing gear rear doors. The door gives access to the refuel/defuel adapter, Refuel/Defuel Indicator (RDI), and switches to carry out pressure refueling/defueling operations.

The dc electrical ground power door gives access to the 28 Vdc ground power receptacle on the left side of the aircraft nose. The door is attached to the nose section by a hinge assembly at the forward end and a latch assembly at the aft end. The ac electrical ground power door gives access to the ac ground power receptacle. The door is attached to the nacelle by a hinge at the top and two latch assemblies at the bottom.

The hydraulic ground power doors, on the right side of both engine nacelles, give access to the hydraulic ground power connections. The doors are attached to the nacelle by a hinge at the top and two latch assemblies at the bottom.

The ground air conditioning door is located flush with the lower right side of the rear fuselage. The door opens outward and downward to give access to the air conditioning duct connection.

The lavatory service door is located flush with the lower right side of the forward fuselage. The door opens outward and downward to give access to the lavatory service port.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The Service Doors System consists of the components that follow:

- Aft Service Door (52–41–00)
- Aft Fuselage Access Door (52–42–00)
- APU Access Door (52–43–00)
- Ground Service Doors (52–44–00)
- Ground–Crew Interphone–Jack Door Assembly (52–45–00)
- Nose Compartment Doors (52–46–00)

Detailed Description

The service door system includes an aft service door, an aft fuselage access door and APU clamshell doors. It also includes several ground service and nose compartment doors.

The ground service doors are identified as follows:

- Refuel/Defuel Panel Door (52–44–01)
- DC Electrical Ground Power Door (52–44–06)
- AC Electrical Ground Power Door (52–44–11)
- Hydraulic Ground Power Door (52–44–16)
- Ground Air Conditioning Connection Door (52–44–21)
- Lavatory Service Door (52–44–26)
- Ground–Crew Interphone–Jack Door (52–45–00)

Aft Service Door

The aft service door is classified as a Type I emergency exit. It is a translating type secured against pressurization loads by five stops on each vertical edge that engage with five related stop brackets installed on the door surround.

Refer to Figures 8 and 9.

When the aft service door is unlocked, both the FUSELAGE DOORS warning light and the MASTER WARNING light flash. The Door Systems page on the Multi–Function Display (MFD) shows the unlocked aft service door as a red–filled rectangle with a SERVICE legend. (Refer to SDS52–70–00).

Aft Fuselage Access Door

The aft fuselage access door is installed in the fuselage structure at the right side by hinges, and at the left side by two latches. The door has a cast aluminum panel with a louvered opening to increase the airflow in the compartment. Moisture drain holes are located in the panel of the door to drain any accumulated water.

APU Access Door

The APU access doors are made up of left and right two access door subassemblies. The left and right doors form a large part of the bottom of the tailcone and together make up a clamshell door system. The doors are hinged on the horizontal plane and open upward and outward, to give access to the APU. The doors are supported in the open position by telescopic struts.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Refuel/Defuel Panel Door

[Refer to Figure 10.](#)

The Refuel/Defuel Panel door is located on the lower side of the No. 2 engine nacelle, aft of the right main landing gear rear doors. The door gives access to the refuel/defuel adapter, Refuel/Defuel Indicator (RDI), and switches to carry out pressure refueling/defueling operations.

The Refuel/Defuel Panel Door operates a door switch installed on the nacelle door structure. When the door is open, the switch causes the amber MASTER VALVE CLOSED light on the Refuel/Defuel panel and the FUELING ON caution light on the Caution and Warning Panel to come on. The service light, installed on the aft vertical section of the Refuel/Defuel compartment, also comes on. When the Refuel/Defuel Panel Door is closed and locked, it operates the door switch, which causes the FUELING ON caution light to go off. The MASTER VALVE CLOSED light and the service light also go off.

Ground Air Conditioning Door

The ground air conditioning door, fabricated from alclad skin and stiffener, is located flush with the lower right side of the rear fuselage. The door opens outward and downward to give access to the air port. The door is secured to the fuselage structure at the lower side by a hinge and at the upper side by a latch. A gasket is fitted around the door cut-out, to provide weather sealing and prevent chafing of the door. The door can be opened and closed from outside the aircraft.

To open the door, press the latch release button to unlock the latch mechanism; the door will swing downward from the hinge on the

door lower edge. To close the door, swing the door to the closed position and ensure the latch mechanism is engaged.

Lavatory Service Door

[Refer to Figure 11.](#)

The lavatory service door is attached to the fuselage structure, on the lower side by a hinge and on the upper side by two latches. The door has a gasket installed around the edge to help prevent chafing.

The latch release buttons are pushed to unlock the latch mechanism and the door swings downward from the hinge on the lower side.

The door can be opened or closed from outside the aircraft.

The door is moved to the closed position and the latch mechanism is engaged to close it.

Ground-Crew Interphone-Jack Door

[Refer to Figure 12.](#)

The ground-crew interphone-jack door gives access to the interphone jack for communication between the ground crew and the pilot. It is a circular door attached to the fuselage by one hinge and is retained in the closed position by a latch assembly.

The ground-crew interphone-jack door is located on the front fuselage, aft of the left lower door of the nose compartment. To open the door, push the latch lever and release the spring loaded safety catch. The door housing has a seal to keep out water and contaminants.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Nose Compartment Doors

There are four nose compartment doors, which give access to the four nose compartments. Two upper nose compartment doors give access to both upper nose compartments. Two lower nose compartment doors give access to both lower nose compartments.

Each nose compartment door has two latches to lock the door. The upper doors are hinged at the top and open upward. The lower doors are hinged at the bottom and open downward.

[Refer to Figures 13 and 14.](#)

Each nose compartment door has a seal around the door structure to keep out water and contaminants.

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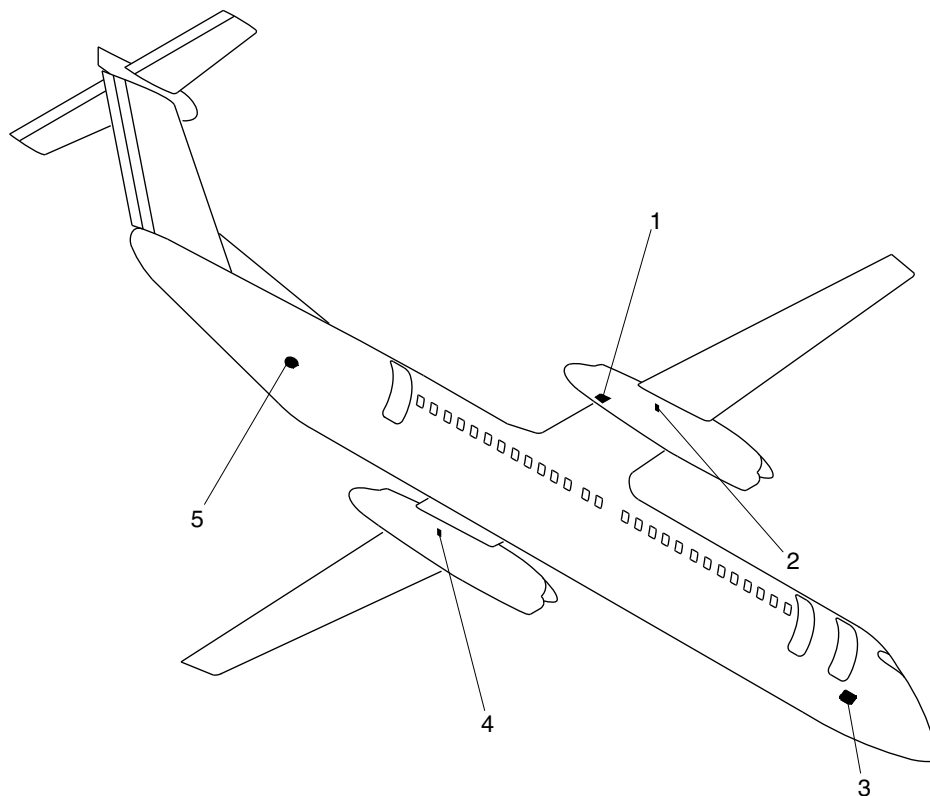
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Refuel/Defuel Panel Door.
2. Right Nacelle Hydraulic Ground Power Door.
3. Lavatory Service Door.
4. Left Nacelle Hydraulic Ground Power Door.
5. Ground Air Conditioning Connection Door.

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SERVICE DOORS LOCATOR
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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Config 001

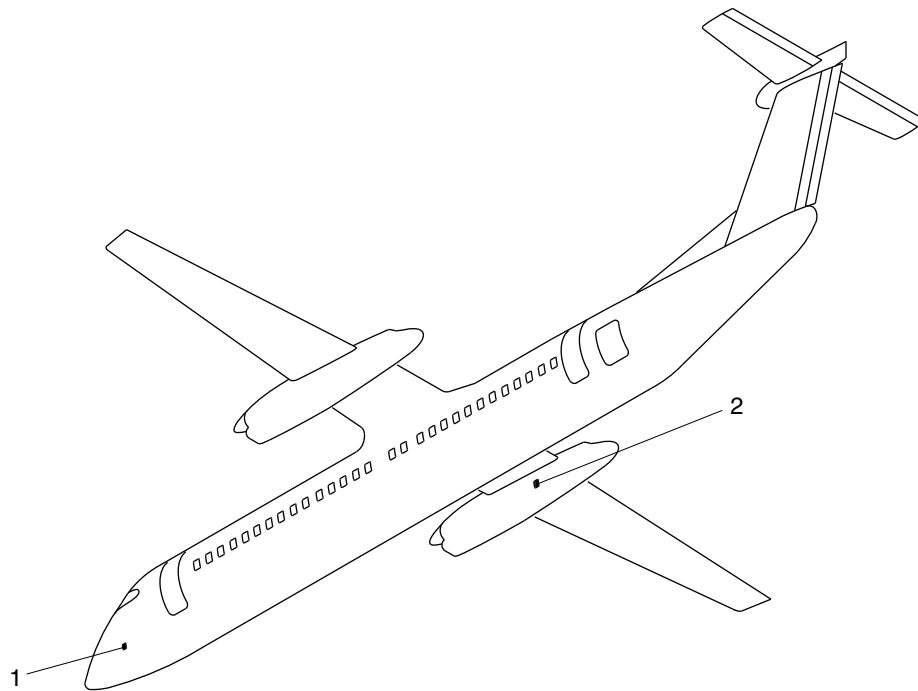
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. DC Electrical Ground Power Door.
- 2. AC Electrical Ground Power Door.

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DC AND AC GROUND POWER DOOR LOCATOR
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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Config 001

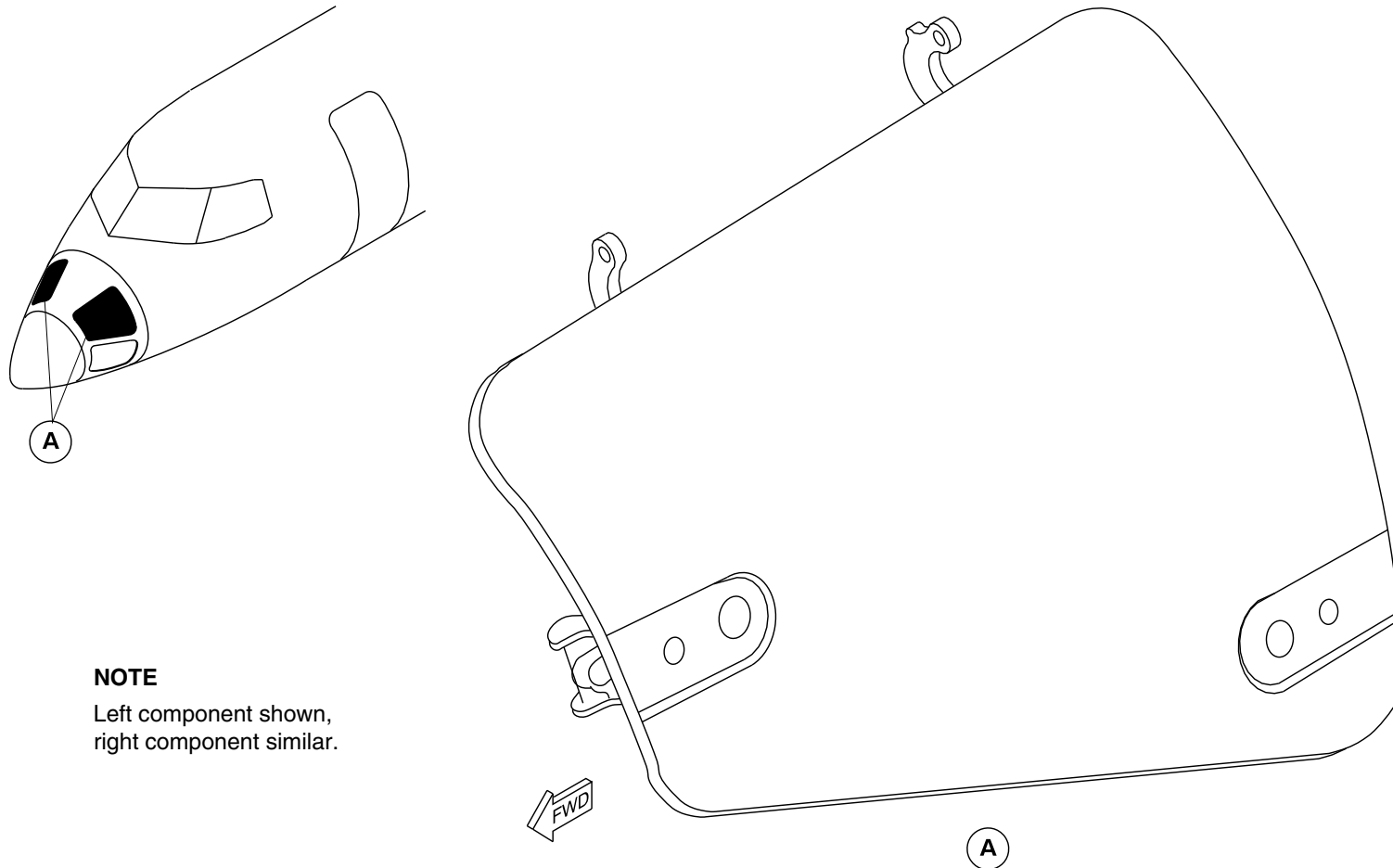
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NOTE

Left component shown,
right component similar.

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Upper Nose Compartment Doors Locator
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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Config 001

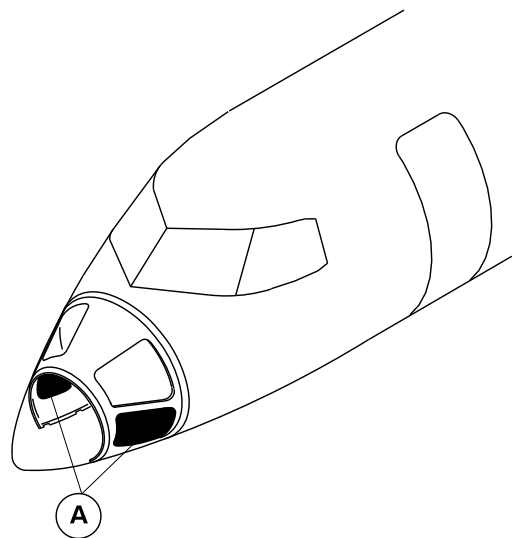
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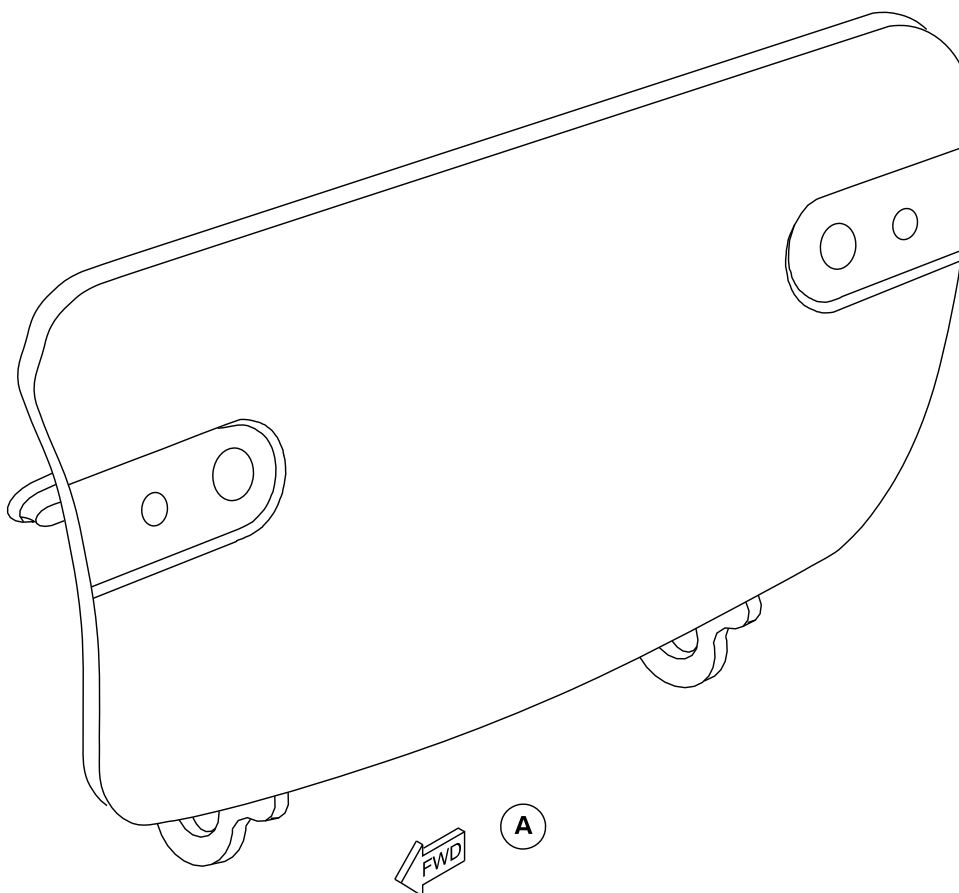
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



NOTE

Left component shown,
right component similar.



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Lower Nose Compartment Doors Locator
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-40-00
Config 001

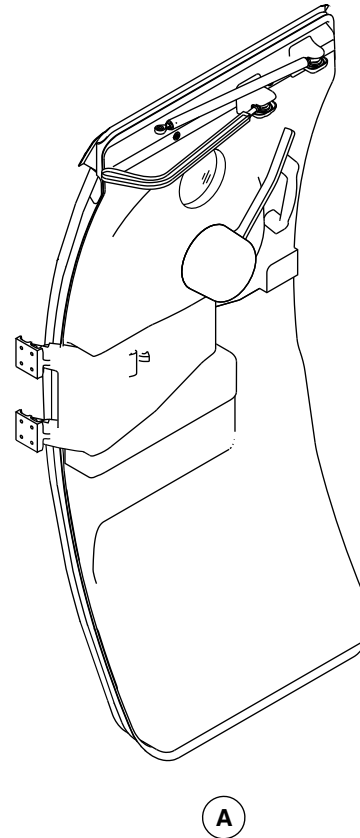
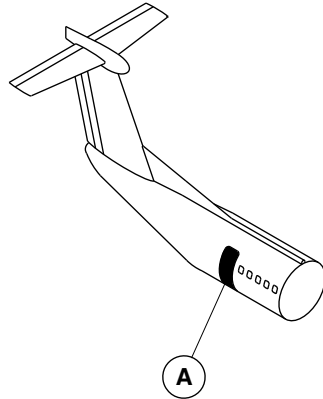
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AFT SERVICE DOOR LOCATOR
Figure 5

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-40-00
Config 001

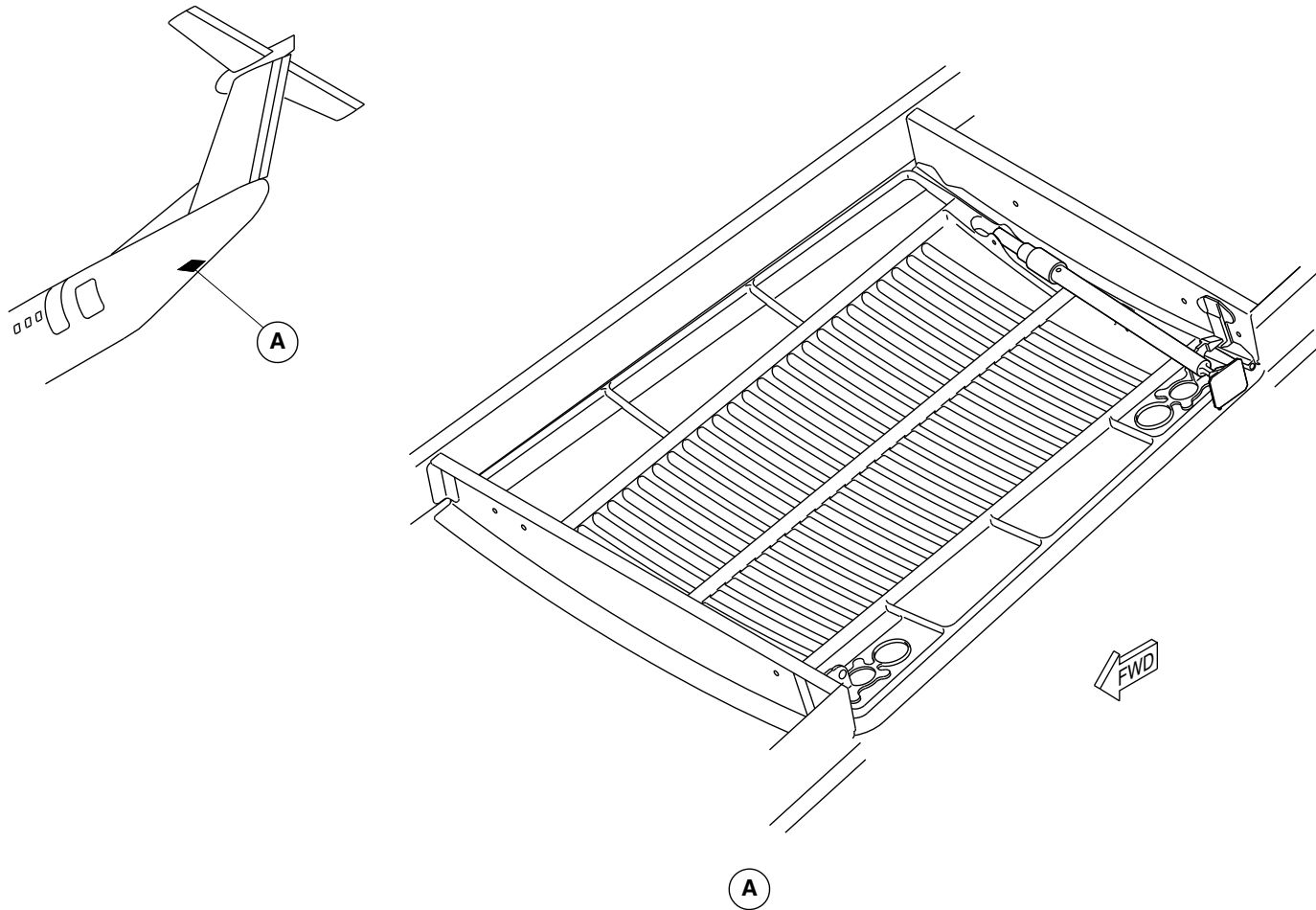
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fs123a01.cgm

AFT FUSELAGE ACCESS DOOR LOCATOR
Figure 6

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-40-00
Config 001

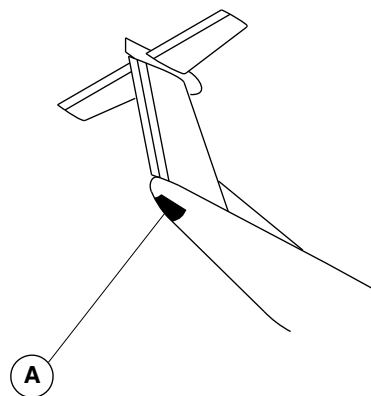
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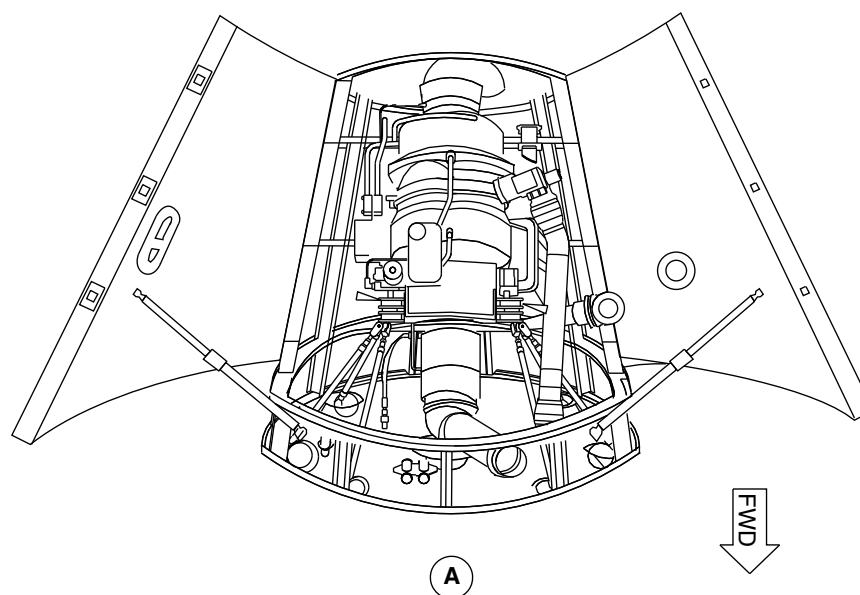


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NOTE
APU Access Doors Open.



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APU ACCESS DOOR LOCATOR
Figure 7

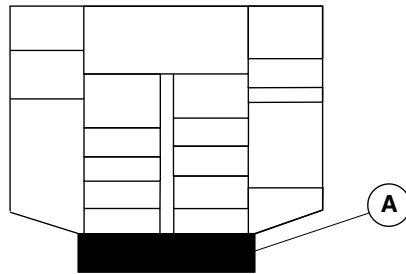
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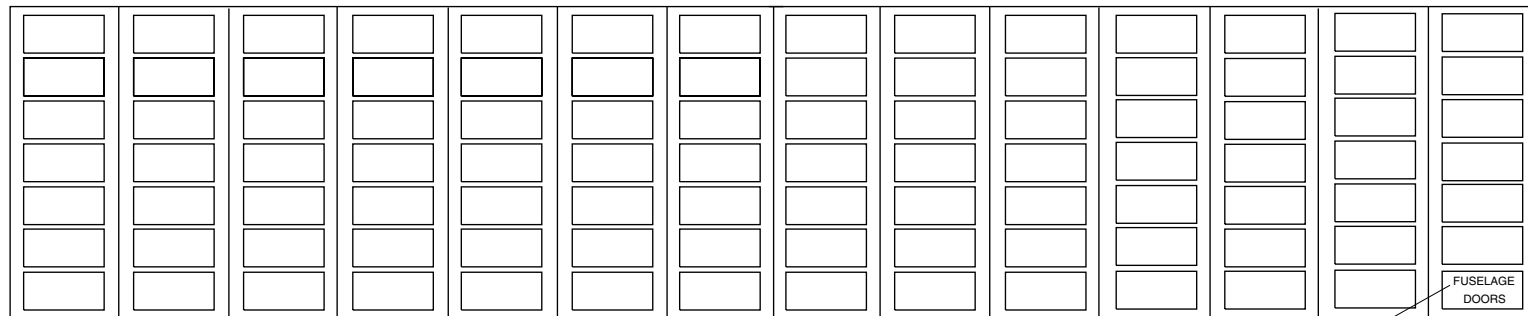
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OVERHEAD CONSOLE

LEGEND

1. Fuselage Doors (Red).



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Fuselage Doors Warning Light
Figure 8

PSM 1-84-2A
EFFECTIVITY:
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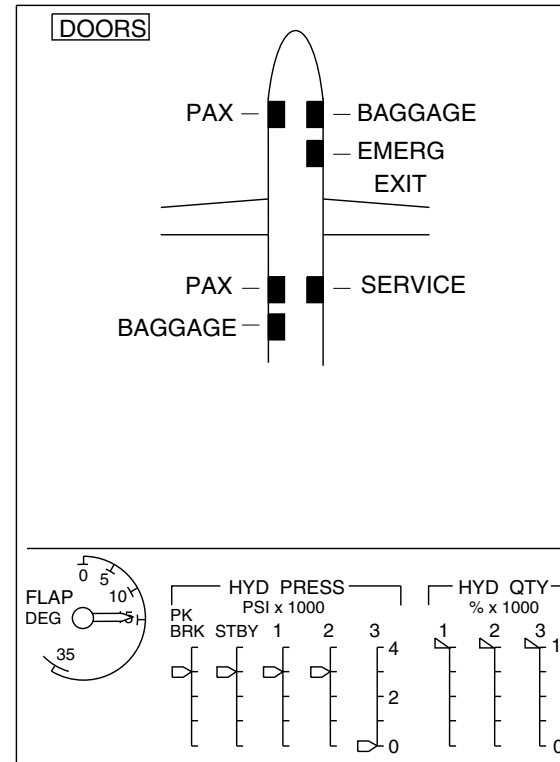
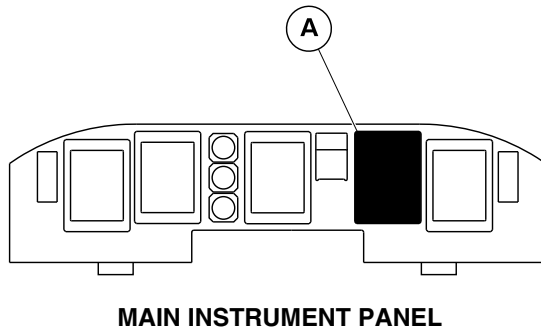
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A

**PRE MODSUM 4-458926
PRE MODSUM 4-458969
PRE MODSUM 4-459339
PRE MODSUM 4-190614
POST MODSUM 4-309306**

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Doors Systems Page
Figure 9 (Sheet 1 of 2)

PSM 1-84-2A
EFFECTIVITY:
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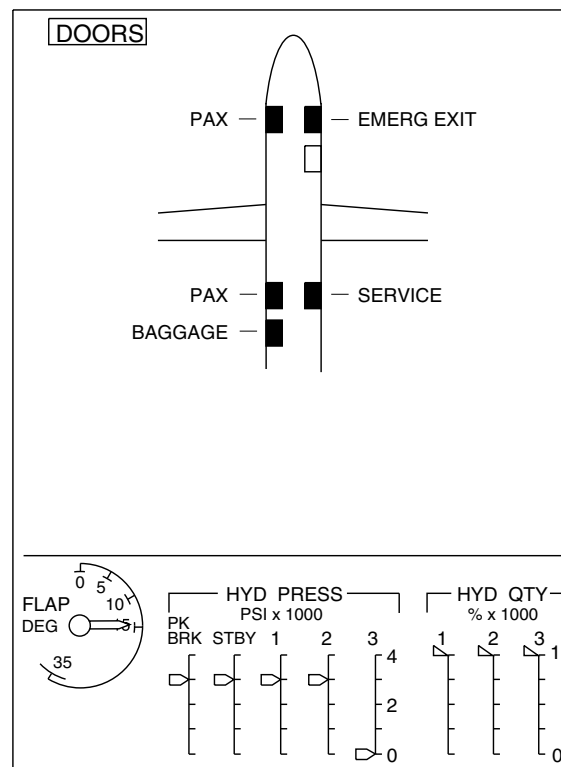
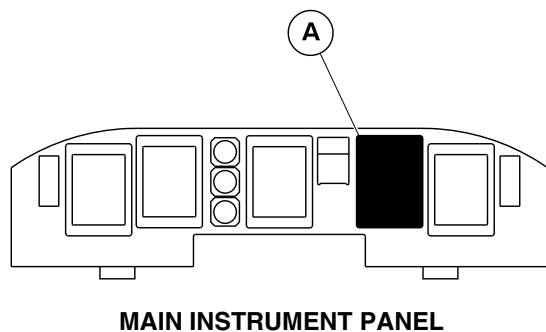
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POST MODSUM 4-201647

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Doors Systems Page
Figure 9 (Sheet 2 of 2)

PSM 1-84-2A
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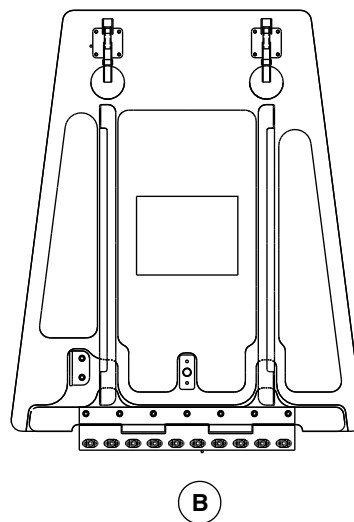
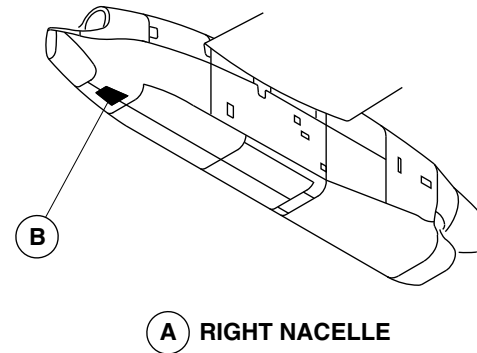
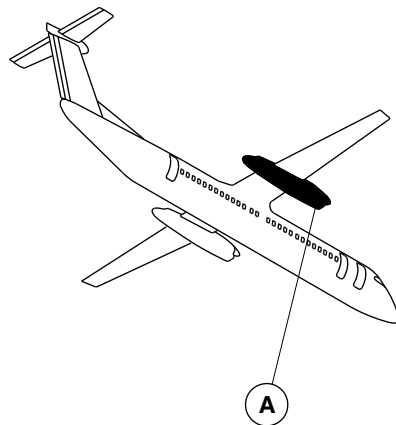
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REFUEL/DEFUEL PANEL DOOR LOCATOR
Figure 10

PSM 1-84-2A
EFFECTIVITY:
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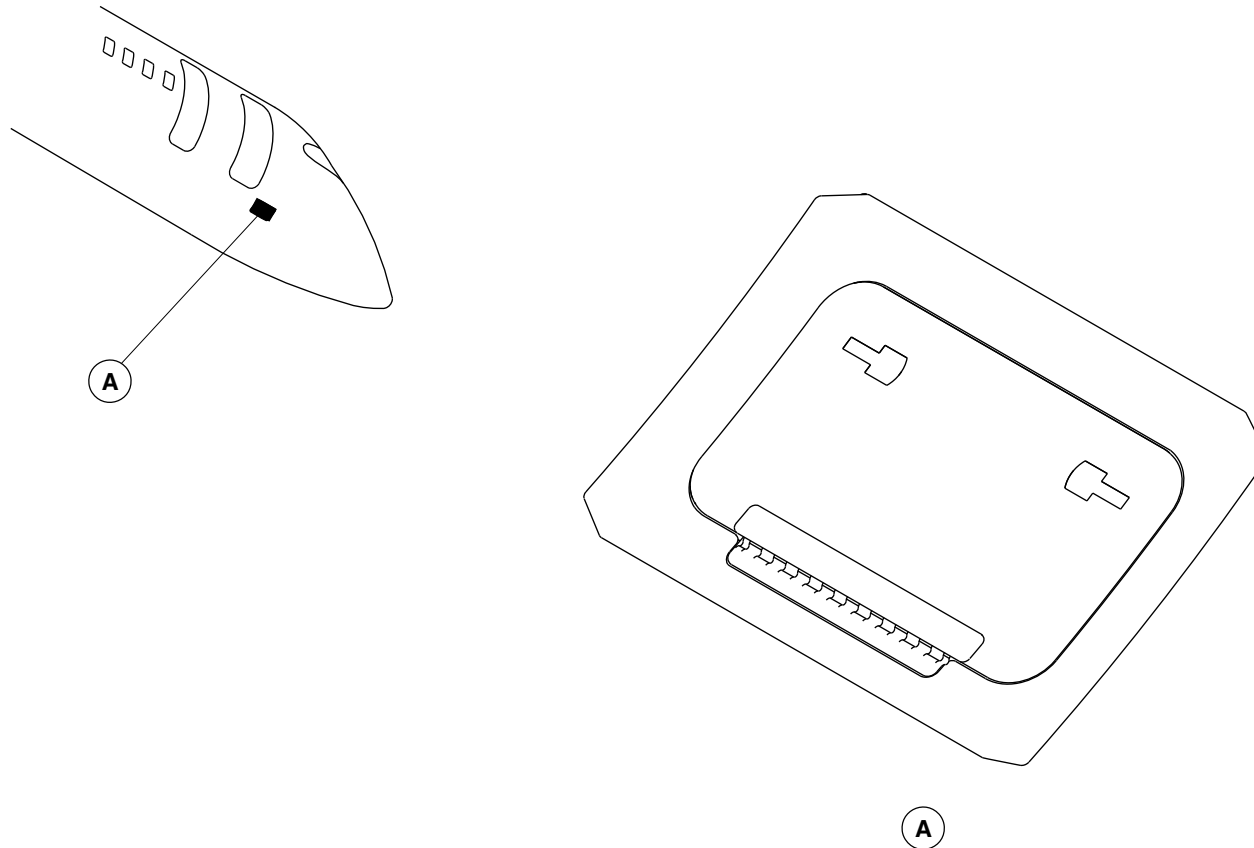
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LAVATORY SERVICE DOOR LOCATOR
Figure 11

PSM 1-84-2A
EFFECTIVITY:
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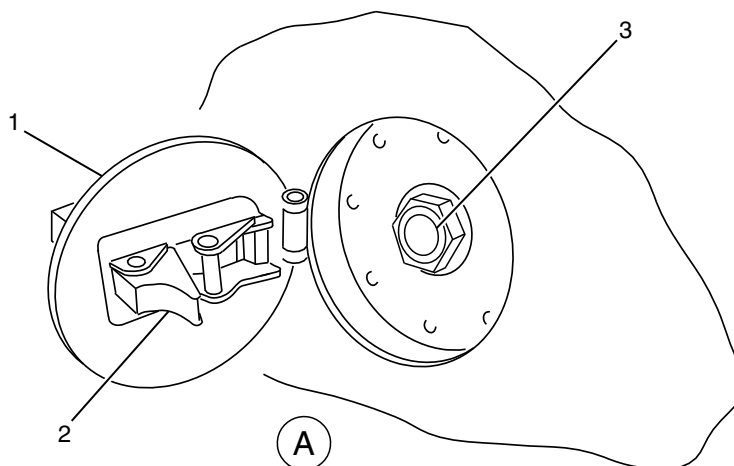
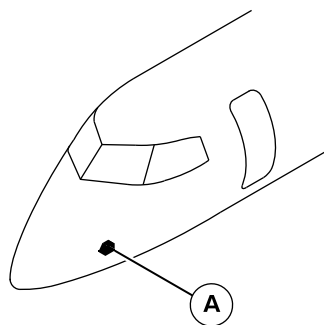
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Door.
- 2. Latch.
- 3. Interphone jack.

VIEW ON OPEN DOOR

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GROUND – CREW INTERPHONE – JACK DOOR LOCATOR
Figure 12

PSM 1-84-2A
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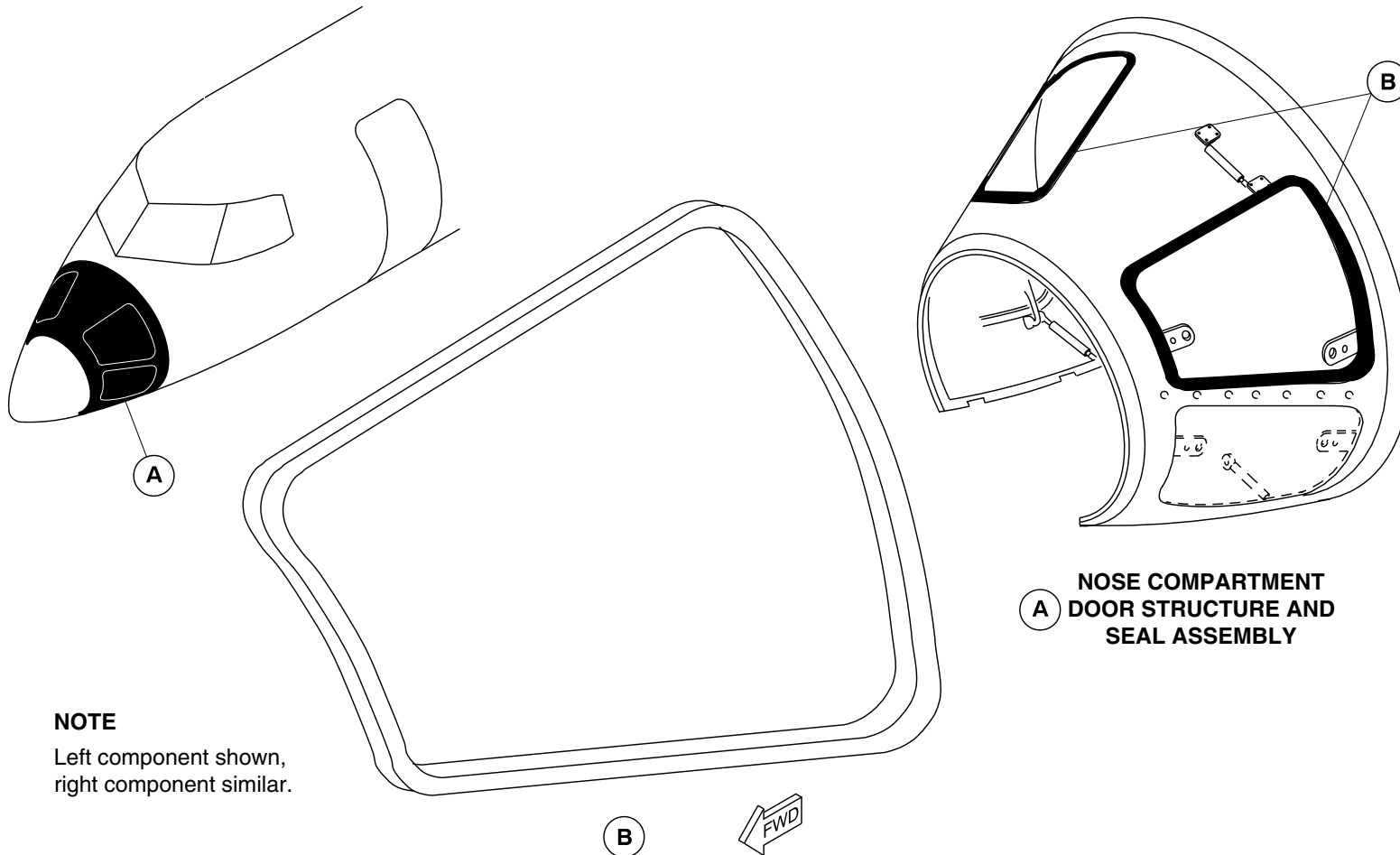
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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Upper Nose Compartment Door Seals
Figure 13

PSM 1-84-2A
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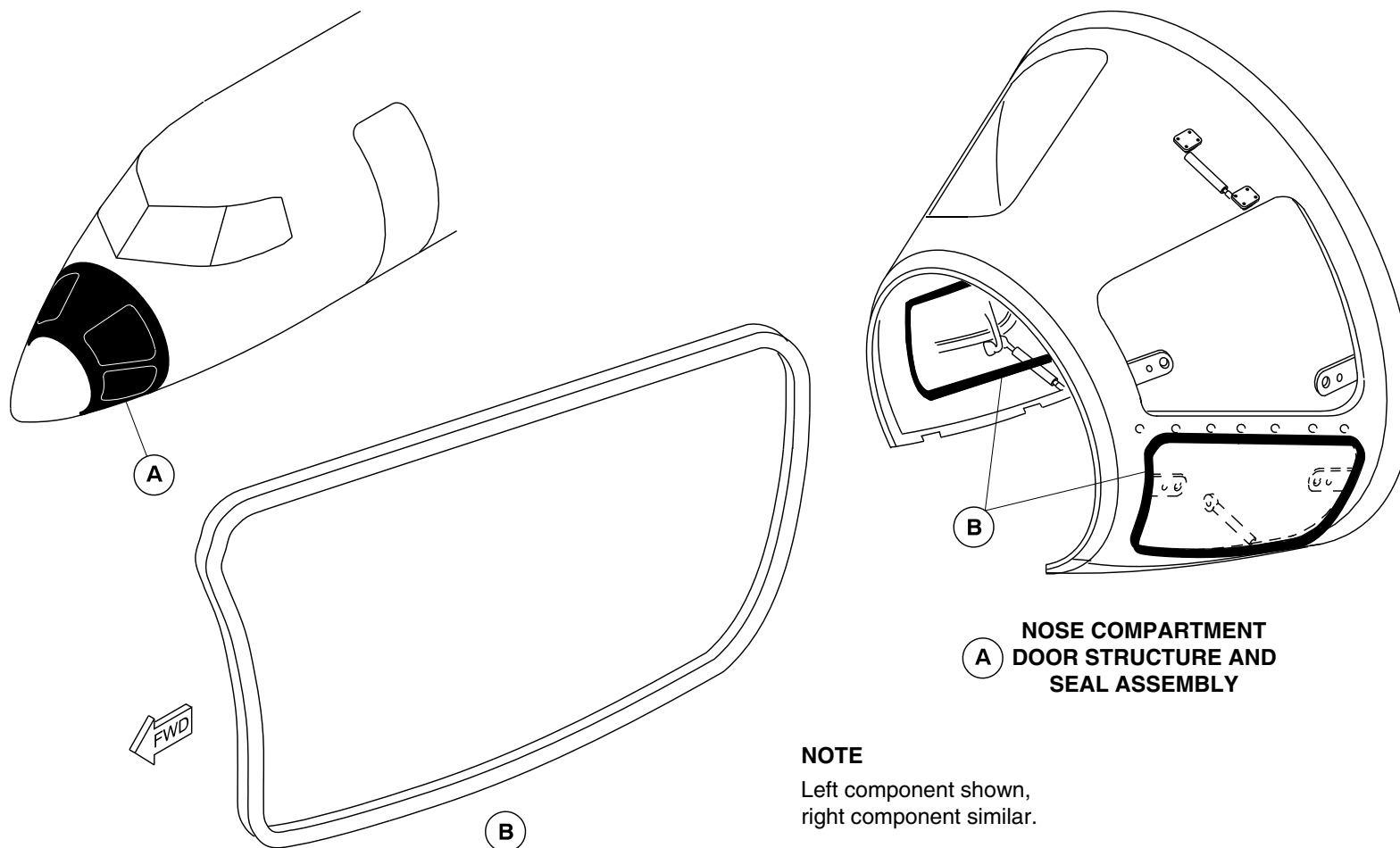
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Lower Nose Compartment Door Seals
Figure 14

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

52-41-00-001

AFT SERVICE DOOR

Introduction

The aft service door gives access to the rear passenger compartment, and also functions as a Type I emergency exit.

On aircraft with modsum 4-459324 incorporated, aft service door is removed and forward baggage door is installed in place of aft service door and permanently locked.

General Description

[Refer to Figure 1.](#)

The aft service door is located on the right side of the aircraft at the rear of the passenger compartment. The aft service door is classified as a Type I emergency exit. It is a translating type secured against pressurization loads by five stops on each vertical edge, that engage with five related stop brackets installed on the door surround.

The proximity sensors monitors the position of the upper fixed rollers and the door handle. A red warning light comes on in the flight compartment when the door is open and unlocked.

The aft service door system has the components that follow:

- Handle mechanism (52-41-01)
- Vent door mechanism (52-41-06)
- Lift and latch mechanism (52-41-11)

- Hinge mechanism (52-41-16)

- Door seal (52-41-21).

The Aft service door interfaces with:

- Door Warning System (52-70-00)
- Proximity Sensors (52-71-00)

Detailed Description

The aft service door gives access to the rear passenger compartment and also function as a Type I emergency exit.

The door operating and retention system is operated by the internal or external handle. The internal handle is pulled inwards and turned counter clockwise to unlock the handle. When the door handle is pulled inwards the vent door opens to release pressure in the cabin and the proximity switch opens. When the handle is turned counter clockwise, the lift/latch shaft pushes the lift/latch rollers down, and the door moves inwards and upwards to clear the door stops.

At the uppermost lifted position, the hold-up hook is connected to the hold-up hook pin of the door hinge, and the door is moved outward. As the door moves outward, the interlock cam connect to a latch crank on the lift/latch shaft, to prevent accidental handle operation.

[Refer to Figure 2.](#)

The door has two fixed rollers located on each side around the upper and lower parts, running in guides on the fuselage which give tangential movement of the door. In pressurized flight, the door is held by the stops against pressurization and vertical loads. In unpressurized flight, the door is kept from upward movement by cam rollers on each side of the lift/latch shaft. If one cam roller assembly

PSM 1-84-2A

EFFECTIVITY:

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

fails, the second cam roller assembly prevents the door from opening in flight. The lift/latch shaft is secured at the closed position by means of both the handle lock mechanism and the overcenter springs.

[Refer to Figure 3.](#)

The door is unlocked by operation of the spring loaded internal or external handle. The door opens or closes with the door face in parallel with the fuselage skin. (This is done by two hinges main and secondary and a stabilizer).

When the door reaches the fully opened position, the door open lock hook on the hinge arm is connected to the door open lock pin on the door to keep the door at its position. An interlock assembly prevents handle operation when the door is opened, preventing the hold-up hook from being unlatched.

[Refer to Figure 4.](#)

The external handle is kept in a recess in the outer skin and is connected to the same handle shaft as that of the internal handle. The handle can be accessed when the flap covers are pushed in. The handle is then pulled out and turned clockwise to open the door.

For safety, a window in the door gives a view of the outside before the door is opened.

[Refer to Figure 3.](#)

To close the door, the release lever adjacent to the lock hook is pulled to release the open lock, and the door is brought to the closed position. The internal handle is turned clockwise to disconnect the door hold-up hook from the hold-up hook pin. The door is moved to the closed position, and the stops align with their related stop bracket lugs.

The handle is moved outward to lock the door and close the proximity switch. The door can also be closed externally by using the external handle.

The main door and vent door are sealed around their periphery. The seal is inflated by cabin pressure through holes in the seal, to keep the gap around the door closed. Cabin pressurization is prevented when the vent door is not fully closed and locked.

Mechanism, Handle

The internal handle operates the lift/latch shaft, and the vent door operating rod. The lower end of the internal handle is connected to the shaft of the handle mechanism. When either the internal or external handle operated, the handle shaft can be turned to operate the lift/latch through cranks, rods and an idler link. The external handle is kept flush with the door skin and locked in the closed position.

Mechanism, Vent

The linkage from the handle transmits the handle input to the vent door mechanism. When the door handle is pulled, the vent door opens to release pressure from the cabin. When the door is in the closed position, the final movement to lock the internal or external handle closes the vent door. The vent door is spring-loaded in the direction of the open position, so that it is kept open in case of fracture or disconnection of the vent door operating mechanism.

The primary function of the vent door is as follows:

- Release residual cabin pressure to the atmosphere before the door is opened, in order to make outward load on the door minimal



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- Prevents pressurization loads on the door in case of manual pressurization before the door is fully closed and locked.

Mechanism, Lift and Latch

The linkage from the handle to the lift/latch shaft transmit handle input to the shaft and locks the lift/latch shaft assembly when the handles are in their locked position.

The bias spring assembly for the latch causes an overcenter against the lift/latch shaft, and locks it at the closed or open position. The overcenter position and linkage from the handle mechanism, gives dual locking of the lift/latch shaft.

The lift/latch shaft is installed to a cam roller assembly at both ends. When the door is opened, the lift/latch shaft turns and let the rollers push down the guide tracks, lifting the door for outward movement. The door operation sequence is reversed at door closing. When the door is in the closed position, the roller center is offset inward against the door lift-up plane, resulting in the rollers being kept in the door closing direction, against the upward movement of the door.

Mechanism, Hinge

The main and secondary hinge arms and the door makes a parallelogram which keeps the door face in parallel with the fuselage skin during door opening or closing. During the travel of the door around the furthest point from the fuselage, the yawing movement of the door is suppressed by the stabilizer rod.

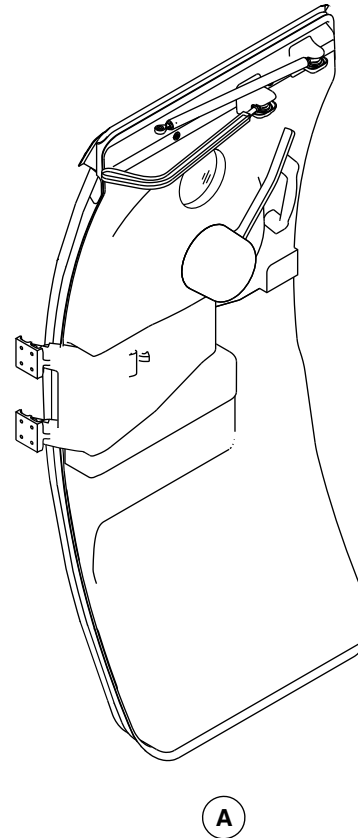
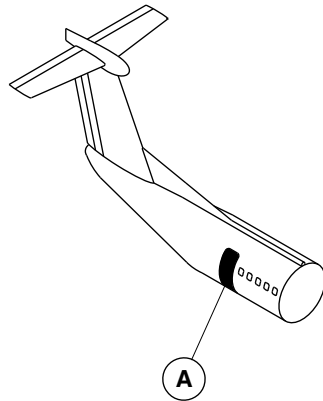
Seal, Door

The door seal is of the passive type which has several holes in the seal and is inflated by cabin pressure through the holes. This type of seal is installed on both the main and vent doors.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AFT SERVICE DOOR LOCATION
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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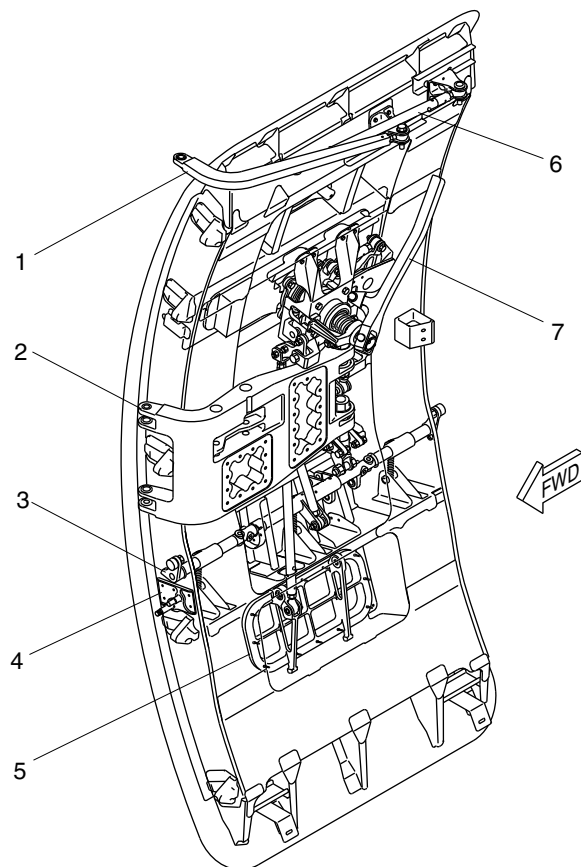
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LEGEND

1. Secondary Hinge.
2. Main Hinge.
3. Lift And Latch Shaft.
4. Interlock Assembly.
5. Vent Door.
6. Stabilizer Rod.
7. Handle Mechanism.

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AFT SERVICE DOOR DETAIL PAGE 1
Figure 2

PSM 1-84-2A
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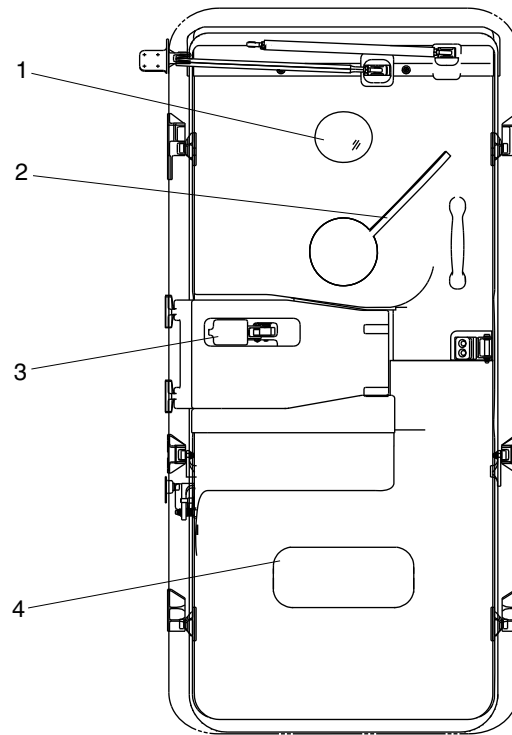
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LEGEND

- 1. Window.
- 2. Internal Handle.
- 3. Open Lock Lever.
- 4. Vent Door.

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AFT SERVICE DOOR INTERNAL VIEW
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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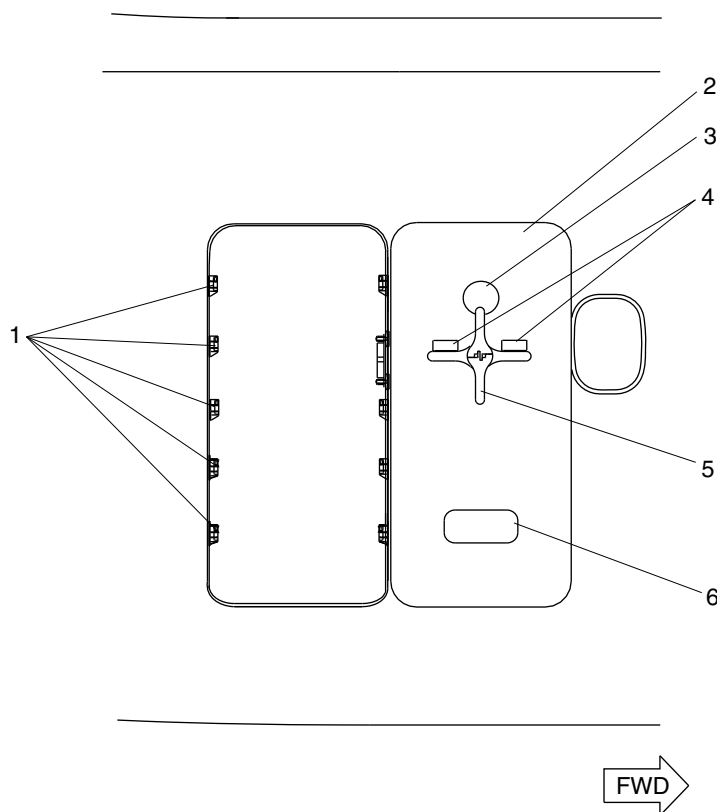
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LEGEND

1. Stops (5 Per Side).
2. Aft Service Door (Open Position).
3. Window.
4. Flap Cover.
5. External Handle (Unlocked).
6. Vent Door (Open).

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AFT SERVICE DOOR DETAIL PAGE 2
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

52-42-00-001

AFT FUSELAGE ACCESS DOOR

Introduction

The aft fuselage access door gives access to equipment in the unpressurized area of the aft fuselage.

General Description

[Refer to Figure 1.](#)

The Aft Fuselage Access Door is located flush with the underside of the rear fuselage, through which the unpressurized equipment compartment is accessed. The door is opened or closed from the exterior of aircraft.

The aft fuselage access door includes the components that follow:

- Latches (52-42-01)
- Strut (52-42-06).

Detailed Description

[Refer to Figure 2.](#)

The door is installed in the fuselage structure at the right side by hinges, and at the left side by two latches. The door has a cast aluminum panel with a louvered opening to increase the airflow in the compartment. Moisture drain holes are located in the panel of the door to drain any accumulated water.

[Refer to Figure 2.](#)

The latch release buttons are pushed to unlock or lock the door. The door movement is outwards and then downwards. A telescopic strut keeps the door in the open position when the screw lock is tightened. A moulded rubber seal around the the door cut out prevents door chafing.

To close the door, the screw lock is loosened to unlock the telescopic strut. The door is then pushed closed and the latch buttons pushed to lock the door.

Latches

The latch mechanism controls the opening and closing of the door. The release buttons are pushed to lock or unlock the latch mechanism.

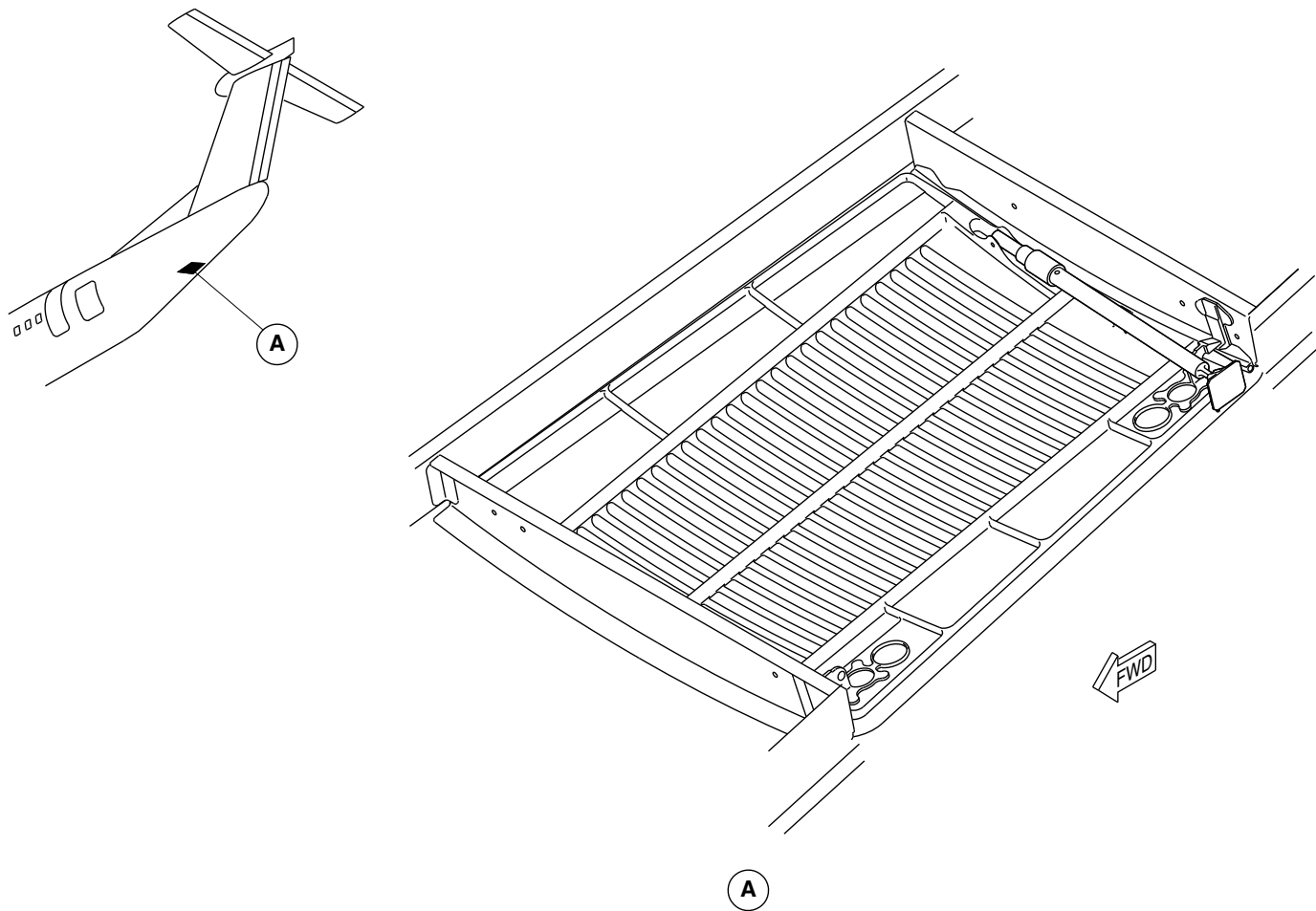
Strut

The telescopic strut keeps the door in the open position.



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AFT FUSELAGE ACCESS DOOR LOCATOR
Figure 1

PSM 1-84-2A
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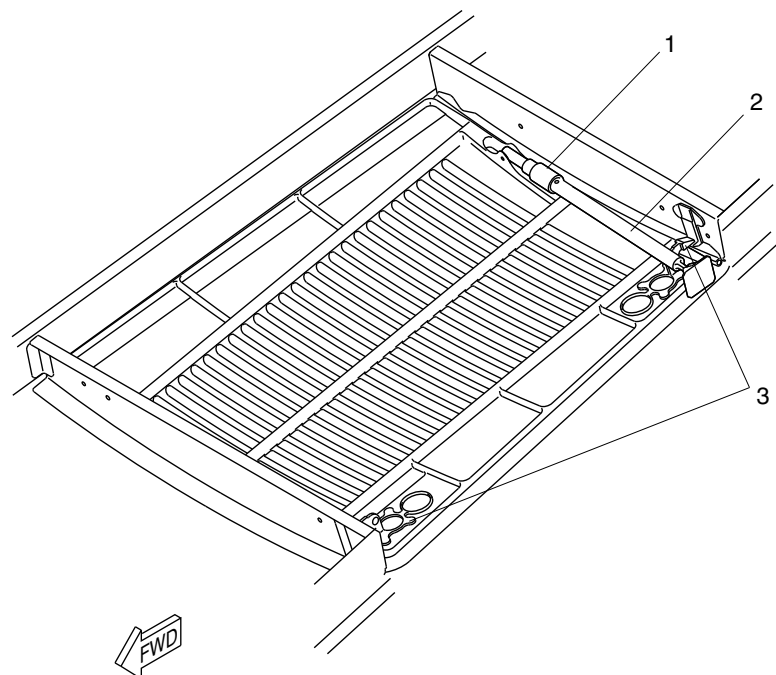
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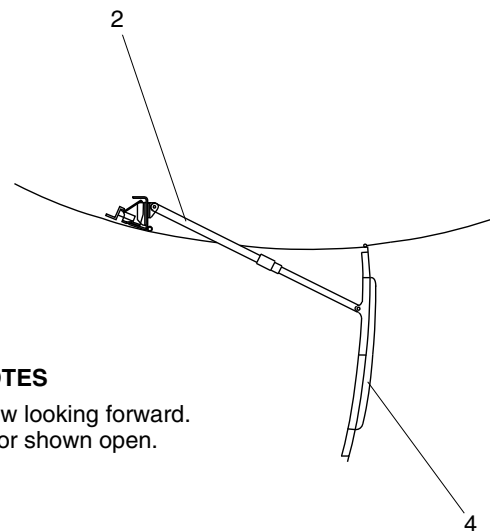
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Screw Lock.
2. Telescopic Strut.
3. Latch.
4. Aft Fuselage Access Door.



NOTES

View looking forward.
Door shown open.

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AFT FUSELAGE ACCESS DOOR
Figure 2

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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

52-43-00-001

APU ACCESS DOOR

Introduction

The Auxiliary Power Unit (APU) access doors give access for ground maintenance of the optional APU.

General Description

The doors are located on the undersurface of the tailcone section on either side of the fuselage centerline. The left and right doors form a large part of the bottom of the tailcone, and together make up a clamshell door system.

The doors are hinged on the horizontal plane and open outward and upward, to give access to the APU engine assembly and components. The doors are held open by telescopic struts.

The APU access doors include the components that follow:

- Two Door Struts (one per door)
- Three Hook Latches
- Three Keepers
- Three Shear Pins
- Two Pin Latches
- Four Hinge Assemblies (two per door)
- Door and APU Drain Mast Seals.

Detailed Description

[Refer to Figure 1.](#)

The APU access doors are located on the bottom of the tailcone section on either side of the fuselage centerline. The doors are hinged horizontally and kept in a locked and closed position by the two pin latches, hook latches and keepers, and the three shear pins.

The APU access door structure is comprised of a thin skin external shell which is supported by longitudinal intercostals and transverse frames. The skin is 0.025 in. (0.0625 mm) thick commercially pure titanium, type CP-1.

The frame and intercostal members are also manufactured from type CP-1 titanium, 0.025 in. (0.0625 mm) thick. Inner skins along the upper and lower edges have been added to increase the torsional rigidity of the doors in the open position.

[Refer to Figure 2.](#)

The two large clamshell-type doors are flush with the tailcone when in the closed and locked position. The doors are opened by unlatching the pin latches and the hook latches. The struts help the upward and outward movement of the doors. Once fully extended, the struts lock the doors in the open position. Closing of the doors is done by unlocking the strut push-release mechanisms, closing the doors and engaging the latches. The door will stay closed if both pin latches, or two of the three hook latches fail.

The fire proof seals give an air tight seal around the APU access doors and drain mast. These seals help prevent leakage of the fire extinguishing agent, if the agent is discharged.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Door Struts

There is one telescoping, maintenance-free strut per door. The struts are attached to the canted station 994.1 frame and installed on each door by adjustable rod ends and stainless steel brackets. The struts are locked at the fully extended position and unlocked by push-release mechanisms. The struts can withstand a 1,000 lb (453.6 kg) axial load or a 150 lb (68 kg) side load.

Hook Latches, Keepers, and Shear Pins

The three hook latches on the left door, and keepers on the right door, keep pre-tension across the two doors at the door centerline. All hook latch components are made from either 17-4PH stainless steel, 17-4PH stainless steel, or CRES 304/305. The main hook assembly is cad plated 4130 steel. A portion of the latch is painted dayglo orange and will be visible if the hook latch is not fully engaged. Three shear pins prevent relative movement between the doors along the door centerline. The shear pins are installed on the right door and engage with related strike plates, installed on the left door, when the doors are closed. The shear pins fit into close tolerance holes in the strike plates.

Pin Latches

The two pin latches are installed on the right door, one forward and one aft. The latches prevent relative movement between the doors and the tailcone structure. All pin latch components are made from either 17-4 CRES or 17-7 CRES cad plated.

Hinge Assemblies

Two hinge assemblies are installed on each door. A hinge assembly has a hinge lug fitting, a hinge arm, and a hinge base fitting. The hinge assembly components are made from 17-4PH stainless steel. The hinge assemblies give outward movement of the doors.

Door and Drain Mast Seals

There are seals on each door and the APU drain mast opening. The seals are made from silicone rubber (ZZ-R-765, Class 3B, Grade 50) covered with a protective nomex weave fabric. The seals meet the fire requirements of 2,000 °F (1080 °C) for 15 minutes. The door seals are kept in place by a titanium strap.

Training Information Points

Do not use permanent or temporary markers on titanium, plastics, composite materials, or painted surfaces. If you do this, you can cause damage.

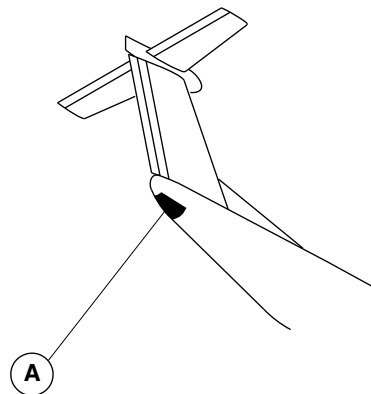
Do not use chlorinated solvents. They can cause a hydrogen embrittlement of the titanium surfaces.

Titanium parts must not touch lead or zinc alloys. This can cause contamination damage to the parts surface.

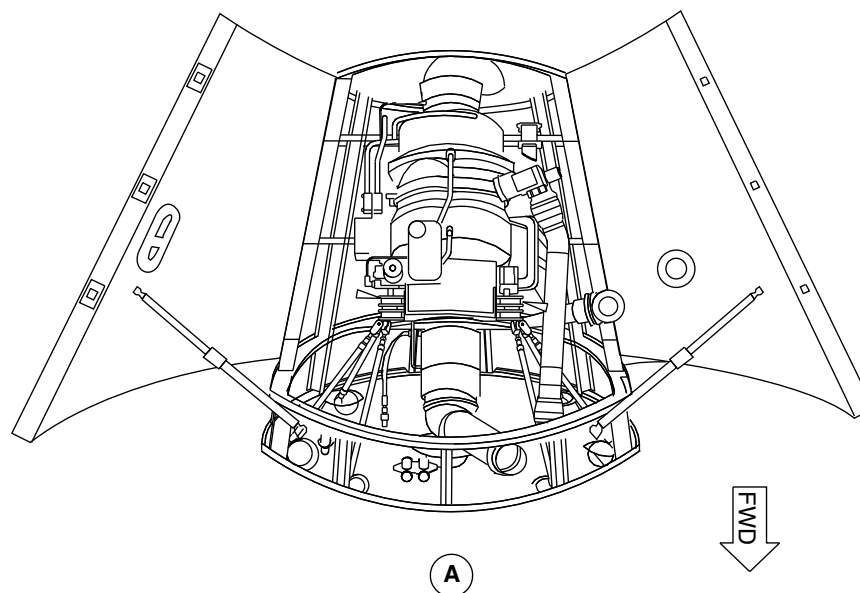


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



NOTE
APU Access Doors Open.



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APU ACCESS DOOR LOCATOR
Figure 1

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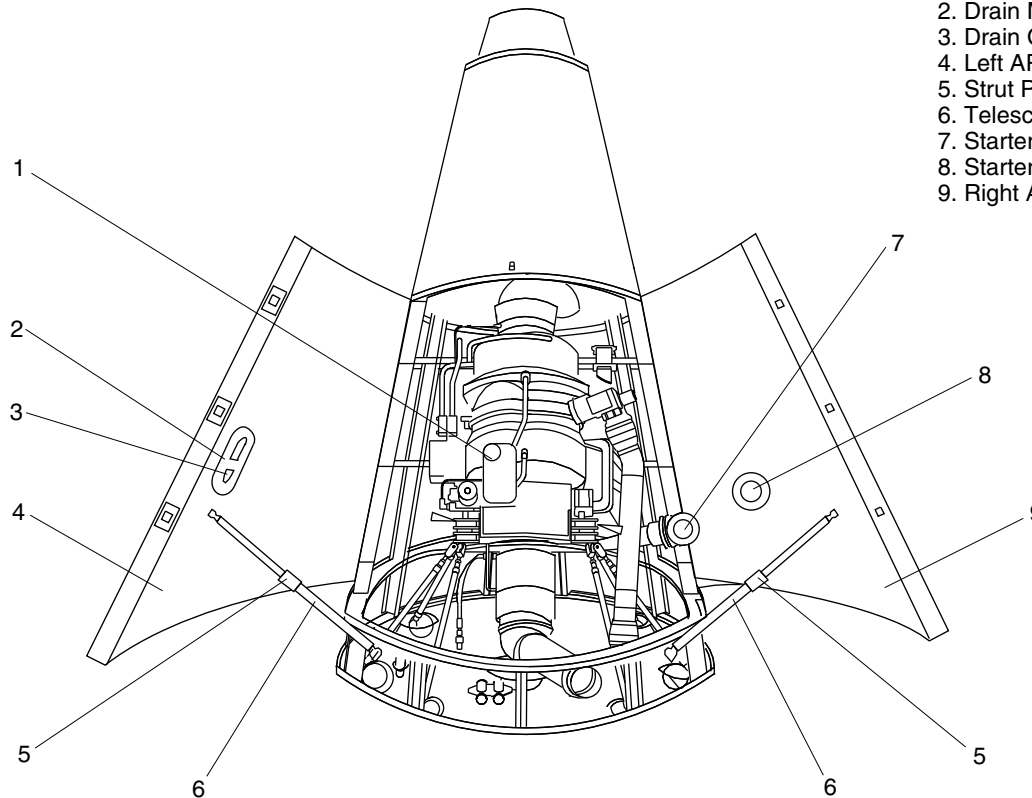


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

1. Drain Mast.
2. Drain Mast Seal.
3. Drain Outlet Opening.
4. Left APU Door.
5. Strut Push-Release.
6. Telescopic Struts.
7. Starter/Generator Exhaust Air Duct.
8. Starter/Generator Exhaust Air Duct Opening.
9. Right APU Door.



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APU ACCESS DOOR DETAIL
Figure 2

PSM 1-84-2A
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**ON A/C ALL

52-44-00-001

GROUND SERVICE DOORS

Introduction

The ground service doors give access for maintenance and servicing purposes.

General Description

The ground service doors are identified as follows:

- Refuel/Defuel Panel Door (52-44-01)
- dc Electrical Ground Power Door (52-44-06)
- ac Electrical Ground Power Door (52-44-11)
- Hydraulic Ground Power Door (52-44-16)
- Ground Air Conditioning Connection Door (52-44-21)
- Lavatory Service Door (52-44-26)

The Refuel/Defuel Panel door is the only ground service door that will give an indication to the flight compartment, if open.

Detailed Description

[Refer to Figures 1 and 2.](#)

Service doors are installed at various locations on the aircraft to give access for servicing and maintenance purposes.

Refuel/Defuel Panel Door

[Refer to Figures 3 and 4.](#)

The Refuel/Defuel Panel door is located on the lower side of the No. 2 engine nacelle, aft of the right main landing gear rear doors. The door gives access to the refuel/defuel adapter, Refuel/Defuel Indicator (RDI), and switches to carry out pressure refueling/defueling operations.

The front end of the door is attached to the lower right engine nacelle by a piano hinge assembly, and two latch assemblies at the aft end. The latch release buttons are pushed to unlock the latch release mechanism and the door moves downwards to the open position. A steel braided cable assembly limits door travel in the open position. The door is moved upwards to engage the latches and to close and lock the door flush with the nacelle.

The Refuel/Defuel Panel Door operates a door switch, installed on the nacelle door structure. When the door is open, the switch causes the amber MASTER VALVE CLOSED light on the Refuel/Defuel panel, and the FUELING ON caution light on the Caution and Warning Panel to come on. The service light, installed on the aft vertical section of the Refuel/Defuel compartment, also comes on. When the Refuel/Defuel Panel Door is closed and locked, it operates the door switch, which causes the FUELING ON caution light to go off. The MASTER VALVE CLOSED light and the service light also go off.

DC Electrical Ground Power Door

[Refer to Figure 5.](#)

The dc electrical ground power door gives access to the 28 Vdc ground power receptacle on the left side of the aircraft nose. The



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door is attached to the nose section by a hinge assembly at the forward end and a latch assembly at the aft end.

To open the door, the latch release button is pushed to release the safety catch and the door is moved to the open position. To close the door, it is moved to the closed position and the latch is engaged.

AC Electrical Ground Power Door

Refer to Figure 6.

The ac electrical ground power door gives access to the ac ground power receptacle on the left side of the No. 2 engine nacelle. The door is attached to the nacelle by a hinge at the top and two latch assemblies at the bottom.

To open the door, the latch release buttons are pushed to release the safety catches. Both latches are engaged to lock the door.

Hydraulic Ground Power Door

Refer to Figure 7.

The hydraulic ground power doors, on the right side of both engine nacelles, give access to the hydraulic ground power connections. The doors are attached to the nacelle by a hinge at the top and two latch assemblies at the bottom.

To open the door, the latch release buttons are pushed to release the safety catches. Both latches are engaged to lock the door.

Ground Air Conditioning Connection Door

Refer to Figures 8 and 9.

The ground air conditioning door is located flush with the lower right side of the rear fuselage. The door opens outward and downward to give access to the air conditioning duct connection.

The door is attached to the fuselage structure at the bottom by a hinge and at the top by a latch. A gasket is installed around the door to help prevent water from entering the area, and to help prevent chafing of the door.

The latch release button is pushed to unlock the latch mechanism and the door moves downward from the hinge on the door lower edge.

The door is moved to the closed position and the latch mechanism is engaged to close it.

Lavatory Service Door

Refer to Figures 10 and 11.

The lavatory service door is located flush with the lower right side of the forward fuselage. The door opens outward and downward to give access to the lavatory service port. The door is attached to the fuselage structure, on the lower side by a hinge and on the upper side by two latches. The door has a gasket installed around the edge to help prevent chafing.

The latch release buttons are pushed to unlock the latch mechanism and the door swings downward from the hinge on the lower side.

The door can be opened or closed from outside the aircraft.



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The door is moved to the closed position and the latch mechanism is engaged to close it.

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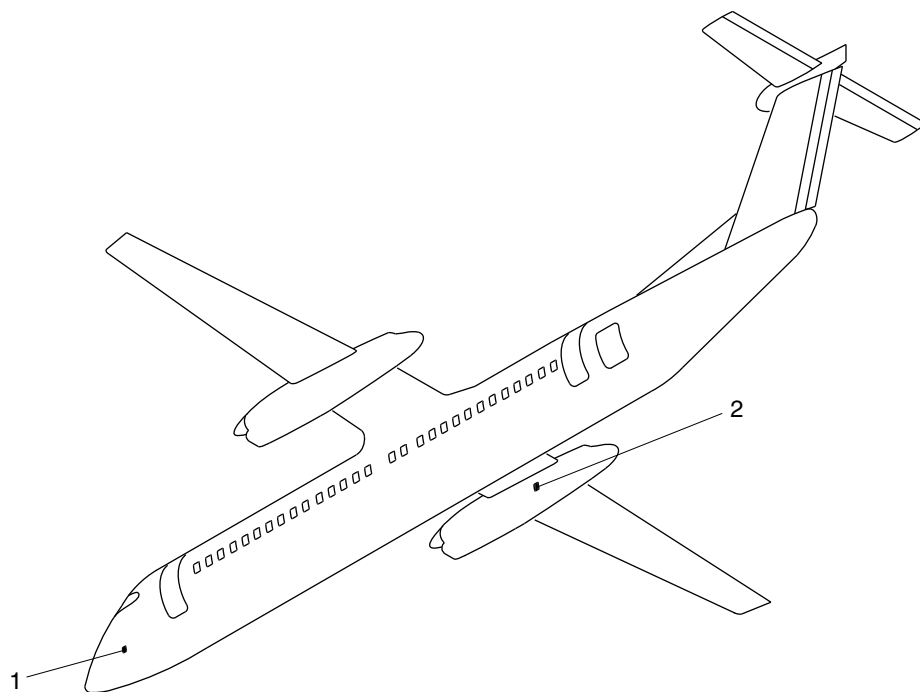
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LEGEND

- 1. DC Electrical Ground Power Door.
- 2. AC Electrical Ground Power Door.

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ELECTRICAL GROUND POWER DOORS LOCATION
Figure 1

PSM 1-84-2A
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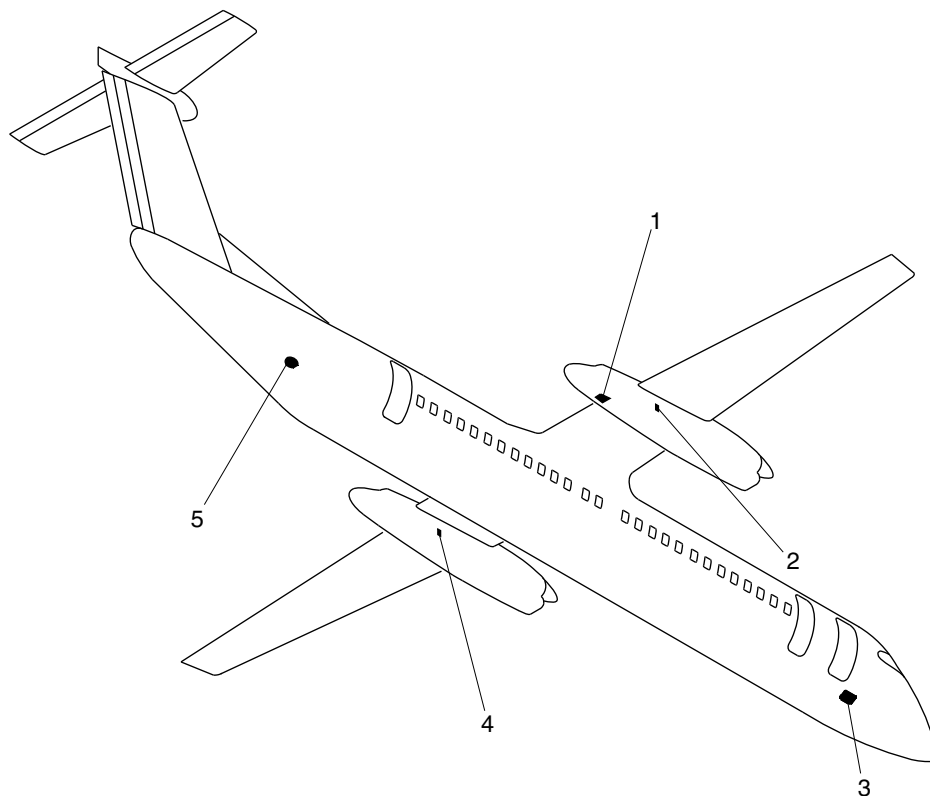
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LEGEND

1. Refuel/Defuel Panel Door.
2. Right Nacelle Hydraulic Ground Power Door.
3. Lavatory Service Door.
4. Left Nacelle Hydraulic Ground Power Door.
5. Ground Air Conditioning Connection Door.

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GROUND SERVICE DOORS LOCATION
Figure 2

PSM 1-84-2A
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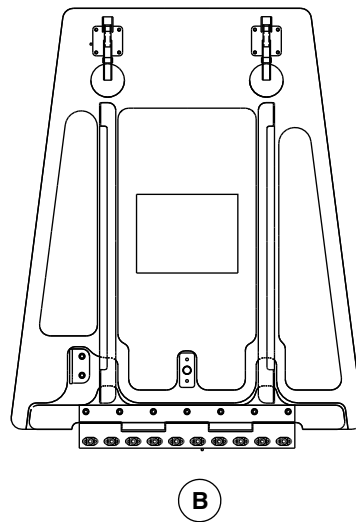
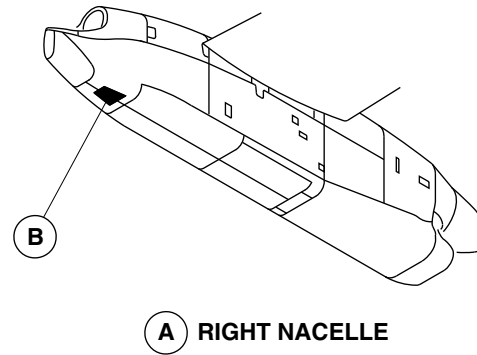
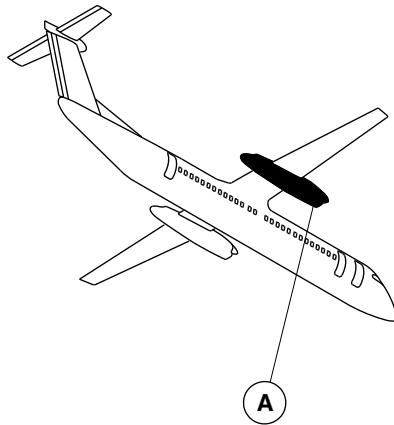
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REFUEL/DEFUEL PANEL DOOR LOCATION
Figure 3

PSM 1-84-2A
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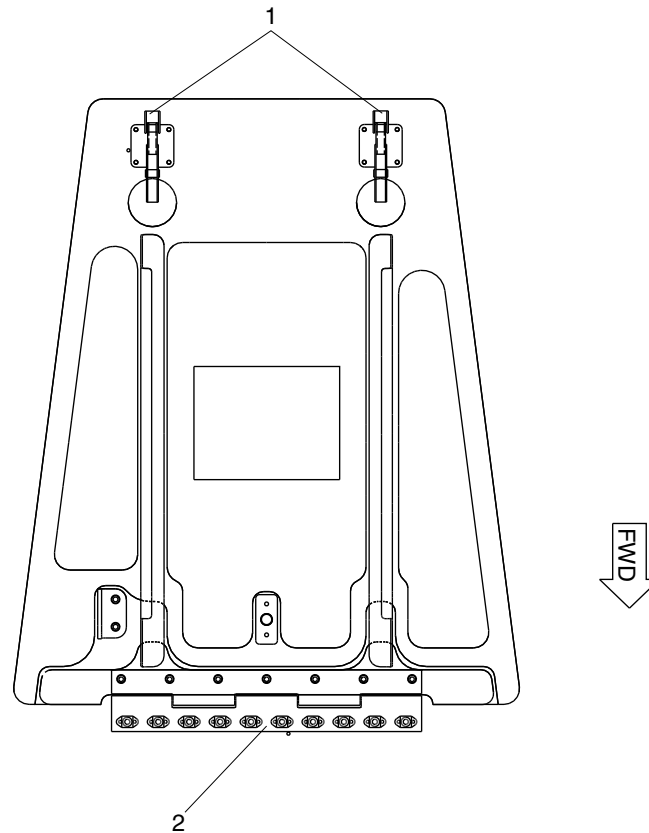
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LEGEND

- 1. Latch.
- 2. Hinge.

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REFUEL/DEFUEL PANEL DOOR
Figure 4

PSM 1-84-2A
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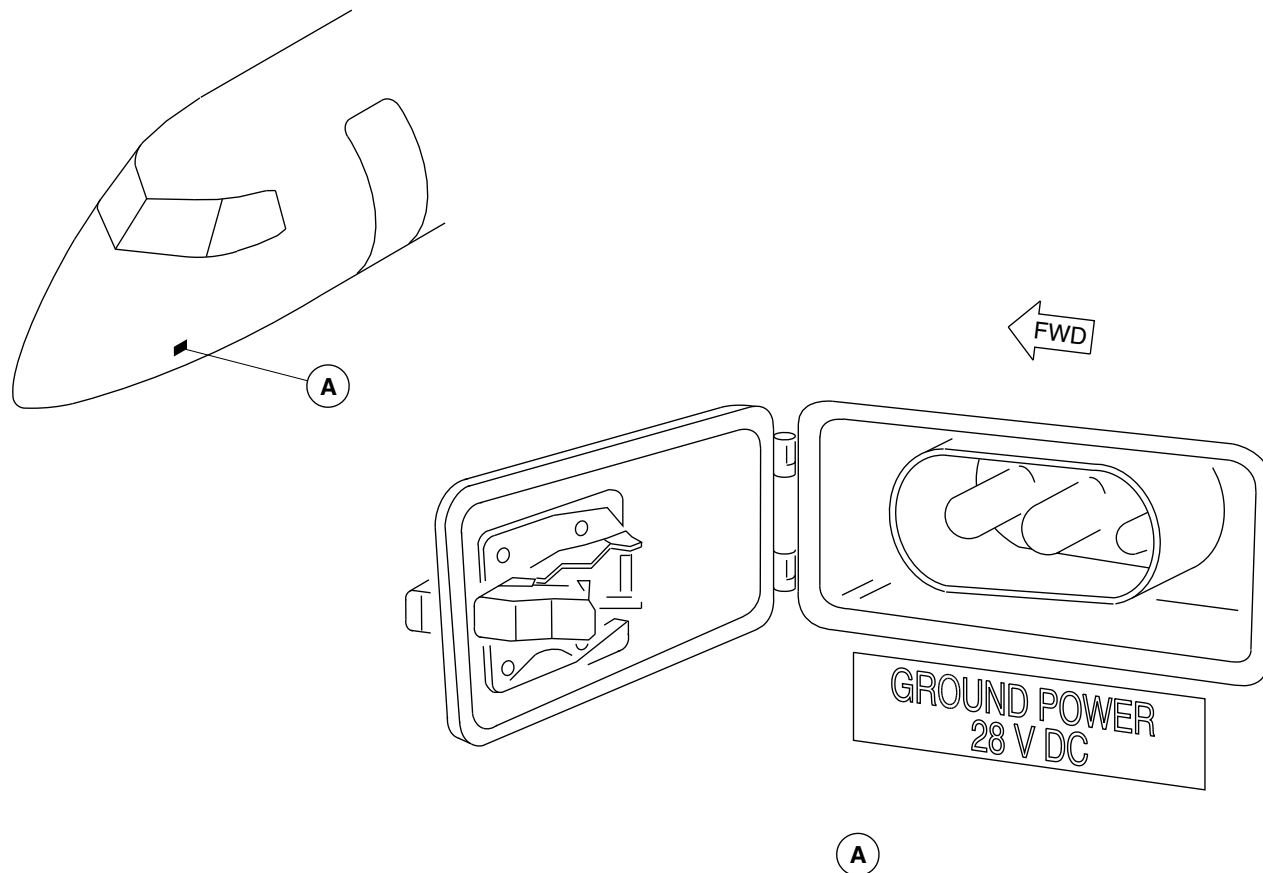
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DC ELECTRICAL GROUND POWER DOOR
Figure 5

PSM 1-84-2A
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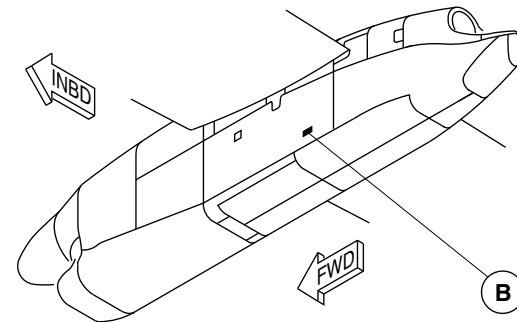
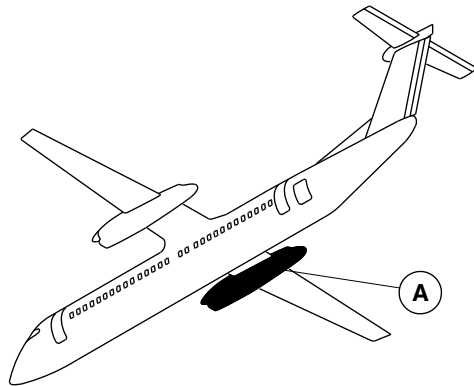
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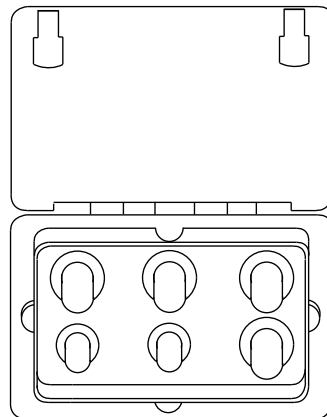


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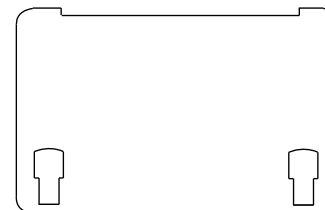
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A RIGHT NACELLE



B OPEN



B CLOSED

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AC ELECTRICAL GROUND POWER DOOR
Figure 6

PSM 1-84-2A
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Config 001

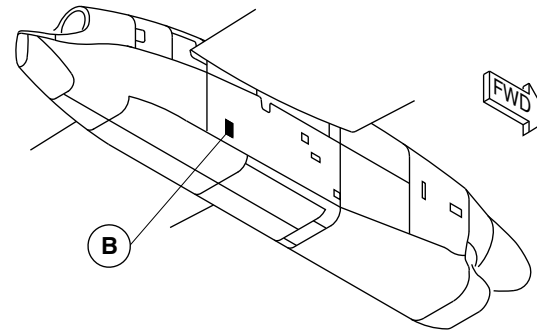
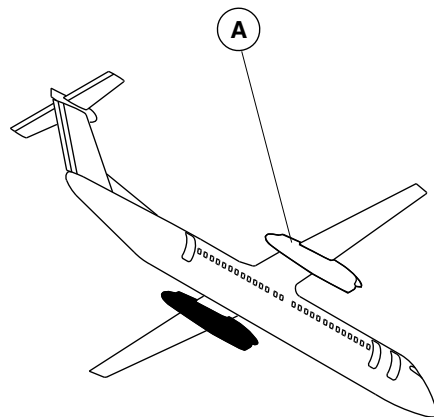
52-44-00

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OF CANADA LIMITED

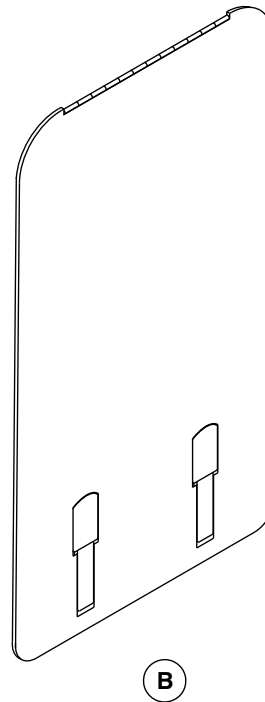
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A LEFT NACELLE

NOTE

Left side shown.
Right side similar.



B

fs847a01.cgm

HYDRAULIC GROUND POWER DOOR
Figure 7

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-44-00
Config 001

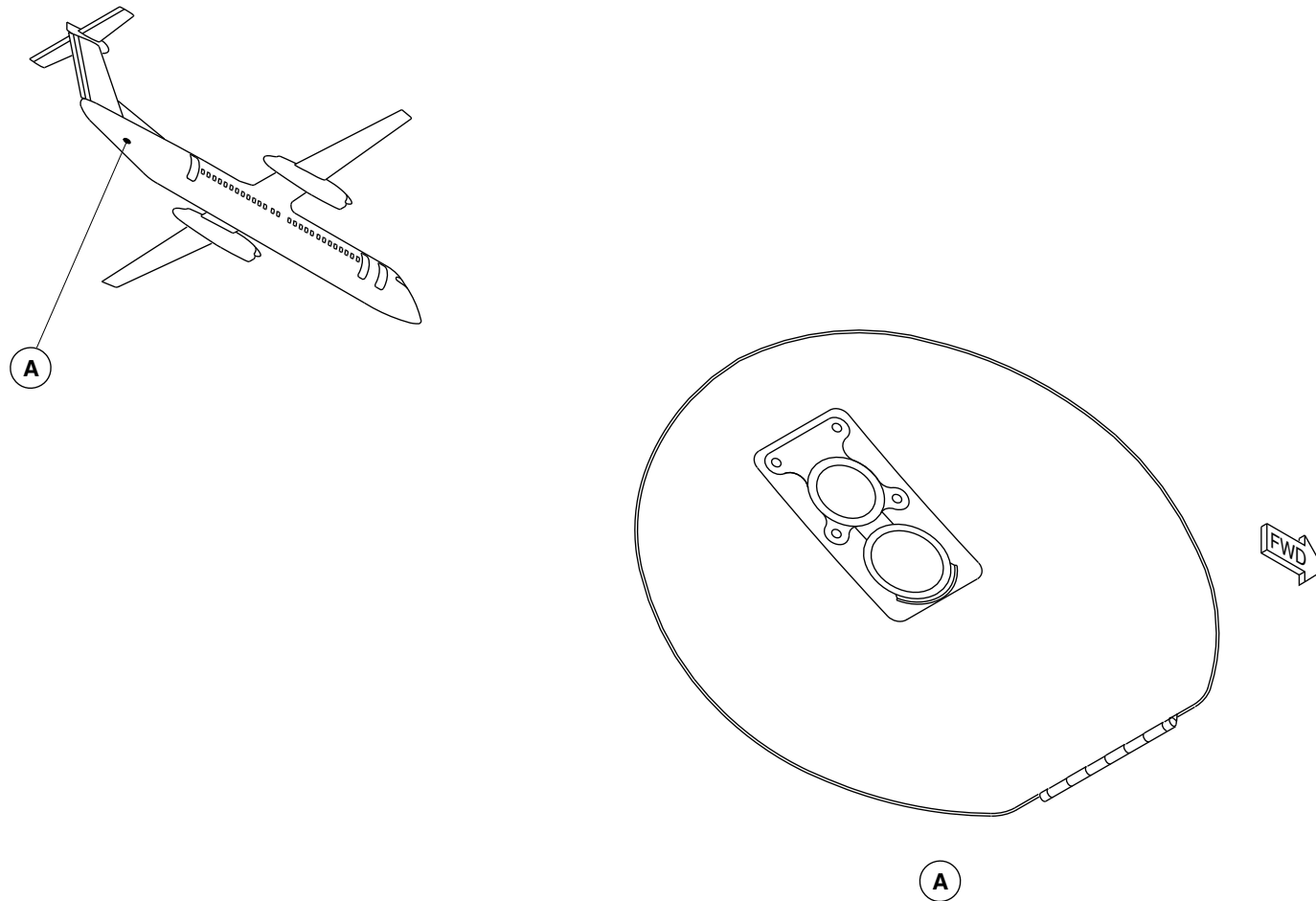
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fs601a01.cgm

GROUND AIR CONDITIONING DOOR LOCATION
Figure 8

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-44-00
Config 001

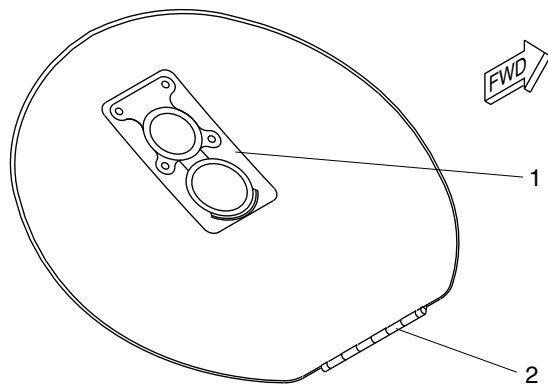
52-44-00

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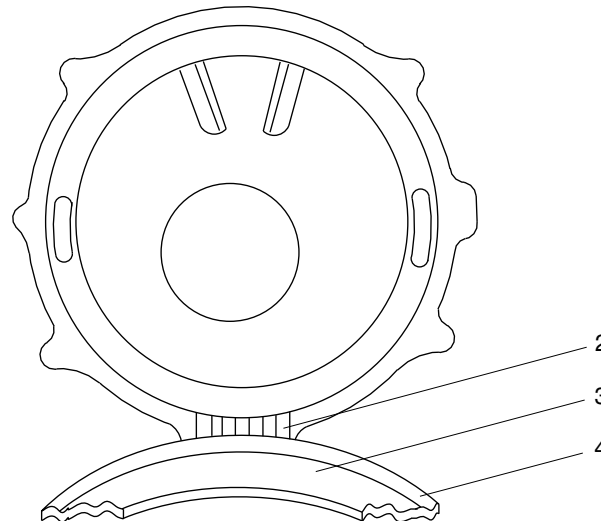
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Latch.
- 2. Hinge.
- 3. Gasket.
- 4. Door.



fs727a01.cgm

GROUND AIR CONDITIONING DOOR
Figure 9

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-44-00
Config 001

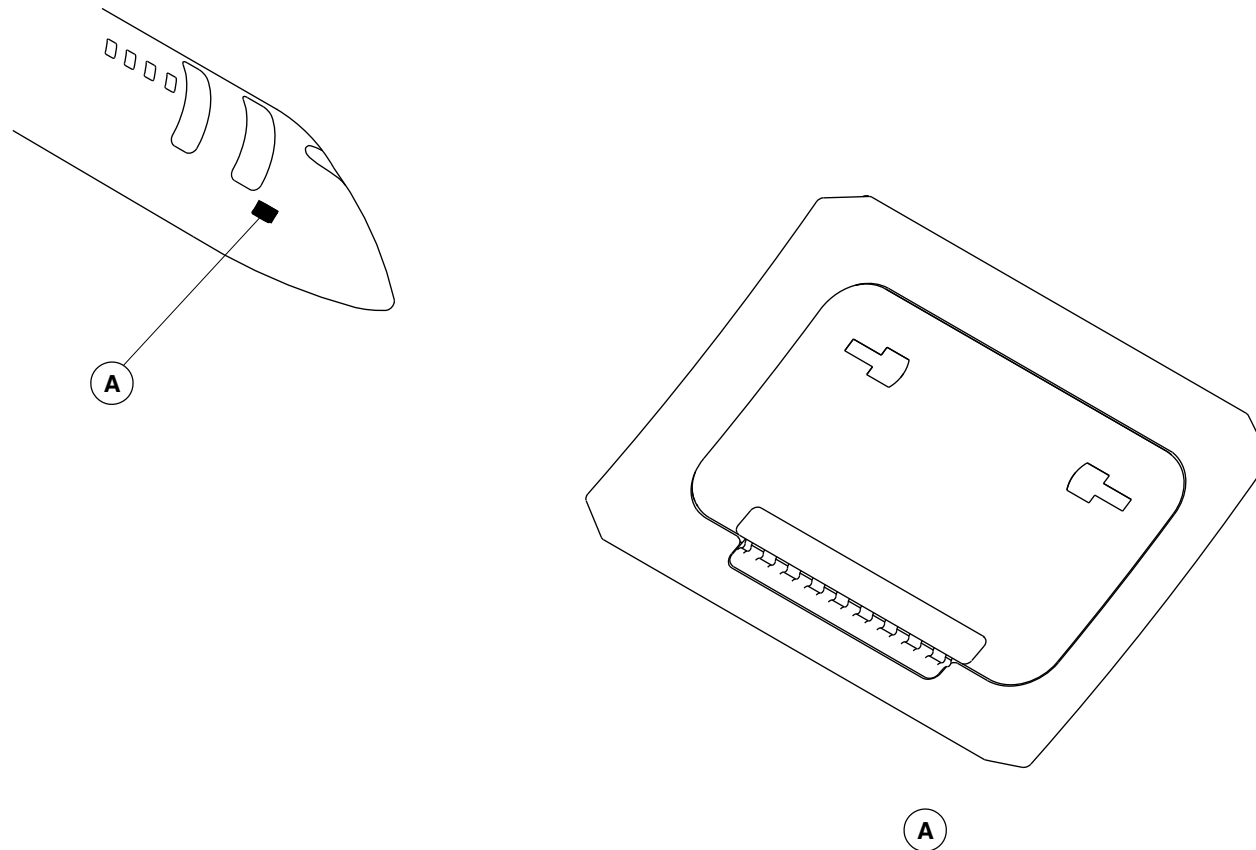
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs611a01.cgm

LAVATORY SERVICE DOOR LOCATION
Figure 10

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-44-00
Config 001

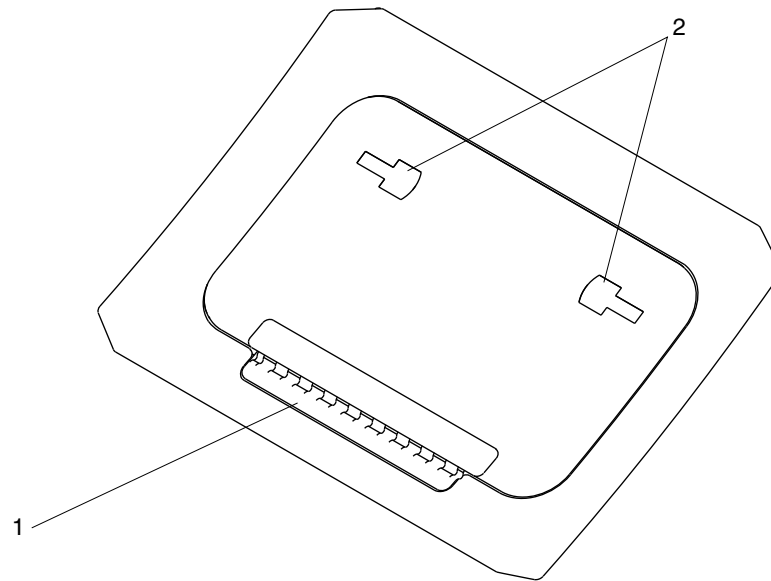
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Hinge.
- 2. Latch.

fs612a01.cgm

LAVATORY SERVICE DOOR
Figure 11

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

52-50-00-001

INTERIOR DOORS

Introduction

The door gives access to the flight compartment from the passenger compartment.

General Description

Refer to Figure 1.

The flight compartment door separates the passenger compartment from the flight compartment.

The door can be locked to prevent passenger entry. It is hinged on the left side and swings aft to open.

The door includes a locking mechanism and a rotating handle on both sides of the door.

Detailed Description

The handle on the flight compartment side opens the door if the door is locked or not locked. The lock is disengaged by operating the small lever on the flight compartment side of the door, or by the using a key on the passenger side. The key can only be removed when the door lock is engaged.

In an emergency, the door can be removed from its hinges, by removing the hinge pins from the upper and center hinges and lifting the door from the bottom hinge.

PSM 1-84-2A
EFFECTIVITY:
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Config 001

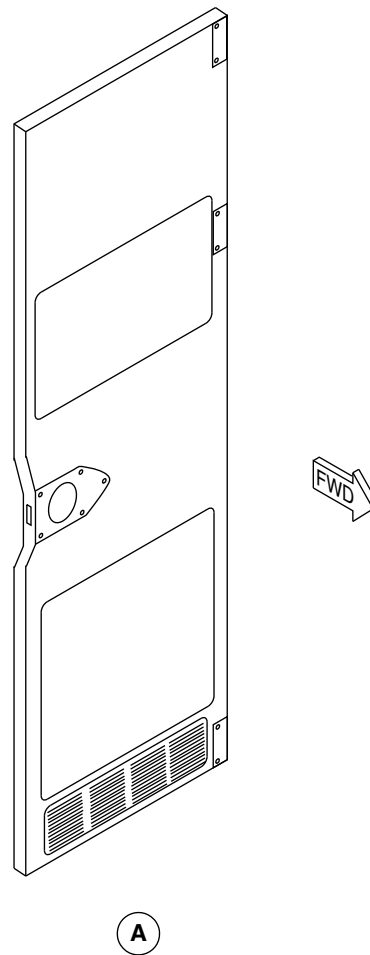
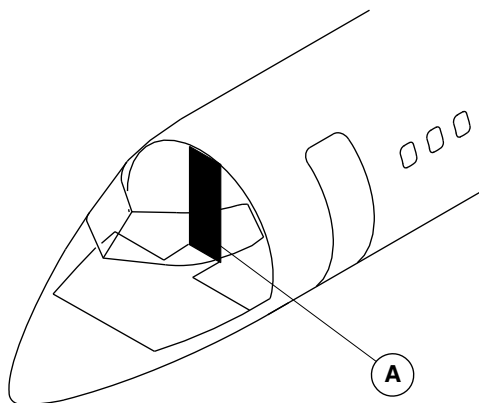
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fs605a01.cgm

Flight Compartment Door
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

52-51-00-001

FORTIFIED FLIGHT COMPARTMENT DOOR AND REMOTE ACCESS SYSTEM

Introduction

The Fortified Flight Compartment Door and Remote Access System are optional and consist of the following major components:

- Fortified Flight Compartment Door
- Internal Doors Control Panel
- Flight Attendant Emergency Access Push-button Annunciator.

General Description

The Fortified Flight Compartment Door and Internal Doors Control Panel provide the flight crew control of access to the flight deck. A Flight Attendant Emergency Access Push-button Annunciator enable the cabin crew to gain access to the flight compartment should the flight crew become incapacitated.

Fortified Flight Compartment Door

The Fortified Flight Compartment Door separates the flight compartment from the passenger compartment. It is hinged on the left side door post on the pilot's bulkhead and opens aft into the passenger cabin.

The Fortified Flight Compartment Door meets ballistic, intrusion and decompression requirements. In the case of decompression, a barometrically actuated lock releases a vent panel located on the lower door allowing the panel to open forward and down.

The door includes a locking mechanism consisting of a slide latch and a deadbolt lock.

Internal Doors Control Panel

The Internal Doors Control Panel is located on the overhead console and has the label "INTERNAL DOORS" inscribed on the face plate. It allows full control and indication of the flight compartment door during flight operations.

The "INTERNAL DOORS" caution light on the Caution and Warning Panel monitors both the internal forward baggage door and the flight compartment door.

Flight Attendant Emergency Access Push-button Annunciator

The emergency access mode consists of a Flight Attendant Emergency Access Push-button Annunciator located on the Maintenance Panel in the forward cabin. It permits access by the cabin crew to the flight compartment in the event that there is no response from the flight crew.

Detailed Description

The Fortified Flight Compartment Door, Internal Doors Control Panel and Flight Attendant Emergency Access Push-button Annunciator functions are described as follows:

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EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Fortified Flight Compartment Door

Refer to Figure 1.

A slide latch, operable from the flight compartment side only, locks the door.

To secure a vacant flight compartment, a deadbolt lock located below the slide latch permits the flight compartment to be locked from the cabin side using a key.

The deadbolt can be operated from the flight compartment by turning a rotary action handle. When the dead bolt rotary knob is set to the unlatch position, two red dots on the dead bolt plate behind the knob will become visible. When the key is rotated in the lock, one of the arms of the rotary action handle will rotate 90 degrees covering half of each red dot. When the rotary knob is set to the latched position, two green dots will become visible and actuation of the key access feature will be disabled.

In the case of a door jam, three door hinge pins accessible only from the flight compartment can be retracted to allow the door to be removed as follows:

- Unlock the hinge by rotating the hinge bail outboard to release it from its lock position and push or step down on bottom hinge pin, then lock it in its retracted position.
- Unlock and pull down upper hinge pin.
- Unlock and lift middle hinge pin.
- Push flight compartment door at hinge side.

NOTE

It may require a large force to open the flight compartment door.

- Rotate the whole flight compartment door counter clockwise and stow against the lavatory wall.

NOTE

Upon forcing the flight compartment door open, it may fall straight aft and lay flat on the cabin floor.

Internal Doors Control Panel

Refer to Figure 2.

The Internal Doors Control Panel is located on the overhead console, directly below the Fire Protection Panel and to the right of the DC Control Panel. This panel consists of the following:

- Baggage Door / Cockpit Door light
- Rotary Knob
- Fail / Auto Unlock light
- Lock Isolate guarded switchlight

The functions of these items are described in the following table:



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

ITEM	DESCRIPTION
1. BAGG DOOR/CKPT DOOR LIGHT (dual lights)	– BAGG DOOR Amber light illuminates if internal forward baggage door is open – CKPT DOOR Amber light illuminates if flight compartment door is unlatched
2. ROTARY KNOB (3 position, momentary action)	– UNLK (momentary action) Push knob in and rotate left Solenoid will de-activate and unlatch flight compartment door

ITEM	DESCRIPTION
	– AUTO Normal position Will enable the auto-unlock sequence to be activated by the cabin crew – DENY (momentary action) Rotate Knob Right Cancels auto-unlock sequence and terminates automatic unlatching of flight compartment door.



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ITEM	DESCRIPTION
3. FAIL/AUTO UNLK LIGHT (dual lights)	– FAIL light illuminates if solenoid failure or LOCK ISOLATE switch is pushed Flight compartment door unlatches – AUTO UNLK Light illuminates when the cabin crew initiates the auto-unlock system from the cabin mounted switch
4. LOCK ISOLATE GUARDED SWITCHLIGHT (two position)	PUSH – Light illuminates and power removed from lock solenoid – Flight compartment door unlatches – CKPT DOOR light illuminates – INTERNAL DOORS light on Caution and Warnings panel illuminates – FAIL light illuminates

Flight Attendant Emergency Access Push-button Annunciator

A Flight Attendant Emergency Access Push-button Annunciator is mounted directly below the optional DPAS (Digital Passenger Address System) on the wardrobe maintenance panel. A white light indicator illuminates on the upper half of the Emergency Access Push-button Annunciator when pressed. A Master Caution tone sounds and the INTERNAL DOORS light on the Caution and Warning Panel illuminates. On the Internal Doors Control Panel, the amber AUTO UNLK advisory light illuminates indicating that the 40 second auto-unlock sequence has been initiated. The flight crew can either select UNLK or DENY. If there is no response by flight crew, the flight compartment door will automatically open after a timed delay of 40 seconds. An amber advisory light will illuminate on the lower half of the Flight Attendant Emergency Access Push-button Annunciator indicating that the flight compartment door is unlatched.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

COCKPIT DOOR EMERGENCY OPENING

1. UNLOCK AND PUSH OR STEP DOWN ON BOTTOM HINGE PIN
2. UNLOCK AND PULL DOWN UPPER HINGE PIN
3. UNLOCK AND LIFT MIDDLE HINGE PIN
4. PUSH DOOR AT HINGE SIDE
5. ROTATE DOOR COUNTER CLOCKWISE AND STOW AGAINST LAVATORY



**CAUTION – RELEASE
THE BOTTOM HINGE PIN FIRST**

**DOOR MUST BE CLOSED
& LOCKED AT ALL TIMES**

FOR GROUND USE ONLY DEADBOLT POSITIONS



UNLOCKED

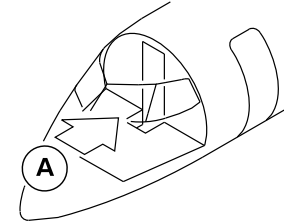
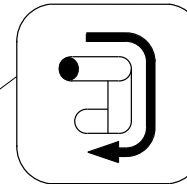
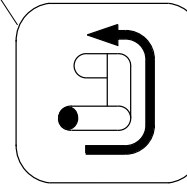
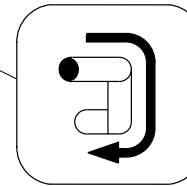
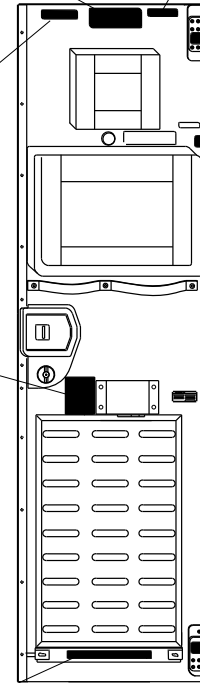


LOCKED
KEY OPERABLE



LOCKED
KEY INOPERABLE

← PINS MUST BE FULLY ENGAGED AT ALL TIMES →



**VIEW LOOKING AFT ON FLIGHT
COMPARTMENT DOOR**

braw33a01.dg, sw, 11/06/04

Fortified Flight Compartment Cockpit Door
Figure 1 (Sheet 1 of 2)

PSM 1-84-2A
EFFECTIVITY:
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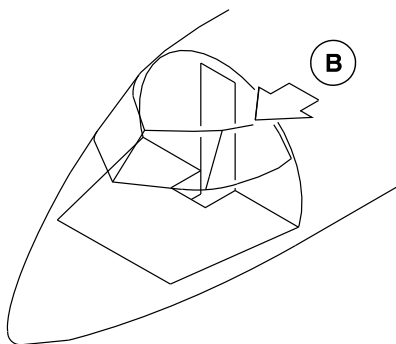
52-51-00

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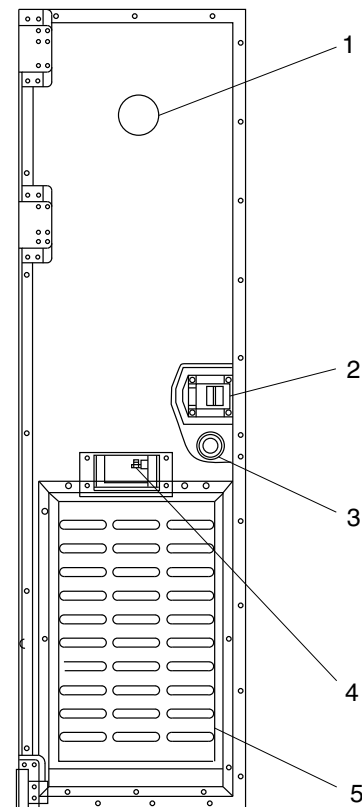
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Spy hole.
2. Slide latch (operable from flight compartment only).
3. Deadbolt lock.
4. Barometric vent panel lock.
5. Vent panel.



(B)

VIEW LOOKING FWD ON FLIGHT
COMPARTMENT DOOR

braw33a02.dg, sw, 11/06/04

Fortified Flight Compartment Cockpit Door
Figure 1 (Sheet 2 of 2)

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-51-00
Config 001

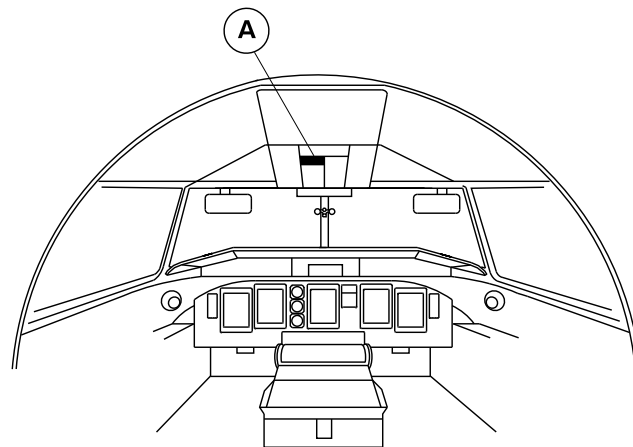
52-51-00

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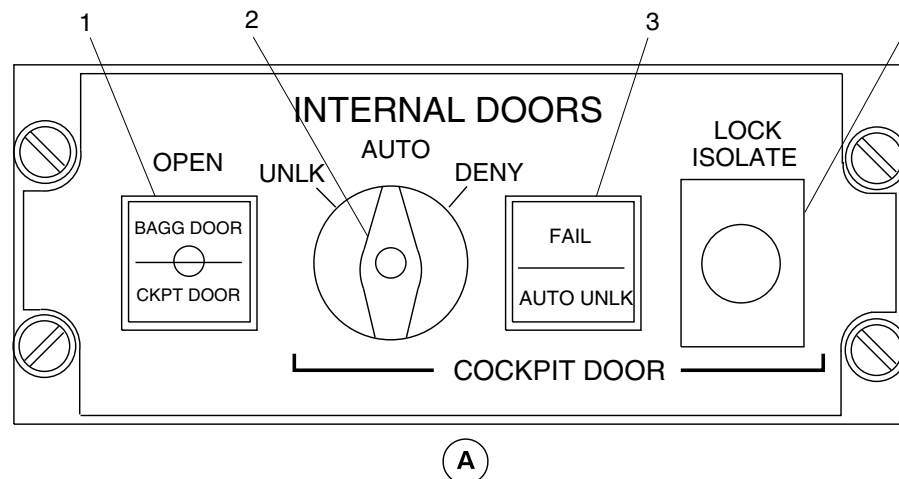
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. BAGG DOOR / CKPT DOOR LIGHT (dual lights).
2. ROTARY KNOB (3 position, momentary action).
3. FAIL / AUTO UNLK LIGHT (dual lights).
4. LOCK ISOLATE GUARDED SWITCH LIGHT (two position).



braw34a01.dg, sw, 04/06/04

Internal Doors Control Panel
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

52-70-00-001

DOOR WARNING SYSTEM

Introduction

The door warning system monitors the position of the fuselage doors.

General Description

The door warning system supplies a visual indication, in the flight compartment, when a fuselage door open or unlocked warning condition occurs.

The door warning system sends door sensor outputs to the Proximity Sensor Electronic Unit (PSEU). The PSEU outputs data to the Input Output Processors (IOP 1 and IOP 2). This data is displayed in the flight compartment and is used by the Cabin Pressure Control System (CPCS).

[Refer to Figure 1.](#)

The door warning system monitors the position of the fuselage doors that follow:

- Forward Passenger Door (52-11-00)
- Aft Passenger Door (52-12-00)
- Forward Baggage Door (52-32-00)
- Aft Baggage Door (52-31-00)
- Aft Service Door (52-41-00)

- Type II Emergency Exit Door (52-22-00) (52-23-00)
- Type III Emergency Exit Door (52-71-01).

The door warning system uses two sensors for each of the fuselage doors, and one sensor for the Type II emergency exit.

The system does not monitor the flight compartment emergency hatch.

Detailed Description

[Refer to Figures 2 , 3 and 4.](#)

The fuselage door positions are monitored by 11 proximity sensors, the outputs are received by the Proximity Sensor Electronic Unit (PSEU). The PSEU door position outputs are sent to the Input Output Processors(IOP 1 & IOP 2) in the Integrated Flight Cabinets (IFC). The data is then sent to the systems that follow:

- The Electronic Flight Information System (EFIS)
- The Caution and Warning Control Panel (CWCP)
- The Cabin Pressure Control System (CPCS)

[Refer to Figure 5.](#)

If a fuselage door is not fully closed and locked, the FUSELAGE DOORS warning light and the MASTER WARNING switchlight start flashing. Three tones also sound over the flight compartment speakers. When the MASTER WARNING switchlight is pushed, the light turns off. The FUSELAGE DOORS warning light remains on until the door is fully closed and locked.

PSM 1-84-2A
EFFECTIVITY:
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Refer to Figures 6 , 7 and 8.

The DOORS system page on the Multi-Function Display (MFD) shows all the fuselage doors and identifies them as:

- PAX for each of the passenger doors
- BAGGAGE for each of the baggage doors
- SERVICE for the aft service door
- EMERG EXIT for the type II emergency exit door
- EMERG EXIT for the type III emergency exit door.

If a fuselage door is not fully closed and locked, the DOORS system page identifies the door and shows it as a red filled rectangle. If the door is fully closed and locked, it is shown only as a green empty rectangle. If the data from the PSEU is invalid, the DOORS system page shows a global white INVALID DATA message.

The PSEU also transmits output signals to the Cabin Pressure Control System (CPCS). If the doors are not closed and locked, the Cabin Pressure Controller (CPC) prevents automatic pressurization of the aircraft.

Forward Passenger Door Monitoring

Refer to Figures 9 , 10 and 11.

The position of the forward passenger door is monitored by two proximity sensors.

One sensor is located on the top aft side of the door, to verify that the door is closed by making sure that the door is fully down. The other is located adjacent to the door seal pressurizing valve and

verifies that the door is locked, when the door closing mechanism is fully overcentred and in safety.

The forward passenger door can be visually inspected to make sure it is closed and locked, by viewing the position of the internal handle. If the door is not locked, the pressurization door seal will not inflate.

Aft Passenger Door Monitoring

Refer to Figures 12 and 13.

The position of the aft passenger door is monitored by two proximity sensors. One sensor is located at the aft upper roller guide and make sure that the door is closed. The other sensor is located at the door handle and make sure that the door is locked. The locking mechanism of the aft passenger door can be inspected to make sure that the door is closed and locked. The door can also be checked to be in the locked position, by seeing that the handle mechanism is pushed fully outward and the vent door is closed, or by checking the lift/latch shaft position indicator on the door.

The location of the two proximity sensors for the aft passenger door is similar to that of the aft service and forward baggage doors.

Forward Baggage Door Monitoring

The position of the forward baggage door is monitored by two proximity sensors. One sensor is located at the upper fixed roller and make sure that the door is closed. The other sensor is located at the handle and makes sure that the door is locked.

The lift/latch shaft position indicator on the door can be visually inspected to make sure that the door is closed and locked (refer to Aft Passenger Door).



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Aft Baggage Door Monitoring

Refer to Figures 14 , 15 , 16 and 17.

The position of the aft baggage door is monitored by two proximity sensors. One sensor is located on the lower forward fuselage frame and verifies that the door is closed. The other sensor, is located at the door handle and verifies that the door is locked. The door must be closed and locked, with the external handle in the recess for the door seal to inflate.

Aft Service Door Monitoring

The position of the aft service door is monitored by two proximity sensors. One sensor is located at the upper fixed roller position and monitors the door closed position. The other sensor located within the handle mechanism verifies that the door is locked (refer to Aft Passenger Door).

Type II Emergency Exit Door Monitoring

Refer to Figures 18 and 19.

The position of the emergency exit door is monitored by one proximity sensor, located on the upper part of the door frame.

The door may be visually confirmed closed and locked, by the position of the internal handle and the physical position of the door.

Type III Emergency Exit Door Monitoring

Refer to Figure 20.

Two proximity sensors monitor the position of the emergency exit door. One sensor is located on the top aft side of the door. It makes

sure that the door is closed by ensuring that the door is fully down. The other sensor is located adjacent to the door seal pressurizing valve. It makes sure that the door is locked, when the door closing mechanism is fully over-centred and in safety.

Door Warning System Input to the Pressurization System (Refer to SDS 21-30-00)

When the aircraft is on the ground, weight on wheels, and all the doors are closed and locked, the CPC starts normal ground sequences. When a door is open or unlocked, the CPC starts a door open sequence, which prevents automatic pressurization. Manual pressurization is still available.

The door position signals are stored in the CPCS, when the aircraft leaves the ground.

If the memorized signal indicates that the doors are closed and locked, the CPC starts the normal flight operational sequence regardless of the door position signal during flight. A door open signal during flight will not affect the CPCS.

If the memorized signal indicates that the doors are not closed and locked, the CPC will start the door open operational sequence regardless of door position signals during flight. Cabin pressurization will not take place when the CPCS is in the automatic mode.

NOTE

Discrete inputs to the CPC are continuously monitored by the CPC when the AUTOMATIC mode is engaged.

On aircraft with SB84-32-66 or ModSum 4-126420 incorporated cabin pressurization inhibit to exclude translating doors which are equipped with vent panels.



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Door Warning System Input to the Caution and Warning Control Panel. (CWCP)

The PSEU transmits an output signal to the Caution and Warning Control Panel (CWCP). The door position signal is transmitted to the CWCP through two independent paths.

The position signals received by the CWCP can be one of the two:

- Electrical ground, door closed and locked
- Open circuit, door open and/or unlocked

Door Warning System input to the Integrated Flight Cabinet

All door proximity sensor output signals are sent to the PSEU to generate one output signal for each door, in the closed and locked position. This signal is sent to the Input Output Processors (IOP 1 and IOP 2) in the Integrated Flight Cabinets (IFC). The data is processed and displayed on the Doors System page.

The output signal is divided to supply each of the IOPs with an independent signal. When the FUSELAGE DOORS warning light is on, the fuselage Doors System page shows any open or unlocked door as a solid red rectangle.

A single failure will not prevent indication of the fuselage door status.

Doors Warning System Indication

When any door is open and unlocked the indications that follow will be shown:

- The Doors Systems page shows the open door as a red filled rectangle

- The FUSELAGE DOORS warning light flashes
- The MASTER WARNING light flashes

A closed and locked door is shown as green empty rectangle on the Doors System page. An open door is shown as a red filled rectangle with a legend identifying the door in red T1 font.

The doors are identified by the legend that follows:

- “PAX” for the Passenger doors
- “BAGGAGE” for the Baggage doors
- “SERVICE” for the Aft Service door
- “EMERG EXIT” for the Type II emergency exit.

If data from both IOPs is invalid, the Doors System page will show a global white “INVALID DATA” message.

When the FUSELAGE DOORS warning light on the Caution and Warning Panel flashes, the MASTER WARNING light also flashes. The MASTER WARNING light is pushed to acknowledge the warning and the MASTER WARNING light goes off. The FUSELAGE DOORS warning light then stays on continuously as long as the door is open or unlocked.

If a subsequent door open/unlocked condition is sensed the FUSELAGE DOORS and MASTER WARNING lights will start flashing again.

Proximity Sensors

The proximity sensors are hermetically sealed two wire devices, used to sense the distance to targets installed on the fuselage doors.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The inductance at the sensor changes with the sensor's proximity to the target. As the target gets closer to the sensor face, the inductance increases and transmits a high signal. High and low signals, for target far or near conditions, are transmitted to the PSEU.

Proximity sensors can fail in one of two modes:

- An open circuit with no output
- A short circuit output

When any proximity sensor fails, a signal is sent to the PSEU, and the FUSELAGE DOORS warning light flashes.

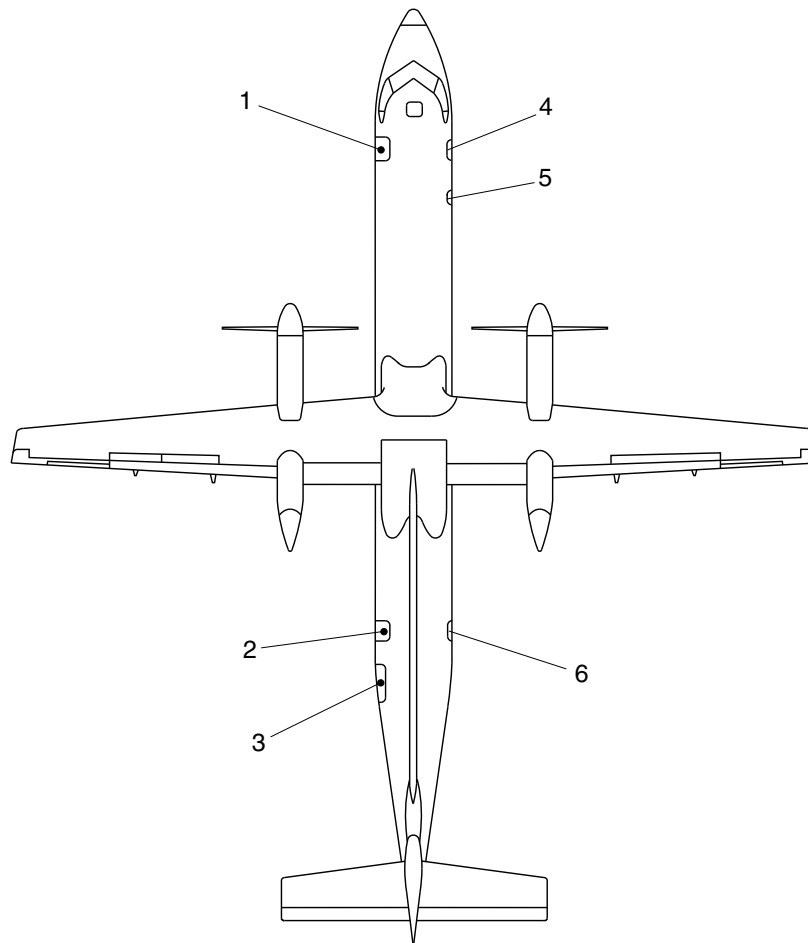
The proximity sensors cannot be repaired if they fail.

Two proximity sensors are located near each door to indicate when the door is closed and when it is locked. One proximity sensor is installed on the door surround structure and senses when the forward roller guide is in the proper position for latching the door. When this occurs, the door is in the closed position. The other proximity sensor is installed in the door handle housing. It senses whether the handle is retracted (door locked) or extended (door unlocked).



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Forward Passenger Door.
"Closed" Sensor and "Locked" Sensor.
2. Aft Passenger Door.
"Closed" Sensor and "Locked" Sensor.
3. Aft Baggage Door.
"Closed" Sensor and "Locked" Sensor.
4. Forward Baggage Door.
"Closed" Sensor and "Locked" Sensor.
5. Upper (Type II) Emergency Exit.
"Locked" Sensor.
6. Aft Service Door.
"Closed" Sensor and "Locked" Sensor.

cg4294a01.dg, ak, dec11/2015

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Aircraft Doors and sensors Page 1
Figure 1 (Sheet 1 of 2)

PSM 1-84-2A
EFFECTIVITY:
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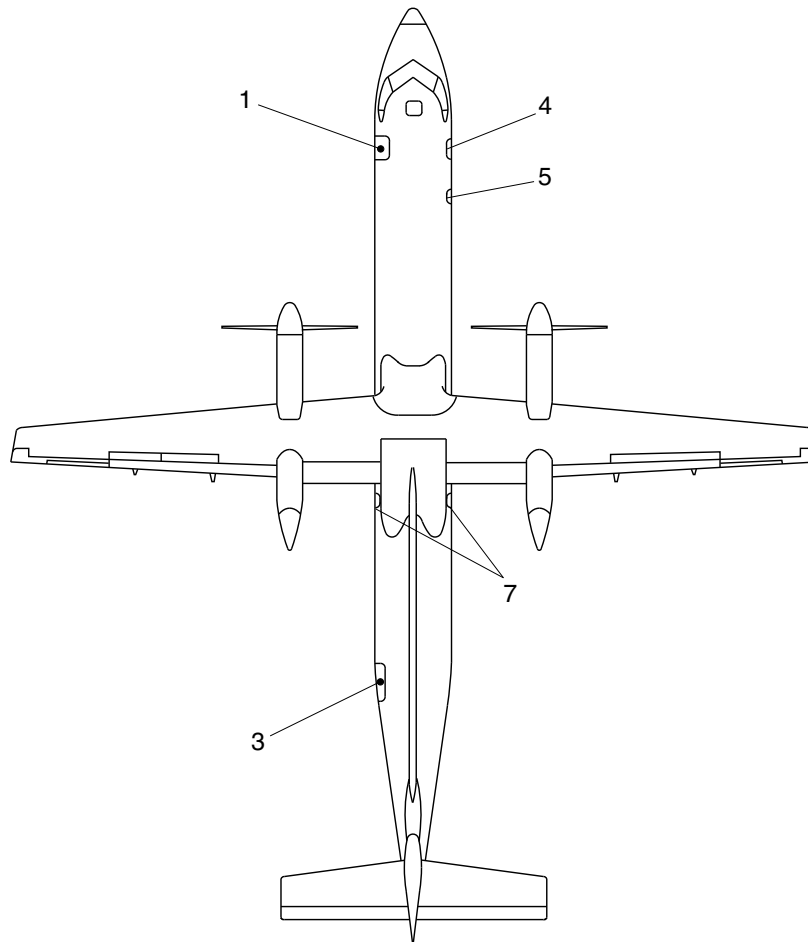
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DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Forward Passenger Door.
"Closed" Sensor and "Locked" Sensor.
3. Aft Baggage Door.
"Closed" Sensor and "Locked" Sensor.
4. Forward Baggage Door.
"Closed" Sensor and "Locked" Sensor.
5. Upper (Type II) Emergency Exit.
"Locked" Sensor.
7. Type III Emergency Exit Door.
"Closed" Sensor and "Locked" Sensor.

cg4294a02.dg, ak, dec11/2015

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Aircraft Doors and sensors Page 1
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PSM 1-84-2A
EFFECTIVITY:
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Config 001

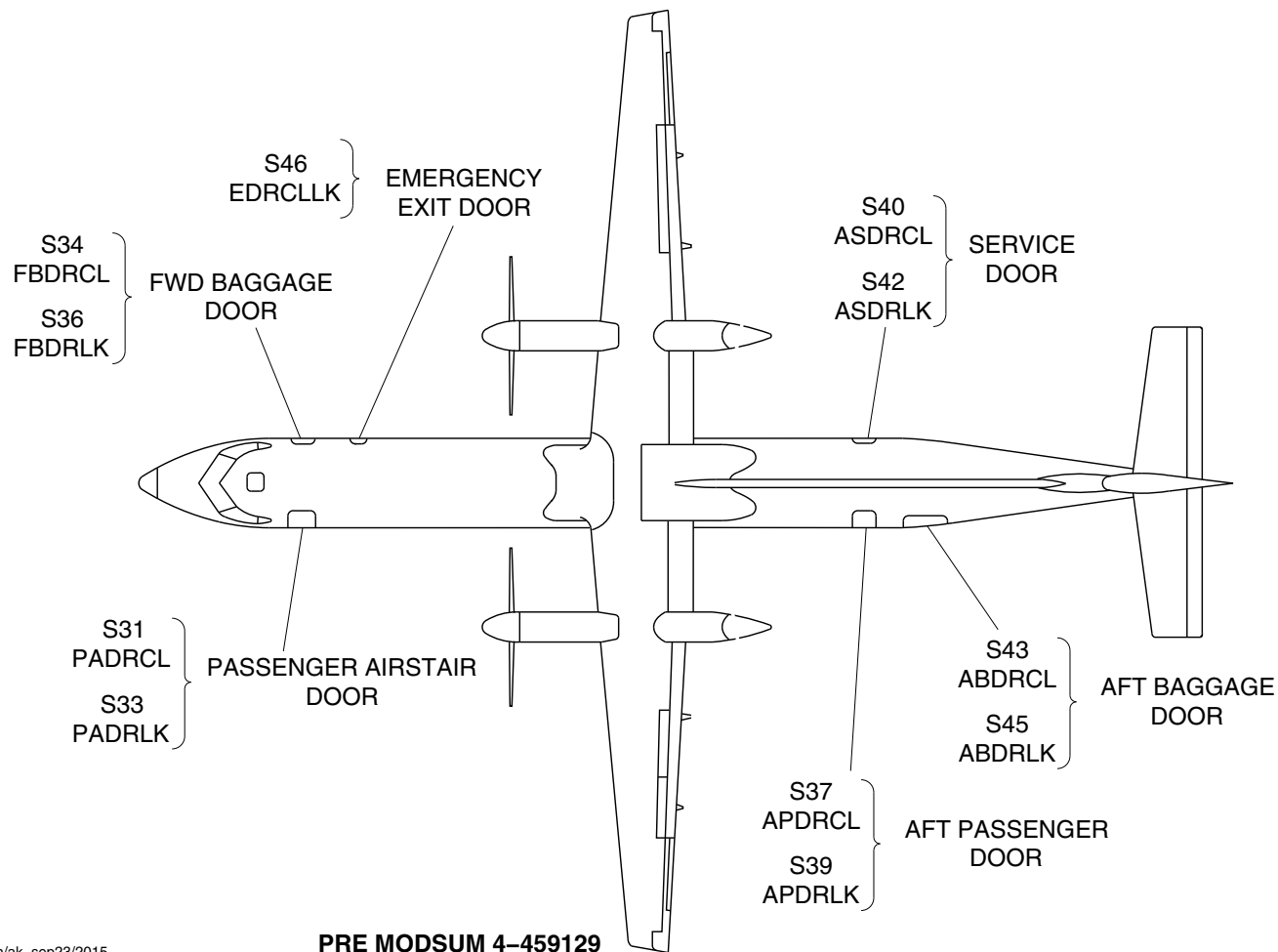
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fsk77a01.dg, vn/ak, sep23/2015

Aircraft Doors and Sensors Page 2
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PSM 1-84-2A
EFFECTIVITY:
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Config 001

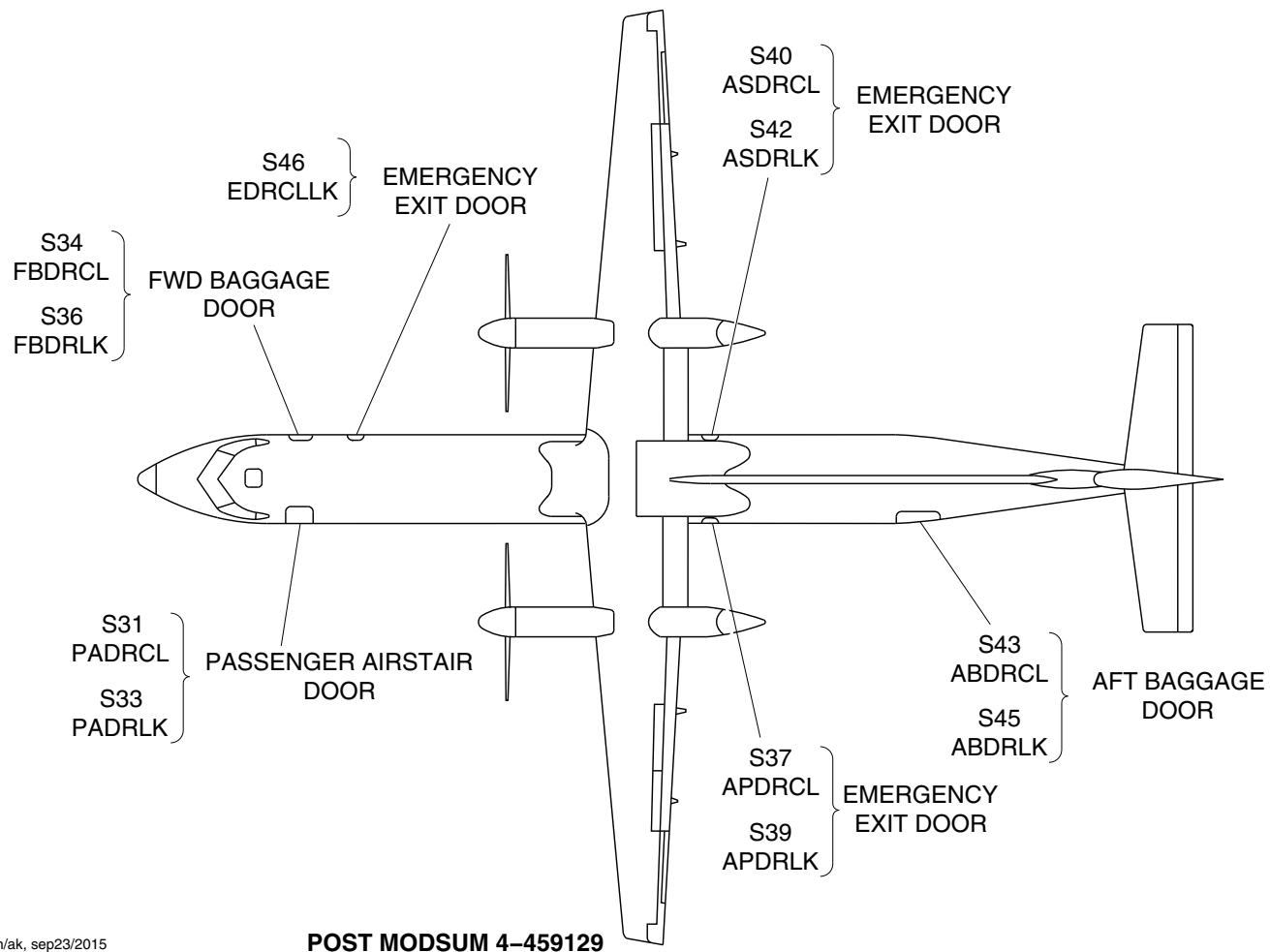
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fsk77a02.dg, vn/ak, sep23/2015

Aircraft Doors and Sensors Page 2
Figure 2 (Sheet 2 of 2)

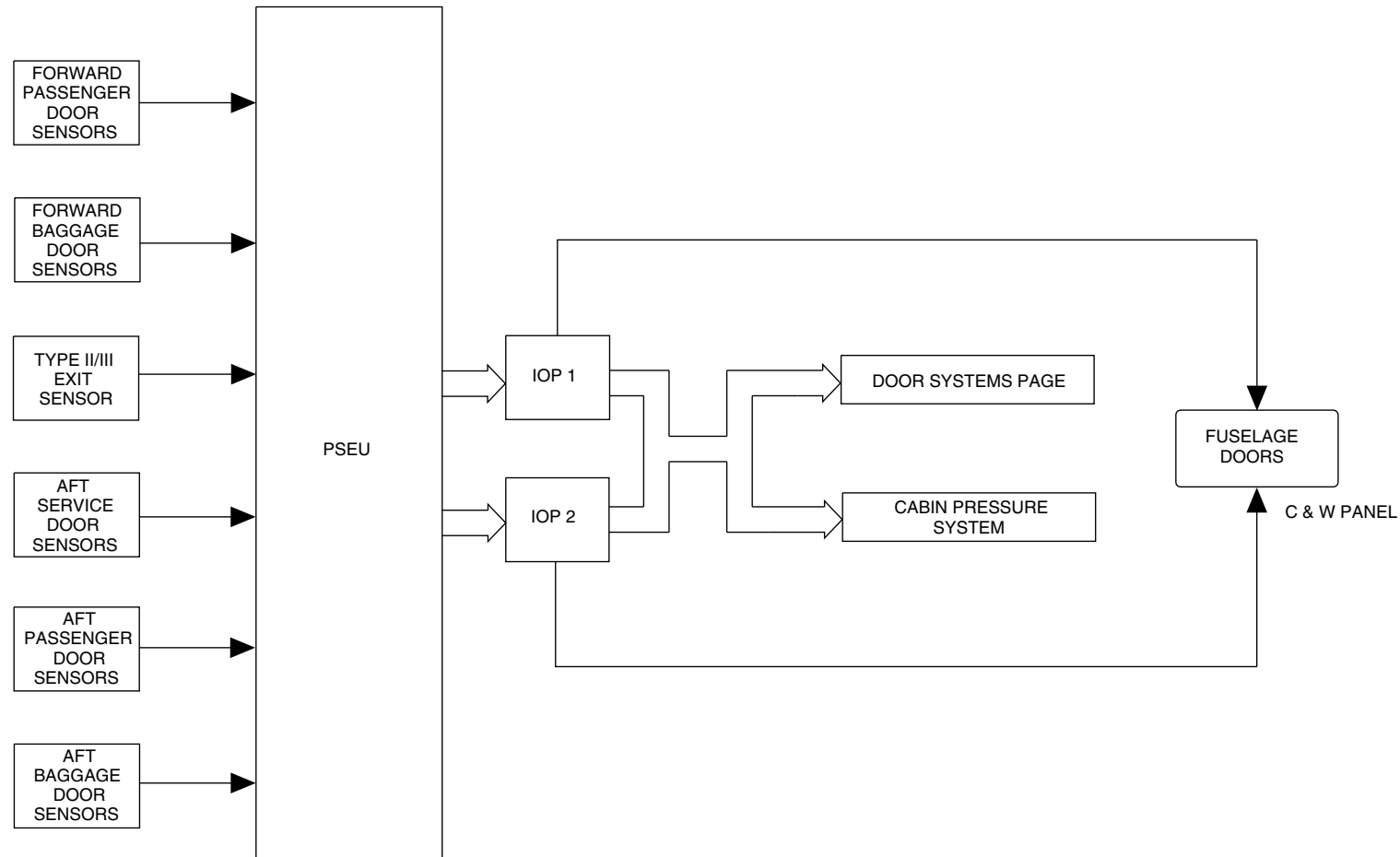
PSM 1-84-2A
EFFECTIVITY:
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fs699a01.cgm

DOOR PROXIMITY SENSOR INPUTS TO PSEU
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 52-70-00
Config 001

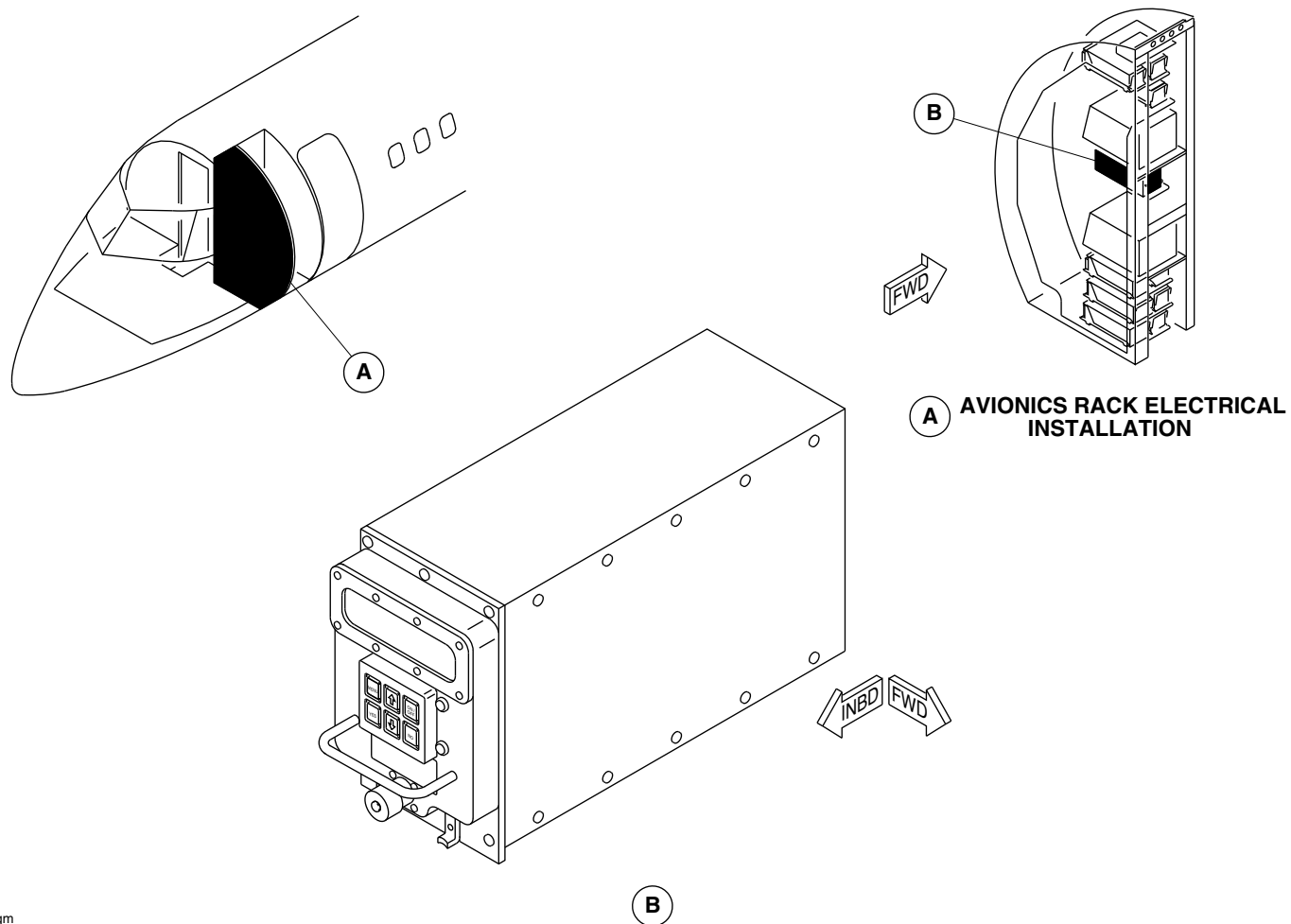
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fs860a01.cgm

PSEU LOCATOR
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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Config 001

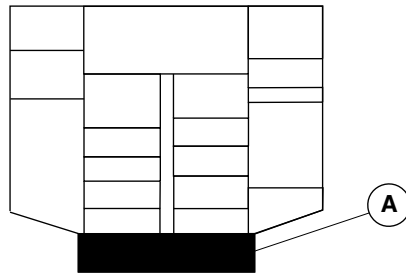
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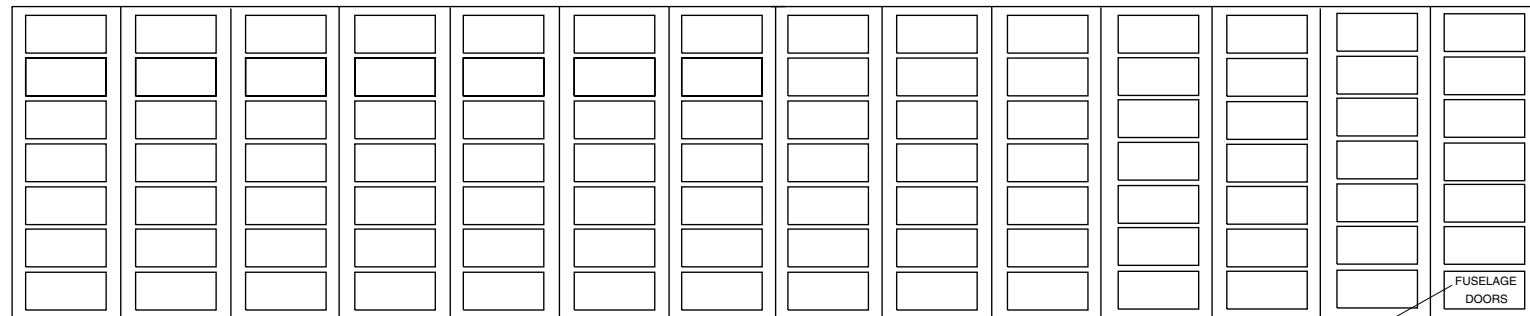
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OVERHEAD CONSOLE

LEGEND

1. Fuselage Doors (Red).



fs805a01.cgm

Fuselage Doors Warning Light
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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Config 001

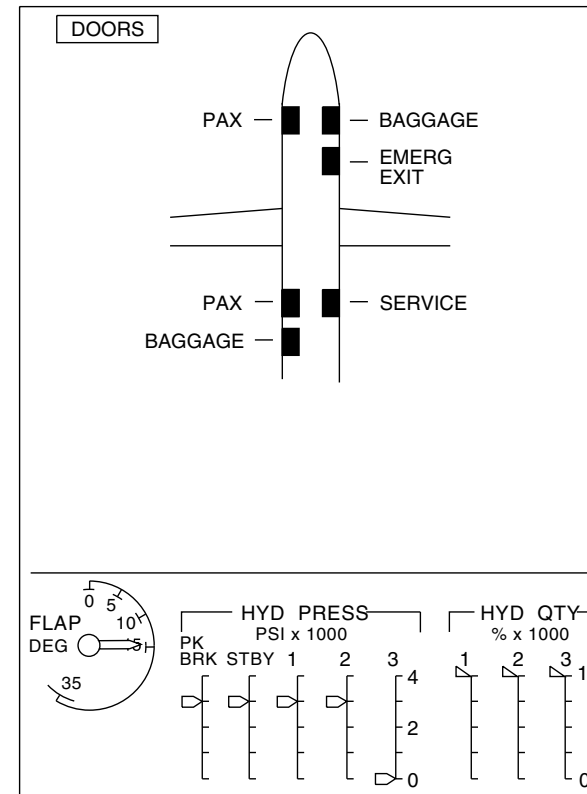
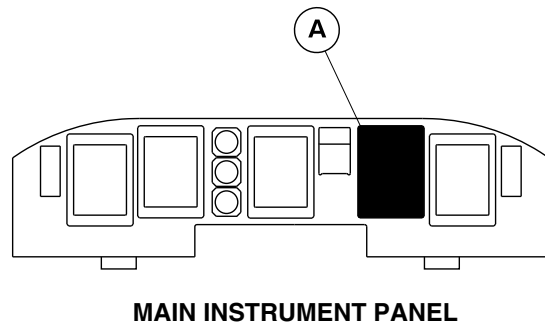
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cg4295a01.dg, ak, sep23/2015

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Doors System Page (Doors Open)
Figure 6 (Sheet 1 of 2)

PSM 1-84-2A
EFFECTIVITY:
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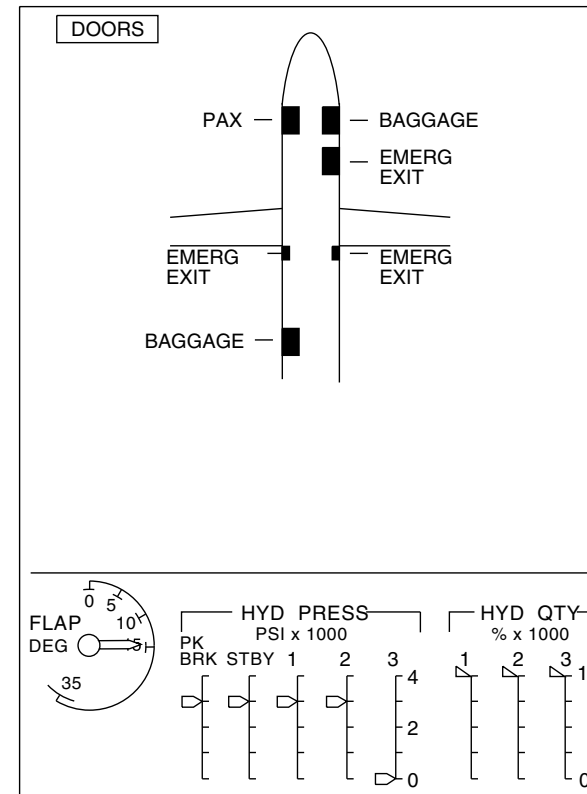
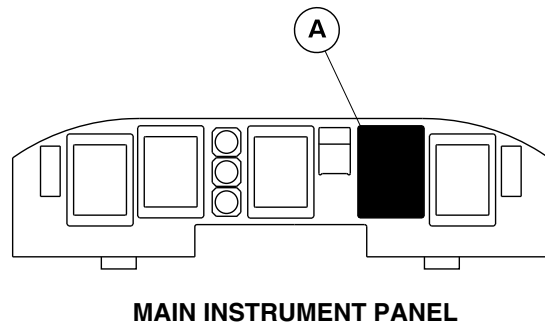
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Doors System Page (Doors Open)
Figure 6 (Sheet 2 of 2)

PSM 1-84-2A
EFFECTIVITY:
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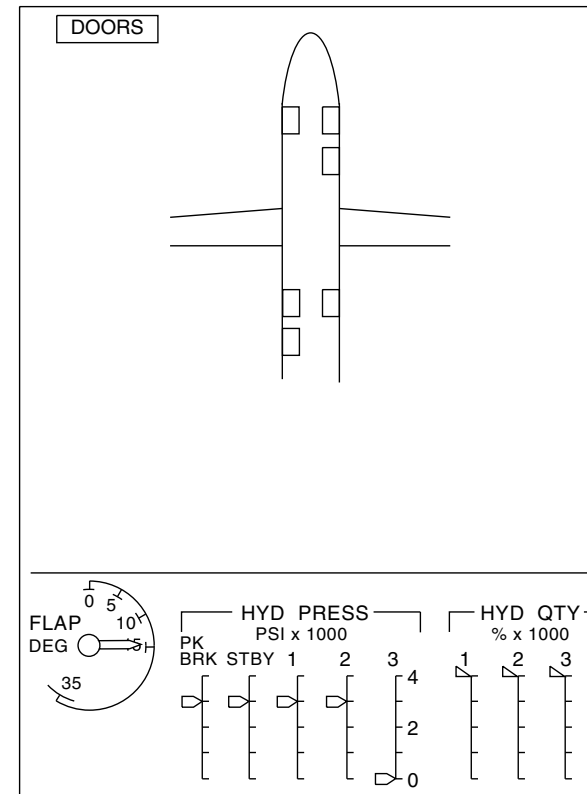
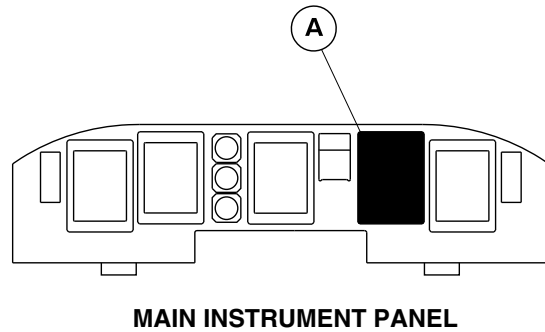
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cg4296a01.dg, ak, sep23/2015

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Door Systems Page (Doors Closed)
Figure 7 (Sheet 1 of 2)

PSM 1-84-2A
EFFECTIVITY:
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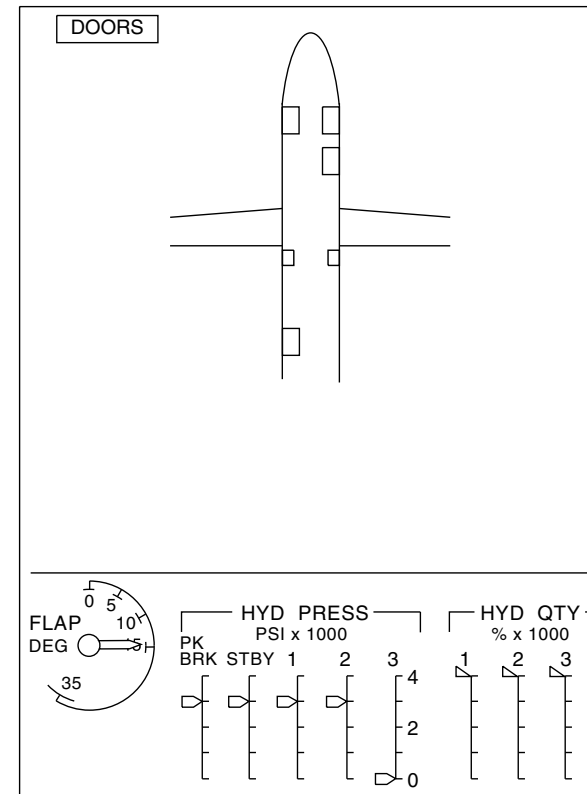
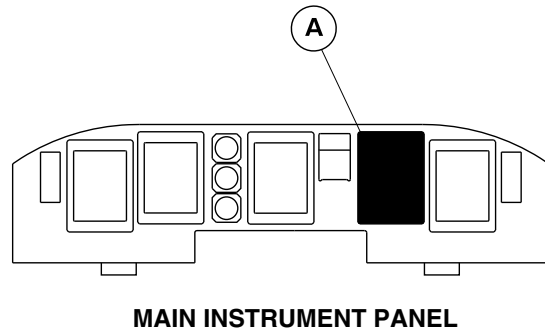
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Door Systems Page (Doors Closed)
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PSM 1-84-2A
EFFECTIVITY:
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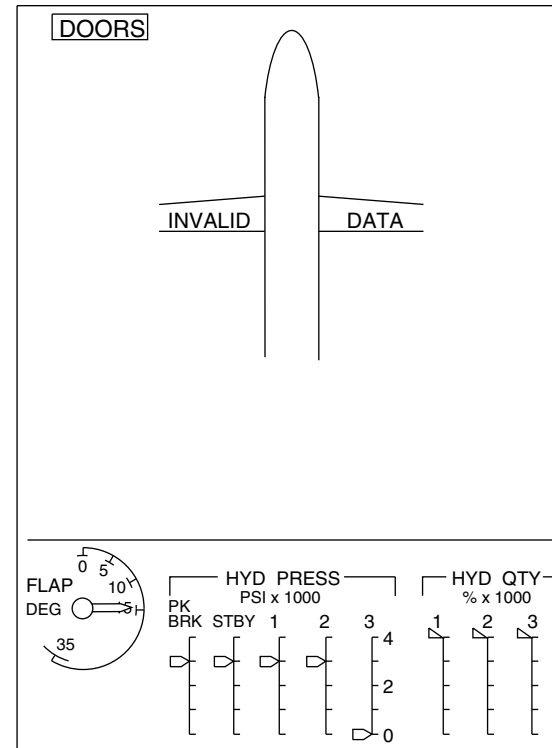
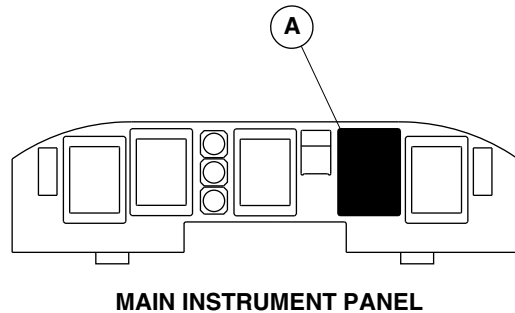
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fs749a01.cgm

DOORS SYSTEM PAGE (MALFUNCTION)
Figure 8

PSM 1-84-2A
EFFECTIVITY:
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Config 001

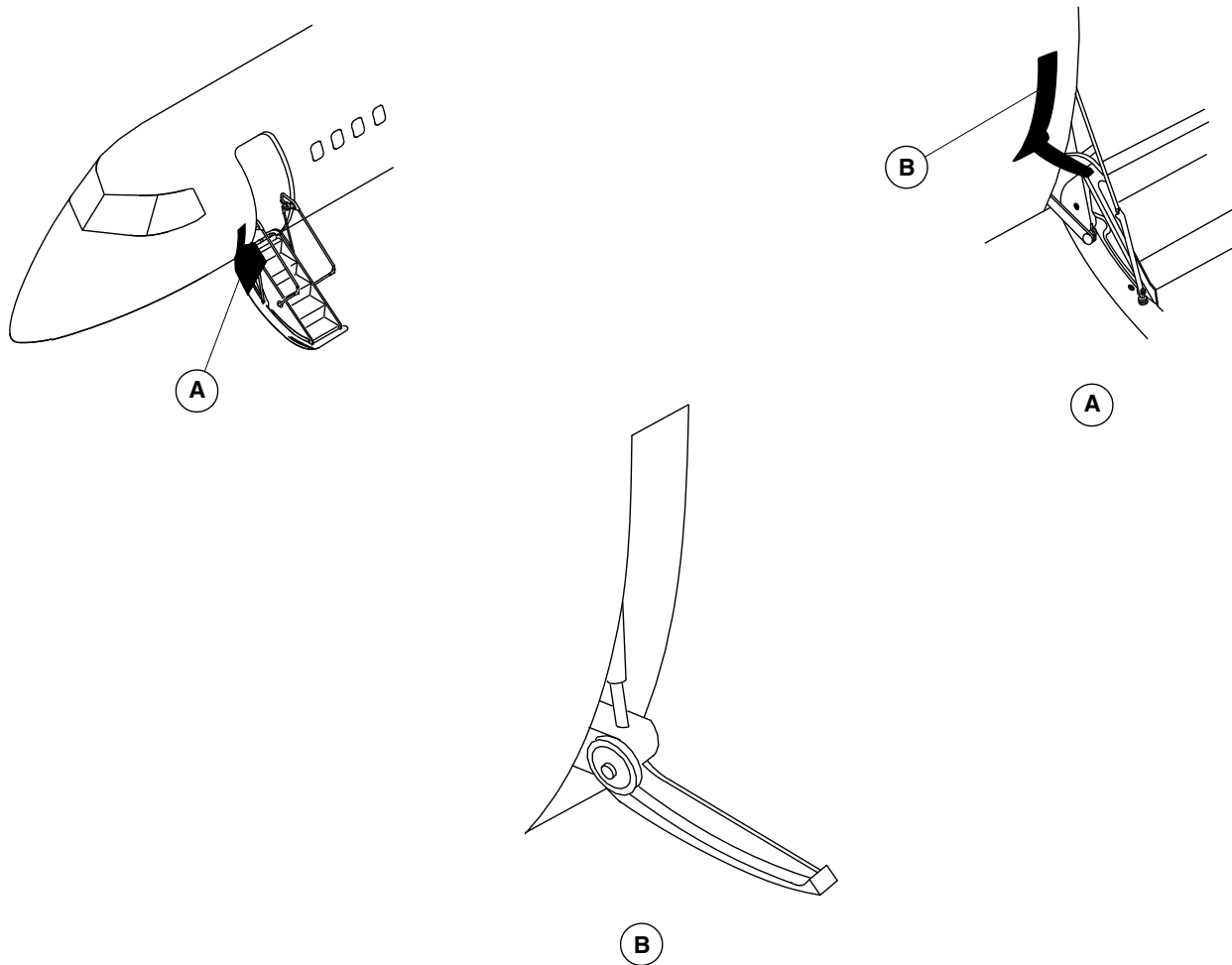
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fs297a01.cgm

FORWARD PASSENGER DOOR EXTERNAL HANDLE LOCATOR
Figure 9

PSM 1-84-2A
EFFECTIVITY:
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Config 001

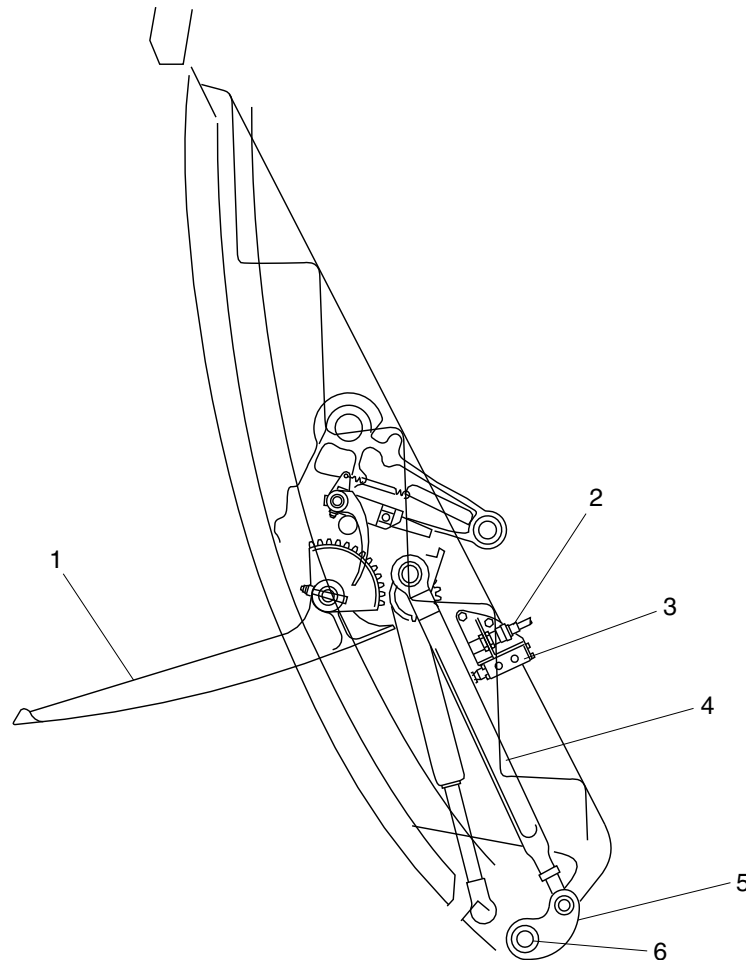
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. External handle.
- 2. Locked sensor.
- 3. Pressurizing valve.
- 4. Actuating rod.
- 5. Crank.
- 6. Door hinge shaft.

fs564a01.dg, vn, aug22/2011

Proximity Sensor Assembly (Locked Sensor)
Figure 10

PSM 1-84-2A
EFFECTIVITY:
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Config 001

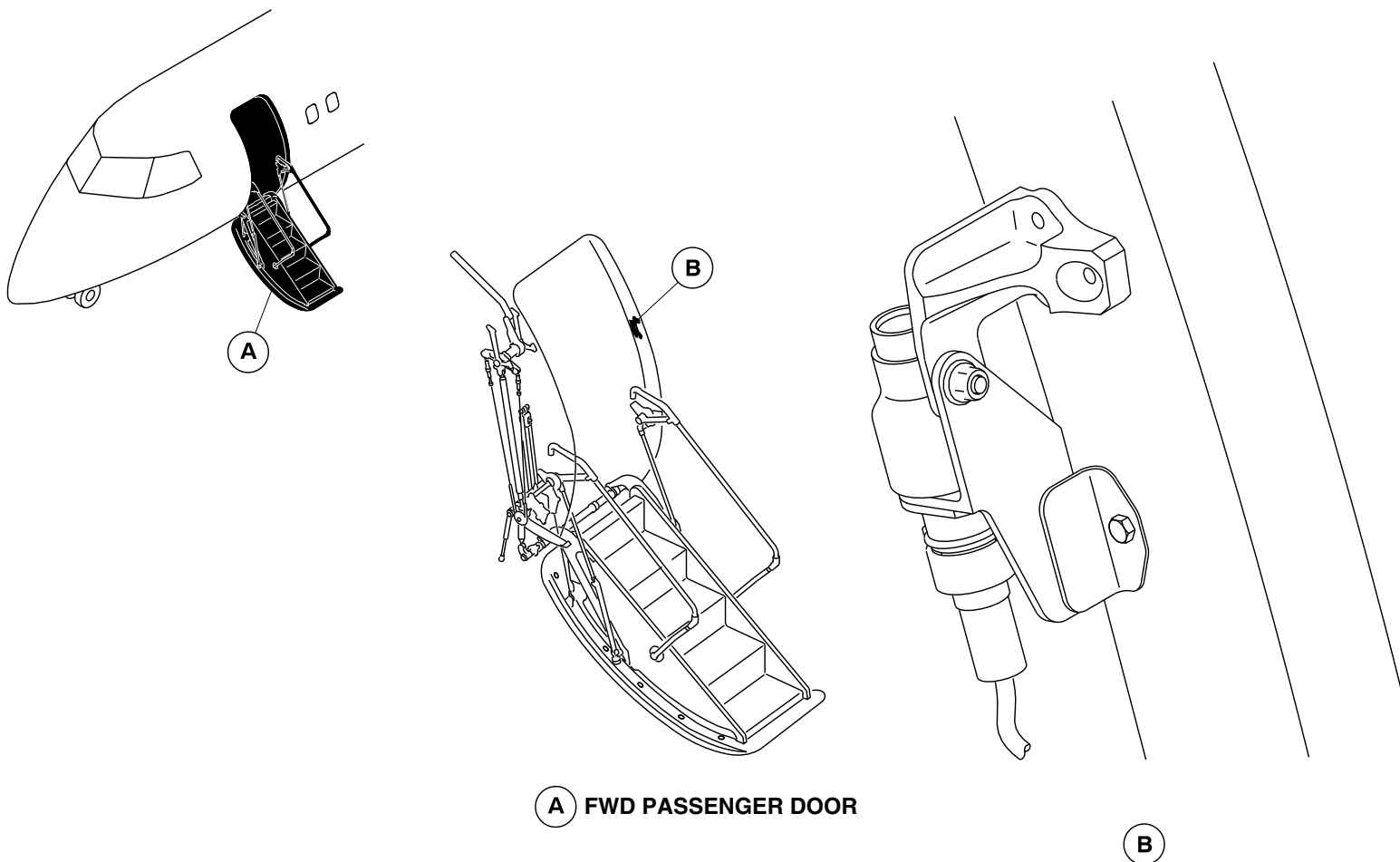
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fsm05a01.dg, vn, aug22/2011

Forward Passenger Door Proximity Sensor Assembly (Closed Sensor)
Figure 11

PSM 1-84-2A
EFFECTIVITY:
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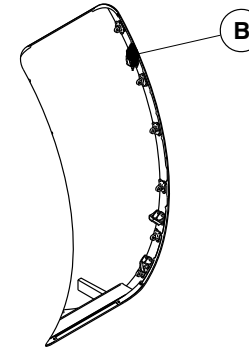
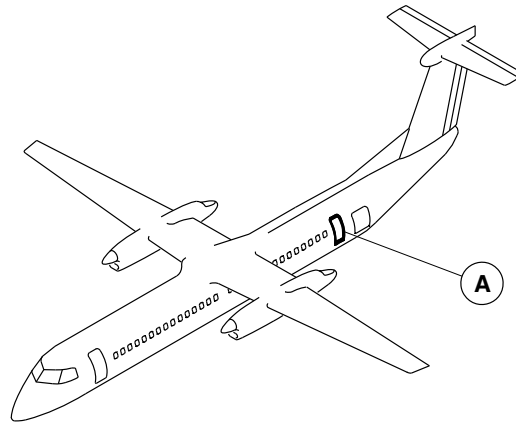
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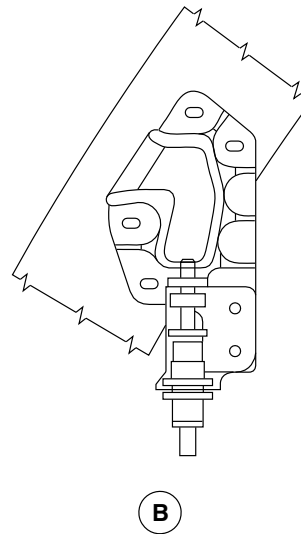


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A AFT PASSENGER DOOR FRAME



B

fsm09a01.cgm

AFT PASSENGER DOOR PROXIMITY (CLOSED) SENSOR LOCATOR
Figure 12

PSM 1-84-2A
EFFECTIVITY:
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Config 001

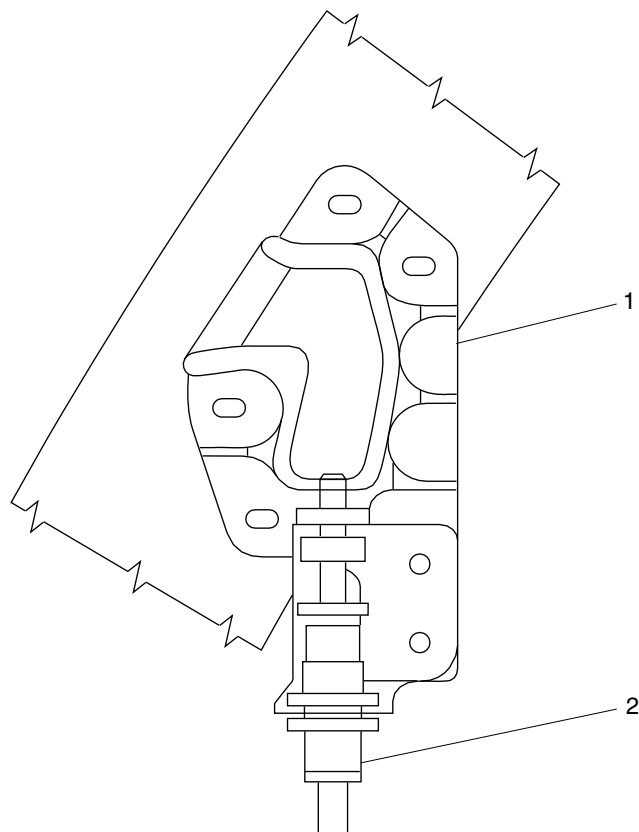
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LEGEND

- 1. Roller Guide.
- 2. "Closed" Proximity Sensor.

fsm11a01.cgm

AFT PASSENGER DOOR PROXIMITY (CLOSED) SENSOR DETAIL
Figure 13

PSM 1-84-2A
EFFECTIVITY:
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Config 001

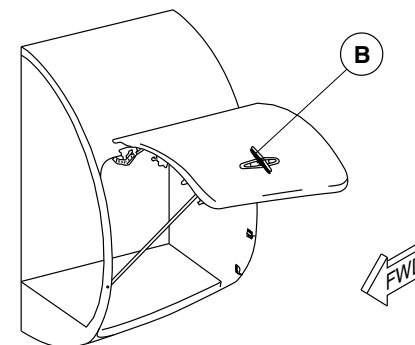
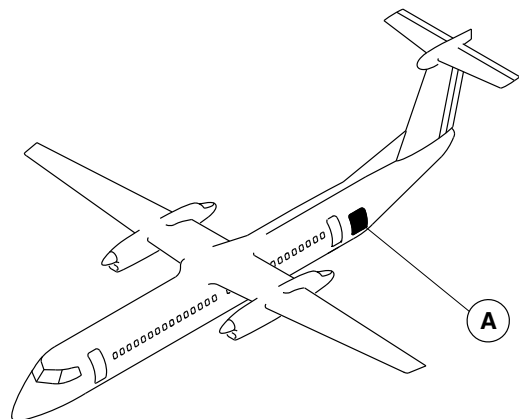
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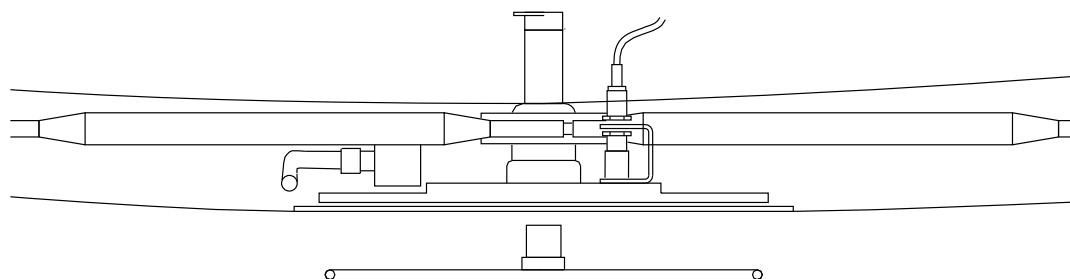


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A AFT BAGGAGE DOOR ASSEMBLY



B

fsk94a01.cgm

AFT BAGGAGE DOOR PROXIMITY (LOCKED) SENSOR LOCATOR
Figure 14

PSM 1-84-2A
EFFECTIVITY:
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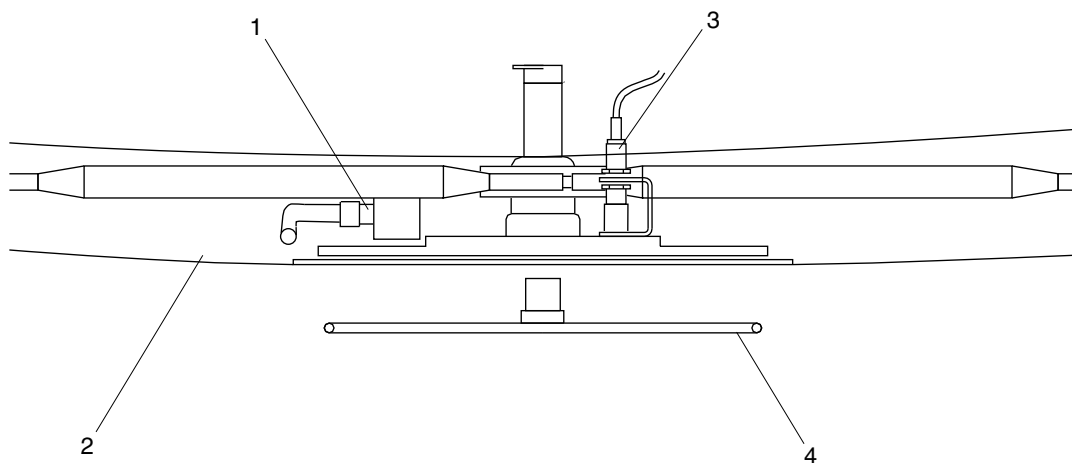


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

1. Pressurizing Valve.
2. Aft Baggage Door.
3. "Locked" Proximity Sensor.
4. External Handle.



fs468a01.cgm

AFT BAGGAGE DOOR PROXIMITY (LOCKED) SENSOR DETAIL
Figure 15

PSM 1-84-2A
EFFECTIVITY:
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Config 001

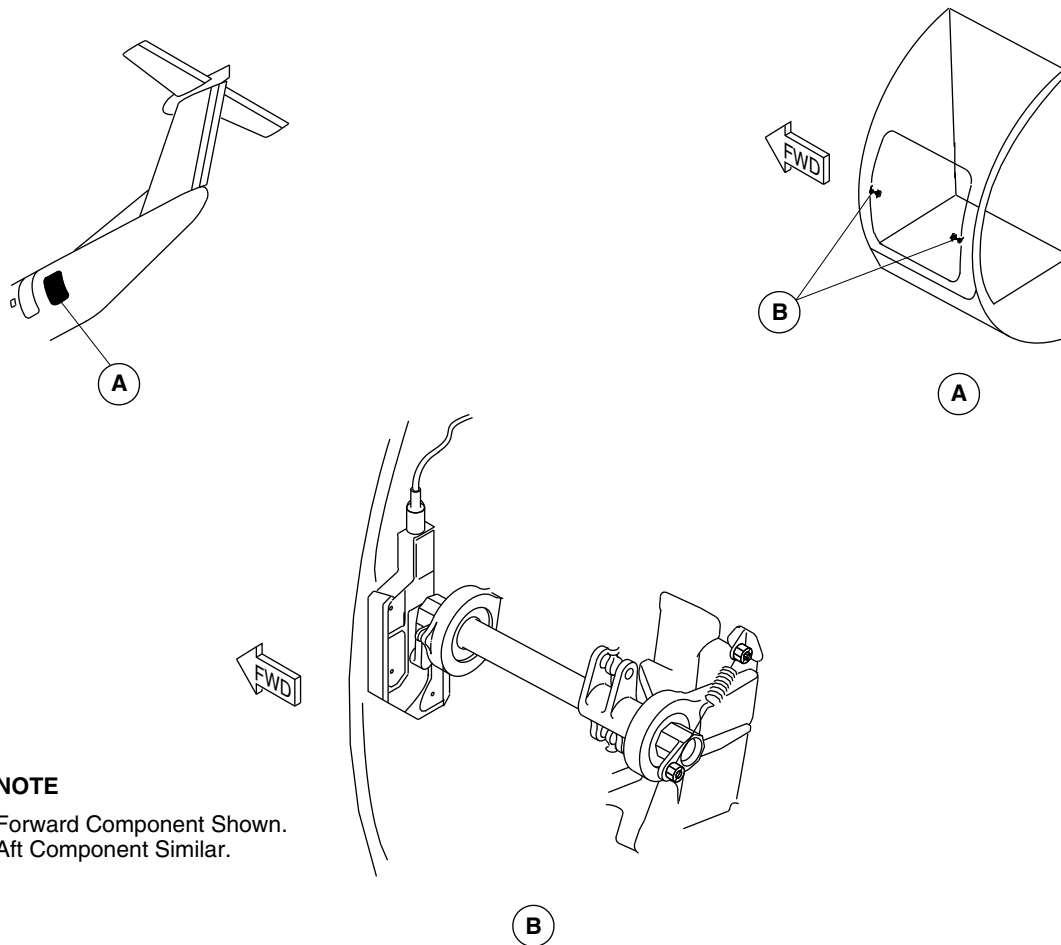
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fs129a01.cgm

AFT BAGGAGE DOOR PROXIMITY (CLOSED) SENSOR LOCATOR
Figure 16

PSM 1-84-2A
EFFECTIVITY:
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Config 001

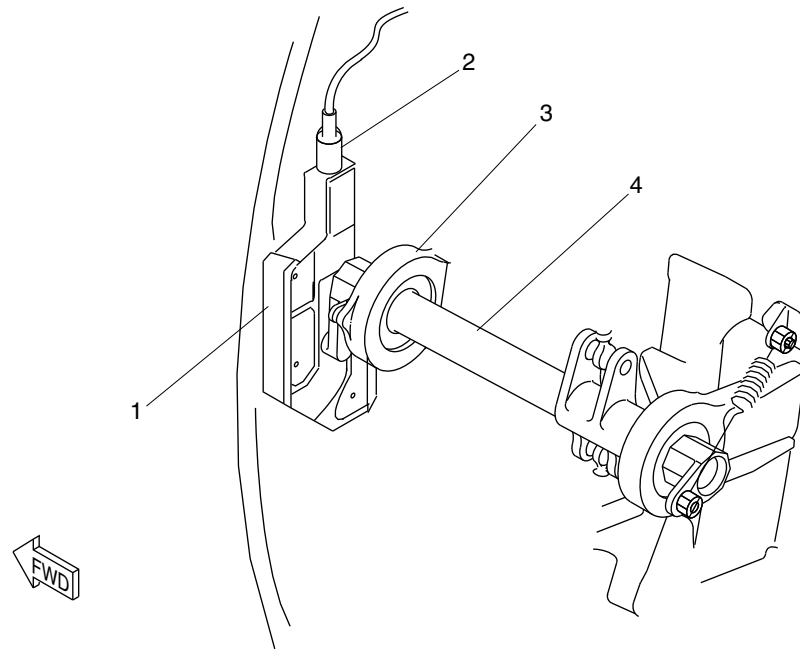
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LEGEND

- 1. Roller Guide.
- 2. "Closed" Proximity Sensor.
- 3. Latch Roller.
- 4. Latch Shaft.

fs120a01.cgm

AFT BAGGAGE DOOR PROXIMITY (CLOSED) SENSOR DETAIL
Figure 17

PSM 1-84-2A
EFFECTIVITY:
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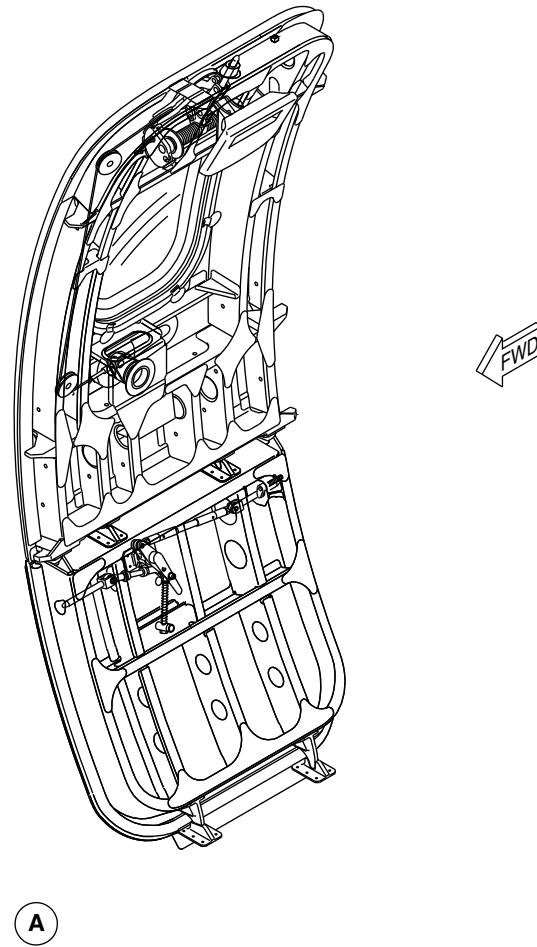
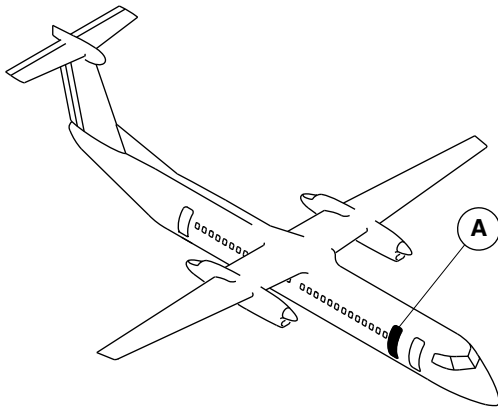
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fs353a01.cgm

TYPE II EMERGENCY EXIT DOOR LOCATOR
Figure 18

PSM 1-84-2A
EFFECTIVITY:
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Config 001

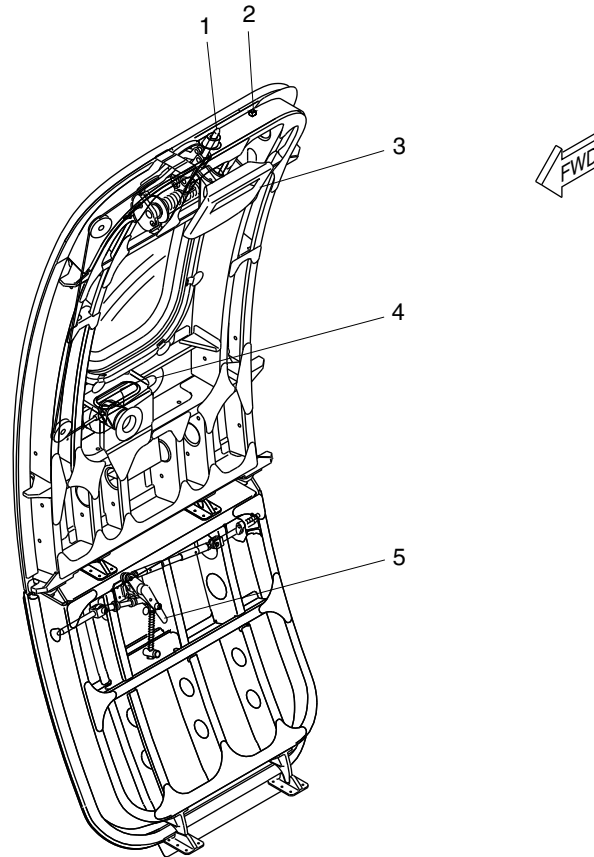
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Lock Pin.
- 2. Proximity Sensor.
- 3. Internal Handle.
- 4. External Handle.
- 5. Lower Door Handle.

fsk97a01.iso

TYPE II EMERGENCY EXIT DOOR PROXIMITY SENSOR
Figure 19

PSM 1-84-2A
EFFECTIVITY:
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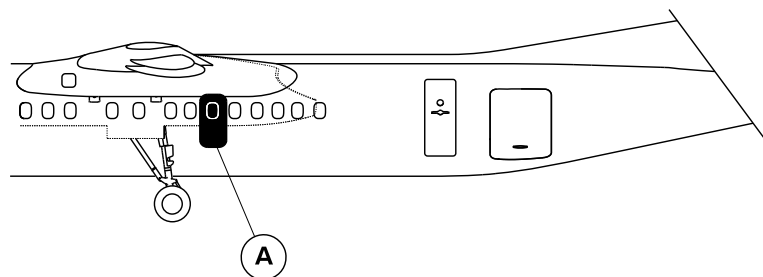
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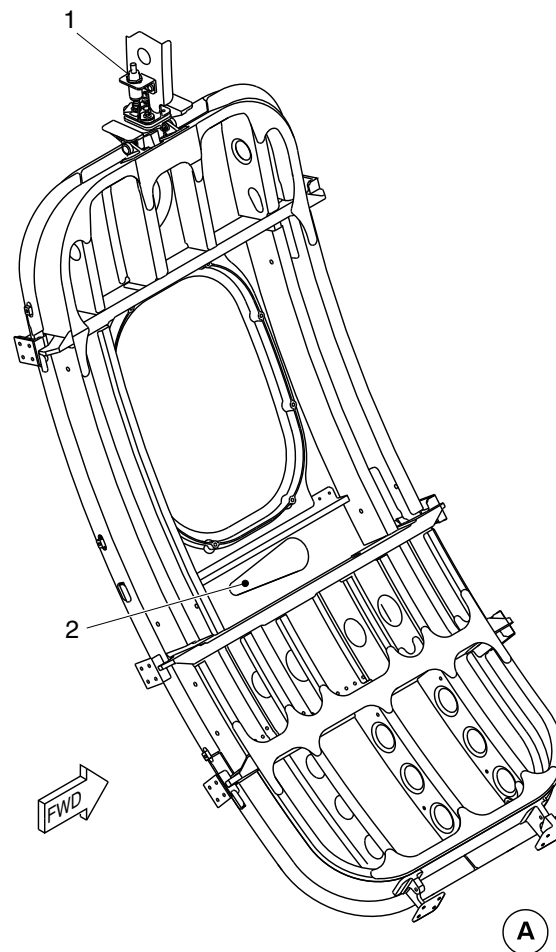
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LEGEND

- 1. Proximity Sensor.
- 2. Internal Handle.



cg4132a01.dg, ak, sep28/2015

Type III Emergency Exit Door Proximity Sensor
Figure 20

PSM 1-84-2A
EFFECTIVITY:
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